Algebraic Geometry

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1 Schemes

1.1 Generalities on sheaves

1.1 Presheaf. Let X be a topological space. Define a category $\mathrm{Top}(X)$ as follows.

— An object of $\mathrm{Top}(X)$ is an open set in X.

— For two objects U, V of $\mathrm{Top}(X)$, set

$$\mathrm{Hom}_{\mathrm{Top}(X)}(U, V) = \begin{cases} \{\iota_{UV} : U \to V\} & \text{, if } U \subseteq V \text{, where } \iota_{UV} : U \to V \text{ denotes the inclusion,} \\ \varnothing & \text{, otherwise.} \end{cases}$$

Let $\mathcal{C}$ be a category. A **$\mathcal{C}$-presheaf** on X is just a contravariant functor $\mathcal{F} : \mathrm{Top}(X) \to \mathcal{C}$. When $\mathcal{C}$ is **Set** (resp. **Ab**, **Ring**, $\mathbf{Mod}_R$), we say $\mathcal{F}$ is a presheaf of sets (resp. abelian groups, rings, R-modules). A morphism between two presheaves is a natural transformation. We denote by $\mathcal{C}_X^{\mathrm{pre}}$ the category of all $\mathcal{C}$-presheaves on X. For $U \subseteq V$, the morphism $\mathcal{F}(\iota_{UV}) : \mathcal{F}(V) \to \mathcal{F}(U)$ is usually called a restriction map. An element in $\mathcal{F}(X)$ is called a **global section** of $\mathcal{F}$, and we usually write $\Gamma(X, \mathcal{F}) = \mathcal{F}(X)$.

1.2 Sheaf. In the following we assume $\mathcal{C}$ is either **Set**, **Ab**,**Ring** or $\mathbf{Mod}_R$. A $\mathcal{C}$-valued presheaf $\mathcal{F}$ is called a **sheaf** if for every open U and every open cover $\{U_i\}_{i\in I}$ of U there is an equalizer diagram

$$\mathcal{F}(U) \longrightarrow \prod_{i\in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$

We denote by $\mathcal{C}_X$ the full subcategory of $\mathcal{C}_X^{\mathrm{pre}}$ consisting of all $\mathcal{C}$-sheaves on X.

It follows from the definition that the empty product is the final object in $\mathcal{C}$. By the sheaf axiom, we see that $\mathcal{F}(\varnothing)$ is the final object in $\mathcal{C}$ as long as $\mathcal{F}$ is a sheaf.

1.3 Sheaf on a basis. Let $\mathcal{B}$ be a basis of open sets of X. We can view $\mathcal{B}$ as a full subcategory of $\mathrm{Top}(X)$. We define a **$\mathcal{C}$-presheaf on the basis** $\mathcal{B}$ to be a contravariant functor $\mathcal{B} \to \mathcal{C}$. A $\mathcal{C}$-presheaf $\mathcal{F}$ on the basis $\mathcal{B}$ is called a sheaf if for any $B \in \mathcal{B}$ and its open cover $\{B_i\}_{i\in I} \subseteq \mathcal{B}$, there exists an equalizer diagram

$$\mathcal{F}(B) \longrightarrow \prod_{i\in I} \mathcal{F}(B_i) \rightrightarrows \prod_{i,j\in I} \prod_{\substack{B' \subseteq B_i \cap B_j \\ B' \in \mathcal{B}}} \mathcal{F}(B')$$

Clearly, every sheaf on X restricts to a sheaf on $\mathcal{B}$. What's more, the converse is also true. Let $\mathcal{F}$ denote a sheaf on $\mathcal{B}$. For $U \in \mathrm{Top}(X)$, define

$$\mathcal{F}'(U) := \varprojlim_{\mathcal{B} \ni B \subseteq U} \mathcal{F}(B).$$

Note that when $U = B \in \mathcal{B}$, the canonical projection $\mathcal{F}'(B) \to \mathcal{F}(B)$ is an isomorphism. For opens $V \subseteq U$, clearly we have $\mathcal{F}'(U) \to \mathcal{F}'(V)$ given by projections. This shows $\mathcal{F}'$ defines a presheaf on X. To show it

is a sheaf, let $U \in \mathrm{Top}(X)$ and $\{U_i\}_{i\in I}$ be an open cover of U. We must show

$$\mathcal{F}'(U) \longrightarrow \prod_{i\in I} \mathcal{F}'(U_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}'(U_i \cap U_j)$$

is an equalizer diagram. A possible way to show this is to express inverse limit as a equalizer of certain arrows, and do some easy diagram chasing; we omit the proof. In fact, $\mathcal{F}$ is unique up to isomorphism, which is easy to see.

If $\mathcal{B}$ is closed under finite intersection, then for a presheaf $\mathcal{F}$ on $\mathcal{B}$, it is a sheaf if and only if for any $B \in \mathcal{B}$ and its open cover $\{B_i\}_{i\in I} \subseteq \mathcal{B}$, there exists an equalizer diagram

$$\mathcal{F}(B) \longrightarrow \prod_{i\in I} \mathcal{F}(B_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}(B_i \cap B_j) \tag{$\star$}$$

This is easily checked from the definition.

Let $\mathcal{B}$, B, $\{B_i\}_{i\in I}$ be as above. Suppose, in addition, that $B \in \mathcal{B}$ is compact. Then to show the exactness of $(\star)$, it suffices to show the exactness of

$$\mathcal{F}(B) \longrightarrow \prod_{i\in J} \mathcal{F}(B_i) \rightrightarrows \prod_{i,j\in J} \mathcal{F}(B_i \cap B_j)$$

for all finite subsets J of I with $B = \bigcup_{i\in J} B_i$. Indeed, the injectivity is clear, and to show the exactness in the middle, consider the diagram

$$\begin{tikzcd}
\mathcal{F}(B) \arrow[r] \arrow[d] & \prod_{i\in I} \mathcal{F}(B_i) \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & \prod_{i,j\in I} \mathcal{F}(B_i \cap B_j) \arrow[d] \\
\prod_{J\subseteq I} \mathcal{F}(B) \arrow[r] & \prod_{J\subseteq I}\prod_{i\in J} \mathcal{F}(B_i) \arrow[r, shift left] \arrow[r, shift right] & \prod_{J\subseteq I}\prod_{i,j\in J} \mathcal{F}(B_i \cap B_j)
\end{tikzcd}$$

An easy diagram chasing and argument then shows the exactness of the above sequence.

1.4 Let X be a topological space and $\mathcal{B}$ an open basis of the topology on X. Let $\mathcal{F}$ and $\mathcal{G}$ be two sheaves on X. Suppose for any $B \in \mathcal{B}$, there exists a map $\mathcal{F}(B) \to \mathcal{G}(B)$ that is compatible with restriction. Then there exists a unique morphism $\mathcal{F} \to \mathcal{G}$ of sheaves extending those maps on $\mathcal{B}$. This is easy to construct and see its uniqueness once we regard $\mathcal{F}$ and $\mathcal{G}$ as constructed from sheaves on $\mathcal{B}$, just like we do in 1.3.

1.5 Stalk and Sheafification. Let $\mathcal{F}$ be a $\mathcal{C}$-presheaf. For each point x of X, the **stalk** of $\mathcal{F}$ at $x \in X$ is the direct limit

$$\mathcal{F}_x := \varinjlim_{\mathrm{Top}(X)\ni U\ni x} \mathcal{F}(U)$$

where the open neighborhoods of x are directed by inclusions. The **étale space** of $\mathcal{F}$ is the set-theoretic disjoint union

$$\operatorname{Et}\mathcal{F} = \bigsqcup_{x\in X} \mathcal{F}_x$$

If $s \in \mathcal{F}(U)$, we use either s_x, $s|_x$ or $\varinjlim_{U\supseteq V\ni x} s|_V$ to denote its image in $\mathcal{F}_x$. For each $s \in \mathcal{F}(U)$, define $s_U : U \to \operatorname{Et}\mathcal{F}$ by $s_U(x) := (x, s_x)$. On $\operatorname{Et}\mathcal{F}$ we install the final topology with respect to the collection of maps $\{s_U \mid U \in \operatorname{Top}(X),\ s \in \mathcal{F}(U)\}$. With this topology the natural projection $\pi : \operatorname{Et}\mathcal{F} \to X$ becomes a local homeomorphism.

Denote by $\mathcal{F}^\dagger$ the sheaf of continuous section of $\pi : \operatorname{Et}\mathcal{F} \to X$. If $\mathcal{F}$ is a $\mathcal{C}$-presheaf, then $\mathcal{F}^\dagger$ is a $\mathcal{C}$-sheaf. This sheaf $\mathcal{F}^\dagger$ is called the **sheafification** of the presheaf $\mathcal{C}$. Clearly $(\cdot)^\dagger$ defines a functor from $\mathcal{C}_X^{\mathrm{pre}}$ to $\mathcal{C}_X$. The map $s \mapsto s_U$ defines a morphism $\iota_{\mathcal{F}} : \mathcal{F} \to \mathcal{F}^\dagger$. This morphism $\iota_{\mathcal{F}}$ enjoys the universal property : there is a bifunctorial bijection

$$\operatorname{Hom}_{\mathcal{C}_X^{\mathrm{pre}}}(\mathcal{F}, \mathcal{G}) \longrightarrow \operatorname{Hom}_{\mathcal{C}_X}(\mathcal{F}^\dagger, \mathcal{G})$$

whose inverse is given by pre-composing with $\iota_{\mathcal{F}}$. Here $\mathcal{G}$ is a $\mathcal{C}$-sheaf, and we view it as a $\mathcal{C}$-presheaf on the left. In other words, the sheafification functor is left adjoint to the forgetful functor $\mathcal{C}_X \to \mathcal{C}_X^{\mathrm{pre}}$. In particular, this shows if $\mathcal{F}$ is already a sheaf, then $\iota_{\mathcal{F}} : \mathcal{F} \to \mathcal{F}^\dagger$ is an isomorphism of sheaves.

1.6 Retain the notation in (1.5). Since each section $s \in \mathcal{F}^\dagger(U)$ is necessarily injective, we obtain a canonical injection

$$\begin{aligned} \mathcal{F}^\dagger(U) &\longrightarrow \prod_{x\in U} \mathcal{F}_x \\ f &\longmapsto (f(x))_{x\in U}. \end{aligned}$$

It is easy to describe the image in $\prod_{x\in U} \mathcal{F}_x$:

$$\mathcal{F}^\dagger(U) = \left\{ (f_x)_{x\in U} \in \prod_{x\in U} \mathcal{F}_x \;\middle|\; \begin{array}{l} \text{for any } x \in U \text{ there exist an open neighborhood } V \subseteq U \text{ of } x \\ \text{and } s \in \mathcal{F}(V) \text{ such that } f_y = s|_y \text{ for all } y \in V \end{array} \right\}.$$

1.7 For any presheaf $\mathcal{F}$ and $x \in X$, the canonical morphism $\iota = \iota_{\mathcal{F}} : \mathcal{F} \to \mathcal{F}^\dagger$ induces a map $\iota_x : \mathcal{F}_x \to \mathcal{F}_x^\dagger$ on the stalk. This is in fact an isomorphism in $\mathcal{C}$. To see this, an element $f_x \in \mathcal{F}_x^\dagger$ is represented by some $f \in \mathcal{F}^\dagger(U)$, where U is a small neighborhood of x in X. By shrinking U further we can assume $f(U) = \{(x, s_x) \mid x \in U\}$ for some $s \in \mathcal{F}(U)$. Then $s_U = f$ since they are sections of a homeomorphism $\pi|^U_{f(U)}$, and thus $(s_U)_x = f_x$. This proves surjectivity, and injectivity can be proved in a similar way.

1.8 Define a category Et_X as follows. An object is a pair $(Y, p_Y : Y \to X)$ consisting of a topological space Y and a local homeomorphism $p_Y : Y \to X$. A morphism between objects is a continuous map compatible

with their projections to the base X. Taking sheaf of continuous sections defines a functor from Et_X to $\mathcal{C}_X$ (operations between sections are defined stalk-wise). In fact, this establishes an equivalence of categories

$$\mathrm{Et}_X \longrightarrow \mathcal{C}_X$$

with inverse given by associating a sheaf $\mathcal{F}$ with its étale space $\mathrm{Et}\,\mathcal{F} \to X$.

1.9 Example : constant sheaf. Let X be a topological space and A a set. The presheaf $U \mapsto A$, $\varnothing \mapsto \{*\}$ is called the **constant presheaf** $\underline{A}_X^{\mathrm{pre}}$, which is usually not a sheaf. Its sheafification is called the constant sheaf, and is denoted by $\underline{A}_X$. There is an explicit description of $\underline{A}_X$: for any open U in X

$$\underline{A}_X(U) = \{f : U \to A \mid f \text{ locally constant}\}.$$

This is clear from the construction of sheafification (1.5). It can be shown that if $\#A \geqslant 2$, then $\underline{A}_X^{\mathrm{pre}}$ is already a sheaf if and only if X is irreducible (1.74).

1.10 Example : skyscraper sheaf. Let X be a topological space and A a set. Let $x \in X$ be a point. Then the presheaf

$$U \mapsto \begin{cases} A & \text{, if } x \in U \\ \{*\} & \text{, if } x \notin U \end{cases}$$

is a sheaf, and is called **the skyscraper sheaf at** x **with value** A. It is so named as if we denote this sheaf by $\mathcal{F}$, then

$$\mathcal{F}_y = \begin{cases} A & \text{, if } y \in \overline{\{x\}} \\ \{*\} & \text{, if } y \notin \overline{\{x\}} \end{cases}$$

Particularly, if $\{x\} = \overline{\{x\}}$ (e.g. if X is a T_1 space), then x is the only point at which $\mathcal{F}$ has nontrivial stalk.

If we view A as a sheaf on the one point space $\{x\}$ and denote by $\iota_x : \{x\} \to X$ the inclusion, then $\mathcal{F} = (\iota_x)_* A$ (1.11).

1.1.1 Adjunction between f^{-1} and f_*.

1.11 Let $f : X \to Y$ be a continuous map. For a $\mathcal{C}$-presheaf $\mathcal{F}$ on X, define the **direct image presheaf** (or the push-forward presheaf) $f_*\mathcal{F}$ on Y by the formula

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$$

where $V \in \mathrm{Top}(Y)$. If $\mathcal{F}$ is a $\mathcal{C}$-sheaf, then $f_*\mathcal{F}$ is also a $\mathcal{C}$-sheaf. If $\mathcal{G}$ is a $\mathcal{C}$-presheaf on Y, define the **inverse image presheaf** (or pull-back presheaf) $f^{\mathrm{pre}}\mathcal{F}$ on X by

$$f^{\mathrm{pre}}\mathcal{F}(U) = \varinjlim_{\mathrm{Top}(Y) \ni V \supseteq f(U)} \mathcal{F}(V)$$

where $U \in \mathrm{Top}(X)$. When $\mathcal{F}$ is a sheaf, $f^{\mathrm{pre}}\mathcal{F}$ may still fail to be a sheaf. Nevertheless, for a sheaf $\mathcal{F}$, we define the **inverse image sheaf** $f^{-1}\mathcal{F}$ of $\mathcal{F}$ to be the sheafification of $f^{\mathrm{pre}}\mathcal{F}$.

We compute the stalk of $f^{\mathrm{pre}}\mathcal{F}$. For $x \in X$,

$$(f^{\mathrm{pre}}\mathcal{F})_x = \varinjlim_{\mathrm{Top}(X)\ni U\ni x} \varinjlim_{\mathrm{Top}(Y)\ni V\supseteq f(U)} \mathcal{F}(V) \cong \varinjlim_{V\ni f(x)} \mathcal{F}(V) = \mathcal{F}_{f(x)}$$

There is no formula for $(f_*\mathcal{F})_y$ in general. Nevertheless, if X is a subspace of Y and f is the inclusion, we have $(f_*\mathcal{F})_y = \mathcal{F}_y$ if $y \in X$. If X is closed, then $(f_*\mathcal{F})_y = 0$ for $y \neq\in X$.

1.12 Adjunction between f^{-1} and f_*. Define a category $\mathcal{C}^{\mathrm{pre}}_{\mathbf{Top}}$ as follows.

— An object in $\mathcal{C}_{\mathbf{Top}}$ is a topological space X together with a $\mathcal{C}$-presheaf $\mathcal{F}$. We denote an object by $(X, \mathcal{F})$.

— For two objects $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ is a continuous map $f : X \to Y$ together with a collection of maps $\left\{\mathcal{G}(V) \overset{T_{U,V}}{\to} \mathcal{F}(U)\right\}_{\substack{f(U)\subseteq V \\ U\in\mathrm{Top}(X),\, V\in\mathrm{Top}(Y)}} \subseteq \mathrm{Mor}(\mathcal{C})$ compatible with the restriction.

Let us have a careful look at the condition imposed on a morphism in $\mathcal{C}^{\mathrm{pre}}_{\mathbf{Top}}$. If $f(U) \subseteq V$, then $U \subseteq f^{-1}(V)$, the latter being open in X since f is continuous. Since a morphism is compatible with inclusions, we see that for a fixed $V \in \mathrm{Top}(Y)$, the subcollection $\{T_{U,V}\}_{\substack{f(U)\subseteq V \\ U\in\mathrm{Top}(X)}}$ is completely determined by a distinguished element, namely $T_{f^{-1}(V),V} : \mathcal{G}(V) \to \mathcal{F}(f^{-1}(V))$. On the other hand, for a fixed $U \in \mathrm{Top}(X)$, the subcollection $\{T_{U,V}\}_{\substack{f(U)\subseteq V \\ V\in\mathrm{Top}(Y)}}$ is packed to a map from the direct limit $\varinjlim_{V\supseteq f(U)} \mathcal{G}(V) \to \mathcal{F}(U)$.

To draw a conclusion, let us consider the forgetful functor $\mathcal{C}^{\mathrm{pre}}_{\mathbf{Top}} \to \mathbf{Top}$. For objects $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ in $\mathcal{C}_{\mathbf{Top}}$, the forgetful functor gives a projection map

$$\Phi : \mathrm{Hom}_{\mathcal{C}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{F}), (Y, \mathcal{G})) \longrightarrow \mathrm{Hom}_{\mathbf{Top}}(X, Y)$$

The above consideration implies the preimage under Φ of a continuous map $f : X \to Y$ has two interpretations, that is,

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{pre}}_Y}(\mathcal{G}, f_*\mathcal{F}) \overset{\sim}{\text{———}} \Phi^{-1}(f) \overset{\sim}{\text{———}} \mathrm{Hom}_{\mathcal{C}^{\mathrm{pre}}_X}(f^{\mathrm{pre}}\mathcal{G}, \mathcal{F})$$

Ignoring the "bridge", we obtain the **adjunction** of f_* and f^{pre}. If we consider the full subcategory $\mathcal{C}_{\mathbf{Top}}$ of $\mathcal{C}^{\mathrm{pre}}_{\mathbf{Top}}$ consisting of $(X, \mathcal{F})$ with $\mathcal{F}$ a $\mathcal{C}$-sheaf on X, by composing with the sheafification we obtain the adjunction of f_* and f^{-1} :

$$\mathrm{Hom}_{\mathcal{C}_Y}(\mathcal{G}, f_*\mathcal{F}) \overset{\sim}{\text{———}} \mathrm{Hom}_{\mathcal{C}_X}(f^{-1}\mathcal{G}, \mathcal{F})$$

For two appropriate continuous map f and g, we clearly have $(fg)_* = f_*g_*$. The adjunction then shows that $(fg)^{-1}$ and $g^{-1}f^{-1}$ are "naturally isomorphic".

1.13 Let us analyze the adjunction (1.12) further in the case $\mathcal{C} = \mathbf{Mod}$. By definition an objective in $\mathcal{C}$ is a pair $A \curvearrowright M$, where A is a ring and M is an A-module. A morphism $A \curvearrowright M \to B \curvearrowright N$ in $\mathbf{Mod}$ is a

ring homomorphism $r : A \to B$ and an abelian group homomorphism $f : M \to N$ with compatible ring action; that is, there is a commutative diagram.

$$\begin{tikzcd} A \times M \arrow[r] \arrow[d, "r \times f"'] & M \arrow[d, "f"] \\ B \times N \arrow[r] & N \end{tikzcd}$$

Let $(X, \mathcal{A} \rightsquigarrow \mathcal{M})$ and $(Y, \mathcal{B} \rightsquigarrow \mathcal{N})$ be in $\mathbf{Mod}^{\mathrm{pre}}_{\mathbf{Top}}$. Forgetful functors gives a chain of projections

$$\mathrm{Hom}_{\mathbf{Mod}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{A} \rightsquigarrow \mathcal{M}), (Y, \mathcal{B} \rightsquigarrow \mathcal{N})) \longrightarrow \mathrm{Hom}_{\mathbf{Ring}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{A}), (Y, \mathcal{B})) \longrightarrow \mathrm{Hom}_{\mathbf{Top}}(X, Y)$$

Let $f : X \to Y$ be a continuous map. As in 1.12, consider the following diagram.

$$\begin{tikzcd} & \mathrm{Hom}_{\mathbf{Mod}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{A} \rightsquigarrow \mathcal{M}), (Y, \mathcal{B} \rightsquigarrow \mathcal{N})) \arrow[d] & \\ \mathrm{Hom}_{\mathbf{Ring}^{\mathrm{pre}}_{X}}(f^{\mathrm{pre}}\mathcal{B}, \mathcal{A}) \arrow[r] & \mathrm{Hom}_{\mathbf{Ring}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{A}) & \mathrm{Hom}_{\mathrm{Ring}^{\mathrm{pre}}_{Y}}(\mathcal{B}, f_*\mathcal{A}) \arrow[l] \end{tikzcd}$$

The fibre of f in the middles gives a bijection between the leftmost set and the rightmost set. Let $\theta : \mathcal{B} \to f_*\mathcal{A}$ be in the rightmost and that $\theta^\sharp$ be the corresponding element in the leftmost; they map to the same element (f, θ) in the middle. We ask what is the fibre of (f, θ) in the upper set. A moment consideration gives the answer :

$$\begin{tikzcd} \mathrm{Hom}_{\mathbf{Mod}^{\mathrm{pre}}_{\mathcal{A}}}(\mathcal{A} \otimes_{f^{\mathrm{pre}}\mathcal{B}} f^{\mathrm{pre}}\mathcal{N}, \mathcal{M}) \arrow[r] \arrow[d] & \mathrm{Hom}_{\mathbf{Mod}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{A} \rightsquigarrow \mathcal{M}), (Y, \mathcal{B} \rightsquigarrow \mathcal{N})) \arrow[d] & \mathrm{Hom}_{\mathbf{Mod}^{\mathrm{pre}}_{\mathcal{B}}}(\mathcal{N}, (f_*\mathcal{M})^{[\theta]}) \arrow[l] \arrow[d] \\ \{\theta^\sharp : f^{\mathrm{pre}}\mathcal{B} \to \mathcal{A}\} \arrow[r] & \mathrm{Hom}_{\mathbf{Ring}^{\mathrm{pre}}_{\mathbf{Top}}}((X, \mathcal{A}) & \{\theta : \mathcal{B} \to f_*\mathcal{A}\} \arrow[l] \end{tikzcd}$$

We need to explain the notations used here.

— For a presheaf (resp. sheaf) of rings $\mathcal{A}$ on X, a presheaf (resp. sheaf) $\mathcal{M}$ of abelian groups on X is called a **presheaf of $\mathcal{A}$-modules** (resp. **sheaf of $\mathcal{A}$-modules**) if $\mathcal{M}(U)$ is an $\mathcal{A}(U)$-module for any open U and the module structure is compatible with the restriction of $\mathcal{A}$. A **morphism** between presheaves of $\mathcal{A}$-modules is a morphism of presheaves T such that $T(U)$ is an $\mathcal{O}_X(U)$-module homomorphism for every open U. We denote by $\mathbf{Mod}^{\mathrm{pre}}_{\mathcal{A}}$ the category of presheaves of $\mathcal{A}$-modules, and when $\mathcal{A}$ is sheaf, we denote by $\mathbf{Mod}_{\mathcal{A}}$ the full subcategory of $\mathbf{Mod}^{\mathrm{pre}}_{\mathcal{A}}$ consisting of sheaves of $\mathcal{A}$-modules.

— For $\mathcal{M}, \mathcal{N} \in \mathbf{Mod}^{\mathrm{pre}}_{\mathcal{A}}$, define their **tensor product** $\mathcal{M} \otimes^{\mathrm{p}}_{\mathcal{A}} \mathcal{N} \in \mathbf{Mod}^{\mathrm{pre}}_{\mathcal{A}}$ by the assignment $U \mapsto \mathcal{M}(U) \otimes_{\mathcal{A}(U)} \mathcal{N}(U)$. When in $\mathbf{Mod}_{\mathcal{A}}$, we define $\mathcal{M} \otimes_{\mathcal{A}} \mathcal{N} \in \mathbf{Mod}_{\mathcal{A}}$ by $(\mathcal{M} \otimes^{\mathrm{p}}_{\mathcal{A}} \mathcal{N})^\dagger$.

— $(f_*\mathcal{M})^{[\theta]})$ means we use $\theta : \mathcal{B} \to f_*\mathcal{A}$ to view $f_*\mathcal{M}$ as a $\mathcal{B}$-module.

— On the left-upper corner, $\mathcal{A}$ is viewed as a $f^{pre}\mathcal{B}$-module via $\theta^\sharp$.

Let us replace everything by sheaves. Suppose that $(X, \mathcal{A} \curvearrowright \mathcal{M})$ and $(Y, \mathcal{B} \curvearrowright \mathcal{N})$ are actually in $\mathbf{Mod}_{\mathbf{Top}}$. Passing to sheafification, the upper part of the diagram above gives a bijection

$$\mathrm{Hom}_{\mathbf{Mod}_{\mathcal{A}}}\left((\mathcal{A} \otimes_{f^{pre}\mathcal{B}} f^{pre}\mathcal{N})^\dagger, \mathcal{M}\right) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Mod}_{\mathcal{B}}}(\mathcal{N}, (f_*\mathcal{M})^{[\theta]})$$

The same consideration, but with each f^{pre} replaced by f^{-1}, gives a bijection

$$\mathrm{Hom}_{\mathbf{Mod}_{\mathcal{A}}}\left(\left(\mathcal{A} \otimes_{f^{-1}\mathcal{B}} f^{-1}\mathcal{N}\right)^\dagger, \mathcal{M}\right) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Mod}_{\mathcal{B}}}(\mathcal{N}, (f_*\mathcal{M})^{[\theta]})$$

As a by-product, we see $(\mathcal{A} \otimes_{f^{pre}\mathcal{B}} f^{pre}\mathcal{N})^\dagger$ and $\left(\mathcal{A} \otimes_{f^{-1}\mathcal{B}} f^{-1}\mathcal{N}\right)^\dagger$ are naturally isomorphic. We give it a new notation : $f^*\mathcal{N} := (\mathcal{A} \otimes_{f^{pre}\mathcal{B}} f^{pre}\mathcal{N})^\dagger \in \mathbf{Mod}_{\mathcal{A}}$. This is called the **inverse image** of $\mathcal{N}$ by (f, θ). Also, we simply put $f_*\mathcal{M} = (f_*\mathcal{M})^{[\theta]}$, and call it the **direct image** of $\mathcal{N}$ by (f, θ). In this way, the adjunction takes the form

$$\mathrm{Hom}_{\mathbf{Mod}_{\mathcal{A}}}(f^*\mathcal{N}, \mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Mod}_{\mathcal{B}}}(\mathcal{N}, f_*\mathcal{M})$$

It is easy to see $f_* : \mathbf{Mod}_{\mathcal{A}} \to \mathbf{Mod}_{\mathcal{B}}$ and $f^* : \mathbf{Mod}_{\mathcal{B}} \to \mathbf{Mod}_{\mathcal{A}}$ define functors, and the bijection above is bifunctorial in $\mathcal{M}$ and $\mathcal{N}$. Concisely, this bijection says that f^* is left adjoint to f_*.

1.14

1.1.2 Gluing process

1.15 Gluing sheaves. Let X_i $(i \in I)$ be a collection of topological space. On each X_i is a sheaf $\mathcal{F}_i$ of abelian groups. Suppose for any $i \in I$ there exist open subspaces $X_{ij} \subseteq X_i$ and $X_{ji} \subseteq X_j$ and an isomorphism $\theta_{ij} : (X_{ij}, \mathcal{F}_i|_{X_{ij}}) \to (X_{ji}, \mathcal{F}_j|_{X_{ji}})$ in $\mathbf{Ab}_{\mathbf{Top}}$. Assume these θ_{ij} satisfy

(a) $\theta_{ii} = \mathrm{id}_{X_i}$ for any $i \in I$;

(b) $\theta_{ik} = \theta_{jk} \circ \theta_{ij}$ on $X_{ij} \cap X_{ik}$ for any $i, j, k \in I$

Then there exists a topological space X containing each X_i as an open subspace with $X_i \cap X_j = X_{ij}$, a sheaf of abelian groups $\mathcal{F}$ on X and isomorphisms $\theta_i : \mathcal{F}|_{X_i} \to \mathcal{F}_i$ of sheaves on X_i $(i \in I)$ satisfying $\theta_i = \theta_{ij} \circ \theta_j$ on $X_i \cap X_j$.

Moreover, the data $(\mathcal{F}, \{\theta_i\}_{i\in I})$ is unique up to a unique isomorphism, in the sense that if $(\mathcal{F}', \{\theta'_i\}_{i\in I})$ is another such data, then there exists a unique isomorphism $\theta : \mathcal{F} \to \mathcal{F}'$ of sheaves on X such that $\theta_i = \theta'_i \circ \theta|_{X_i}$ for any $i \in I$.

Proof. We first construct the topological space X. Consider the topological disjoint union $\widetilde{X} := \bigsqcup_{i\in I} X_i$. On this space define a relation $\sim$:

$$(i, p) \sim (j, q) \text{ if and only if } p \in X_{ij},\ q \in X_{ji} \text{ and } \theta_{ij}(p) = q.$$

(a) shows that $\sim$ is symmetric, and (b) shows $\sim$ is transitive. (a) and (b) implies $\theta_{ij} = \theta_{ji}^{-1}$, so $\sim$ is reflexive. In sum, $\sim$ defines an equivalence relation on $\widetilde{X}$, so we may form the quotient space

$$X := \widetilde{X}/\sim .$$

It is easy to see that the canonical map $X_i \to X$ is an open embedding of topological spaces. We identify each X_i with its image in X. Then by definition we have $X_{ij} = X_i \cap X_j = X_{ji}$ in X for any $i, j \in I$. Now the existence of $(X, \mathcal{F}) \in \mathbf{Ab_{Top}}$ reduces to the following situation described in the next paragraph. □

1.16 Let X be a topological with an open cover $\mathcal{U} = \{U_i\}_{i\in I}$. Suppose that on each U_i there is a sheaf $\mathcal{F}_i$ of abelian groups. For any $i, j \in I$, suppose there is an isomorphism $\theta_{ij} : \mathcal{F}_i|_{U_i\cap U_j} \to \mathcal{F}_j|_{U_i\cap U_j}$ of sheaves on $U_i \cap U_j$. Assume these θ_{ij} satisfy

(a) θ_{ii} is the identity map for any $i \in I$;

(b) $\theta_{ik}|_{U_i\cap U_j\cap U_k} = \theta_{jk}|_{U_i\cap U_j\cap U_k} \circ \theta_{ij}|_{U_i\cap U_j\cap U_k}$.

Then there exists a unique sheaf of abelian groups $\mathcal{F}$ on X and isomorphisms $\theta_i : \mathcal{F}|_{U_i} \to \mathcal{F}_i$ of sheaves on U_i such that $\theta_j|_{U_i\cap U_j} = \theta_{ij} \circ \theta_i|_{U_i\cap U_j}$ for any $i, j \in I$.

Proof. Define a presheaf $\mathcal{F}$ on X by setting

$$\mathcal{F}(V) = \left\{ (s_i)_{i\in I} \in \prod_{i\in I} \mathcal{F}_i(V \cap U_i) \mid \theta_{ij}(V \cap U_i \cap U_j)(s_i|^{V\cap U_i}_{V\cap U_i\cap U_j}) = s_j|^{V\cap U_j}_{V\cap U_i\cap U_j} \text{ for any } i, j \in I \right\}$$

By (b), this really defines a presheaf on X. We can show $\mathcal{F}$ is a sheaf of abelian groups by a way similar to (1.3). □

1.17 Let X, Y be two topological spaces, and $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X and Y, respectively. Suppose there are an open cover $\{U_i\}_{i\in I}$ and a collection of morphisms $f_i : (U_i, \mathcal{F}|_{U_i}) \to (Y, \mathcal{G})$ in $\mathbf{Ab_{Top}}$ such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$ for any $i, j \in I$. Then there exists a unique morphism $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ in $\mathbf{Ab_{Top}}$ extending the f_i.

1.18 Let us mention some categorical limit and colimit objects in $\mathcal{C}_X^{\text{pre}}$ and $\mathcal{C}_X$.

— Direct product. If $\{\mathcal{F}_i\}_{i\in I}$ is a family of presheaves (resp. sheaves), then $U \mapsto \prod_{i\in I} \mathcal{F}_i(U)$ defines a presheaves (resp. sheaves), and is the categorical product in either category.

— Finite direct sum. Suppose $\mathcal{C} = \mathbf{Ab}$. Then finite direct sum coincides with finite direct product.

— Inverse limit. Let $\{\mathcal{F}_i\}_{i\in I}$ be an inverse system of presheaves (resp. sheaves). Then the assignment $U \mapsto \varprojlim_{i\in I} \mathcal{F}_i(U)$ defines a presheaf (resp. sheaf), and is the categorical limit in either category.

— Direct limit. Let $\{\mathcal{F}_i\}_{i\in I}$ be a direct system of presheaves (resp. sheaves). Then $U \mapsto \varinjlim_{i\in I} \mathcal{F}_i(U)$ is a presheaf, but fails to be a sheaf even if each $\mathcal{F}_i$ is a sheaf. For this, for a direct system of sheaves, we denote by $\varinjlim_{i\in I} \mathcal{F}_i$ the sheafification of the above direct limit presheaf. Both are categorical directed limit in respective category.

1.19 Let $\mathcal{C}$ be either **Ab**, **Ring** or $\mathbf{Mod}_R$. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves (resp. sheaves), then $U \mapsto \ker \varphi(U)$ defines a presheaves (resp. sheaves), called the **kernel** of $\varphi : \mathcal{F} \to \mathcal{G}$, and is denoted by $\ker \varphi$. When $\mathcal{C}$ is **Ab** or $\mathbf{Mod}_R$, the assignment $U \mapsto \mathcal{G}(U)/\mathcal{F}(U) = \operatorname{coker} \varphi(U)$ only defines a presheaf, so if we are discussing in $\mathcal{C}_X$, we define the **cokernel** $\operatorname{coker} \varphi$ of $\varphi : \mathcal{F} \to \mathcal{G}$ to be the sheafification of the previously mentioned presheaf. Similarly, we define the **image** $\operatorname{Im} \varphi$ to be the sheafification of $U \mapsto \operatorname{Im} \varphi(U)$. Both kernel and cokernel in $\mathcal{C}_X^{\mathrm{pre}}$ (resp. $\mathcal{C}_X$) satisfy the usual universal properties.

1.20 Let us assume $\mathcal{C} = \mathbf{Ab}$. We say a sequence $\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$ is **exact** in $\mathbf{Ab}_X^{\mathrm{pre}}$ (resp. $\mathbf{Ab}_X$) if $\operatorname{Im} \alpha = \ker \beta$. Thus a sequence of sheaves may be exact in $\mathbf{Ab}_X$ while fails to be exact in $\mathbf{Ab}_X^{\mathrm{pre}}$. A useful criterion for exactness is the following :

Lemma. A sequence $\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$ of sheaves of abelian groups is exact if and only if the induced map on the stalk $\mathcal{F}_x \xrightarrow{\alpha_x} \mathcal{G}_x \xrightarrow{\beta_x} \mathcal{H}_x$ for every $x \in X$.

Proof.

□

1.21 As a consequence, we see the sheafification functor $(\cdot)^\dagger : \mathbf{Ab}_X^{\mathrm{pre}} \to \mathbf{Ab}_X$ is exact, in the sense that it sends short exact sequences to short exact sequences. To see this, let $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ be a short exact sequence in $\mathbf{Ab}_X^{\mathrm{pre}}$. By (1.7) and (1.20), we only need to show $0 \to \mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x \to 0$ is exact in **Ab** for every $x \in X$. This is indeed the case, which can be seen from the proof of (1.20).

1.22 Let $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ be a morphism in $\mathbf{Ab}_{\mathbf{Top}}$. Suppose Y admits an open cover $\mathcal{U}$ such that the induced morphism $f|^U_{f^{-1}(U)} : (f^{-1}(U), \mathcal{F}|_{f^{-1}(U)}) \to (U, \mathcal{G}|_U)$ is an isomorphism for any $U \in \mathcal{U}$. Then f is an isomorphism. The map on topological spaces are clearly a homeomorphism. For the sheaf map, one can use 1.20 to show $\mathcal{G} \to f_*\mathcal{F}$ is an isomorphism.

1.2 Local-ringed spaces

1.23 Ringed space. An object in $\mathbf{Ring}_{\mathbf{Top}}$ is called a **ringed space**. For simplicity, put $\mathbf{RS} = \mathbf{Ring}_{\mathbf{Top}}$. Explicitly, a ringed space morphism from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ consists of a continuous map $f : X \to Y$ and a morphism $f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ of sheaves of rings on Y.

1.24 The morphism $f^\sharp$ induces maps of stalks. To be precise, for $x \in X$ and V an open neighborhood of $y = f(x)$ in Y, we have a map $f^\sharp_V : \mathcal{O}_Y(V) \to f_*\mathcal{O}_X(V) = \mathcal{O}_X(f^{-1}(V))$. Post-composing with inclusion into direct limit, we obtain a map $\mathcal{O}_Y(V) \to \varinjlim_{\mathrm{Top}(X) \ni U \ni x} \mathcal{O}_X(U) = \mathcal{O}_{X,x}$. Letting V varying and passing to limit, we obtain a map $f^\sharp_x : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$.

On the other hand, write $f^\flat : f^{-1}\mathcal{O}_Y \to \mathcal{O}_X$ to be the morphism obtained via (1.12). Let $x \in X$. By

(1.11) and (1.7), we have a map

$$\varinjlim_{\mathrm{Top}(X)\ni U\ni x} f^\flat : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}.$$

By construction in (1.12), this map coincides with $f^\sharp_x$.

1.25 Local-ringed space. A ringed space $(X, \mathcal{O}_X)$ is called a **local-ringed space** (or locally ringed space) if each stalk $\mathcal{O}_{X,x}$ is a local ring. A **morphism** between local-ringed spaces is a morphism of ringed spaces such that each stalk map is a **local homomorphism** of local rings, which is to be explained. For a ring homomorphism $\varphi : (A, \mathfrak{m}_A) \to (B, \mathfrak{m}_B)$ of local rings, we say it is a **local homomorphism** when $\varphi^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$. Note that in general we only have $\varphi^{-1}(\mathfrak{m}_B) \subseteq \mathfrak{m}_A$. The category of local-ringed spaces is denoted by **LRS**.

1.26 Let k be a field. A **ringed space over** k is a ringed space $(X, \mathcal{O}_X)$ such that $\mathcal{O}_X$ is a sheaf of k-algebras. The category $\mathbf{RS}_k$ of ringed spaces is the full subcategory of $(\mathbf{Alg}_k)_{\mathbf{Top}}$ whose objects consist of ringed spaces over k. A **local-ringed space over** k is a local-ringed space $(X, \mathcal{O}_X)$ that is also a ringed-space over k, and a **morphism of local-ringed spaces of** k is both a morphism of local-ringed spaces and ringed space over k. Denote by $\mathbf{LRS}_k$ the category of local-ringed space over k.

1.27 Let $(X, \mathcal{O}_X)$ be a local-ringed space. For a point $x \in X$, the quotient $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is called the **residue field** at x. For open U, $x \in U$ and $f \in \mathcal{O}_X(U)$, we sometimes write $f(x)$ to denote the class of f at the residue field $\kappa(x)$. For an open subset U of X and $f \in \mathcal{O}_X(U)$, put

$$\begin{aligned} U_f &:= \{x \in U \mid f(x) \neq 0 \text{ in } \kappa(x)\} \\ &= \{x \in U \mid f_x \in \mathcal{O}_{X,x}^\times\}. \end{aligned}$$

Then U_f is an open set in U. For if $f_x \in \mathcal{O}_{X,x}^\times$, then we can find some neighborhood V of x and $g \in \mathcal{O}_X(V)$ such that $f_x g_x = 1$. But this means $fg = 1$ in $\mathcal{O}_X(W)$ for a smaller neighborhood $W \subseteq V$ of x, and thus $x \in U \cap W \subseteq U$. Note that since $\mathcal{O}_X$ is a sheaf, for $f \in \mathcal{O}_X(X)$, we in fact have $f \in \mathcal{O}_X(X_f)^\times$. To construct an inverse, we do it locally and patch them together to a section over X_f.

1.28 Topological embedding. Let X be a topological space and $Y \subseteq X$ an subspace. We denote by $\iota : Y \to X$ the inclusion map. For a sheaf $\mathcal{F}$ on X, we denote the inverse image sheaf $\iota^{-1}\mathcal{F}$ by $\mathcal{F}|_Y$. From the adjunction we also have a morphism $\rho : \mathcal{F} \to \iota_*(\mathcal{F}|_Y)$. We call the morphism $(\iota, \rho) : (Y, \mathcal{F}|_Y) \to (X, \mathcal{F})$ the **topological embedding**. It enjoys the following universal property similar to that of the subspace topology : if $(f, f^\sharp) : (T, \mathcal{G}) \to (X, \mathcal{F})$ is a morphism with $f(T) \subseteq Y$, then there exists a unique morphism

$(T, \mathcal{G}) \to (Y, \mathcal{F}|_Y)$ making the following triangle commute

$$\begin{tikzcd}
(T, \mathcal{G}) \arrow[rr, "(f, f^\sharp)"] \arrow[dr] & & (X, \mathcal{F}) \\
& (Y, \mathcal{F}|_Y) \arrow[ur, "(\iota, \rho)"'] &
\end{tikzcd}$$

The only candidate for the map on topological spaces are $f|^Y$. The morphism between sheaves is best defined using the very definition of a morphism in $\mathcal{C}^{\mathrm{pre}}_{\mathrm{Top}}$ (1.12) then passing to sheafification. In this way the uniqueness is also evident. Another way is to define it via adjunction. To be specific, the morphism $f^\sharp : \mathcal{F} \to f_*\mathcal{G} = \iota_*(f|^Y)_*\mathcal{G}$ induces, by adjunction, another morphism $\iota^*\mathcal{F} \to (f|^Y)_*\mathcal{G}$, which is what we want. The commutativity is proved using functoriality of adjunction.

Let $\mathcal{G}$ be a sheaf on Y. By adjunction the identity morphism $\iota_*\mathcal{G} \to \iota_*\mathcal{G}$ induces a morphism $\iota^{-1}\iota_*\mathcal{G} \to \mathcal{G}$. This is in fact an isomorphism as long as ι is an embedding. It is enough to check on the stalks, and this follows from 1.11.

1.29 Let us consider the above situation in **LRS**. Let $(X, \mathcal{O}_X)$ be a local-ringed space and $Y \subseteq X$ a subspace. We check $(Y, \mathcal{O}_X|_Y)$ is also a local-ringed space, and the topological embedding $(Y, \mathcal{O}_X|_Y) \to (X, \mathcal{O}_X)$ is a morphism in **LRS**. The first is easy, for we have

$$(\mathcal{O}_X|_Y)_y = \varinjlim_{\mathrm{Top}(X) \ni U \supseteq W} \varinjlim_{\mathrm{Top}(Y) \ni W \ni y} \mathcal{O}_X(U) = \mathcal{O}_{X,y}.$$

To see the stalk map is a local homomorphism, since the stalk of a presheaf coincides with that of its sheafification (1.7), we can replace $(Y, \mathcal{O}_X|_Y)$ by $(Y, \iota^{\mathrm{pre}}\mathcal{O}_X)$, and the computation above indicates the stalk map is simply the identity map.

Now let $(\psi, \theta) : (X', \mathcal{O}_{X'}) \to (X, \mathcal{O}_X)$ be a morphism in **LRS** and let Y be a subspace of X. Put $Y' = \psi^{-1}(Y)$ and give it subspace topology of X'. Then we have a cartesian square in **LRS**

$$\begin{tikzcd}
(Y', \mathcal{O}_{X'}|_{Y'}) \arrow[r] \arrow[d] & (Y, \mathcal{O}_X|_Y) \arrow[d] \\
(X', \mathcal{O}_{X'}) \arrow[r, "(\psi, \theta)"] & (X, \mathcal{O}_X)
\end{tikzcd}$$

The two vertical morphisms are the canonical embedding, and the upper horizontal map follows from the universal property for $(Y, \mathcal{O}_X|_Y)$; to see the so obtained map is a morphism in **LRS**, one can argue as in 1.28.

1.30 Ideal sheaf. Let $(X, \mathcal{O}_X)$ be a ringed space. An **ideal sheaf** $\mathcal{J}$ of $\mathcal{O}_X$ is a sheaf on X such that $\mathcal{J}(U)$ is an ideal of $\mathcal{O}_X(U)$ for any open U. Symbolically we write $\mathcal{J} \trianglelefteq \mathcal{O}_X$. The ringed space $(X, \mathcal{O}_X/\mathcal{J})$

is called the ringed space associated to $\mathcal{J}$. Note that $\mathcal{O}_X/\mathcal{J}$ means the sheafification of the presheaf $U \mapsto \mathcal{O}_X(U)/\mathcal{J}(U)$.

The natural map $\iota_{\mathcal{J}} : (X, \mathcal{O}_X/\mathcal{J}) \to (X, \mathcal{O}_X)$ enjoys the following universal property : if $(f, f^\sharp) : (T, \mathcal{G}) \to (X, \mathcal{O}_X)$ is a morphism in **RS** with $\mathcal{J} \subseteq \ker f^\sharp$, then there exists a unique morphism $(T, \mathcal{G}) \to (X, \mathcal{O}_X/\mathcal{J})$ making the following triangle commute

$$\begin{tikzcd}
(T, \mathcal{G}) \arrow[rr, "(f, f^\sharp)"] \arrow[dr] & & (X, \mathcal{O}_X) \\
& (X, \mathcal{O}_X/\mathcal{J}) \arrow[ur, "\iota_{\mathcal{J}}"'] &
\end{tikzcd}$$

1.31 Let $(f, \theta) : (Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ be a morphism in **RS** and $\mathcal{J}$ an ideal sheaf of $\mathcal{O}_X$. Note that $f^*\mathcal{O}_X = \mathcal{O}_Y$ (1.13). For each $\mathcal{O}_Y$-module $\mathcal{N}$, define

$$\mathcal{J}\mathcal{N} = \operatorname{im}\left(f^*\mathcal{J} \otimes_{\mathcal{O}_Y} \mathcal{N} \to f^*\mathcal{O}_X \otimes_{\mathcal{O}_Y} \mathcal{N} \cong \mathcal{N}\right) \in \mathbf{Mod}_{\mathcal{O}_Y}$$

When $\mathcal{N} = \mathcal{O}_Y$, we see $\mathcal{J}\mathcal{O}_Y = \operatorname{Im}(f^*\mathcal{J} \to \mathcal{O}_Y)$. Put $\theta^\sharp : f^{-1}\mathcal{O}_X \to \mathcal{O}_Y$. Then $\theta^\sharp$ induces a map

$$f^{-1}\left(\mathcal{O}_X/\mathcal{J}\right) \cong f^{-1}\mathcal{O}_X/f^{-1}\mathcal{J} \to \mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y$$

By adjunction we obtain a morphism $\overline{\theta} : \mathcal{O}_X/\mathcal{J} \to f_*\left(\mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y\right)$. In sum, we obtain a map

$$(f, \overline{\theta}) : (Y, \mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y) \longrightarrow (X, \mathcal{O}_X/\mathcal{J})$$

in **RS**. In fact, it fits into a Cartesian square in **RS**

$$\begin{tikzcd}
(Y, \mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y) \arrow[r, "(f, \overline{\theta})"] \arrow[d] & (X, \mathcal{O}_X/\mathcal{J}) \arrow[d] \\
(Y, \mathcal{O}_Y) \arrow[r, "(f, \theta)"] & (X, \mathcal{O}_X)
\end{tikzcd}$$

1.32 Let $(X, \mathcal{O}_X)$ be a local-ringed space and $\mathcal{J} \trianglelefteq \mathcal{O}_X$ an ideal ideal. For $x \in X$, it may happen that the stalk $(\mathcal{O}_X/\mathcal{J})_x$ is not a local ring. Taking stalk is exact, so $(\mathcal{O}_X/\mathcal{J})_x \cong \mathcal{O}_{X,x}/\mathcal{J}_x$. Thus it is not a local ring if and only if it is zero, or equivalently, $\mathcal{O}_{X,x} = \mathcal{J}_x$. Define

$$V(\mathcal{J}) = \{x \in X \mid \mathcal{J}_x \subsetneq \mathcal{O}_{X,x}\}$$

This is a closed subset of X, for if $x \notin V(\mathcal{J})$, then $\mathcal{O}_{X,x} = \mathcal{J}_x$, so there is a neighborhood U of x and $f \in \mathcal{J}(U)$ such that $f_x = 1 \in \mathcal{O}_{X,x}$. Shrinking U further shows that $f|_U = 1 \in \mathcal{J}(U)$, so $U \subseteq X \backslash V(\mathcal{J})$.

Let $j : V(\mathcal{J}) \to X$ denote the inclusion. Then $V(\mathcal{J})$ together with the sheaf of rings $j^{-1}\left(\mathcal{O}_X/\mathcal{J}\right) = \mathcal{O}_X/\mathcal{J}\big|_{V(\mathcal{J})}$ becomes a local-ringed space, called the **closed local-ringed subspace of** $(X, \mathcal{O}_X)$ **associated to the ideal sheaf** $\mathcal{J}$**.**

Denote by $i_{\mathcal{J}} : (V(\mathcal{J}), j^{-1}(\mathcal{O}_X/\mathcal{J})) \xrightarrow{\iota_{\mathcal{J}}} (X, \mathcal{O}_X/\mathcal{J}) \to (X, \mathcal{O}_X)$ the composition (1.30). This morphism in **LRS** enjoys the following universal property : if $(f, f^\sharp) : (T, \mathcal{G}) \to (X, \mathcal{O}_X)$ is a morphism in **LRS** with $\mathcal{J} \subseteq \ker f^\sharp$, then there exists a unique morphism $(T, \mathcal{G}) \to (V(\mathcal{J}), j^{-1}(\mathcal{O}_X/\mathcal{J}))$ making the following triangle commute

$$\begin{tikzcd}
(T, \mathcal{G}) \arrow[rr, "(f, f^\sharp)"] \arrow[dr] & & (X, \mathcal{O}_X) \\
& (V(\mathcal{J}), j^{-1}(\mathcal{O}_X/\mathcal{J})) \arrow[ur, "i_{\mathcal{J}}"'] &
\end{tikzcd}$$

We first show $f(t) \in V(\mathcal{J})$. Since taking stalk at $f(t)$ is exact, we see $(\ker f^\sharp)_{f(t)} = \ker(\mathcal{O}_{X,f(t)} \to \mathcal{G}_t)$. Since the stalk map is a local homomorphism, we see $(\ker f^\sharp)_{f(t)}$ is contained in the maximal ideal of $\mathcal{O}_{X,f(t)}$, whence $\mathcal{J}_{f(t)} \subsetneq \mathcal{O}_{X,f(t)}$. Similar to (1.30), we have a morphism $\mathcal{O}_X/\mathcal{J} \to f_*\mathcal{G} = j_* \left(f|^{V(\mathcal{J})}\right)_* \mathcal{G}$, so adjunction gives $j^{-1}(\mathcal{O}_X/\mathcal{J}) \to \left(f|^{V(\mathcal{J})}\right)_* \mathcal{G}$. The resulting map is a morphism in **LRS** and makes this triangle commutative, as one can argue as in 1.28.

1.33 Let $(f, \theta) : (Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ be a morphism in **LRS**. Let $\mathcal{N} \in \mathbf{Mod}_{\mathcal{O}_Y}$ and $\mathcal{J} \trianglelefteq \mathcal{O}_X$ an ideal sheaf. By definition there exists a commutative diagram

$$\begin{tikzcd}
f^*\mathcal{J} \otimes_{\mathcal{B}} \mathcal{N} = f^{-1}\mathcal{J} \otimes_{f^{-1}\mathcal{O}_X} \mathcal{O}_Y \otimes_{\mathcal{O}_Y} \mathcal{N} \arrow[rr] \arrow[dr, two heads] & & \mathcal{N} \\
& \mathcal{J}\mathcal{N} \arrow[ur, hook] &
\end{tikzcd}$$

Taking stalk at $y \in Y$ gives (c.f. (1.14))

$$\begin{tikzcd}
\mathcal{J}_{f(y)} \otimes_{\mathcal{O}_{X,f(y)}} \mathcal{O}_{Y,y} \otimes_{\mathcal{O}_{Y,y}} \mathcal{N}_y \arrow[rr] \arrow[dr, two heads] & & \mathcal{N}_y \\
& (\mathcal{J}\mathcal{N})_y \arrow[ur, hook] &
\end{tikzcd}$$

This proves that

$$(\mathcal{J}\mathcal{N})_y = \operatorname{Im}\left(\mathcal{J}_{f(y)} \otimes_{\mathcal{O}_{X,f(y)}} \mathcal{N}_y \to \mathcal{N}_y\right) = \mathcal{J}_{f(y)}\mathcal{N}_y.$$

1.34 Lemma. $V(\mathcal{J}\mathcal{O}_Y) = f^{-1}(V(\mathcal{J}))$.

Proof. Recall that for $y \in Y$, $(\mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y)_y \cong \mathcal{O}_{Y,y}/\mathcal{J}_{f(y)}\mathcal{O}_{Y,y}$. Let $\mathfrak{m}_{f(y)}$ and $\mathfrak{m}_y$ be the maximal ideals of $\mathcal{O}_{X,f(y)}$ and $\mathcal{O}_y$ respectively; note that $f_y(\mathfrak{m}_{f(y)}) \subseteq \mathfrak{m}_y$.

— If $y \in V(\mathcal{J}\mathcal{O}_Y)$, then $\mathcal{J}_{f(y)}\mathcal{O}_{Y,y} \subsetneq \mathcal{O}_{Y,y}$. A fortiori we have $\mathcal{J}_{f(y)} \subsetneq \mathcal{O}_{X,x}$, so $y \in f^{-1}(V(\mathcal{J}))$.

— If $y \in f^{-1}(V(\mathcal{J}))$, then $\mathcal{J}_{f(y)} \subseteq \mathfrak{m}_{f(y)}$. But

$$\mathcal{J}_{f(y)}\mathcal{O}_{Y,y} \subseteq \mathfrak{m}_{f(y)}\mathcal{O}_{Y,y} \subseteq \mathfrak{m}_y \subsetneq \mathcal{O}_{Y,y}$$

so $y \in V(\mathcal{J}\mathcal{O}_Y)$.

$\square$

As as corollary, we see from 1.32 and this lemma that a morphism $(Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ in **LRS** induces a morphism

$$(V(\mathcal{J}\mathcal{O}_Y), (\mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y)\big|_{V(\mathcal{J}\mathcal{O}_Y)}) \to (V(\mathcal{J}), (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})})$$

Moreover, it fits into a Cartesian square in **LRS**.

$$\begin{tikzcd}
(V(\mathcal{J}\mathcal{O}_Y), (\mathcal{O}_Y/\mathcal{J}\mathcal{O}_Y) \arrow[r] \arrow[d] & (V(\mathcal{J}), (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})}) \arrow[d] \\
(Y, \mathcal{O}_Y) \arrow[r] & (X, \mathcal{O}_X)
\end{tikzcd}$$

This follows from 1.31 and 1.32.

1.35 Definition. Let $(X, \mathcal{O}_X)$ be a local-ringed space.

1. For open U in X, $(U, \mathcal{O}_X\big|_U)$ is called an **open local-ringed subspace** of $(X, \mathcal{O}_X)$.
2. For $\mathcal{J} \trianglelefteq \mathcal{O}_X$, $(V(\mathcal{J}), (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})})$ is called a **closed local-ringed space** of $(X, \mathcal{O}_X)$ associated to $\mathcal{J}$.
3. For open U and $J \trianglelefteq \mathcal{O}_X\big|_U$, $(V(\mathcal{J}), (\mathcal{O}_X\big|_U/\mathcal{J})\big|_{V(\mathcal{J})})$ is called a **locally closed local ringed subspace** of $(X, \mathcal{O}_X)$.

A morphism $f : (Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ in **LRS** is called an **open immersion / closed immersion / immersion** if there exists an open / closed / locally closed local-ringed subspace $(Z, \mathcal{O}_Z)$ of $(X, \mathcal{O}_X)$ and an isomorphism $(Y, \mathcal{O}_y) \to (Z, \mathcal{O}_Z)$ of **LRS** making the following diagram commute

$$\begin{tikzcd}
(Y, \mathcal{O}_Y) \arrow[rr, "f"] \arrow[dr, "\sim"'] & & (X, \mathcal{O}_X) \\
& (Z, \mathcal{O}_Z) \arrow[ur] &
\end{tikzcd}$$

1.36 Proposition. Let $f : (Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ be a morphism in **LRS**.

1. f is an open immersion if and only if f is a topological open embedding and for any $y \in Y$, the stalk map $f_y : \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is an isomorphism.
2. f is a closed immersion if and only if f is a topological closed embedding and for any $y \in Y$, the stalk map $f_y : \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is surjective.
3. f is an immersion if and only if f is a topological locally closed embedding and for any $y \in Y$, the stalk map $f_y : \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is surjective.

Proof.

1. The only if part is clear. For the if part, let U be the image of f. It suffices to consider $(U, \mathcal{O}_X\big|_U)$.

2. To show the only if part, say

$$\begin{tikzcd}
(Y, \mathcal{O}_Y) \arrow[rr, "f"] \arrow[dr, "\sim"'] & & (X, \mathcal{O}_X) \\
& (V(\mathcal{J}), (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})}) \arrow[ur] &
\end{tikzcd}$$

Taking stalk at $y \in Y$ gives

$$\begin{tikzcd}
\mathcal{O}_{Y,y} \arrow[rr] \arrow[dr, "\sim"'] & & \mathcal{O}_{X,f(y)} \\
& \mathcal{O}_{X,f(y)}/\mathcal{J}_{f(y)} \arrow[ur] &
\end{tikzcd}$$

and this proves the surjectivity.

Now we consider the if part. Let $\mathcal{J}$ be the kernel of the morphism $\mathcal{O}_X \to f_*\mathcal{O}_Y$. By the universal property 1.32 there exists a morphism $(Y, \mathcal{O}_Y) \to (V(\mathcal{J}), (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})})$ making the triangle commute

$$\begin{tikzcd}
(Y, \mathcal{O}_Y) \arrow[rr, "f"] \arrow[dr] & & (X, \mathcal{O}_X) \\
& (V(\mathcal{J}), (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})}) \arrow[ur] &
\end{tikzcd}$$

We claim that $f(Y) = V(\mathcal{J})$, and $\mathcal{O}_Y \cong (\mathcal{O}_X/\mathcal{J})\big|_{V(\mathcal{J})}$. For $x \in X$, since f is a closed embedding, we have

$$(f_*\mathcal{O}_Y)_x = \begin{cases} \mathcal{O}_{Y,y} & \text{, if } x = f(y) \\ 0 & \text{, otherwise} \end{cases}$$

For convenience, we write $\operatorname{supp} f_*\mathcal{O}_X$ to denote the set of those points in X at which the stalk of $f_*\mathcal{O}_X$ does not vanish. Then we see $\operatorname{supp} f_*\mathcal{O}_Y = f(Y)$.

On the other hand, since f is an embedding (and by the fact mentioned in the last part of 1.28), we have an exact sequence

$$0 \longrightarrow f^{-1}\mathcal{J} \longrightarrow f^{-1}\mathcal{O}_X \longrightarrow f^{-1}f_*\mathcal{O}_Y \cong \mathcal{O}_Y$$

We contend the last morphism is surjective. This follows from our assumption once we look at the stalks. Hence

$$(f_*\mathcal{O}_Y)_x = (f^{-1}f_*\mathcal{O}_Y)_x = \begin{cases} (\mathcal{O}_X/\mathcal{J})_x & \text{, if } x = f(y) \\ 0 & \text{, otherwise} \end{cases}$$

and this shows $\operatorname{supp} f_*\mathcal{O}_Y \subseteq V(\mathcal{J}) = \operatorname{supp} \mathcal{O}_X/\mathcal{J}$ Finally, taking stalk directly to the exact sequence $0 \to \mathcal{J} \to \mathcal{O}_X \to f_*\mathcal{O}_Y \to 0$ gives $(f_*\mathcal{O}_Y)_x \subseteq (\mathcal{O}_X/\mathcal{J})_x$, showing that $V(\mathcal{J}) \subseteq \operatorname{supp} f_*\mathcal{O}_Y$. Hence the equality $f(Y) = V(\mathcal{J})$ is proved. The above computation of stalks also proves the assertion for sheaves.

3. This follow from 1. and 2. (and perhaps their proofs).

$\square$

1.37 The definition of locally closed subspaces seems to depend on the choice of open sets we choose. In fact it does not, in the following sense. Let U be an open subspace of X and $\mathcal{J} \trianglelefteq \mathcal{O}_X\big|_U$. Denote by $(Y, \mathcal{O}_Y)$ the associated locally closed subspace. Let U_0 be the largest open subspace of X containing Y as a closed subset and let $j : U \to U_0$ be the inclusion. Let ι_U and ι_{U_0} be the inclusion of Y into U and U_0, respectively. Let us write $\mathcal{O}_U$ and $\mathcal{O}_{U_0}$ for brevity to mean the sheaves of rings for their open local-ringed subspace structures. By definition we have an exact sequence

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_U \longrightarrow (\iota_U)_*\mathcal{O}_Y \longrightarrow 0$$

Applying j_* this sequence (j_* is exact in this case), we obtain an exact sequence

$$\begin{tikzcd}
0 \arrow[r] & j_*\mathcal{J} \arrow[r] & j_*\mathcal{O}_U \arrow[r] & j_*(\iota_U)_*\mathcal{O}_Y = (\iota_{U_0})_*\mathcal{O}_Y \arrow[r] & 0 \\
 & & \mathcal{O}_{U_0} \arrow[u] \arrow[r] & (\iota_{U_0})_*\mathcal{O}_Y \arrow[u, equal] &
\end{tikzcd}$$

The middle vertical arrow is surjective, and hence so is the lower horizontal one. Denote by $\mathcal{I}$ the kernel of the lower horizontal arrow. Then we have a commutative diagram with exact rows.

$$\begin{tikzcd}
0 \arrow[r] & j_*\mathcal{J} \arrow[r] & j_*\mathcal{O}_U \arrow[r] & j_*(\iota_U)_*\mathcal{O}_Y \arrow[r] & 0 \\
0 \arrow[r] & \mathcal{I} \arrow[r] \arrow[u] & \mathcal{O}_{U_0} \arrow[r] \arrow[u] & (\iota_{U_0})_*\mathcal{O}_Y \arrow[r] \arrow[u, equal] & 0
\end{tikzcd}$$

We claim $V(\mathcal{I}) = V(\mathcal{J}) = Y$ and the chain on the bottom induces an isomorphism $(\mathcal{O}_{U_0}/\mathcal{I})\big|_{V(\mathcal{I})} \cong \mathcal{O}_Y$. The first is easy, as taking stalks we see $V(\mathcal{I}) = \operatorname{supp}(\iota_{U_0})_*\mathcal{O}_Y$, and for $x \in U_0$,

$$((\iota_{U_0})_*\mathcal{O}_Y)_x = \begin{cases} \mathcal{O}_{Y,x} & \text{, if } x \in V(\mathcal{J}) \\ 0 & \text{, otherwise} \end{cases} .$$

So $V(\mathcal{I}) = V(\mathcal{J})$. Now the assertion for sheaves is clear.

1.3 Affine schemes

1.38 Let A be a (unital commutative) ring. The set of all prime ideals of A is called the **spectrum** of A, and is denoted by $\operatorname{Spec} A$. For a subset $S \subseteq A$, put $V(S) = \{\mathfrak{p} \in \operatorname{Spec} A \mid S \subseteq \mathfrak{p}\}$. One checks easily that for ideals I, J and I_α, we have

(i) $\bigcap_\alpha V(I_\alpha) = V\left(\bigcup_\alpha I_\alpha\right)$,

(ii) $V(I) \cup V(J) = V(I \cap J) = V(IJ)$,

(iii) $V(A) = \varnothing$, $V((0)) = \operatorname{Spec} A$,

(iv) $V(I) = V(\sqrt{I})$,

(v) $V(S) = V(\langle S \rangle)$, where $S \subseteq A$ is a subset and $\langle S \rangle$ denotes the ideal generated by S.

(vi) $V(I) \subseteq V(J)$ if and only if $J \subseteq \sqrt{I}$.

In particular, by (i), (ii), (iii), the $V(S)$ define a topology of closed sets on $\operatorname{Spec} A$. For each $f \in A$, the open set $D(f) = \operatorname{Spec} A \backslash V((f)) = \{\mathfrak{p} \mid f \notin \mathfrak{p}\}$ is called a **principal open set**. The collection of all principal open sets form a basis for this topology.

The topological space $\operatorname{Spec} A$ is compact. More generally, each principal open set $D(f)$ is compact. To see this, say $D(f) = \bigcup_i D(g_i)$. Taking complement, we see $V((f)) = \bigcap_i V((g_i)) = V((g_i)_{i \in I})$. By (vi), this means $f \in \sqrt{(g_i)_{i \in I}}$. Thus we can find $i_1, \ldots, i_n \in I$ such that $f \in \sqrt{(g_{i_1}, \ldots, g_{i_n})}$, and by (vi) again we see $V((f)) \supseteq V((g_{i_1}, \ldots, g_{i_n}))$. Taking complements, we see $D(f) \subseteq \bigcup_{k=1}^{n} D(g_{i_k})$, and hence $D(f) = \bigcup_{k=1}^{n} D(g_{i_k})$.

1.39 Affine schemes. For a principal open set $D(f)$, define

$$\mathcal{O}_A(D(f)) = \mathcal{O}_{\operatorname{Spec} A}(D(f)) = A_f.$$

If $D(f) \subseteq D(g)$, i.e., $f \in \sqrt{(g)}$ by 1.38.(vi), then $f^n = gh$ for some $n \in \mathbb{N}$ and $h \in A$. This gives rises to a map $A_g \to A_f$, by sending $\frac{1}{g}$ to $\frac{h}{f^n}$. The resulting map is easily seen to be independent of the choice of n and h. This independence also shows that $\mathcal{O}_A$ defines a presheaf of rings on the principal open sets. To show this is a sheaf, since $D(f) \cap D(g) = D(fg)$, by 1.3 and 1.38 we need to check to exactness of the following sequence

$$0 \longrightarrow A_f \longrightarrow \prod_{i=1}^{n} A_{f_i} \longrightarrow \prod_{i,j=1}^{n} A_{f_i f_j} \qquad (\star)$$

where $f, f_i \in A$ with $D(f) = \bigcup_{i=1}^{n} D(f_i)$. The argument is the same as the one in the next paragraph, so we defer our proof. By 1.3 we then obtain a sheaf of rings $\mathcal{O}_A = \mathcal{O}_{\operatorname{Spec} A}$ defines on the whole $\operatorname{Spec} A$. The ringed space $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ is called the **affine scheme**. If no confusion will occur, we write $\operatorname{Spec} A$ to denote the pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$.

We compute the stalk of $\mathcal{O}_{\operatorname{Spec} A}$ at a prime $\mathfrak{p}$. Since the $D(f)$ form a basis, we have

$$\mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} = \varinjlim_{\operatorname{Top}(\operatorname{Spec} A) \ni U \ni \mathfrak{p}} \mathcal{O}_A(U) = \varinjlim_{D(f) \ni \mathfrak{p}} \mathcal{O}_A(D(f)) = \varinjlim_{f \notin \mathfrak{p}} A_f = A_{\mathfrak{p}}.$$

The last isomorphism is given by the natural maps $A_f \to A_{\mathfrak{p}}$. Hence $\operatorname{Spec} A$ is in fact a local-ringed space.

1.40 Affine tilde. Let A be a ring and $(X, \mathcal{O}_X) = (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$. For an A-module M, we construct an $\mathcal{O}_X$-module $\widetilde{M}$ as follows. For each principal open set $D(f)$, define

$$\widetilde{M}(D(f)) = M_f = M \otimes_A A_f.$$

This defines a presheaf on principal open sets. To show this really defines a sheaf on them, and hence on X, by 1.3 we must show the sequence

$$0 \longrightarrow M_f \longrightarrow \prod_{i=1}^{n} M_{f_i} \longrightarrow \prod_{i,j=1}^{n} M_{f_i f_j} \tag{$\heartsuit$}$$

is exact for every $f, f_1, \ldots, f_n \in A$ with $D(f) = D(f_1) \cup \cdots \cup D(f_n)$. Since $D(f) = D(f_1) \cup \cdots \cup D(f_n)$, $V(f) = V((f_1, \ldots, f_n))$, so there exist some $k \geqslant 0$ and $a_1, \ldots, a_n \in A$ such that $f^k = \sum_{i=1}^{n} a_i f_i$. Raising to arbitrary powers, we see for each $m \in \mathbb{N}$ we can find $M \geqslant 0$ and $a_1, \ldots, a_n \in A$ such that

$$f^M = \sum_{i=1}^{n} a_i f_i^m$$

We first show the exactness at the first place. Suppose $\frac{a}{f^\ell} \in M_f$ is mapped to zero in each M_{f_i}. By definition this means $f_i^{r_i} a = 0$ in M for some $r_i \geqslant 0$. If we take $m \geqslant \max\{r_1, \ldots, r_n\}$, we see $f_i^m a = 0$ in M. Choose $M \geqslant 0$ and $a_1, \ldots, a_n \in A$ corresponding to m as above. Then

$$0 = a(a_1 f_1^m + \cdots + a_n f_n^m) = a f^M,$$

and this means $\frac{a}{f^\ell} = 0$ in M_f.

Next we show the exactness at the middle place. Suppose $\left(\frac{b_i}{f_i^{\ell_i}}\right) \in \prod_{i=1}^{n} M_{f_i}$ satisfies $\frac{b_i}{f_i^{\ell_i}} - \frac{b_j}{f_j^{\ell_j}} = 0$ in $M_{f_i f_j}$ for all $1 \leqslant i, j \leqslant n$; we may assume each ℓ_i is the same, say equal to ℓ. A similar argument as above show that we can find $m \geqslant 0$ such that

$$(b_i f_j^\ell - b_j f_i^\ell)(f_i f_j)^m = 0 \text{ in } M$$

for all i, j. If we put $b_i' = b_i f_i^m$, the above identities become $b_i' f_j^{\ell+m} = b_i' f_j^{\ell+m}$. Take $M \geqslant 0$ and $a_1, \ldots, a_n \in A$ with respect to $\ell + m$. We claim $\frac{a_1 b_1' + \cdots + a_n b_n'}{f^M}$ is mapped to $\left(\frac{b_i}{f_i^\ell}\right) = \left(\frac{b_i'}{f_i^{\ell+m}}\right) \in \prod_{i=1}^{n} M_{f_i}$. Indeed,

$$(a_1 b_1' + \cdots + a_n b_n') f_j^{\ell+m} = \sum_{i=1}^{n} a_i b_i' f_j^{\ell+m} = \sum_{i=1}^{n} a_i b_j' f_i^{\ell+m} = \left(\sum_{i=1}^{n} a_i f_i^{\ell+m}\right) b_j' = f^M b_j'.$$

This show $\widetilde{M}$ is really a sheaf on X. In fact, as one easily can see, $\widetilde{M}$ is an $\mathcal{O}_X$-module, and $M \mapsto \widetilde{M}$ defines a functor $\mathrm{Mod}_A \to \mathrm{Mod}_{\mathcal{O}_X}$. The computation of stalks in 1.39 implies that $\widetilde{M}_\mathfrak{p} \cong M \otimes A_\mathfrak{p} = M_\mathfrak{p}$ for all primes $\mathfrak{p}$.

1.41 Lemma. For a complex $M \to N \to L$ in Mod_A, it is exact if and only if $\widetilde{M} \to \widetilde{N} \to \widetilde{L}$ is an exact sequence in $\mathrm{Mod}_{\mathcal{O}_X}$.

Proof. Suppose $M \to N \to L$ is exact. Since localization is an exact functor, for each $\mathfrak{p} \in \operatorname{Spec} A$, there exists an exact sequence $M_{\mathfrak{p}} \to N_{\mathfrak{p}} \to L_{\mathfrak{p}}$. Moreover, this is the same as the sequence $\widetilde{M}_{\mathfrak{p}} \to \widetilde{N}_{\mathfrak{p}} \to \widetilde{L}_{\mathfrak{p}}$, so by 1.20 we see $\widetilde{M} \to \widetilde{N} \to \widetilde{L}$ is exact. For the converse, it suffices to show $M \xrightarrow{\alpha} N \xrightarrow{\beta} L$ is exact if each localization sequence $M_{\mathfrak{p}} \to N_{\mathfrak{p}} \to L_{\mathfrak{p}}$ is exact. That is, we must show $\ker \beta / \operatorname{Im} \alpha = 0$ if $(\ker \beta / \operatorname{Im} \alpha)_{\mathfrak{p}} = 0$ for all $\mathfrak{p} \in \operatorname{Spec} A$. If $D := \ker \beta / \operatorname{Im} \alpha$ is nontrivial, take $m \in D - \{0\}$ with $\operatorname{ann}_A(m) \subsetneq A$ and consider a maximal ideal $\mathfrak{m}$ containing $\operatorname{ann}_A(m)$. The image of m in $D_{\mathfrak{m}}$ is then nonzero, for otherwise $ma = 0$ for some $a \in A - \mathfrak{m}$, a contradiction to our choice of $\mathfrak{m}$. Hence $D = 0$, and the sequence $M \to N \to L$ is exact.

□

1.42 Definition. A **scheme** is a local-ringed space $(X, \mathcal{O}_X)$ admitting an open cover $\mathcal{U}$ such that for any $U \in \mathcal{U}$, there is an isomorphism $(U, \mathcal{O}_X|_U) \cong (\operatorname{Spec} A_U, \mathcal{O}_{\operatorname{Spec} A_U})$ in **LRS** for some ring A_U.

1.43 Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be local-ringed spaces. There exists a canonical map

$$\operatorname{Hom}_{\mathbf{LRS}}(X, Y) \longrightarrow \operatorname{Hom}_{\mathbf{Ring}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)).$$

Moreover, this map is bifunctorial in X and Y. More generally, there is a bifunctorial map

$$\operatorname{Hom}_{\mathbf{Mod}_{\mathbf{LRS}}}((X, \mathcal{F}), (Y, \mathcal{G})) \longrightarrow \operatorname{Hom}_{\mathbf{Mod}}(\mathcal{O}_Y(Y) \curvearrowright \mathcal{G}(Y), \mathcal{O}_X(X) \curvearrowright \mathcal{F}(X)).$$

1.44 Theorem. Let A be a ring and $(X, \mathcal{O}_X) \in \mathbf{LRS}$. Then the map in 1.43

$$\operatorname{Hom}_{\mathbf{LRS}}(X, \operatorname{Spec} A) \longrightarrow \operatorname{Hom}_{\mathbf{Ring}}(A, \mathcal{O}_X(X)).$$

is a bijection.

Proof. Let $(f, f^\sharp) \in \operatorname{Hom}_{\mathbf{LRS}}(X, \operatorname{Spec} A)$. Then for any $x \in X$, if we put $y = f(x) = \mathfrak{p} \in \operatorname{Spec} A$, then we have a commutative diagram

$$\begin{tikzcd}
\mathcal{O}_X(X) \arrow[d, "\mathrm{res}"] & A \arrow[l, "f^\sharp_{\operatorname{Spec} A}"'] \arrow[d] \\
\mathcal{O}_{X,x} & A_{\mathfrak{p}} \arrow[l, "f_x"]
\end{tikzcd}$$

Write $\mathfrak{m}_x$ the unique maximal ideal of $\mathcal{O}_{X,x}$. Then since f_x is a local homomorphism, we see $\mathfrak{p} = (f^\sharp_{\operatorname{Spec} A})^{-1} \operatorname{res}^{-1}(\mathfrak{m}_x)$. On the other hand, for any $g \in A$, there is a commutative diagram

$$\begin{tikzcd}
\mathcal{O}_X(X) \arrow[d, "\mathrm{res}"] & A \arrow[l, "f^\sharp_{\operatorname{Spec} A}"'] \arrow[d] \\
f_*\mathcal{O}_X(D(g)) & A_g \arrow[l, "f^\sharp_{D(g)}"]
\end{tikzcd}$$

The right vertical arrow is localization, so $f^\sharp_{D(g)}$ is in fact uniquely determined by $\mathrm{res} \circ f^\sharp_{\operatorname{Spec} A}$. Since the $D(g)$ form an open basis for $\operatorname{Spec} A$, this shows $f^\sharp$ is uniquely determined by $f^\sharp_{\operatorname{Spec} A}$. This proves the injectivity.

For the surjectivity, let $\theta \in \mathrm{Hom}_{\mathbf{Ring}}(\mathcal{O}_X(X), A)$. We define a map $f : X \to \operatorname{Spec} A$ by setting $f(x) = \theta^{-1}(\mathrm{res}^X_x)^{-1}(\mathfrak{m}_x)$, where $\mathrm{res}^X_x : \mathcal{O}_X(X) \to \mathcal{O}_{X,x}$ is the restriction and $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. To show this is continuous, we claim

$$f^{-1}(D(g)) = X_{\theta(g)}$$

Indeed, $x \in f^{-1}(D(g)) \Leftrightarrow f(x) \in D(g) \Leftrightarrow g \notin f(x) = \theta^{-1}(\mathrm{res}^X_x)^{-1}(\mathfrak{m}_x) \Leftrightarrow \mathrm{res}^X_x(\theta(g)) \notin \mathfrak{m}_x \Leftrightarrow x \in X_{\theta(g)}$.

For $g \in A$, consider the diagram

$$\begin{tikzcd}
\mathcal{O}_X(X) \arrow[d, "\mathrm{res}^X_{X_{\theta(g)}}"] & A \arrow[l, "\theta"'] \arrow[d] \\
f_*\mathcal{O}_X(D(g)) & A_g
\end{tikzcd}$$

To construct a morphism in the bottom so that the square is commutative, we use the universal property of localization. To this end, we need to show $\mathrm{res}^X_{X_{\theta(g)}}(\theta(g)) \in (f_*\mathcal{O}_X(D(g)))^\times = \mathcal{O}_X(X_{\theta(g)})^\times$. But this is the content of 1.27, thus there exists a unique arrow $f_*\mathcal{O}_X(D(g)) \leftarrow A_g$ making the above square commuting. For arbitrary U, the map can be defined using 1.3. Finally, by construction we see each stalk map is a local homomorphism. □

1.45 Corollary. Let A be a ring, M an A-module. and $(X, \mathcal{O}_X) \in \mathbf{LRS}$. Then the map in 1.43

$$\mathrm{Hom}_{\mathbf{Mod}_{\mathbf{LRS}}}((X, \mathcal{F}), (\operatorname{Spec} A, \widetilde{M})) \longrightarrow \mathrm{Hom}_{\mathbf{Ring}}(A \curvearrowright M, \mathcal{O}_X(X) \curvearrowright \mathcal{F}(X)).$$

is a bijection.

1.46 Corollary. Let A be a ring. Then $M \mapsto \widetilde{M}$ defines a fully faithful exact functor $\mathrm{Mod}_A \to \mathrm{Mod}_{\mathcal{O}_{\operatorname{Spec} A}}$.

1.47 Denote by **Sch** (resp. **AffSch**) the full subcategory of **LRS** whose objects are schemes (resp. affine schemes). Then the bijection in (1.44) implies that the functor $\operatorname{Spec} : \mathbf{Ring} \to \mathbf{AffSch}$ defines an anti-equivalence of categories

$$\begin{aligned}
\operatorname{Spec} : \mathbf{Ring} &\longrightarrow \mathbf{AffSch} \\
A &\longmapsto \operatorname{Spec} A \\
\mathcal{O}_X(X) &\longmapsfrom X
\end{aligned}$$

1.48 The bifunctorial maps in 1.43 fit into the commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbf{Mod_{LRS}}}((X,\mathcal{F}),(Y,\mathcal{G})) \arrow[r] \arrow[d] & \mathrm{Hom}_{\mathbf{Mod}}(\mathcal{O}_Y(Y) \curvearrowright \mathcal{G}(Y), \mathcal{O}_X(X) \curvearrowright \mathcal{F}(X)) \arrow[d] \\
\mathrm{Hom}_{\mathbf{LRS}}(X,Y) \arrow[r] & \mathrm{Hom}_{\mathbf{Ring}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)).
\end{tikzcd}$$

Choosing $f \in \mathrm{Hom}_{\mathbf{LRS}}(X,Y)$ and taking its preimages of vertical maps we obtain a functorial bijection

$$\mathrm{Hom}_{\mathrm{Mod}_{\mathcal{O}_{\mathrm{Spec}\,A}}}(\mathcal{G}, f_*\mathcal{F}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_Y(Y)}(\mathcal{G}(Y), \mathcal{F}(X)^{[f^\sharp_{\mathrm{Spec}\,A}]})$$

where $\mathcal{F}(X)^{[f^\sharp_Y]}$ means that we are regarding $\mathcal{F}(X)$ as an $\mathcal{O}_Y(Y)$-module via the map $f^\sharp_Y : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$. By 1.44 and 1.45, we know when $(Y,\mathcal{G}) = (\mathrm{Spec}\,A, \widetilde{M})$, this is a bijection :

$$\mathrm{Hom}_{\mathrm{Mod}_{\mathcal{O}_{\mathrm{Spec}\,A}}}(\widetilde{M}, f_*\mathcal{F}) \longrightarrow \mathrm{Hom}_A(M, \mathcal{F}(X)^{[f^\sharp_{\mathrm{Spec}\,A}]})$$

1.49 Let $\varphi : A \to B$ be a ring homomorphism. Then it induces a continuous map $\mathrm{Spec}\,\varphi : \mathrm{Spec}\,B \to \mathrm{Spec}\,A$ between affine schemes. We list some properties of $\mathrm{Spec}\,\varphi$. For brevity, we write $f = \mathrm{Spec}\,\varphi$.

(i) $f^{-1}(D(f)) = D(\varphi(f))$ for any $f \in A$.

(ii) $f^{-1}(V(I)) = V(IB)$ for any ideal I of A.

(iii) $\overline{f(V(J))} = V(J \cap A)$ for any ideal J of B.

(iv) $f(\mathrm{Spec}\,B)$ is dense in $\mathrm{Spec}\,A$ if and only if $\ker\varphi \subseteq \sqrt{0}$.

(v) f is a homeomorphism onto its image if for all $b \in B$ there exists some $u \in B^\times$ such that $ub \in f(A)$.

Proof.

(v) For $b \in B$, we can find $u \in B^\times$ and $a \in A$ such that $ub = \varphi(a)$. Thus

$$D(b) = D(ub) = D(\varphi(a)) = f^{-1}(D(a)).$$

Thus we are left to prove the injectivity of f. Let $\mathfrak{p} \neq \mathfrak{q} \in \mathrm{Spec}\,B$. Say we can pick $b \in \mathfrak{p}\backslash\mathfrak{q}$. Choose $a \in A$ with $D(b) = f^{-1}(D(a))$. Then $\mathfrak{q} \in D(b) \not\ni \mathfrak{p}$, and thus $f(\mathfrak{q}) \in D(a) \not\ni f(\mathfrak{p})$.

□

1.50 Let S be a multiplicatively closed subset of A. As a consequence of 1.49.(v), we see the map $\mathrm{Spec}\,S^{-1}A \to \mathrm{Spec}\,A$ induced by the canonical map $A \to S^{-1}A$ is a homeomorphism onto its image. Recall from algebra the image of $\mathrm{Spec}\,S^{-1}A \to \mathrm{Spec}\,A$ is $\{\mathfrak{p} \in \mathrm{Spec}\,A \mid \mathfrak{p} \cap S = \varnothing\}$.

If we consider the case $S = \{f^n\}_{n \geqslant 0}$, we then obtain a homeomorphism $h : \operatorname{Spec} A_f \to D(f)$. For $D(g) \subseteq D(f)$, we have a map $A_f \to A_g$. Denote by $\overline{g}$ its image in A_f; we then have an isomorphism $(A_f)_{\overline{g}} \to A_g$ given by the universal property of localization. Then

$$\mathcal{O}_A(D(g)) = A_g \cong (A_f)_{\overline{g}} = \mathcal{O}_{\operatorname{Spec} A_f}(D(\overline{g})) \overset{1.49.(i)}{=} h_* \mathcal{O}_{\operatorname{Spec} A_f}(D(g))$$

This isomorphism is compatible with the restriction of principal open sets, so by 1.4 we obtain an isomorphism $\mathcal{O}_{\operatorname{Spec} A}\big|_{D(f)} \cong h_* \mathcal{O}_{\operatorname{Spec} A_f}$. In sum, the natural homomorphism $A \to A_f$ induces an open immersion $\operatorname{Spec} A_f \to \operatorname{Spec} A$ of **LRS** with image being the principal open set $D(f)$.

In fact, the above computation shows that $\widetilde{M}|_{D(f)} \cong h_* \widetilde{M_f}$, where in the right hand side, M_f is viewed as an A_f-module.

1.51 The morphism $D(f) \to \operatorname{Spec} A_f$ is in fact the same as the morphism coming from the bijection 1.44 and id_{A_f}. To be specific, we have a bijection

$$\operatorname{Hom}_{\mathbf{LRS}}(D(f), \operatorname{Spec} A_f) \longrightarrow \operatorname{Hom}_{\mathbf{Ring}}(A_f, A_f).$$

Then the morphism corresponds to the identity map $\mathrm{id} : A_f \to A_f$ is exactly the same as the one we construct in 1.50. This follows from the construction of the bijection 1.44.

1.52 The morphism $D(f) \to \operatorname{Spec} A_f$ is compatible with the restriction. Suppose $D(g) \subseteq D(f)$, so we have a map $M_f \to M_g$. Then we have a commutative diagram

$$\begin{tikzcd}
D(f) \arrow[r] & \operatorname{Spec} A_f \\
D(g) \arrow[r] \arrow[u] & \operatorname{Spec} A_g \arrow[u]
\end{tikzcd}$$

The map on topological spaces are clearly commutative. For sheaves, let $D(h) \subseteq D(g)$. Then we have a commutative diagram

$$\begin{tikzcd}
\widetilde{M}(D(h)) = M_h \arrow[d, equal] & (M_f)_{\overline{h}} \arrow[l] \arrow[d] \\
\widetilde{M}(D(h)) & (M_g)_{\overline{h}} \arrow[l]
\end{tikzcd}$$

each arrow given by the localization. This tells us the commutativity of the sheaf maps.

1.53 Let A be a ring and I an ideal. Then the morphism

$$\iota : \operatorname{Spec}(A/I) \to \operatorname{Spec} A$$

induced by the quotient map $A \to A/I$ is a closed immersion in the sense of (1.35). Indeed, the map on topological spaces is obviously a homeomorphism onto the closed subset $V(I)$ of $\operatorname{Spec} A$. To see it is a closed immersion, we use (1.36). For a prime $\mathfrak{p}$ of A containing I, the stalk map is

$$A_{\mathfrak{p}} \to (A/I)_{\mathfrak{p}/I}$$

given by $\dfrac{a}{s} \mapsto \dfrac{a \bmod I}{s \bmod I}$. By the universal property of localization, it induces a map $\varphi : A_{\mathfrak{p}}/I_{\mathfrak{p}} \to (A/I)_{\mathfrak{p}/I}$. On the other hand, since $I \subseteq I_{\mathfrak{p}} \cap A$, we have a map $A/I \to A_{\mathfrak{p}}/I_{\mathfrak{p}}$, and the universal property again gives $\psi : (A/I)_{\mathfrak{p}/I} \to A_{\mathfrak{p}}/I_{\mathfrak{p}}$. It is clear that φ and ψ are inverses to each other, and this show the stalk map is surjective.

Let us write down the image of ι. Following the proof of (1.36), we need to compute the kernel $\mathcal{I}$ of the sheaf map $\mathcal{O}_{\operatorname{Spec} A} \to \iota_* \mathcal{O}_{\operatorname{Spec}(A/I)}$. Let $f \in A$. On the open set $D(f)$, this map is $A_f \to (A/I)_{f \bmod I}$. By a similar argument as above, we see the kernel is I_f, so that $\mathcal{I}(D(f)) = I_f$. Therefore, $\mathcal{I}$ is the affine tilde $\widetilde{I}$, and thus the image of ι is $V(I)$ with sheaf $(\mathcal{O}_{\operatorname{Spec} A}/\widetilde{I})|_{V(I)} \cong \widetilde{A/I}|_{V(I)}$.

1.3.1 Quasi-coherent sheaves

1.54 Definition. Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{F} \in \operatorname{Mod}_{\mathcal{O}_X}$.

1. $\mathcal{F}$ is **quasi-coherent** if every point in X admits an open neighborhood U such that there exists an exact sequence
$$(\mathcal{O}_X|_U)^{\oplus I} \longrightarrow (\mathcal{O}_X|_U)^{\oplus J} \longrightarrow \mathcal{F}|_U \longrightarrow 0$$
for some index sets I and J, depending on x.
2. $\mathcal{F}$ is **coherent** if every point in X admits an open neighborhood U such that there exists an exact sequence
$$(\mathcal{O}_X|_U)^{n} \longrightarrow (\mathcal{O}_X|_U)^{m} \longrightarrow \mathcal{F}|_U \longrightarrow 0$$
for some integers n, m depending on x. That is, $\mathcal{F}$ is coherent if it is finitely presented over $\mathcal{O}_X$.
3. $\mathcal{F}$ is **locally free** if every point in X admits an open neighborhood U such that $\mathcal{F}|_U = (\mathcal{O}_X|_U)^{\oplus I}$ for some index set I, depending on x.
4. $\mathcal{F}$ is **locally free of rank** n if the index set I in the 3. can be chosen to be $\{1, \dots, n\}$ for any $x \in X$.
5. An **invertible sheaf** is a locally free sheaf of rank 1.

1.55 Lemma. Let A, B be rings and $f : \operatorname{Spec} B \to \operatorname{Spec} A$ be any morphism. We know $f = (\varphi^a, \varphi^\sharp)$ for a unique ring homomorphism $\varphi : A \to B$.

1. $f_* \widetilde{N} = \widetilde{N^{[\varphi]}}$ for any B-module N.
2. $f^* \widetilde{M} = \widetilde{B \otimes_A M}$ for any A-module M.

Proof.

1. Let $h \in A$. By (1.49), we have

$$f_*\widetilde{N}(D(h)) = \widetilde{N}(f^{-1}(D(h))) = \widetilde{N}(D(\varphi(h))) = N_{\varphi(h)} = (N^{[\varphi]})_h = \widetilde{N^{[\varphi]}}(D(h))$$

   Now 1. follows from (1.4).

2.

□

1.56 Theorem. Let $\mathcal{F}$ be a quasi-coherent sheaf on $X = \operatorname{Spec} A$. Then $\mathcal{F} \cong \widetilde{M}$ for some A-module M.

Proof. Let $U = D(f)$ be an open set such that there exist I, J and an exact sequence

$$\begin{tikzcd}
(\mathcal{O}_X|_U)^{\oplus I} \arrow[r] \arrow[d, equal] & (\mathcal{O}_X|_U)^{\oplus J} \arrow[r] \arrow[d, equal] & \mathcal{F}|_U \arrow[r] \arrow[d, equal] & 0 \\
\widetilde{A_f}^{\oplus I} \arrow[r] & \widetilde{A_f}^{\oplus J} \arrow[r] & \mathcal{F}|_U \arrow[r] & 0
\end{tikzcd}$$

where $\widetilde{\cdot}$ is taken from Mod_{A_f}. Consider the corresponding map $A_f^{\oplus I} \to A_f^{\oplus J}$ of A-modules and let K denote its cokernel. Then $\widetilde{K} \cong \mathcal{F}|_U$ as they are the cokernels of the same map.

Since X is compact, we can find $D(f_1), \ldots, D(f_n)$ such that $X = D(f_1) \cup \cdots \cup D(f_n)$ and

$$(D(f_i), \mathcal{F}|_{D(f_i)}) \cong (\operatorname{Spec} A_{f_i}, \widetilde{M_i})$$

for some $M_i \in \mathrm{Mod}_{A_{f_i}}$, where the isomorphism here is as in 1.50. For each open U, we have an exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(U) \arrow[r] \arrow[d, equal] & \prod_{i=1}^n \mathcal{F}(U \cap D(f_i)) \arrow[r] \arrow[d, equal] & \prod_{i,j=1}^n \mathcal{F}(U \cap D(f_i) \cap D(f_j)) \arrow[d, equal] \\
0 \arrow[r] & \mathcal{F}(U) \arrow[r] & \prod_{i=1}^n (\iota_{D(f_i)})_* \left(\mathcal{F}|_{D(f_i)}\right)(U) \arrow[r] & \prod_{i,j=1}^n (\iota_{D(f_i)\cap D(f_j)})_* \left(\mathcal{F}|_{D(f_i)\cap D(f_j)}\right)(U)
\end{tikzcd}$$

where for an open U, $\iota_U : U \to X$ denotes the inclusion, so we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \prod_{i=1}^n (\iota_{D(f_i)})_* \left(\mathcal{F}|_{D(f_i)}\right) \longrightarrow \prod_{i,j=1}^n (\iota_{D(f_i)\cap D(f_j)})_* \left(\mathcal{F}|_{D(f_i)\cap D(f_j)}\right)$$

Now Lemma 1.55 implies the latter two $\mathcal{O}_X$-modules arise from some A-modules, which in turns says that $\mathcal{F}$ arises from the A-module. □

1.57 From 1.45, for $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_A}$ we have a canonical morphism $\alpha : \mathcal{F}(\widetilde{\operatorname{Spec} A}) \to \mathcal{F}$ in $\mathbf{Mod}_{\mathcal{O}_A}$ induced by the identity map $\mathrm{id}_{\mathcal{F}(X)}$. The content of Theorem 1.56 can be made more formal. That is,

Lemma. $\alpha : \mathcal{F}(\widetilde{\operatorname{Spec} A}) \to \mathcal{F}$ is an isomorphism if $\mathcal{F}$ is $\mathcal{O}_A$-quasi-coherent.

Proof. Let $U = D(f)$ be a principal affine open subset of $\operatorname{Spec} A$. By construction, α_U is given by the localization. Precisely, we have a commutative triangle

$$\begin{tikzcd} & \mathcal{F}(X) \arrow[dl] \arrow[dr] & \\ \mathcal{F}(X)_f \arrow[rr, "\alpha_U"] & & \mathcal{F}(D(f)) \end{tikzcd}$$

each map being canonical. We claim α_U is an isomorphism. This amounts to show that

(i) If $s \in \mathcal{F}(X)$ restricts to 0 in $\mathcal{F}(D(f))$, then $f^n s = 0$ for some $n > 0$.

(ii) Given $t \in \mathcal{F}(D(f))$, there exists some n such that $f^n t \in \mathcal{F}(X)$.

Note that (i) and (ii) imply injectivity and surjectivity, respectively. We first show (i). Let $s \in \mathcal{F}(X)$ with $s|_{D(f)} = 0$. Let $D(f_i)$ be as in the proof of Theorem 1.56. Then

$$0 = s|_{D(f_i) \cap D(f)} \in \mathcal{F}(D(f_i) \cap D(f)) = \mathcal{F}|_{D(f_i)}(D(f_i f)) \cong (M_i)_f$$

(note that f_i acts invertibly on M_i). This means $f^{n_i}(s|_{D(f_i)}) = 0$ in M_i for some $n_i \geqslant 1$, and hence an $n \geqslant 1$ such that $(f^n s)|_{D(f_i)} = 0$ for each i. But this means $f^n s = 0$.

For (ii), let $t \in \mathcal{F}(D(f))$. Then $t|_{D(ff_i)} \in \mathcal{F}(D(ff_i)) \cong (M_i)_f$, so we can find $t_i \in M_i \cong \mathcal{F}(D(f_i))$ and $n_i \geqslant 1$ such that $f^{n_i} t = t_i$ on $D(ff_i)$. Again pick $n \gg 0$ so that $f^n t = t_i$ on $D(ff_i)$ for each i. On the intersection $D(f_i) \cap D(f_j) = D(f_i f_j)$, we have

$$t_i|_{D(ff_i f_j)} = (f^n t)|_{D(ff_i f_j)} = t_j|_{D(ff_i f_j)}$$

so by (i) (applied to $X = D(f_i f_j)$) we can find $m_{ij} \geqslant 1$ such that $f^{m_{ij}}(t_i - t_j) = 0$. Again we can take $m \gg 0$ such that $f^m(t_i - t_j) = 0$ for any i, j. Now $f^m t_i \in \mathcal{F}(D(f_i))$ patches to a global section $s \in \mathcal{F}(X)$ whose restriction to $D(f)$ is $f^{n+m} t$. $\square$

1.58 Corollary. Let X be a scheme and $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. TFAE :

1. $\mathcal{F}$ is quasi-coherent.
2. For any affine open $U \subseteq X$, $\mathcal{F}|_U \cong \widetilde{M}$ for some $\mathcal{O}_X(U)$-module M.
3. There exists an affine open cover $\mathcal{U}$ of X such that for any $U \in \mathcal{U}$, $\mathcal{F}|_U \cong \widetilde{M}$ for some $\mathcal{O}_X(U)$-module M.

1.59 Corollary. Let X be a scheme and $f : \mathcal{F} \to \mathcal{G}$ be a morphism in $\mathbf{Mod}_{\mathcal{O}_X}$. If $\mathcal{F}$ and $\mathcal{G}$ are $\mathcal{O}_X$-quasi-coherent modules, then so are $\ker f$, $\operatorname{coker} f$ and $\operatorname{Im} f$.

1.60 Let $f : X \to Y$ be a morphism of schemes. If $\mathcal{F}$ is an $\mathcal{O}_Y$-quasi-coherent module, then the pullback $f^*\mathcal{F}$ is also $\mathcal{O}_X$-quasi-coherent. Indeed, this follows from the definition and the fact that tensor product is right exact. For the pushforward, we have the following

Lemma. Let $\mathcal{F}$ be an $\mathcal{O}_X$-quasi-coherent module. Then $f_*\mathcal{F}$ is $\mathcal{O}_Y$-quasi-coherent if there exist

(i) an affine open cover $\mathcal{U} = \{U_i\}_{i\in I}$ of Y, and

(ii) for each $i \in I$, a finite affine open cover $\{X_{i,j}\}_{j\in J_i}$ of $f^{-1}(U_i)$ such that $X_{i,j} \cap X_{i,j'}$ is compact for every $j, j' \in J_i^2$.

Proof. For each $i \in I$, put $f_i : f^{-1}(U_i) \to U_i$ to be the map induced by f. Then $(f_*\mathcal{F})|_{U_i} = (f_i)_*(\mathcal{F}|_{f^{-1}(S_i)})$. Since quasi-coherence is a local property, we may then assume that $Y = \operatorname{Spec} A$ for some ring A and X is covered by finite affine opens X_j $(j \in J)$ with $X_j \cap X_{j'}$ compact for any $j, j' \in J$.

For any $j, j' \in J$ by compactness we may fix a finite affine open cover $X_{j,j',k}$ $(k \in K_{j,j'})$; let $\iota_j : X_j \to X$ and $\iota_{j,j',k} : X_{j,j',k} \to X$ be the inclusions. Then we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \prod_{j\in J}(\iota_j)_*\left(\mathcal{F}|_{X_j}\right) \longrightarrow \prod_{j,j'\in J}\prod_{k\in K_{j,j'}}(\iota_{j,j',k})_*\left(\mathcal{F}|_{X_{j,j',k}}\right)$$

Since f_* is left exact, we have the following exact sequence

$$0 \longrightarrow f_*\mathcal{F} \longrightarrow \prod_{j\in J}(f\circ\iota_j)_*\left(\mathcal{F}|_{X_j}\right) \longrightarrow \prod_{j,j'\in J}\prod_{k\in K_{j,j'}}(f\circ\iota_{j,j',k})_*\left(\mathcal{F}|_{X_{j,j',k}}\right)$$

Note that $f \circ \iota_j$ and $f \circ \iota_{j,j',k}$ are morphisms between affine schemes, so by Lemma 1.55, the last two sheaves above are $\mathcal{O}_Y$-quasi-coherent. Thus by Corollary 1.59 $f_*\mathcal{F}$ is $\mathcal{O}_Y$-coherent as well. □

1.61 Let X be a local ringed space. Recall in 1.27 for any $g \in \mathcal{O}_X(X)$ we defined the open subset X_g of X. Then by patching we can construct an inverse of $g|_{X_g}$, so we actually have $g|_{X_g} \in \mathcal{O}_X(X_g)^\times$. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then g acts on $\mathcal{F}(X_g)$ in an invertible manner, so the universal property of localization gives rise to a commutative triangle

$$\begin{tikzcd} & \mathcal{F}(X) \arrow[dl] \arrow[dr] & \\ \mathcal{F}(X)_g \arrow[rr] & & \mathcal{F}(X_g) \end{tikzcd}$$

Now let $f : X \to S$ be a morphism in **LRS** with S affine. Let $h \in \mathcal{O}_S(S)$ and $g = f_S^\sharp(h) \in \mathcal{O}_X(X)$. Suppose $f_*\mathcal{F}$ is $\mathcal{O}_S$-quasi-coherent, then the above map

$$\mathcal{F}(X)_g \longrightarrow \mathcal{F}(X_g)$$

is in fact an isomorphism. The first step to see this is the equality

$$X_g = f^{-1}(S_h).$$

For $x \in X_g$ if and only if $g_x \neq 0$ in $\kappa(x)$. But $g_x = f_x(h_{f(x)})$ and $f_x : \kappa(f(x)) \hookrightarrow \kappa(x)$, so this is equivalent to saying that $h_{f(x)} \neq 0$ in $\kappa(f(x))$, which is the same as saying that $x \in f^{-1}(S_h)$. Since S is affine, by 1.56 we know $f_*\mathcal{F} = \widetilde{M}$ for some $\mathcal{O}_S(S)$-module. We have a similar commutative triangle

$$\begin{tikzcd} & (f_*\mathcal{F})(S) \arrow[dl] \arrow[dr] & \\ (f_*\mathcal{F})(S)_h \arrow[rr] & & (f_*\mathcal{F})(S_h) \end{tikzcd}$$

In this time the lower horizontal arrow is an isomorphism, for they are in fact both M_h. But

$$(f_*\mathcal{F})(S)_h = \mathcal{F}(X)_h = \mathcal{F}(X)_g$$

where h acts on $\mathcal{F}(X)$ via $f_S^\sharp : \mathcal{O}_S(S) \to \mathcal{O}_X(X)$. Thus the horizontal arrow becomes the arrow that we focus on, and this proves it is an isomorphism.

1.62 We apply 1.61 to the case $\mathcal{F} = \mathcal{O}_A$ for some scheme X. We claim that X is affine if and only if $\alpha : X \to \operatorname{Spec} \mathcal{O}_X(X)$, which is induced from the identity map $\mathrm{id}_{\mathcal{O}_X(X)}$ and 1.44, is an isomorphism in **LRS**. Only "only if" part needs a proof. For the map on topological space, we easily see from the construction that there is a commuting square

$$\begin{tikzcd} X \arrow[r, "\alpha"] \arrow[d, "\wr"] & \operatorname{Spec} \mathcal{O}_X(X) \arrow[d, "\wr"] \\ \operatorname{Spec} A \arrow[r, equal] & \operatorname{Spec} A \end{tikzcd}$$

where A is a ring such that $X \cong \operatorname{Spec} A$ as **LRS**, so α is a homeomorphism. For the sheaf map, for each $g \in \mathcal{O}_X(X)$, $\alpha_*\mathcal{O}(D(g)) \leftarrow \mathcal{O}_{\operatorname{Spec}\mathcal{O}_X(X)}(D(g))$ is induced by the localization $\mathcal{O}_X(X_g) \leftarrow \mathcal{O}_X(X)_g$ (here we use $\alpha^{-1}(D(g)) = X_g$), and thus by 1.61 it is an isomorphism. Since the $D(g)$ form a basis of the topology on $\operatorname{Spec} \mathcal{O}_X(X)$, so $\alpha_*\mathcal{O}_X \leftarrow \mathcal{O}_{\operatorname{Spec}\mathcal{O}_X(X)}$ is an isomorphism.

1.63 For another application of 1.61, we introduce

Definition. A morphism $f : X \to S$ of schemes is called **affine** if it satisfies the following equivalent conditions.

(i) S admits an affine open cover $\mathcal{V}$ such that $f^{-1}(V)$ is affine for each $V \in \mathcal{V}$.

(ii) For any affine open $V \subseteq S$, $f^{-1}(V)$ is affine.

Clearly (ii) implies (i). To see (i) implies (ii), first note that if we write $\mathcal{V} = \{V_i\}_{i \in I}$ to be a cover satisfying (i) for f, then the preimage of principal affine open subsets of each V_i is also affine. This means f admits an open basis consisting of affine opens whose preimages under f are affine. This means

for any affine open V, the induced morphism $f^{-1}(V) \to V$ again satisfies (i). Thus we may replace S by $V \cong \operatorname{Spec} A$ and X by $f^{-1}(V)$, and we must show, in this case, that X is affine. By 1.44 we have a commuting square

$$\begin{tikzcd} X \arrow[r, "\alpha_X"] \arrow[d, "f"'] & \operatorname{Spec} \mathcal{O}_X(X) \arrow[d, "\operatorname{Spec}(f^\sharp_S)"] \\ S \arrow[r, "\alpha_S"'] & \operatorname{Spec} \mathcal{O}_S(S) \end{tikzcd}$$

where the horizontal arrows are canonical; note that α_S is an isomorphism by 1.62. As said above, we can find $\{h_i\}_{i \in I} \subseteq A$ such that $X_{g_i} = f^{-1}(D(h_i))$ is affine, where $g_i = f^\sharp_S(h_i)$. We see the $D(g_i) = (\operatorname{Spec} f^\sharp_S)^{-1}(D(h_i))$ covers $\operatorname{Spec} \mathcal{O}_X(X)$ and $(\alpha_X)^{-1}(D(g_i)) = f^{-1}(D(h_i)) = X_{g_i}$. Hence

$$(\alpha_X)^{-1}(D(g_i)) = X_{g_i} \xrightarrow[\sim]{\alpha_{X_{g_i}}} \operatorname{Spec} \mathcal{O}_X(X_{g_i}) \xrightarrow[\sim]{1.61} \operatorname{Spec} \mathcal{O}_X(X)_{g_i} \cong D(g_i)$$

Note that this is simply $\alpha_X|^{D(g_i)}_{X_{g_i}}$ by 1.50, so this proves α_X is an isomorphism (1.22).

1.64 Closed subscheme. Let X be a scheme, and let $\mathcal{I}$ be an ideal sheaf of $\mathcal{O}_X$. Following (1.32), we may construct a closed local-ringed subspace $(V(\mathcal{I}), (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})})$ of $(X, \mathcal{O}_X)$. Since we are discussing schemes, a natural question is whether $(V(\mathcal{I}), (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})})$ is itself a scheme.

Suppose $Y = (Y, \mathcal{O}_Y) = (V(\mathcal{I}), (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})})$ is a scheme. Pick $y \in Y$ and let U' be an affine open neighborhood of y in X. Since we assume Y is a scheme, we can find an affine open neighborhood V' of y in Y contained in $U' \cap Y$. The topology on Y is given by subspace topology, so we can find $f \in \mathcal{O}_X(U')$ with $D(f) \cap Y \subseteq V'$. Note that we can restrict f to $Y \cap U'$, and $D(f) \cap Y = (U' \cap Y)_{f|_{Y \cap U'}} = (V')_{f|_{Y \cap U'}}$ which is still affine in Y. In sum we find an affine neighborhood U of y in X such that $U \cap Y$ remains affine in Y. In other words, the closed immersion $Y \to X$ is an affine morphism (1.63).

Suppose $U = \operatorname{Spec} A$, $U \cap Y = \operatorname{Spec} B$ and let $\phi : A \to B$ be the homomorphism corresponding to $U \cap Y \subseteq U$. We content that $\mathcal{I}|_U \cong \widetilde{\ker \phi}$. In fact, for any $f \in A$,

$$\widetilde{\ker \phi}(U_f) = (\ker \phi)_f \cong \ker(A_f \to B_f) \cong \ker(\mathcal{O}_X(U_f) \to \mathcal{O}_Y(Y \cap U_f)) = \mathcal{I}(U_f).$$

The isomorphisms involved are functorial in U_f, so this proves our contention. In particular, this shows $\mathcal{I}$ is quasi-coherent at $y \in Y$. Furthermore, the inclusion $\mathcal{I}|_{X \setminus Y} \to \mathcal{O}_X|_{X \setminus Y}$ is an isomorphism, which shows that $\mathcal{I}$ is in fact quasi-coherent on the whole X.

Conversely, suppose $\mathcal{I}$ is quasi-coherent. Let U be an affine open set of X. Then $\mathcal{I}|_U$ is an quasi-coherent ideal sheaf of $\mathcal{O}_X|_U$. Clearly, $V(\mathcal{I}) \cap U = V(\mathcal{I}|_U)$, and

$$(\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})}|_{V(\mathcal{I}) \cap U} \cong (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I}) \cap U} \cong (\mathcal{O}_X|_U/\mathcal{I}|_U)|_{V(\mathcal{I}|_U)}.$$

Thus we can assume $X = \operatorname{Spec} A$ is affine. By (1.57), there is a commutative diagram with vertical arrows

being isomorphisms

$$\begin{tikzcd} \mathcal{I} \arrow[r] & \mathcal{O}_X \\ \widetilde{\mathcal{I}(X)} \arrow[u] \arrow[r] & \widetilde{A} \arrow[u] \end{tikzcd}$$

so that $\mathcal{O}_X/\mathcal{I} \cong \widetilde{A/\mathcal{I}(X)}$. A easy computation shows that the closed subset $V(\mathcal{I}(X))$ of $\operatorname{Spec} A$ coincides with $V(\mathcal{I})$. From (1.53) we can conclude that $(V(\mathcal{I}), (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})})$ is isomorphic to the affine scheme $\operatorname{Spec} A/\mathcal{I}(X)$. We summarize what we obtain and give some consequences in the following theorem.

1.65 Theorem. Let X be a scheme and $\mathcal{I}$ an ideal sheaf of X.

(i) Every closed immersion of schemes is an affine morphism.

(ii) The closed local-ringed subspace $(V(\mathcal{I}), (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})})$ is a scheme if and only if $\mathcal{I}$ is $\mathcal{O}_X$-quasi-coherent. In this case, we say $(V(\mathcal{I}), (\mathcal{O}_X/\mathcal{I})|_{V(\mathcal{I})})$ is a **closed subscheme** of X.

(iii) If $X = \operatorname{Spec} A$ is affine, then every closed subscheme of X has the form $(V(I), \widetilde{A/I}|_{V(I)})$ for some ideal I of A.

1.66 Let A be a ring. For an ideal I, we can equip the closed subset $Z := V(I)$ with a scheme structure, making it the image of the closed immersion $\operatorname{Spec} A/I \to \operatorname{Spec} A$. However, there are many possible closed subscheme structures on Z. For instance, if J is another ideal such that $\sqrt{J} = \sqrt{I}$, then $V(J) = Z$, so we can also equip Z with the scheme structure defined by J. Nevertheless, among all possible closed subscheme structures on Z we see the one determined by $\sqrt{I}$ is the most natural one. Moreover, any ideal with $V(J) = Z$ has a inclusion $J \subseteq \sqrt{I}$, which induces a closed immersion

$$(Z, \widetilde{A/\sqrt{I}}|_Z) \twoheadrightarrow (Z, \widetilde{A/J}|_Z)$$

of schemes.

For an ideal I, there is an ideal sheaf $I\mathcal{O}_{\operatorname{Spec} A}$ defined by $U \mapsto I\mathcal{O}_{\operatorname{Spec} A}(U)$. The ideal sheaf $\sqrt{I}\mathcal{O}_{\operatorname{Spec} A}$ has the following description :

$$\sqrt{I}\mathcal{O}_{\operatorname{Spec} A}(U) = \{s \in \mathcal{O}_{\operatorname{Spec} A}(U) \mid s|_x = 0 \text{ for all } x \in U \cap Z\}.$$

Indeed, this follows as $\sqrt{IA_f} = \sqrt{I}A_f$ for any $f \in A$.

1.3.2 Invertible sheaves

1.67 Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}^{\mathrm{pre}}_{\mathcal{O}_X}$. We define the **hom sheaf** as follows. For any open $U \subseteq X$, let

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})(U) = \operatorname{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$$

and for any $V \subseteq U$, the restriction is given by the obvious arrow $\mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U) \to \mathrm{Hom}_{\mathcal{O}_X|_V}(\mathcal{F}|_V, \mathcal{G}|_V)$. This makes $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ a presheaf of abelian groups on X. We also write $\mathcal{E}nd_{\mathcal{O}_X}\mathcal{F} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F})$.

When $\mathcal{G}$ is a sheaf, we easily check that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is a sheaf of abelian groups. Also, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is naturally, as in the case of modules, a (left) $\mathcal{E}nd_{\mathcal{O}_X}\mathcal{G}$-module and a right $\mathcal{E}nd_{\mathcal{O}_X}\mathcal{F}$-module.

1.68 Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{L}$ be an invertible sheaf (1.54). Define its **dual sheaf**

$$\mathcal{L}^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$$

For any index set I and $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$, one has

$$\mathrm{Hom}_{\mathcal{O}_X}((\mathcal{O}_X)^{\oplus I}, \mathcal{F}) \cong \mathcal{F}(X)^{\oplus I}.$$

the map being given by evaluation at $(e_i)_{i \in I}$, where $e_i \in \mathcal{O}_X(X)^{\oplus I}$ is $e_i = \delta_{ij}(j)$. In particular, $\mathcal{E}nd_{\mathcal{O}_X}\mathcal{O}_X = \mathcal{O}_X$, and $\mathcal{L}^\vee$ is thus an $\mathcal{O}_X$-module.

Lemma. Let $\mathcal{L}$ be as above.

(i) If $\mathcal{L}_1$ and $\mathcal{L}_2$ are invertible sheaves, so is $\mathcal{L}_1 \otimes_{\mathcal{O}_X} \mathcal{L}_2$, and $\mathcal{L}_1 \otimes_{\mathcal{O}_X} \mathcal{L}_2 \cong \mathcal{L}_2 \otimes_{\mathcal{O}_X} \mathcal{L}_1$.

(ii) $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{O}_X \cong \mathcal{L} \cong \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{L}$.

(iii) $\mathcal{L}^\vee$ is also an invertible sheaf.

(iv) $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$.

Proof.

(iv) Define $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \to \mathcal{O}_X$ as follows. We define it on $\mathcal{L} \otimes^p_{\mathcal{O}_X} \mathcal{L}^\vee$ (1.13). For any open U, define

$$\begin{aligned} \mathcal{L}(U) \otimes_{\mathcal{O}_X(U)} \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{L}|_U, \mathcal{O}_X|_U) &\longrightarrow \mathcal{O}_X(U) \\ (f, T) &\longmapsto T_U(f) \end{aligned} \tag{$\star$}$$

This defines a morphism of presheaves, so by passing to sheafification we obtain a morphism $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \to \mathcal{O}_X$ of sheaves. To show this is an isomorphism, it suffices to check that $(\star)$ is an isomorphism for arbitrary small open set. Let U be an open set such that $\mathcal{L}|_U \cong \mathcal{O}_X|_U$. Then we have a commutative diagram

$$\begin{tikzcd}
 & \mathcal{L}(U) \otimes_{\mathcal{O}_X(U)} \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{L}|_U, \mathcal{O}_X|_U) \arrow[r] \arrow[d, "\wr"'] & \mathcal{O}_X(U) & ab \\
(f, T) \arrow[d, mapsto] & \mathcal{O}_X(U) \otimes_{\mathcal{O}_X(U)} \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{O}_X|_U, \mathcal{O}_X|_U) \arrow[ur] \arrow[d, "\wr"'] & & \\
(f, T_U(1)) & \mathcal{O}_X(U) \otimes_{\mathcal{O}_X(U)} \mathcal{O}_X(U) \arrow[uur, bend right] & & \\
 & (a, b) \arrow[uuur, mapsto, bend left] & &
\end{tikzcd}$$

(this is commutative since T is an $\mathcal{O}_X|_U$-module homomorphism). The arrow $(a, b) \mapsto ab$ is an isomorphism, hence so is the arrow on the top.

$\square$

In particular, this shows the isomorphism classes of invertible sheaves on X form an abelian group (multiplication being tensor product). We denote by this group $\mathrm{Pic}(X)$, called the **Picard group** of the ringed space X.

1.69 Let $f : X \to Y$ be a morphism in **RS**. If $\mathcal{L}$ is an invertible sheaf on Y, then $f^*\mathcal{L}$ is an invertible sheaf on X. Indeed, if U is an open set in Y such that $\mathcal{L}|_U \cong \mathcal{O}_Y|_U$, we have

$$(f^*\mathcal{L})|_{f^{-1}(U)} \cong \left(f^{-1}\mathcal{L}\right)\big|_{f^{-1}(U)} \otimes_{(f^{-1}\mathcal{O}_Y)|_{f^{-1}(U)}} \mathcal{O}_X\big|_{f^{-1}(U)}$$

If we put $g = f|^U_{f^{-1}(U)}$, then the above is isomorphic to

$$g^{-1}(\mathcal{L}|_U) \otimes_{g^{-1}(\mathcal{O}_Y|_U)} \mathcal{O}_X|_{f^{-1}(U)} \cong \mathcal{O}_X|_{f^{-1}(U)}.$$

Thus f^* induces a map $\mathrm{Pic}(Y) \to \mathrm{Pic}(X)$. Moreover, since $\otimes$ commutes with $\varinjlim$, we see $f^* : \mathrm{Pic}(Y) \to \mathrm{Pic}(X)$ is a group homomorphism.

1.70 Let $(X, \mathcal{O}_X)$ be a ringed space. For an invertible sheaf $\mathcal{L}$ and $n \geqslant 0$, denote $\mathcal{L}^n = \mathcal{L}^{\otimes n}$, and for $n \leqslant 0$, denote $\mathcal{L}^{-n} = (\mathcal{L}^\vee)^{\otimes n}$. Then for $n, m \in \mathbb{Z}$, we have $\mathcal{L}^n \otimes \mathcal{L}^m \cong \mathcal{L}^{n+m}$.

For any open U, by definition we have a canonical bilinear map $\mathcal{L}^n(U) \times \mathcal{L}^m(U) \to (\mathcal{L}^n \otimes \mathcal{L}^m)(U) \cong \mathcal{L}^{n+m}(U)$. This makes

$$S(U) := \bigoplus_{n \in \mathbb{Z}} \mathcal{L}^n(U)$$

a graded ring, and $U \mapsto S(U)$ is a presheaf of graded rings on X.

Now suppose $(X, \mathcal{O}_X)$ is a local-ringed space. For each $g \in \mathcal{L}(X)$, define

$$X_g = X_g^{\mathcal{L}} = \{x \in X \mid g_x \notin \mathfrak{m}_x \mathcal{L}_x\}$$

If U is an open set such that $\mathcal{L}|_U \cong \mathcal{O}_X|_U$, and if $g|_U$ corresponds to $a \in \mathcal{O}_X(U)$, then $X_g \cap U = U_a$ in the sense of 1.27. In particular, this shows X_g is an open subset of X. If $U \subseteq X_g$ is an open set such that $\mathcal{L}|_U \cong \mathcal{O}_X|_U$, then $X_g \cap U = U_a$ and we can find a unique $h \in \mathcal{L}^\vee|_U$ such that under the isomorphism $\mathcal{L}^\vee|_U \cong \mathcal{O}_X|_U$, it corresponds to the inverse of a. This means $g|_U h = 1 \in \mathcal{O}_X(U)$; let us denote $h = (g|_U)^{-1}$. If V is another open set trivializing $\mathcal{L}$ and intersecting with U nontrivially, and $U \cap V$ also trivializes $\mathcal{L}$, then clearly $(g|_V)^{-1} = (g|_{U\cap V})^{-1} = (g|_U)^{-1}$, so the $(g|_U)^{-1}$ patch to a section $(g|_{X_f})^{-1} \in \mathcal{L}^\vee(X_f)$, satisfying $g(g|_{X_f})^{-1} = 1 \in \mathcal{O}_X(X_f)$. If there is no confusion, we simply put $g^{-1} = (g|_{X_f})^{-1}$.

For any $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$ and any open U, define

$$T(\mathcal{F}, \mathcal{L})(U) = \bigoplus_{n \in \mathbb{Z}} (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(U)$$

This is naturally a graded $S(U)$-module, and $T(\mathcal{F},\mathcal{L})$ is a presheaf of S-modules. Note that every $g \in \mathcal{L}^n(U)$ acts on $T(\mathcal{F},\mathcal{L})(U_g)$ invertibly, for the presence of $g^{-1} \in \mathcal{L}^{-n}(U_g)$. Thus we have a canonical commuting triangle

$$\begin{tikzcd} & T(\mathcal{F},\mathcal{L})(U) \arrow[dl, "\text{localization}"'] \arrow[dr, "\text{restriction}"] & \\ T(\mathcal{F},\mathcal{L})(U)_g \arrow[rr, "\alpha(g)"] & & T(\mathcal{F},\mathcal{L})(U_g) \end{tikzcd}$$

1.71 Lemma. Let X be a scheme, $\mathcal{L}$ an invertible sheaf and $\mathcal{F}$ a $\mathcal{O}_X$-quasi-coherent sheaf. Then for any $g \in \mathcal{L}^d(X)$, we have the following.

(1) If X is compact, then $\alpha(g)$ is injective.

(2) If X admits a finite affine open cover $\{U_i\}_{i\in I}$ such that $\mathcal{L}|_{U_i} \cong \mathcal{O}_X(U_i)$ and $U_i \cap U_j$ is compact for any $i, j \in I$, then $\alpha(g)$ is surjective.

Proof. This is a generalization of Lemma 1.57, and can be proved in a similar way. Here we use a slightly different way which is essentially the same. Cover X be a family of affine open subsets U_i $(i \in I)$ such that

$$(U_i,\ \mathcal{L}|_{U_i},\ \mathcal{F}|_{U_i},\ g|_{U_i}) \cong (\operatorname{Spec} A_i,\ \widetilde{A_i},\ \widetilde{M_i},\ a_i)$$

for some ring A_i, $a_i \in A_i$ and $M_i \in \mathbf{Mod}_{A_i}$. Consider the following commutative diagram

$$\begin{tikzcd} 0 \arrow[d] & & 0 \arrow[d] \\ T(\mathcal{F},\mathcal{L})(X)_g \arrow[rr, "\alpha(g)"] \arrow[d] & & T(\mathcal{F},\mathcal{L})(X_g) \arrow[d] \\ \prod_{i\in I} T(\mathcal{F},\mathcal{L})(U_i)_g \arrow[rr, "\prod_j \alpha(g|_{U_i})"] \arrow[d] & & \prod_{i\in I} T(\mathcal{F},\mathcal{L})(X_g \cap U_j) \arrow[d] \\ \prod_{i,j\in I} T(\mathcal{F},\mathcal{L})(U_i \cap U_j)_g \arrow[rr, "\prod_{i,j} \alpha(g|_{U_i\cap U_j})"] & & \prod_{i,j\in I} T(\mathcal{F},\mathcal{L})(X_g \cap U_i \cap U_j) \end{tikzcd} \tag{$\star$}$$

Note that the middle horizontal arrow is bijective, since there exists a commutative diagram

$$\begin{tikzcd} T(\mathcal{F},\mathcal{L})(U_i)_g \arrow[r, "\alpha(g|_{U_i})"] \arrow[d, "\wr"] & T(\mathcal{F},\mathcal{L})(X_g \cap U_i) \arrow[d, "\wr"] \\ T(\widetilde{M_i},\widetilde{A_i})(\operatorname{Spec} A_i)_a \arrow[r] \arrow[d, "\wr"] & T(\widetilde{M_i},\widetilde{A_i})(D(a_i)) \arrow[d, "\wr"] \\ \bigoplus_{n\in\mathbb{Z}} (M_i \otimes_{A_i} A_i^{\otimes n})_{a_i} \arrow[r, "\sim"] & \bigoplus_{n\in\mathbb{Z}} \left((M_i)_{a_i} \otimes_{(A_i)_{a_i}} (A_i)_{a_i}^{\otimes n} \right) \end{tikzcd}$$

For (1), since X is compact, we can assume $\#I < \infty$. In this case, there exists an exact sequence

$$0 \longrightarrow T(\mathcal{F}, \mathcal{L})(X) \longrightarrow \prod_{i \in I} T(\mathcal{F}, \mathcal{L})(U_i) \longrightarrow \prod_{i,j \in I} T(\mathcal{F}, \mathcal{L})(U_i \cap U_j)$$

To be specific, this is obtained by taking direct sum of exact sequences

$$0 \to (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(X) \to \prod_i (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(U_i) \to \prod_{i,j} (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(U_i \cap U_j).$$

The finiteness of I is used here for then the product and direct sum commute. In particular, this means the two vertical sequences in $(\star)$ are exact. In particular, the left middle arrow in $(\star)$ is injective, so $\alpha(g)$ is injective.

For (2), we can still assume $\#I < \infty$ so the above discussion is valid. Moreover, since $U_i \cap U_j$ is compact, the middle bottom arrow in $(\star)$ is injective by (1). As this stage, the surjectivity of $\alpha(g)$ follows from a simple diagram chasing. $\square$

1.3.3 Cartier divisors

1.72 Localization. Let $(X, \mathcal{O}_X)$ be a ringed space. A subsheaf $\mathcal{S}$ of $\mathcal{O}_X$ is called **multiplicatively closed** if $\mathcal{S}(U) \subseteq \mathcal{O}_X(U)$ is a submonoid for any $U \in \mathrm{Top}(X)$. In this situation, any restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$ naturally gives rises to a ring homomorphism $\mathcal{S}(U)^{-1}\mathcal{O}_X(U) \to \mathcal{S}(V)^{-1}\mathcal{O}_X(V)$. We denote by $\mathcal{S}^{-1}\mathcal{O}_X$ the sheafification of the presheaf $U \mapsto \mathcal{S}(U)^{-1}\mathcal{O}_X(U)$.

It is clear that for any $x \in X$, the stalk $\mathcal{S}_x$ is a multiplicatively closed subset of $\mathcal{O}_{X,x}$, so we can form the localization $\mathcal{S}_x^{-1}\mathcal{O}_{X,x}$. The natural map

$$\mathcal{S}(U)^{-1}\mathcal{O}_X(U) \to \mathcal{S}_x^{-1}\mathcal{O}_{X,x}$$

then induces a ring isomorphism $(\mathcal{S}^{-1}\mathcal{O}_X)_x \cong \mathcal{S}_x^{-1}\mathcal{O}_{X,x}$.

1.73 Total quotient sheaf. For a ring A, an element $r \in A$ is **regular** if the multiplication $A \to A$ by r is injective. Denote by A_{reg} the set of all regular elements in A, which is a submonoid of A. The localization $\mathrm{Frac}\, A := A_{\mathrm{reg}}^{-1}A$ is called the **total quotient ring** of A.

Let $(X, \mathcal{O}_X)$ be a ringed space. For any $U \in \mathrm{Top}(X)$, define

$$\mathcal{O}_{X,\mathrm{reg}}(U) := \{s \in \mathcal{O}_X(U) \mid s|_x \in (\mathcal{O}_{X,x})_{\mathrm{reg}} \text{ for any } x \in U\}.$$

It is clear from its local nature that $\mathcal{O}_{X,\mathrm{reg}}$ is a multiplicatively closed subsheaf of $\mathcal{O}_X$. The sheaf

$$\mathrm{Frac}\, \mathcal{O}_X := \mathcal{O}_{X,\mathrm{reg}}^{-1}\mathcal{O}_X$$

defined as in (1.72) is called the **total quotient sheaf** of $\mathcal{O}_X$.

1.74 Definition. Let X be a topological space.

1. X is called **irreducible** if it is nonempty and for any closed subspaces C_1, C_2 of X, if $C_1 \cup C_2 = X$, then either $C_1 = X$ or $C_2 = X$. Equivalently, X is irreducible if every nonempty open subspace is dense in X.
2. If Z is an irreducible closed subset of X, a **generic point** of Z is a point $\eta \in Z$ such that $Z = \overline{\{\eta\}}$.
3. It is easy to see that a union of two irreducibles is irreducible. An **irreducible component** of X is then a maximal irreducible subset in X with respect to inclusion.

It is easy to see that the closure of an irreducible subset is again irreducible. It follows that an irreducible component is a closed subset in X. Also, a continuous image of an irreducible subset is irreducible.

1.75 Lemma. Let X be an irreducible topological space. Then the constant presheaf defined in (1.9) is already a sheaf.

1.76 Lemma. Let X be a scheme and let Z be an irreducible closed subset of X. Then Z admits a unique generic point. In particular, this establishes a bijection

$$\begin{aligned} X &\longrightarrow \{\text{irreducible closed subsets of } X\} \\ x &\longmapsto \overline{\{x\}}. \end{aligned}$$

Proof. Assume first that $X = \operatorname{Spec} A$ is affine. Then $Z = V(I)$ for some ideal I, and by irreducibility $I = \mathfrak{p}$ is a prime ideal. Then $V(\mathfrak{p}) = \overline{\{\mathfrak{p}\}}$. Indeed, if $\mathfrak{p}' \in V(\mathfrak{p})$, then for any $V(J)$ containing $\mathfrak{p}$, we have $J \subseteq \mathfrak{p} \subseteq \mathfrak{p}'$, so $\mathfrak{p}' \in V(J)$. This proves $\mathfrak{p}' \in \overline{\{\mathfrak{p}\}}$, so $V(\mathfrak{p}) = \overline{\{\mathfrak{p}\}}$.

Now consider the general case. Let U be an affine open set that meets Z nontrivially. Then $U \cap Z$ is an irreducible closed subset of U, so by the preceding case $U \cap Z$ admits a generic point η. Since $U \cap Z$ is an open dense subset of Z, it follows that $\overline{\eta} = \overline{U \cap Z} = Z$. Let η' be another generic point of Z and pick an affine open neighborhood W of η' in X. Then $W \cap Z$ is a nontrivial open set in Z, so $\eta \in W \cap Z$ as $\{\eta\}$ is dense in Z. The uniqueness in the preceding case then implies that $\eta' = \eta$. $\square$

1.77 Definition. Let X be a scheme.

1. X is call **reduced** if $\mathcal{O}_{X,x}$ is a reduced ring for every $x \in X$.
2. X is called **irreducible** if the underlying topological space is irreducible.
3. X is called **integral** if it is both reduced and irreducible.

1.78 Lemma. Let A be a ring and let X be a scheme.

(i) $\operatorname{Spec} A$ is reduced if and only if A is reduced.

(ii) $\operatorname{Spec} A$ is irreducible if and only if the nilradical $\sqrt{0}$ of A is a prime.

(iii) $\operatorname{Spec} A$ is integral if and only if A is an integral domain.

(iv) X is reduced if and only if $\mathcal{O}_X(U)$ is a reduced ring for any open set U of X. If it is the case, $\mathcal{O}_X(U)$ is reduced for any open U in X.

(v) X is integral if and only if $\mathcal{O}_X(U)$ is an integral domain for any open set U of X.

Proof.

(i) Let $N = \sqrt{0}$ be the nilradical of A. It suffices to show $N_{\mathfrak{p}}$ is the nilradical of $A_{\mathfrak{p}}$ for any $\mathfrak{p} \in \operatorname{Spec} A$. One containment is clear. For the other way around, let $\frac{x}{s} \in A_{\mathfrak{p}}$ ($x \in A$, $s \in A - \mathfrak{p}$) be nilpotent. Then $x^n s' = 0$ for $n \gg 0$ and $s' \in A - \mathfrak{p}$, so $(xs')^n = 0$. Then we are done, as $\frac{x}{s} = \frac{xs'}{ss'}$.

(ii) First note that for $f \in A$, we have $D(f) = \varnothing$ if and only if $f \in \mathfrak{p}$ for any $\mathfrak{p} \in \operatorname{Spec} A$, i.e., $f \in \sqrt{0}$. Assume $\operatorname{Spec} A$ is irreducible. Let $f, g \in A$ with $fg \in \sqrt{0}$ and $f \notin \sqrt{0}$. Then $D(f) \cap D(g) = D(fg) = \varnothing$, and since $D(f)$ is dense (by irreducibility), this forces that $D(g) = \varnothing$, i.e., $g \in \sqrt{0}$. Conversely, assume $\sqrt{0}$ is a prime. Let $f \notin \sqrt{0}$. We must show $D(f)$ is dense. If $g \notin \sqrt{0}$, then $fg \notin \sqrt{0}$ as $\sqrt{0}$ is assumed to be a prime, which implies $D(f) \cap D(g) = D(fg) \neq \varnothing$. This proves that $D(f)$ is dense.

(iii) This follows from (i) and (ii).

(iv) This follows from the local natural of being reduced and (i). The last assertion follows from the sheaf axiom and the fact that a product of reduced rings is reduced.

(v) By (iv) we may assume in the first place that X is reduced. If X is irreducible, then every nonempty open set U of X is itself irreducible, and hence reduced. Replacing U by X, it suffices to prove $\mathcal{O}_X(X)$ is integral. Let $f, g \in \mathcal{O}_X(X)$ with $fg = 0$. Let U be an affine open set in X. Then $V(f|_U) \cup V(g|_U) = V((fg)|_U) = U$, so by irreducibility we have, say $V(f|_U) = U$. Let V be any other affine open set in X. Then $V \cap U \subseteq V(f|_V)$ with $V \cap U$ dense in V, so $V(f|_V) = V$, or $f|_V = 0$. This implies $f = 0$ in $\mathcal{O}_X(X)$. Conversely, assume that $\mathcal{O}_X(U)$ is an integral domain for any open in X. Let U, V be two nonempty open set in X. If $U \cap V = \varnothing$, then the sheaf axiom implies that

$$\mathcal{O}_X(U \cup V) \cong \mathcal{O}_X(U) \oplus \mathcal{O}_X(V)$$

as rings, which is a contradiction as this is not an integral domain.

□

1.79 Definition. Let X be a scheme. The total quotient sheaf of $\mathcal{O}_X$ is called the **sheaf of rational functions** of X, which we denote by $\mathcal{K}_X$.

1.80 Example : integral schemes. Let X be an integral scheme. By (1.76), X admits a unique generic point η. Let U be an affine open subset of X; note that $\eta \in U$. The natural map $\mathcal{O}_X(U) \to \mathcal{O}_{X,\eta}$ induces an isomorphism

$$\operatorname{Frac}(\mathcal{O}_X(U)) \cong \mathcal{O}_{X,\eta} \cong \kappa(\eta).$$

Indeed, if we write $U = \operatorname{Spec} A$, then η corresponds to the zero ideal, and $\operatorname{Frac} A$ is by definition $A_{(0)} \cong \mathcal{O}_{X,\eta} \cong \kappa(\eta)$. This implies that $\mathcal{K}_X$ is isomorphic to the constant sheaf $\underline{\kappa(\eta)}_X$. In this case, we call

$K(X) := \kappa(\eta)$ the **(rational) function field** of X. Since A is an integral domain, we see the affine tilde $\widetilde{\operatorname{Frac} A}$ is the constant sheaf $\underline{\operatorname{Frac} A}_U$; in particular, we see $\mathcal{K}_X$ is a quasi-coherent $\mathcal{O}_X$-algebra.

1.81 Definition. A morphism $f : X \to Y$ of schemes is called **dominant** if the image $f(X)$ is dense in Y.

1.82 Example. Suppose $f : X \to Y$ is a morphism between irreducible schemes. By (1.76), X (resp. Y) admits a unique generic point η_X (resp. η_Y). Then f is dominant if and only if $f(X)$ is dense in Y. Since $X = \overline{\{\eta_X\}}$ and f is continuous, the latter happens if and only if $\overline{\{f(\eta_X)\}} = Y$. By uniqueness, it is equivalent to saying that $f(\eta_X) = \eta_Y$, or equivalently $\eta_Y \in f(X)$.

Assume further that X, Y are integral. If $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, then $f : X \to Y$ is dominant if and only if the corresponding ring homomorphism $A \to B$ is injective. This is clear for the generic point corresponds to the zero ideal. It follows that for arbitrary integral schemes X, Y, the morphism $f : X \to Y$ is dominant if and only if the sheaf map $f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is injective. Note that since X is integral, by reducing to the affine cases, the restriction map $\mathcal{O}_X(U) \to \mathcal{O}_X(U')$ is injective. Thus $f^\sharp$ is injective implies that $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is injective for any open $V \subseteq Y$ and open $U \subseteq f^{-1}(V)$. It follows that if f is dominant, the sheaf map $f^\sharp$ *extends* to a field homomorphism $\kappa(\eta_Y) \to \kappa(\eta_X)$, or $\mathcal{K}_Y \to f_*\mathcal{K}_X$.

1.83 Definition.

1. A ring homomorphism $\varphi : A \to B$ is **flat** if $B^{[\varphi]}$ a flat A-module.
2. A morphism $f : X \to Y$ of schemes is **flat at** $x \in X$ if the induced stalk map $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is flat.
3. A morphism $f : X \to Y$ of schemes is **flat** if it is flat at every point of X.

1.84 In (1.82) we see if $f : X \to Y$ is a dominant morphism between integral schemes, then the sheaf map $\mathcal{O}_Y \to f_*\mathcal{O}_X$ extends to a morphism of $\mathcal{K}_Y \to f_*\mathcal{K}_X$. In particular, they fit into a commutative diagram

$$\begin{tikzcd} \mathcal{O}_Y \arrow[r] \arrow[d] & f_*\mathcal{O}_X \arrow[d] \\ \mathcal{K}_Y \arrow[r] & f_*\mathcal{K}_X. \end{tikzcd}$$

It is natural to ask whether such extension can be defined for other types of morphisms. It is the case for any flat morphism $f : X \to Y$ between arbitrary schemes, which we now prove. It suffices to show for any affine open $V = \operatorname{Spec} A$ in Y and affine open $U = \operatorname{Spec} B$ in X with $f(U) \subseteq V$, the corresponding homomorphism $\varphi : A = \mathcal{O}_Y(V) \to \mathcal{O}_X(U) = B$ sends regular elements to regular elements. Let $a \in A_{\text{reg}}$. The multiplication $B \to B$ by $\varphi(a)$ is obtained by tensoring with B the multiplication $A \to A$ by a. Since φ is flat, this shows $\varphi(a) \in B_{\text{reg}}$.

1.85 Definition. Let X be a scheme. A global section

$$\Gamma(X, \mathcal{K}_X^\times/\mathcal{O}_X^\times) = (\mathcal{K}_X^\times/\mathcal{O}_X^\times)(X)$$

of the quotient sheaf $\mathcal{K}_X^\times/\mathcal{O}_X^\times$ is called a **Cartier divisor** on X. Here for a sheaf of rings $\mathcal{A}$, we denote by $\mathcal{A}^\times$ the sheaf $U \mapsto \mathcal{A}(U)^\times$ of invertible elements.

Unwinding the definition (c.f. (1.6)), we see a Cartier divisor is represented by a collection of pairs $\{(f_U, U)\}_{U\in\mathcal{U}}$, where $\mathcal{U}$ is an open cover of X and $f \in \mathcal{K}(U)^\times$ such that for any $U, V \in \mathcal{U}$ with $U \cap V \neq \varnothing$, we have $f_U f_V^{-1} \in \mathcal{O}_X(U \cap V)^\times$. Two such collections $\{(f_U, U)\}_{U\in\mathcal{U}}$ and $\{(g_V, V)\}_{V\in\mathcal{V}}$ represent the same Cartier divisor if and only if $f_U g_V^{-1} \in \mathcal{O}_X(U \cap V)$ for any $U \in \mathcal{U}$, $V \in \mathcal{V}$ with $U \cap V \neq \varnothing$.

This gives an alternative way to define a Cartier divisor : it is a maximal collection of pairs $\{(f_U, U)\}_{U\in\mathcal{U}}$, in the sense that if (g, V) with $V \underset{\text{open}}{\subseteq} X$, $g \in \mathcal{K}(V)^\times$ satisfies $f_U g^{-1} \in \mathcal{O}_X(U \cap V)^\times$ for any $U \in \mathcal{U}$ with $U \cap V \neq \varnothing$, then in fact $(g, V) \in \{(f_U, U)\}_{U\in\mathcal{U}}$.

1.86 Cartier class group. Taking global sections of the exact sequence

$$0 \longrightarrow \mathcal{O}_X^\times \longrightarrow \mathcal{K}_X^\times \longrightarrow \mathcal{K}_X^\times/\mathcal{O}_X^\times \longrightarrow 0$$

we obtain

$$0 \longrightarrow \mathcal{O}_X(X)^\times \longrightarrow \mathcal{K}_X(X)^\times \longrightarrow (\mathcal{K}_X^\times/\mathcal{O}_X^\times)(X).$$

An element in the image of the last arrow is called a **principal Cartier divisor**. A Cartier divisor is called **effective** if it lies in the image of $\Gamma(X, \mathcal{O}_X \cap \mathcal{K}_X^\times) \to \Gamma(X, \mathcal{K}_X^\times/\mathcal{O}_X^\times)$. We write $D \geqslant 0$ if D is an effective Cartier divisor.

There is an obvious abelian group structure on $\Gamma(X, \mathcal{K}_X^\times/\mathcal{O}_X^\times)$, which we will write additively. We say two Cartier divisors D_1, D_2 on X is **linearly equivalent** if $D_1 - D_2$ is principal. Denote by $\mathrm{CaCl}(X)$ the group of **Cartier divisors** on X modulo the linear equivalence relation. Equivalently, it is the quotient group of $\Gamma(X, \mathcal{K}_X^\times/\mathcal{O}_X^\times)$ mod out the group of principal Cartier divisor. The group $\mathrm{CaCl}(X)$ is called the **Cartier class group** of X.

1.87 Invertible sheaf associated to a Cartier divisor. Let X be a scheme and let D be a Cartier divisor on X. Define the subsheaf $\mathcal{O}_X(D)$ of $\mathcal{K}_X$ as follows. Let $\{(f_U, U)\}_{U\in\mathcal{U}}$ be the maximal collection of compatible pairs that represents D. We take $\mathcal{O}_X(D)|_U = f_U^{-1}\mathcal{O}_X|_U \subseteq \mathcal{K}_X|_U$. This is well-defined, as on $U \cap V$, we have

$$(\mathcal{O}_X(D)|_U)|_{U\cap V} = f_U^{-1}(\mathcal{O}_X|_U)|_{U\cap V} = f_V^{-1}(f_V f_U^{-1})\mathcal{O}_X|_{U\cap V} = f_V^{-1}(\mathcal{O}_X|_V)|_{U\cap V} = (\mathcal{O}_X(D)|_V)|_{U\cap V}.$$

The sheaf $\mathcal{O}_X(D)$ is an invertible sheaf of X. If E be a principal Cartier divisor defined by $g \in \Gamma(X, \mathcal{K}_X^\times)$, then there is an isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D+E)$ given by multiplication by g^{-1}. Precisely, on any open U of X such that D is represented by f_U, the isomorphism is given by $s \mapsto s(g|_U)^{-1}$. This isomorphism is independent of the choice of g representing E, so this gives a well-defined map

$$\begin{aligned} \mathrm{CaCl}(X) &\longrightarrow \mathrm{Pic}(X) \\ D &\longmapsto \mathcal{O}_X(D). \end{aligned}$$

If A is a ring and $f, g \in A_{\mathrm{reg}}$, then the multiplication gives an isomorphism $fA \otimes_A gA \to fgA$ of A-modules. This shows $D \mapsto \mathcal{O}_X(D)$ is also a group homomorphism (c.f. (1.68)).

1.4 Functor of points

1.88 For a scheme X, we can associate it with a functor $h_X : \mathbf{Sch} \to \mathbf{Set}$ by $h_X(T) := \mathrm{Hom}_{\mathbf{Sch}}(T, X)$. It is the content of Yoneda's lemma that there is a functorial bijection

$$\begin{aligned} \mathrm{Hom}(h_X, h_Y) &\longrightarrow \mathrm{Hom}_{\mathbf{Sch}}(X, Y) \\ T &\longmapsto T_X(\mathrm{id}_X). \end{aligned}$$

From this we see that $X \cong Y$ if and only if $h_X \cong h_Y$ as functors. This more or less says that the scheme X is uniquely determined by the family $\{\mathrm{Hom}_{\mathbf{Sch}}(T, X)\}_{T \in \mathbf{Sch}}$. In fact, we have more.

1.89 Lemma. For a scheme X, let $h_X : \mathbf{Ring} \to \mathbf{Set}$ be the functor defined by $h_X(R) = \mathrm{Hom}_{\mathbf{Sch}}(\mathrm{Spec}\, R, X)$. There is a bijection

$$\mathrm{Hom}_{\mathbf{Sch}}(X, Y) \cong \mathrm{Hom}(h_X, h_Y)$$

natural in X, $Y \in \mathbf{Sch}$. This is an incarnation of the local nature of a scheme.

Proof. Let $F \in \mathrm{Hom}(h_X, h_Y)$. Let $\mathcal{U}$ be an affine open cover of X. For each $U \in \mathcal{U}$, let $\iota_U : U = \mathrm{Spec}\, A_U \to X$ be the inclusion. Then $\iota_U \in h_X(A_U)$, so applying F_U we obtain $f_U = F_U(\iota_U) \in h_Y(A_U)$. We claim the $f_U : U \to Y$ glue to a global morphism $X \to Y$. Let $U, V \in \mathcal{U}$, and let $W = \mathrm{Spec}\, B \subseteq U \cap V$ be an affine open set. Put $\alpha : W = \mathrm{Spec}\, B \to X$ be the inclusion. Applying F to the diagram

$$\begin{tikzcd} & U = \mathrm{Spec}\, A_U \arrow[dr, "\iota_U"] & \\ W = \mathrm{Spec}\, B \arrow[ur, "\text{inclusion}"] \arrow[rr, "\alpha"] \arrow[dr, "\text{inclusion}"'] & & X \\ & V = \mathrm{Spec}\, A_V \arrow[ur, "\iota_V"'] & \end{tikzcd}$$

we obtain

$$\begin{tikzcd} & U = \mathrm{Spec}\, A_U \arrow[dr, "f_U"] & \\ W = \mathrm{Spec}\, B \arrow[ur, "\text{inclusion}"] \arrow[rr, "F_W(\alpha)"] \arrow[dr, "\text{inclusion}"'] & & Y \\ & V = \mathrm{Spec}\, A_V \arrow[ur, "f_V"'] & \end{tikzcd}$$

In particular, this shows $f_U|_W = F_W(\alpha) = f_V|_W$. Hence we obtain a well-defined morphism $f : X \to Y$ extending the f_U's. It is easy to see the resulting f is independent of the choice of $\mathcal{U}$; in fact, we can use all the affine open sets in X in the first place. This defines a map $\mathrm{Hom}(h_X, h_Y) \to \mathrm{Hom}_{\mathbf{Sch}}(X, Y)$. The map the other way around is defined by composition.

We claim they are mutually inverse. One direction is clear. For the other, let $F \in \mathrm{Hom}(h_X, h_Y)$ and let f be the corresponding morphism $X \to Y$. Let R be a ring and $g \in h_X(R)$. Let U be an affine open set in X and pick $W \subseteq \operatorname{Spec} R$ an affine open set such that $g(W) \subseteq U$. Then

$$(f \circ g)|_W = F_V(\iota_V) \circ g|_W^V = F_W(\iota_V \circ g|_W^V) = F_W(g|_W) = F_U(g)|_W,$$

where the second equality follows from naturality of F. Hence $f \circ (\cdot) = F$. $\square$

1.90 Definition. Let S be a scheme. A scheme X together with a morphism $X \to S$, called the **structure morphism**, is called a **S-scheme / scheme over** S. A **morphism between S-schemes** is a morphism of schemes that commutes with the structure morphisms to S. Such a morphism is called an **S-morphism**. Denote by $\mathbf{Sch}_S$ the category of S-schemes. If $S = \operatorname{Spec} A$, we write $\mathrm{Sch}_S = \mathrm{Sch}_A$, and simply call an S-morphism as an A-morphism.

1.91 Example. The ring $\mathbb{Z}$ is the initial object in **Ring**, so by (1.44) the affine scheme $\operatorname{Spec} \mathbb{Z}$ is the final object in **Sch**. In particular, the categories Sch and $\mathrm{Sch}_{\mathbb{Z}}$ are isomorphic.

1.92 Let A be a ring. Using the forgetful functors $\mathbf{Alg}_A \to \mathbf{Ring}$ and $\mathbf{AffSch}_A \to \mathbf{AffSch}$, we deduce from the anti-equivalence of categories $\operatorname{Spec} : \mathbf{Ring} \to \mathbf{AffSch}$ in (1.47) gives rise to the anti-equivalence of categories

$$\begin{aligned} \operatorname{Spec} : \mathbf{Alg}_A &\longrightarrow \mathbf{AffSch}_A \\ R &\longmapsto \operatorname{Spec} R \end{aligned}$$

1.93 Let S be a scheme. Let X be an S-scheme and denote by $f : X \to S$ the structure morphism of X. For an open set U of S, we sometimes use $X|_U$ to denote the U-scheme $f|^U_{f^{-1}(U)} : f^{-1}(U) \to U$ so as to prevent ourselves from cumbersome notation.

1.94 Let S be a scheme and Y, X be two S-schemes. Denote by $f : Y \to S$ and $g : X \to S$ the structure morphisms. Consider the commutative diagram

$$\begin{tikzcd} \mathrm{Hom}_{\mathbf{Sch}}(Y, X) \arrow[r] \arrow[d, "g\circ(-)"] & \mathrm{Hom}_{\mathbf{Ring}}(\mathcal{O}_X(X), \mathcal{O}_Y(Y)) \arrow[d, "(-)\circ\theta_S"] \\ \mathrm{Hom}_{\mathbf{Sch}}(Y, S) \arrow[r] & \mathrm{Hom}_{\mathbf{Ring}}(\mathcal{O}_S(S), \mathcal{O}_Y(Y)) \end{tikzcd}$$

where $\theta : \mathcal{O}_S \to f_*\mathcal{O}_X$ is the sheaf map of f. Taking the preimage of f along the vertical maps we obtain

$$\mathrm{Hom}_{\mathbf{Sch}_S}(Y, X) \longrightarrow \mathrm{Hom}_{\mathbf{Alg}_{\mathcal{O}_S}}(\mathcal{O}_X(X), \mathcal{O}_Y(Y)).$$

By (1.44), this is an isomorphism if S and X are affine.

1.95 Functor of points. Let S be a scheme and let X be an S-scheme. For another S-scheme T, put

$$X(T) := \mathrm{Hom}_{\mathbf{Sch}_S}(T, X)$$

and call its element a T**-valued point** of $X_{/S}$. If $T = \operatorname{Spec} R$ (where R is an S-algebra), we call it an R**-valued point**, and simply write $X(T) =: X(R)$.

1.96 Example. Let $f_1, \dots, f_m \in \mathbb{Z}[x_1, \dots, x_n]$ and let $X = \operatorname{Spec} \mathbb{Z}[x_1, \dots, x_n]/(f_1, \dots, f_m)$. Let R be a ring. Then there is a natural bijection

$$X(R) \xrightarrow{\sim} \{a = (a_1, \dots, a_n) \in R^n \mid f_1(a) = \cdots = f_m(a) = 0\}.$$

Indeed, by (1.44) we have

$$\operatorname{Hom}_{\mathbf{Sch}}(\operatorname{Spec} R, X) \cong \operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x_1, \dots, x_n]/(f_1, \dots, f_m), R)$$

There is a canonical inclusion

$$\operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x_1, \dots, x_n]/(f_1, \dots, f_m), R) \xrightarrow{\circ p} \operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Z}[x_1, \dots, x_n], R) \cong R^n$$

where $p : \mathbb{Z}[x_1, \dots, x_n] \to \mathbb{Z}[x_1, \dots, x_n]/(f_1, \dots, f_m)$ is the quotient map. The last isomorphism is the evaluation at $x_1, \dots, x_n$. The image of the first set in R^n is then the common zeros of $f_1, \dots, f_m$ in R^n.

Inspired by this bijection, we can understand the R-valued points of a scheme as a generalization of solving equations in the ring R. In this way the functor of points is quite a natural concept.

1.97 Example. Let k be a field and V a finite dimensional vector space over k. Consider the affine scheme $\overline{V} := \operatorname{Spec} \operatorname{Sym} V^\vee$, where $\operatorname{Sym} V^\vee$ means the symmetric algebra of $V^\vee$. By the universal property of symmetric algebras, we have functorial bijections

$$\overline{V}(k) = \operatorname{Hom}_{\mathbf{Sch}_k}(\operatorname{Spec} k, \overline{V}) \cong \operatorname{Hom}_{\mathbf{Alg}_k}(\operatorname{Sym} V^\vee, k) \cong \operatorname{Hom}_k(V^\vee, k) = (V^\vee)^\vee \cong V$$

Hence we can always think of a finite dimensional vector space V over k as the k-valued point of the affine space $\overline{V} \cong \mathbb{A}_k^{\dim_k V}$ (non-canonically) over k.

1.98 Denote by **LocRing** the category of local rings. By definition, a morphism of local rings is a local homomorphism (1.25). Let A be a local ring and $\mathfrak{m}$ be the unique maximal ideal. We show that there is a bijection

$$\operatorname{Hom}_{\mathbf{Sch}}(\operatorname{Spec} A, X) \longrightarrow \bigsqcup_{x \in X} \operatorname{Hom}_{\mathbf{LocRing}}(\mathcal{O}_{X,x}, A).$$

First, let $f : \operatorname{Spec} A \to X$ be a morphism of scheme and put $x = f(\mathfrak{m})$. Choose an affine open neighborhood $U = \operatorname{Spec} B$ of x in X, and let $\mathfrak{p} \in \operatorname{Spec} B$ be the point corresponding to x. Then $f(\operatorname{Spec} A) \subseteq U$. Indeed, $f^{-1}(U)$ is an open set in $\operatorname{Spec} A$ containing $\mathfrak{m}$, and there exists $h \in A$ such that $f^{-1}(U) \supseteq D(h) \ni \mathfrak{m}$, which implies $h \notin \mathfrak{m}$ and, thus, $D(h) = \operatorname{Spec} A$. In fact, this shows that $f(\operatorname{Spec} A)$ is contained in any open neighborhood of x in X. The map f then factors through $f|^U : \operatorname{Spec} A \to U = \operatorname{Spec} B$, so it gives a map $(f|^U)^\sharp : B \to A$ with $((f|^U)^\sharp)^{-1}(\mathfrak{m}) = \mathfrak{p}$. This means it factors through the localization $B \to B_\mathfrak{p}$, giving a

commutative triangle

$$\begin{tikzcd} B \arrow[rr, "(f|^U)^\sharp"] \arrow[dr] & & A \\ & B_\mathfrak{p} \arrow[ur] & \end{tikzcd}$$

Since $B_\mathfrak{p} \cong \mathcal{O}_{X,x}$, we have a homomorphism $\theta_f : \mathcal{O}_{X,x} \to A$. Taking spec gives a morphism $\widetilde{\theta_f} : \operatorname{Spec} A \to \operatorname{Spec} \mathcal{O}_{X,x}$. At the same time, we obtain another morphism $\varphi_x : \operatorname{Spec} \mathcal{O}_{X,x} \to \operatorname{Spec} B = U \to X$.

The constructions of $\widetilde{\theta_f}$ and φ_x depend on the choice of U. Nevertheless, the morphisms themselves do not, which we now prove. Let $V = \operatorname{Spec} C$ be another affine open neighborhood of x and let $W = \operatorname{Spec} D \subseteq U \cap V$ be still another affine open neighborhood of x. To distinguish, let $\mathfrak{p}_W, \mathfrak{p}_U, \mathfrak{p}_V$ denote the corresponding primes of x in D, B, C. Now we have ring homomorphism $B, C \to D \to A$, and since $\mathfrak{m} \mapsto \mathfrak{p}_W \mapsto \mathfrak{p}_V, \mathfrak{p}_U$, the natural diagram

$$\begin{tikzcd} & & A & & \\ & & & D_{\mathfrak{p}_W} \arrow[ul, "\sim"'] & \\ C_{\mathfrak{p}_V} \arrow[uurr, "\sim"] \arrow[urrr] & & D \arrow[uu] \arrow[ur] & & B_{\mathfrak{p}_U} \arrow[ul, "\sim"] \arrow[uull, bend right] \\ & C \arrow[ul] \arrow[ur] & & B \arrow[ul] \arrow[ur] & \end{tikzcd}$$

commutes everywhere. Taking spec gives

$$\begin{tikzcd} & & \operatorname{Spec} A \arrow[d] & & \\ & & \operatorname{Spec} D_{\mathfrak{p}_W} \arrow[dl, "\sim"'] \arrow[d, "\wr"] \arrow[dr, "\sim"] \arrow[ddl, bend left, "\sim"] & & \\ & \operatorname{Spec} C_{\mathfrak{p}_V} \arrow[dl] & \operatorname{Spec} \mathcal{O}_{X,x} \arrow[l, "\sim"] \arrow[r, "\sim"] & \operatorname{Spec} B_{\mathfrak{p}_U} \arrow[dr] & \\ & & \operatorname{Spec} D \arrow[dl] \arrow[dr] \arrow[dd] & & \\ \operatorname{Spec} C \arrow[dr] & & & & \operatorname{Spec} B \arrow[dl] \\ & & X & & \end{tikzcd}$$

This tells the independence. Hence $f : \operatorname{Spec} A \to X$ admits a *canonical* factorization

$$\begin{tikzcd} \operatorname{Spec} A \arrow[rr, "f"] \arrow[dr, "\widetilde{\theta_f}"'] & & X \\ & \operatorname{Spec} \mathcal{O}_{X,x} \arrow[ur, "\varphi_x"'] & \end{tikzcd}$$

Recall that $\widetilde{\theta}_f$ corresponds to the homomorphism $\theta_f : \mathcal{O}_{X,x} \to A$.

Finally we can define the maps

$$\begin{aligned} \mathrm{Hom}_{\mathbf{Sch}}(\operatorname{Spec} A, X) &\longrightarrow \bigsqcup_{x \in X} \mathrm{Hom}_{\mathbf{LocRing}}(\mathcal{O}_{X,x}, A) \\ f &\longmapsto \theta_f : \mathcal{O}_{X,x} \to A \text{ with } x = f(\mathfrak{m}) \\ \varphi_x \circ \operatorname{Spec} \theta &\longleftarrow\!\shortmid \theta : \mathcal{O}_{X,x} \to A \end{aligned}$$

By checking on any affine open neighborhood of x, we easily see that these are mutually inverse.

1.99 Example. Let X be a scheme over a field k. Then there is a bijection

$$\begin{aligned} X(k) = \mathrm{Hom}_{\mathbf{Sch}_k}(\operatorname{Spec} k, X) &\xrightarrow{\sim} \{x \in X \mid \kappa(x) = k\} \\ f &\longmapsto f(\mathrm{pt}) \end{aligned}$$

Here pt denotes the unique point in $\operatorname{Spec} k$. Note that a scheme over k is the same as a scheme that is a local-ringed space over k (1.26). Each $\kappa(x)$ is naturally a k-algebra, so the equality $\kappa(x) = k$ makes sense.

More generally, if K/k is a field extension, then there is a bijection

$$X(K) \xrightarrow{\sim} \bigsqcup_{x \in X} \mathrm{Hom}_{\mathbf{Alg}_k}(\kappa(x), K).$$

This follows from (1.98) : the kernel of a local k-homomorphism $\mathcal{O}_{X,x} \to K$ is precisely its unique maximal ideal, so such a map is the same as a k-algebra homomorphism $\kappa(x) \to K$.

1.100 Galois action. Let X be a scheme over a field k. Let K/k be a Galois extension with Galois group G. If $s : \operatorname{Spec} K \to X$ and $\sigma \in G$, it is clear that $s \circ \operatorname{Spec} \sigma$ is again a morphism of k-scheme, so that $s \circ \operatorname{Spec} \sigma \in X(K)$. This defines a (left) G-action on $X(K)$.

Consider the (set-theoretic) map

$$\begin{aligned} \phi : X(K) &\longrightarrow X \\ s &\longmapsto s(\operatorname{Spec} K). \end{aligned}$$

By (1.99), the fibre of each point $x \in X$ is identified as

$$\phi^{-1}(x) \cong \mathrm{Hom}_{\mathbf{Alg}_k}(\kappa(x), K).$$

It follows from the construction this bijection is G-equivariant, where G acts on $\mathrm{Hom}_{\mathbf{Alg}_k}(\kappa(x), K)$ naturally. As K/k is Galois, it follows that the G-action on $\phi^{-1}(x)$ is transitive, and hence ϕ induces an injective map

$$G \backslash X(K) \hookrightarrow X.$$

On the other hand, since the bijection

$$X(K) \xrightarrow{\sim} \bigsqcup_{x \in X} \mathrm{Hom}_{\mathbf{Alg}_k}(\kappa(x), K).$$

is also G-equivariant, taking invariants gives

$$X(K)^G \xrightarrow{\sim} \bigsqcup_{x \in X} \mathrm{Hom}_{\mathbf{Alg}_k}(\kappa(x), K)^G = \bigsqcup_{x \in X} \mathrm{Hom}_{\mathbf{Alg}_k}(\kappa(x), k) \xrightarrow{\sim} X(k).$$

To summarize,

Lemma. Let X be a scheme over a field k and let K/k be a Galois extension with Galois group G.

(i) The natural map $X(K) \to X$ induces an injection $G\backslash X(K) \to X$.

(ii) The injection $X(k) \to X(K)$ gives a bijection $X(k) \cong X(K)^G$.

1.4.1 Tangent spaces

1.101 Definition. Let $(X, \mathcal{O}_X)$ be a local-ringed space. For a point $x \in X$, the **(Zariski) cotangent space** $T^*_{X,x}$ of X at the point x is defined as the quotient $\mathfrak{m}_{X,x}/\mathfrak{m}^2_{X,x}$. The **(Zariski) tangent space** $T_{X,x}$ is defined as the linear dual

$$T_{X,x} := \mathrm{Hom}_{\kappa(x)}(\mathfrak{m}_{X,x}/\mathfrak{m}^2_{X,x}, \kappa(x))$$

of the cotangent space $T^*_{X,x}$.

1.102 Example. Let M be a smooth manifold. Write $C^\infty := C^\infty_M$ for the sheaf of real-valued smooth functions on M. Then (M, C^∞) is a local-ringed space over $\mathbb{R}$. For each $p \in M$, the stalk C^∞_p is the collection of smooth functions defined near p. The unique maximal ideal $\mathfrak{m}_p$ of C^∞_p is the kernel of the evaluation map

$$\begin{aligned} C^\infty_p &\longrightarrow \mathbb{R} \\ f &\longmapsto f(p) \end{aligned}$$

Denote by $D_p = \mathrm{Der}(C^\infty_p)$ the space of all point derivations on C^∞_p. Then there is an isomorphism

$$\begin{aligned} D_p &\longrightarrow \mathrm{Hom}_{\mathbb{R}}(\mathfrak{m}_p/\mathfrak{m}^2_p, \mathbb{R}) \\ X &\longmapsto f \mapsto X(f). \end{aligned}$$

This is well-defined, as for $f, g \in \mathfrak{m}_p$, clearly $X(fg) = X(f)g(p) + f(p)X(g) = 0$. For $f \in C^\infty_p$, $f - f(p)$ lies in $\mathfrak{m}_p$, so $X(f) = X(f - f(p)) + X(f(p)) = X(f - f(p))$; this shows the injectivity. For $T \in \mathrm{Hom}_{\mathbb{R}}(\mathfrak{m}_p/\mathfrak{m}^2_p, \mathbb{R})$, define $X_T : C^\infty_p \to \mathbb{R}$ by $X_T(f) = T(f - f(p))$. This is clearly linear, and for $f, g \in C^\infty_p$,

$$\begin{aligned} X_T(fg) &= T(fg - f(p)g(p)) \\ &= T((f - f(p))(g - g(p)) + (f - f(p))g(p) + f(p)(g - g(p))) \\ &= T(f - f(p))g(p) + f(p)T(g - g(p)) = X_T(f)g(p) + f(p)X_T(g) \end{aligned}$$

showing that $X_T \in D_p$, whence the surjectivity. Now since M is smooth, D_p is naturally identified as the tangent space of M at p. This somewhat gives the intuition that why (co)tangent spaces of local-ringed spaces are defined so.

1.103 Let X be a scheme over a field k.

1.5 Fibre products

1.104 Let S be a scheme. Let $f : X \to S$ and $g : Y \to S$ be two schemes over S. A **fibre product** of f and g is a scheme $X \times_S Y$ together with two morphisms $X \times_S Y \to X$ and $X \times_S Y \to Y$ that represents the functor

$$T \mapsto \mathrm{Hom}_{\mathbf{Sch}}(T, X) \times_{\mathrm{Hom}_{\mathbf{Sch}}(T,S)} \mathrm{Hom}_{\mathbf{Sch}}(T, Y).$$

By universal property nonsense, if a fibre product exists, it is unique up to a unique isomorphism.

1.105 Let C be a ring and let A, B be two C-algebras. The tensor product ring $A \otimes_C B$ is the fibre coproduct of $A \leftarrow C \to B$ in the category of rings. This means we have a bijection

$$\mathrm{Hom}_{\mathbf{Ring}}(A \otimes_C B, D) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Ring}}(A, D) \times_{\mathrm{Hom}_{\mathbf{Ring}}(C,D)} \mathrm{Hom}_{\mathbf{Ring}}(B, D)$$

functorial in $D \in \mathbf{Ring}$. It then follows from (1.44) that $\operatorname{Spec} A \otimes_C B$ is the fibre product of $\operatorname{Spec} A$ and $\operatorname{Spec} B$ over $\operatorname{Spec} C$ in the category of schemes (in fact, also in **LRS**).

1.106 Base change and functor of points. Let S be a scheme and X, Y be two S-schemes. By (1.104), for any S-scheme Z, there is a canonical bijection

$$\mathrm{Hom}_{\mathrm{Sch}_S}(Z, X \times_S Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Sch}_S}(Z, X) \times \mathrm{Hom}_{\mathrm{Sch}_S}(Z, Y)$$

By taking $Z = Y$, we obtain

$$\begin{tikzcd} \mathrm{Hom}_{\mathrm{Sch}_S}(Y, X \times_S Y) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathrm{Sch}_S}(Y, X) \times \mathrm{Hom}_{\mathrm{Sch}_S}(Y, Y) \\ & \mathrm{Hom}_{\mathrm{Sch}_S}(Y, X) \times \{\mathrm{id}_Y\}. \arrow[u, hook] \end{tikzcd}$$

Taking preimage of the subset $\mathrm{Hom}_{\mathrm{Sch}_S}(Y, X) \times \{\mathrm{id}_Y\}$, we see this induces a bijection

$$\mathrm{Hom}_{\mathrm{Sch}_Y}(Y, X \times_S Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Sch}_S}(Y, X)$$

1.5.1 Scheme-theoretic fibre

1.107 In (1.50) we see that if A is a ring and S is a submonoid of A, the natural map $f : \operatorname{Spec} S^{-1}A \to \operatorname{Spec} A$ is a homeomorphism onto its image. In fact, this is also a topological embedding (1.28). To start

with, put

$$Y = \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \cap S = \varnothing\}$$

to be the image. By the universal property (1.28), we then have a commutative triangle

$$\begin{tikzcd}
(\operatorname{Spec} S^{-1}A, \mathcal{O}_{\operatorname{Spec} S^{-1}A}) \arrow[rr, "f"] \arrow[dr, "f|^Y"'] & & (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}) \\
& (Y, \mathcal{O}_{\operatorname{Spec} A}|_Y) \arrow[ur] &
\end{tikzcd}$$

We need to show the morphism on the left is an isomorphism in **LRS**, and it remains to show the sheaf map

$$\mathcal{O}_{\operatorname{Spec} A}|_Y \to (f|^Y)_* \mathcal{O}_{\operatorname{Spec} S^{-1}A}$$

is an isomorphism. This is clear.

1.108 Let us talk a little more about the morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ constructed in (1.98). Pick an affine open subset $U = \operatorname{Spec} A$ that covers the image of $\operatorname{Spec} \mathcal{O}_{X,x}$. Then the morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to \operatorname{Spec} A$ corresponds to the ring homomorphism $A \to A_{\mathfrak{p}} \cong \mathcal{O}_{X,x}$, where $\mathfrak{p}$ is the prime corresponding to the point x. By (1.107), the morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to \operatorname{Spec} A$ is then a topological embedding in **LRS**. Hence the morphism $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ identifies $\operatorname{Spec} \mathcal{O}_{X,x}$ with $(Y, \mathcal{O}_X|_Y)$, where

$$Y = \{x' \in X \mid x \in \overline{\{x'\}}\}$$

1.109 Definition. Let $(A, \mathfrak{m})$ be a local ring. An ideal $I \trianglelefteq A$ is called a **defining ideal** if there exists $n \in \mathbb{Z}_{\geqslant 1}$ such that $\mathfrak{m}^n \subseteq I \subseteq \mathfrak{m}$.

In other words, I is an defining ideal if and only if the I-adic topology on A is the same as the $\mathfrak{m}$-adic topology on A.

1.110 Let $f : X \to S$ be a morphism of schemes, $s \in S$, and $\mathfrak{a}$ a defining ideal of $\mathcal{O}_{S,s}$. Consider the fibre squares :

$$\begin{tikzcd}
X \arrow[d, "f"'] & X \times_S \operatorname{Spec} \mathcal{O}_{S,s} \arrow[l] \arrow[d] & X \times_S \operatorname{Spec} \mathcal{O}_{S,s}/I \arrow[l] \arrow[d] \\
S & \operatorname{Spec} \mathcal{O}_{S,s} \arrow[l] & \operatorname{Spec} \mathcal{O}_{S,s}/I \arrow[l]
\end{tikzcd}$$

By (1.29) and (1.108) we see that $X \times_S \operatorname{Spec} \mathcal{O}_{S,s}$ is identified with $(Y, \mathcal{O}_X|_Y)$, where

$$Y = f^{-1}\left(\{s' \in S \mid s \in \overline{\{s'\}}\}\right).$$

Note that $\operatorname{Spec} \mathcal{O}_{S,s}/I$ is the singleton $\{\mathfrak{m}/I\}$, for if $\mathfrak{p}$ is a prime containing I, then since I is an defining ideal, taking radicals gives $\mathfrak{m} \subseteq \mathfrak{p} \subseteq \mathfrak{m}$. This implies that $X \times_S \operatorname{Spec} \mathcal{O}_{S,s}/I \to X$ is a homeomorphism onto its image which is $f^{-1}(s)$, the *set-theoretic fibre*.

1.111 Definition. For a morphism $f : X \to S$ of schemes and $s \in S$, we call the fibre product

$$X \times_S \operatorname{Spec} \kappa(s)$$

the **scheme-theoretic fibre** of f at s. In (1.110), we see that as a topological space it is homeomorphism to the set-theoretic fibre $f^{-1}(s)$.

From now on we regard $f^{-1}(s)$ as a scheme by identifying $f^{-1}(s)$ with $X \times_S \operatorname{Spec} \kappa(s)$. In this way there is always a fibre square

$$\begin{tikzcd} X \arrow[d, "f"'] & f^{-1}(s) \arrow[l] \arrow[d] \\ S & \operatorname{Spec} \kappa(s) \arrow[l] \end{tikzcd}$$

1.112 Underlying space of a fibre product. We can describe the underlying set of $X \times_S Y$ as follows. Let $s \in S$ and let $x \in X$, $y \in Y$ lie over s. Then we have fibre squares

$$\begin{tikzcd} \operatorname{Spec} \kappa(x) \arrow[d] & \operatorname{Spec} \kappa(x) \times_S Y \arrow[l] \arrow[d] & \operatorname{Spec}(\kappa(x) \otimes_{\kappa(s)} \kappa(y)) \arrow[l] \arrow[d] \\ X \arrow[d] & X \times_S Y \arrow[l, "\pi_X"'] \arrow[d, "\pi_Y"] & X \times_S \operatorname{Spec} \kappa(y) \arrow[l] \arrow[d] \\ S & Y \arrow[l] & \operatorname{Spec} \kappa(y). \arrow[l] \end{tikzcd}$$

For a scheme T, denote by $\underline{T}$ the underlying topological space. The universal property of $\underline{X} \times_{\underline{S}} \underline{Y}$, the fibre product in **Top**, gives a continuous map

$$\underline{X \times_S Y} \to \underline{X} \times_{\underline{S}} \underline{Y}.$$

The above fibre squares read that each fibre at $(s, x, y) \in \underline{X} \times_{\underline{S}} \underline{Y}$ is homeomorphic to $\operatorname{Spec}(\kappa(x) \otimes_{\kappa(s)} \kappa(y))$.

1.6 Proj

1.113 Let $S = \bigoplus_{n \geqslant 0} S_0$ be a graded ring, i.e., $S_n \cdot S_m \subseteq S_{n+m}$ for any $n, m \geqslant 0$ and each S_n is an abelian subgroup of S. An element f in S_n ($n \geqslant 0$) is said to be **homogeneous of degree** n, and in this case we put $n = \deg f$. For an element $x \in S$, we can write $x = \sum_{n \geqslant 0} x_n$ with $x_n \in S_n$ in a unique way; the x_n

are called the **homogeneous parts** of x. An ideal I of S is called **homogeneous** if it is generated by its homogeneous elements, i.e.,

$$I = \sum_{n \geqslant 0} (I \cap S_n)$$

Put $S_+ = \bigoplus_{n \geqslant 1} S_0$. This is the ideal of elements of positive degrees. Now define

$$\operatorname{Proj} S := \{\mathfrak{p} \in \operatorname{Spec} S \mid \mathfrak{p} \not\supseteq S_+,\ \mathfrak{p} \text{ is homogeneous}\}$$

For a subset $T \subseteq S$, define $V_+(T) := \{\mathfrak{p} \in \operatorname{Proj} S \mid T \subseteq \mathfrak{p}\}$. By definition we have $V_+(T) = V(T) \cap \operatorname{Proj} S$. We equip $\operatorname{Proj} S$ with the subspace topology inherited from $\operatorname{Spec} S$. Note that $V_+(T) = V_+(I)$, where I is the homogeneous ideal generated by the homogeneous part of all $x \in T$. Thus all closed sets of $\operatorname{Proj} S$ have the form $V_+(I)$ for some homogeneous ideals I of S. For any homogeneous f, define the **principal open set**

$$D_+(f) = \{\mathfrak{p} \in \operatorname{Proj} S \mid f \notin \mathfrak{p}\} = D(f) \cap \operatorname{Proj} S = \operatorname{Proj} S \backslash V_+(f).$$

The principal open sets form a basis for the topology on $\operatorname{Proj} S$. In fact, homogeneous elements of positive degree suffices to produce a basis. Indeed, for $f \in S_0$, one has

$$D_+(f) = \bigcup_{n \geqslant 1} \bigcup_{g \in S_n} D_+(fg)$$

$\supseteq$ is clear. To see $\subseteq$ it suffices to recall $\mathfrak{p} \in \operatorname{Proj} S$ do not contain whole S_+ by definition. Because of this fact, in the following by a principal open set we always refer to the one given by homogeneous of positive degree.

1.114 Lemma. For an homogeneous ideal I, we have

$$\sqrt{I} = \bigcap_{\mathfrak{p} \in V_+(I)} \mathfrak{p}$$

Proof. $\subseteq$ is clear. For the another containment, suppose $f \notin \sqrt{I}$. Choose from the family

$$\{J : \text{homogeneous} \mid J \supseteq I,\ f^n \notin J \text{ for all } n \geqslant 1\}$$

a maximal element $\mathfrak{q}$ by Zorn's lemma. Then $\mathfrak{q}$ is a homogeneous prime not containing f. $\square$

1.115 Let $S = \bigoplus_{n \geqslant 0} S_n$ be a graded ring and let $M = \bigoplus_{n \in \mathbb{Z}} M_n$ be a $\mathbb{Z}$-graded S-module, i.e., $S_n \cdot M_m \subseteq M_{n+m}$ for any $n \geqslant 0, m \in \mathbb{Z}$. For a multiplicatively closed subset T of S consisting of homogeneous elements, the localization $T^{-1}S$ is naturally a $\mathbb{Z}$-graded ring, and $T^{-1}M$ is naturally a $\mathbb{Z}$-graded $T^{-1}S$-module, i.e.,

$$(T^{-1}M)_n = \left\{ \frac{x}{t} \mid x \in M_{m+n},\ t \in T \cap S_m \text{ for some } m \geqslant 0 \right\}.$$

For any homogeneous $f \in S_+$ of positive degree, put

$$M_{(f)} = (M_f)_0$$

to be the degree 0 part of the localization M_f.

1.116 $\mathcal{O}_{\operatorname{Proj} S}$ **and projective tilde.** For a graded ring S and homogeneous f of positive degree, define

$$\mathcal{O}_{\operatorname{Proj} S}(D_+(f)) = S_{(f)}.$$

If $D_+(f) \supseteq D_+(g)$, then $V_+((f)) \subseteq V_+((g))$, and $(g) \subseteq \sqrt{(f)}$ by Lemma 1.114. This means $g^n = sf$ for some $n \geqslant 1$ and $s \in S$; in particular s is homogeneous. The map $S_f \to S_g$ given by $\dfrac{1}{f} \mapsto \dfrac{s}{g^n}$ is independent of the choice of n and s, and is degree-preserving, so it gives a map $S_{(f)} \to S_{(g)}$. This shows $\mathcal{O}_{\operatorname{Proj} S}$ defines a presheaf of rings on the principal open sets. To show it is a sheaf, we use 1.3, and prove more generally that there is an exact sequence

$$0 \longrightarrow M_{(f)} \longrightarrow \prod_{i=1}^{n} M_{(g_i)} \longrightarrow \prod_{i,j=1}^{n} M_{(g_i g_j)}.$$

where M is a $\mathbb{Z}$-graded S-module, and $f, g_i \in S_+$ are homogeneous with $D_+(f) = \bigcup_{i=1}^{n} D_+(g_i)$. We use the isomorphism $\varphi : D_+(f) \to \operatorname{Spec} S_{(f)}$ in the next paragraph 1.117. The identity $D_+(fg_i) = \varphi^{-1}(D(g_i^{\deg f} f^{-\deg g_i}))$ then implies

$$\operatorname{Spec} S_{(f)} = \bigcup_{i=1}^{n} D(g_i^{\deg f} f^{-\deg g_i}) \tag{$\spadesuit$}$$

The natural map $M_f \to M_g$ induces a map $M_{(f)} \to M_{(g)}$, and an isomorphism $(M_{(f)})_{g_i^{\deg f} f^{-\deg g_i}} \cong M_{(g_i)}$, where we regard them all as $S_{(f)}$-modules. Similarly $(M_{(f)})_{(g_i g_j)^{\deg f} f^{-\deg g_i g_j}} \cong M_{(g_i g_j)}$. So the sequence we are concerning becomes

$$0 \longrightarrow M_{(f)} \longrightarrow \prod_{i=1}^{n} (M_{(f)})_{g_i^{\deg f} f^{-\deg g_i}} \longrightarrow \prod_{i,j=1}^{n} (M_{(f)})_{(g_i g_j)^{\deg f} f^{-\deg g_i g_j}}.$$

In view of ($\spadesuit$), this reduces to 1.40.($\heartsuit$) (with $A = S_{(f)}$, $M = M_{(f)}$, $f = 1$, $f_i = g_i^{\deg f} f^{-\deg g_i}$). This finishes the proof.

For a $\mathbb{Z}$-graded S-algebra M, we define

$$\widetilde{M}(D_+(f)) = M_{(f)}$$

Then we have showed that this defines a sheaf on principal open sets, and hence a sheaf on the ringed space $(\operatorname{Proj} S, \mathcal{O}_{\operatorname{Proj} S})$.

1.117 We prove there exists an isomorphism in $\mathbf{Ring}_{\mathbf{Top}}$

$$(D_+(f), \mathcal{O}_{\operatorname{Proj} S}|_{D_+(f)}) \longrightarrow (\operatorname{Spec} S_{(f)}, \mathcal{O}_{\operatorname{Spec} S_{(f)}}).$$

In particular, this shows $\operatorname{Proj} S$ is a scheme.

We first construction a continuous map on topological spaces. Recall in 1.50 we have a homeomorphism $D(f) \to \operatorname{Spec} S_f$. We claim the composition $\varphi : D_+(f) \subseteq D(f) \to \operatorname{Spec} S_f \to \operatorname{Spec} S_{(f)}$ is again a homeomorphism. Explicitly, $\varphi(\mathfrak{p}) = \mathfrak{p}S_f \cap S_{(f)}$. We define an inverse map $\psi : \operatorname{Spec} S_{(f)} \to D_+(f)$ by

$$\psi(\mathfrak{p}') = \bigoplus_{n \geqslant 0} \left\{ a \in S_n \mid \frac{a^{\deg f}}{f^n} \in \mathfrak{p}' \right\}$$

Each summand is an abelian subgroup of S_n, for if $a^{\deg f} f^{-n}$, $b^{\deg f} f^{-n} \in \mathfrak{p}'$, then $(a-b)^{2\deg f}/f^{-2n} \in \mathfrak{p}'$, and thus $(a-b)^{\deg f}/f^{-n} \in \mathfrak{p}'$ as $\mathfrak{p}'$ is a prime. This is an homogeneous ideal for if $a^{\deg f} f^{-n} \in \mathfrak{p}'$ and $s \in S_m$, then $(as)^{\deg f} f^{-(n+m)} = (a^{\deg f} f^{-n})(s^{\deg f} f^{-m}) \in \mathfrak{p}'$. This is a prime ideal, for if $a \in S_n$ and $b \in S_m$ with $(ab)^{\deg f} f^{-(n+m)} \in \mathfrak{p}'$, then either $a^{\deg f} f^{-n} \in \mathfrak{p}'$ or $b^{\deg f} f^{-m} \in \mathfrak{p}'$ as the fractions are of degree 0 and $\mathfrak{p}'$ is a prime. Lastly, $f \notin \psi(\mathfrak{p}')$ for otherwise we would have $1 = f^{\deg f} f^{-\deg f} \in \mathfrak{p}'$, which is absurd. We must show φ and ψ are mutually inverses. For $g \in S_n$, we have

$$g \in \psi(\varphi(\mathfrak{p})) \Leftrightarrow g^{\deg f} f^{-n} \in \varphi(\mathfrak{p}) \subseteq \mathfrak{p}S_f \Rightarrow g \in \mathfrak{p}S_f \cap S = \mathfrak{p} \Rightarrow g^{\deg f} f^{-n} \in \mathfrak{p}S_f \cap S_{(f)} = \varphi(\mathfrak{p}) \Leftrightarrow g \in \psi(\varphi(\mathfrak{p})$$

so $\psi(\varphi(\mathfrak{p})) = \mathfrak{p}$. For $g \in S_{(f)}$, if $g \in \varphi(\psi(\mathfrak{p}'))$, we can write $g = \dfrac{a}{f^n}$ for some homogeneous $a \in \psi(\mathfrak{p}')$ with $\deg a = n \deg f$. But these imply $\mathfrak{p}' \ni a^{\deg f} f^{-\deg a} = g^{\deg f}$, so $g \in \mathfrak{p}'$.

Conversely, if $g \in \mathfrak{p}'$, write $g = \dfrac{a}{f^n}$ for some homogeneous $a \in S$ with $\deg a = n \deg f$. The above argument shows $a \in \psi(\mathfrak{p}')$, so $g \in \psi(\mathfrak{p}')S_f \cap S_{(f)} = \varphi(\psi(\mathfrak{p}'))$. To show φ is a homeomorphism, note that for $g^{\deg f} f^{-\deg g} \in S_{(f)}$ with $g \in S_+$ homogeneous, we have

$$\varphi^{-1}(D(g^{\deg f} f^{-\deg g})) = D_+(gf). \tag{$\star$}$$

For if $g^{\deg f} f^{-\deg g} \notin \varphi(\mathfrak{p})$, then $g^{\deg f} f^{-\deg g} \notin \mathfrak{p}S_f$, so $g \notin \mathfrak{p}$. This implies $gf \notin \mathfrak{p}$. Note that is $g^{\deg f} f^{-\deg g} \in \mathfrak{p}S_f$, then $g^{\deg f} f^{-\deg g} = af^{-m}$ for some homogeneous $a \in \mathfrak{p}$, so $g \in \mathfrak{p}$ since $\mathfrak{p}$ is a prime. Thus the above argument is reversible, so this proves the equality. To conclude our assertion, it suffices to note that $D_+(gf)$ forms an open basis of $D_+(f)$ when g runs over S_+ and is homogeneous.

To show $\mathcal{O}_{\operatorname{Spec} S_{(f)}} \cong \varphi_* \mathcal{O}_{\operatorname{Proj} S}|_{D_+(f)}$, we check there are compatible isomorphisms on basis elements of the form $(\star)$. One has

$$\mathcal{O}_{\operatorname{Spec} S_{(f)}}(D(g^{\deg f} f^{-\deg g})) = (S_{(f)})_{g^{\deg f} f^{-\deg f}} \cong S_{(gf)} = \mathcal{O}_{\operatorname{Proj} S}(D_+(gf)) = \varphi_* \mathcal{O}_{\operatorname{Proj} S}(D(g^{\deg f} f^{-\deg g})),$$

the isomorphism being given by the universal property of localization. This isomorphism is compatible with the restriction, so by 1.4 it patches to an isomorphism $\mathcal{O}_{\operatorname{Spec} S_{(f)}} \to \varphi_* \left(\mathcal{O}_{\operatorname{Proj} S}|_{D_+(f)}\right)$.

In fact, the same argument also shows that $\widetilde{M_{(f)}} \cong \varphi_* \left(\widetilde{M}|_{D_+(f)}\right)$, so in fact we have an isomorphism in $\mathbf{Mod}_{\mathbf{LRS}}$

$$(D_+(f), \mathcal{O}_{\operatorname{Proj} S}|_{D_+(f)}, \widetilde{M}|_{D_+(f)}) \longrightarrow (\operatorname{Spec} S_{(f)}, \mathcal{O}_{\operatorname{Spec} S_{(f)}}, \widetilde{M_{(f)}}).$$

Be aware of the difference of two tildes : one is projective, and another is affine.

1.118 The map $D_+(f) \to \operatorname{Spec} S_{(f)}$ in 1.117 is compatible with the restriction, as one can argue as in 1.52. Composing with $\operatorname{Spec} S_{(f)} \to \operatorname{Spec} S_0$ (coming from the natural map $S_0 \to S_{(f)}$), we obtain a family of morphisms $D_+(f) \to \operatorname{Spec} S_0$, compatible with the restriction. By 1.17 they give rises to a morphism $\operatorname{Proj} S \to \operatorname{Spec} S_0$.

Similarly, let A be a ring and let S be a graded A-algebra. Here we assume the image of A lies in S_0. With this assumption, the homomorphism $A \to S \to S_f$ stabilizes $S_{(f)}$, making $S_{(f)}$ an A-algebra, which gives rise to a morphism $\operatorname{Spec} S_{(f)} \to \operatorname{Spec} A$. Composing with $D_+(f) \to \operatorname{Spec} S_{(f)}$ and gluing, we obtain $\operatorname{Proj} S \to \operatorname{Spec} A$ so that $\operatorname{Proj} S$ is naturally an A-scheme.

1.119 Let S and S' be two graded rings and $\varphi : S \to S'$ a ring homomorphism such that $\varphi(S_n) \subseteq S'_{nd}$ for any $n \in \mathbb{Z}$ and a fixed $d \in \mathbb{N}$. For any $\mathfrak{p}' \in \operatorname{Proj} S'$, we have

$$\varphi^{-1}(\mathfrak{p}') \in \operatorname{Proj} S \Leftrightarrow S_+ \nsubseteq \varphi^{-1}(\mathfrak{p}') \Leftrightarrow \varphi(S_+) \nsubseteq \mathfrak{p}'$$

or $\mathfrak{p}' \in \operatorname{Proj} S' \backslash V_+(\varphi(S_+))$. If we put $G(\varphi) = \operatorname{Proj} S' \backslash V_+(\varphi(S_+))$, we see φ induces a continuous map

$$\begin{aligned} \Phi : G(\varphi) &\longrightarrow \operatorname{Proj} S \\ \mathfrak{p}' &\longmapsto \varphi^{-1}(\mathfrak{p}'). \end{aligned}$$

Note that $G(\varphi) = \bigcup\{D_+(\varphi(f)) \mid f \in S_d,\ d \geqslant 1\}$ and $\Phi^{-1}(D_+(f)) = D_+(\varphi(f))$ for homogeneous $f \in S_+$. Observe that φ induces a homomorphism $S_f \to S'_{\varphi(f)}$ of *graded rings*, i.e., $\varphi((S_f)_n) \subseteq (S'_{\varphi(f)})_n$ for any $n \geqslant 0$, so it further induces a ring homomorphism $S_{(f)} \to S_{(\varphi(f))}$. This gives a morphism $D_+(\varphi(f)) \to D_+(f)$ of schemes (1.117). For any homogeneous $g \in S_+$, it is easy to see there is a commuting square

$$\begin{tikzcd} S'_{\varphi(f)} \arrow[d] & S_{(f)} \arrow[l] \arrow[d] \\ S'_{\varphi(fg)} & S_{(fg)} \arrow[l] \end{tikzcd}$$

so the $D_+(\varphi(f)) \to D_+(f)$ glue to a morphism $G(\varphi) \to \operatorname{Proj} S$ of schemes. We denote this morphism by $\operatorname{Proj} \varphi$. By construction we see $\operatorname{Proj} \varphi$ is an affine morphism (1.63).

1.120 Lemma. Let A be a ring, B a graded A-algebra and C an A-algebra. Then there is a canonical isomorphism

$$\operatorname{Proj}(B \otimes_A C) \cong \operatorname{Proj} B \times_{\operatorname{Spec} A} \operatorname{Spec} C$$

Proof. Put $E = B \otimes_A C$. The canonical A-algebra homomorphism $\iota : B \to E$ satisfies $\iota(B_n) \subseteq E_n = B_n \otimes_A C$ for every n, so $\iota(B_+) \subseteq E_+$; in fact, $\iota(B_+)E = E_+$. Then $V_+(\iota(B_+)) = V_+(E_+) = \varnothing$. By (1.119) and (1.118)

we thus obtain an A-morphism $\psi : \operatorname{Proj} E \to \operatorname{Proj} B$. Also, by (1.118) the natural map $C \to E$ gives an A-morphism $\operatorname{Proj} E \to \operatorname{Spec} C$. From the definition of fibre product these two morphisms give

$$\varphi : \operatorname{Proj} E \to \operatorname{Proj} B \times_{\operatorname{Spec} A} \operatorname{Spec} C$$

For homogeneous $f \in B_+$, we have

$$\varphi^{-1}(D_+(f) \times_{\operatorname{Spec} A} \operatorname{Spec} C) = \psi^{-1}(D_+(f)) = D_+(\iota(f))$$

so we only need to show $D_+(\iota(f)) \to D_+(f) \times_{\operatorname{Spec} A} \operatorname{Spec} C$ is an isomorphism for every such f. Back to algebra, we need to show $B_{(f)} \otimes_A C \to E_{(\iota(f))}$ is an isomorphism. Clearly, this map is given by

$$\frac{b}{f^n} \otimes c \mapsto \frac{b \otimes c}{\iota(f)^n}$$

which is clearly surjective. To show the injectivity, we only need to show $B_{(f)} \otimes_A C \to B_f \otimes_A C \to E_{\iota(f)}$ is injective. This is clear. $\square$

1.121 Let S be a graded ring. Then S can be viewed as an S_0-algebra. Let $E \subseteq S_+$ be a subset consisting of homogeneous elements.

Lemma. $S_+ = SE$ if and only if $S = S_0[E]$.

Proof. Considering the grading, we see if part holds obviously. For the only if part, we prove $S_n \subseteq S_0[E]$ inductively on $n \geqslant 1$. For $s \in S_n$, write $s = \sum_{i=1}^{m} s_i e_i$, where $s_i \in S$ and $e_i \in E \cap S_{n_i}$ for some $n_i \geqslant 1$. Writing s_i as the sum of its homogeneous part, we can assume $s_i \in S_{n-n_i}$. Since $n - n_i < n$, we can apply induction hypothesis to see $s \in S_0[E]$. $\square$

1.122 Suppose S be a graded ring that is generated by S_1 as an S_0-algebra. Concisely, $S = S_0[S_1]$. Then $\operatorname{Proj} S$ is covered by those principal open sets of the form $D_+(f)$ with $f \in S_1$. Indeed, for any $f \in S_d$, write $f = \sum_{i=1}^{m} s_i f_i$, where f_i is a product of elements in S_1. Then $(f) \subseteq (f_1, \ldots, f_m)$, so $V_+(f) \supseteq V_+(f_1) \cap \cdots \cap V_+(f_m)$, and hence $D_+(f) \subseteq D_+(f_1) \cup \cdots \cup D_+(f_m)$. But $D_+(f_i) \subseteq D_+(g_i)$ if $g_i \in S_1$ is any element appearing in the product f_i, so $D_+(f)$ is covered by those principal open sets given by elements in S_1. Since the $D_+(f)$ form a basis of topology of $\operatorname{Proj} S$, this proves the assertion.

In fact, the above argument shows that if $E \subseteq S_+$ consists of homogeneous elements such that $S = S_0[E]$, then S can be cover by those principal open sets given by elements in E.

1.123 Let S be a graded ring. For each $n \in \mathbb{Z}$, denote by $S(n)$ the $\mathbb{Z}$-graded S-module whose graded pieces are given by

$$S(n)_d = S_{n+d}.$$

We call this a **twist of** S **by** n. The $\mathcal{O}_{\operatorname{Proj} S}$-module $\widetilde{S(n)}$ is denoted by $\mathcal{O}_{\operatorname{Proj} S}(n)$. Particularly, the sheaf $\mathcal{O}_{\operatorname{Proj} S}(1)$ is called the **twisting sheaf of Serre**. For any $\mathcal{O}_{\operatorname{Proj} S}$-module $\mathcal{F}$, we denote by $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_{\operatorname{Proj} S}} \mathcal{O}_{\operatorname{Proj} S}(n)$, and called this $\mathcal{F}$ **twisted by** n.

1.124 Lemma. Let S be a graded ring and $X = \operatorname{Proj} S$. Assume that $S = S_0[S_1]$.

(i) The sheaf $\mathcal{O}_X(n)$ is an invertible sheaf on X.

(ii) For any graded S-module M, $\widetilde{M}(n) \cong \widetilde{M(n)}$.

(iii) $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$.

Proof.

(i) Let $f \in S_1$, and consider $\mathcal{O}_X(n)|_{D_+(f)}$. By 1.117, there is an isomorphism

$$(D_+(f), \mathcal{O}_X(n)|_{D_+(f)}) \longrightarrow (\operatorname{Spec} S_{(f)}, \widetilde{S(n)_{(f)}}).$$

in **LRS**. Thus we only need to show $\widetilde{S(n)_{(f)}}$ is invertible. In fact, we will show $S(n)_{(f)}$ is free over $S_{(f)}$ of rank 1. Consider the $S_{(f)}$-module isomorphism

$$\begin{aligned} S_{(f)} &\longrightarrow S(n)_{(f)} \\ s &\longmapsto f^n s \end{aligned}$$

This is well-defined for any n since f is invertible, and $S(n)_{(f)}$ in fact consists of elements of degree n in S_f. This proves that $S(n)_{(f)}$ is free of rank 1. Since $S = S_0[S_1]$, by 1.122 X is covered by $D_+(f)$, $f \in S_1$, so this proves that $\mathcal{O}_X(n)$ is invertible.

(ii) For any $f \in S_d$ and graded S-module M, N, the natural isomorphism $M_f \otimes_{S_f} N_f \cong (M \otimes_S N)_f$ is grading preserving. In particular, we have a canonical map $(*) : M_{(f)} \otimes_{S_{(f)}} N_{(f)} \to (M \otimes_S N)_{(f)}$; explicitly, this map takes the form

$$\frac{m}{f^a} \otimes \frac{n}{f^b} \mapsto \frac{m \otimes n}{f^{a+b}}$$

Moreover, for any other homogeneous $g \in S_+$, we have a commutative diagram

$$\begin{tikzcd} M_{(f)} \otimes_{S_{(f)}} N_{(f)} \arrow[r] \arrow[d] & (M \otimes_S N)_{(f)} \arrow[d] \\ M_{(fg)} \otimes_{S_{(fg)}} N_{(fg)} \arrow[r] & (M \otimes_S N)_{(fg)} \end{tikzcd}$$

This implies we have a morphism $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} \to \widetilde{M \otimes_S N}$ in $\mathbf{Mod}_{\mathcal{O}_X}$. Note that when $f \in S_1$, $(*)$ is in fact an isomorphism. Indeed, define $M \times N \to M_{(f)} \otimes_{S_{(f)}} N_{(f)}$ by $(\sum_i m_i, \sum_j n_j) \mapsto \sum_{i,j} \frac{m_i}{f^i} \otimes \frac{n_j}{f^j}$. If we let S acts on the codomain via the ring homomorphism $S \to S_{(f)}$ defined by $\sum_k s_k \mapsto \sum_k \frac{s_k}{f^k}$,

then this is an S-bilinear map, so it gives rise to a map $M \otimes_S N \to M_{(f)} \otimes_{S_{(f)}} N_{(f)}$. Since $\deg f = 1$, f acts on the codomain invertibly, so we obtain a map $(M \otimes_S N)_f \to M_{(f)} \otimes_{S_{(f)}} N_{(f)}$, where S_f acts on the codomain via the map $S_f \to S_{(f)}$ induced by $S \to S_{(f)}$. Composing with the inclusion gives the inverse of $(*)$. Thus again by 1.122, $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} \cong \widetilde{M \otimes_S N}$.

To show (ii), note that by above we have $\widetilde{M}(n) \cong \widetilde{M \otimes_S S}(n)$, so it remains to show $M \otimes_S S(n) \cong M(n)$ as graded S-modules, which is clear after a moment consideration.

(iii) This is a particular case of (ii).

□

1.125 Let $\varphi : S \to T$ be a homomorphism of graded rings, and suppose $S = S_0[S_1]$ and $T = T_0[T_1]$. Let $U = G(\varphi)$ and $f = \operatorname{Proj} \varphi : U \to \operatorname{Proj} S$ be the morphism associated with φ (1.119). We have an analog of Lemma 1.55, namely

(i) $f^*\widetilde{M} \cong \left(\widetilde{M \otimes_S T}\right)|_U$ for any graded S-module M.

(ii) $f_*(\widetilde{N}|_U) \cong \widetilde{N^{[\varphi]}}$ for any graded T-module N.

In particular, if we put $X = \operatorname{Proj} S$ and $Y = \operatorname{Proj} T$, this shows that $f^*(\mathcal{O}_X(n)) \cong \mathcal{O}_Y(n)|_U$ and $f_*(\mathcal{O}_Y(n)|_U) \cong (f_*(\mathcal{O}_X|_U))(n)$.

1.126 Let S be a graded ring, $X = \operatorname{Proj} S$ and $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Define (cf. 1.70)

$$\Gamma_*(\mathcal{F}) = \bigoplus_{m \in \mathbb{Z}} \mathcal{F}(m)(X)$$

This is naturally a graded S-module; for $s \in S_d$, it determines a unique global section $\mathcal{O}_X(d)(X)$. Then for $t \in \mathcal{F}(n)(X)$, define $st \in \mathcal{F}(n+d)$ by the map $\mathcal{F}(n) \otimes \mathcal{O}_X(d) \to \mathcal{F}(n+d)$.

1.127 Lemma. Let A be a ring, $n \in \mathbb{Z}_{\geqslant 1}$, $S = A[x_0, \ldots, x_n]$ and $X = \operatorname{Proj} S$. Then $\Gamma_*(\mathcal{O}_X) \cong S$ as graded rings.

Proof. Cover X be the affine open $D_+(x_i)$ (1.122). For any $m \in \mathbb{Z}$, by (1.117) we have a commutative diagram with the first row exact

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}_X(m)(X) \arrow[r] & \prod_{i=0}^{n} \mathcal{O}_X(m)(D_+(x_i)) \arrow[r] \arrow[d, "\wr"] & \prod_{i,j=0}^{n} \mathcal{O}_X(m)(D_+(x_i x_j)) \arrow[d, "\wr"] \\
& & \prod_{i=0}^{n} S(m)_{(x_i)} \arrow[r] \arrow[d, equal] & \prod_{i,j=0}^{n} S(m)_{(x_i x_j)} \arrow[d, equal] \\
& & \prod_{i=0}^{n} S_{x_i}(m) \arrow[r] & \prod_{i,j=0}^{n} S_{x_i x_j}(m)
\end{tikzcd}$$

Since the x_i are non-zero divisors in S, the localization maps $S \to S_{x_i}$ and $S_{x_i} \to S_{x_i x_j}$ are injective, and these rings can be viewed as subrings of $S_{x_0 \cdots x_n}$. Hence the above diagram reads $\mathcal{O}_X(m)(X) = \bigcap_{i=0}^{n} S_{x_i}(m)$, and thus by degree consideration

$$\Gamma_*(\mathcal{O}_X) = \bigcap_{i=0}^{n} S_{x_i} \text{ in } S_{x_0 \cdots x_n}.$$

It remains to show this intersection is precisely S. An element in $S_{x_0 \cdots x_n}$ can be written uniquely as $x_0^{\alpha_0} \cdots x_n^{\alpha_n} f(x_0, \ldots, x_n)$ with $\alpha_j \in \mathbb{Z}$ and $f \in S$ homogeneous. Such an element lies in S_{x_i} if and only if $\alpha_j \geqslant 0$ for any $j \neq i$. Now the result follows. □

1.6.1 Projective spaces

1.128 Let A be a ring. For an integer $n \geqslant 1$, we can form the polynomial ring $S = A[x_0, x_1, \ldots, x_n]$ with n-variables and coefficient in A. If we view each variable x_i as a degree one element in S, it naturally becomes a $\mathbb{Z}_{\geqslant 0}$-graded ring generated by all degree one elements. The scheme

$$\mathbb{P}^n_A = \operatorname{Proj} A[x_0, x_1, \ldots, x_n]$$

is called the **n-dimensional projective space over** A. By (1.122), we have

$$\mathbb{P}^n_A = D_+(x_0) \cup D_+(x_1) \cup \cdots \cup D_+(x_n)$$

and each $D_+(x_i)$ is isomorphic to the affine scheme (1.117)

$$(D_+(x_i), \mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_i)}) \cong (\operatorname{Spec} A[x_0, \ldots, x_n]_{x_i}, \mathcal{O}_{\operatorname{Spec} A[x_0, \ldots, x_n]_{x_i}})$$

The variables $x_0, \ldots, x_n$ are called the **homogeneous coordinates** of $\mathbb{P}^n_A$.

1.129 Definition. Let X be a scheme and $\mathcal{F}$ be a $\mathcal{O}_X$-module.

1. $\mathcal{F}$ is **generated by global sections at** $x \in X$ if $\mathcal{F}(X) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_{X,x} \to \mathcal{F}_x$ is surjective.
2. $\mathcal{F}$ is **generated by global sections** if it is so at every point of X.
3. Let $S \subseteq \mathcal{F}(X)$ be a subset. $\mathcal{F}$ is **generated by** S if $S \otimes_{\mathcal{O}_X(X)} \mathcal{O}_{X,x} \to \mathcal{F}_x$ is surjective for any $x \in X$.

1.130 Let X be a ringed space and $\mathcal{F}$ a $\mathcal{O}_X$-module. Then there is a bijection

$$\begin{aligned} \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) &\longrightarrow \mathcal{F}(X) \\ T &\longrightarrow T_X(1). \end{aligned} \tag{♣}$$

Via this, we can **pullback a global section**, an operation we now define. Let $f : X \to Y$ be a morphism of ringed space and $\mathcal{F}$ a $\mathcal{O}_Y$-module. Let $s \in \mathcal{F}(Y)$ and identify it with the corresponding $\mathcal{O}_Y$-morphism

$s : \mathcal{O}_Y \to \mathcal{F}$. Define $f^*s \in (f^*\mathcal{F})(X)$ to be the global section corresponding to the morphism $\mathcal{O}_X \to f^*\mathcal{F}$ obtain by adjunction (1.13) (here we use $\mathcal{O}_Y = f_*\mathcal{O}_X$). One liner : f^*s is the image of s under the composition

$$\mathcal{F}(Y) \xrightarrow{\clubsuit} \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{O}_Y, \mathcal{F}) \xrightarrow{(1.13)} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, f^*\mathcal{F}) \xrightarrow{\clubsuit} (f^*\mathcal{F})(X)$$

A careful computation using the construction in (1.12) and (1.14) shows that

$$(f^*s)|_x = s|_y \otimes 1 \in \mathcal{F}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \cong (f^*\mathcal{F})_x$$

for any $x \in X$ and $y = f(x) \in Y$. In particular, this implies $X_{f^*s} = f^{-1}(Y_s)$. Furthermore, we have

$$f^*s = \varinjlim_U s|_U \otimes 1 \in (f^*\mathcal{O}_Y)(X)$$

1.131 The homogeneous coordinates $x_0, \ldots, x_n$ of $\mathbb{P}^n_A$, by (1.127), give rise to elements in $\mathcal{O}_{\mathbb{P}^n_A}(1)(\mathbb{P}^n_A)$. In fact, the sheaf $\mathcal{O}_{\mathbb{P}^n_A}(1)$ is generated by the x_i's. To see this, note that $\mathcal{O}_{\mathbb{P}^n_A}(1)(D_+(x_i)) = S(1)_{(x_i)} = x_i S_{(x_i)}$ is a free $S_{(x_i)}$-module of rank 1. Then for any $p = \mathfrak{p} \in \operatorname{Spec} S_{(x_i)} \cong D_+(x_i)$, the stalk at p is $x_i(S_{(x_i)})_{\mathfrak{p}} = x_i\mathcal{O}_{\mathbb{P}^n_A,p}$, which is generated by x_i over $\mathcal{O}_{\mathbb{P}^n_A,p}$, as said.

1.132 Let A be a ring and X be a scheme over A. Let $f : X \to \mathbb{P}^n_A$ be a morphism. By (1.69), the pullback sheaf $f^*\mathcal{O}_{\mathbb{P}^n_A}(1)$ is an invertible sheaf; let us put $\mathcal{L} = f^*(\mathcal{O}_{\mathbb{P}^n_A}(1))$. For $0 \leqslant i \leqslant n$, let $s_i := f^*x_i \in \mathcal{L}(X)$ (1.130). The last assertion in (1.130) implies that the s_i's generate the sheaf $\mathcal{L}$. In fact, the datum $(\mathcal{L}, s_0, \ldots, s_n)$ characterizes the morphism f.

1.133 Theorem. Let A be a ring and X be a scheme over A.

(i) If $f : X \to \mathbb{P}^n_A$ is an A-morphism, then $f^*(\mathcal{O}_{\mathbb{P}^n_A}(1))$ is an invertible sheaf on X generated by the global sections f^*x_i $(0 \leqslant i \leqslant n)$.

(ii) If $\mathcal{L}$ is an invertible sheaf on X generated by the global sections $s_0, \ldots, s_n \in \mathcal{L}(X)$, then there exists a unique A-morphism $f : X \to \mathbb{P}^n_A$ with $\mathcal{L} \cong f^*(\mathcal{O}_{\mathbb{P}^n_A}(1))$ as $\mathcal{O}_X$-modules and $s_i = f^*x_i$ $(0 \leqslant i \leqslant n)$ under this isomorphism.

Proof. It remains to prove (b). Let $X_i = X^{\mathcal{L}}_{s_i}$ be the open set defined as in (1.70). Since the s_i's generate $\mathcal{L}$, it follows at once that $(X_i)_{i=0}^n$ covers X. Define the morphism $f : X \to \mathbb{P}^n_A$ as follows. Define an A-algebra homomorphism

$$\begin{aligned} A_i := A\left[\frac{x_0}{x_i}, \ldots \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right] &\longrightarrow \mathcal{O}_X(X_i) \\ \frac{x_j}{x_i} &\longmapsto \frac{s_j}{s_i} \end{aligned}$$

This is well-defined as on X_i, there exists $s_i^{-1} \in \mathcal{L}^\vee(X_i)$ and so $s_j s_i^{-1} := (s_j|_{X_i}) \otimes s_i^{-1} \in (\mathcal{L} \otimes \mathcal{L}^\vee)(x_i) \cong \mathcal{O}_X(X_i)$ (1.70). By (1.94), this gives an A-morphism $f_i : X_i \to D_+(x_i) \subseteq \mathbb{P}^n_A$. These morphisms glue for an obvious reason, so we obtain a global A-morphism $f : X \to \mathbb{P}^n_A$. By construction and (1.130), we have

$$(f^*(\mathcal{O}_{\mathbb{P}^n_A}(1)))|_{X_i} = f_i^*(\mathcal{O}_{\mathbb{P}^n_A}(1))|_{D_+(x_i)}) \cong f_i^*(x_i\mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_i)}) = (f^*x_i)f_i^*(\mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_i)}) = (f^*x_i)\mathcal{O}_X|_{X_i}$$

On the other hand, we have $\mathcal{L}|_{X_i} \cong s_i\mathcal{O}_X|_{X_i}$. By viewing $\mathcal{O}_{\mathbb{P}^n_A}(1)|_{X_i \cap X_j}$ as subsheaves of $\mathcal{O}_{\mathbb{P}^n_A}(1)|_{D_+x_i}$ and $\mathcal{O}_{\mathbb{P}^n_A}(1)|_{D_+(x_j)}$ respectively, we obtain a *transition map*

$$\begin{aligned} \frac{x_i}{x_j} : x_i\mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_ix_j)} &\longrightarrow x_j\mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_ix_j)} \\ x_i &\longmapsto x_j \end{aligned}$$

On the other hand, the transition map for $\mathcal{L}|_{X_i}$ and $\mathcal{L}|_{X_j}$ is $\dfrac{s_i}{s_j} : s_i\mathcal{O}_X|_{X_i \cap X_j} \to s_j\mathcal{O}_X|_{X_i \cap X_j}$. This implies the isomorphisms $(f^*x_i)\mathcal{O}_X|_{X_i} \to s_i\mathcal{O}_X|_{X_i}$ defined by $f^*x_i \mapsto s_i$ glue to an isomorphism $f^*(\mathcal{O}_{\mathbb{P}^n_A}(1)) \cong \mathcal{L}$

For the uniqueness, let $g : X \to \mathbb{P}^n_A$ be an A-morphism satisfying (ii). By (1.130), the morphism $g_i = g|_{X_i}^{D_+(x_i)}$ is well-defined. It suffices to show the corresponding homomorphism $\theta_i : A_i \to \mathcal{O}_X(X_i)$ satisfies $\theta_i(x_jx_i^{-1}) = s_js_i^{-1}$. Restricting, we have $\mathcal{L}|_{X_i} \cong (g^*\mathcal{O}_{\mathbb{P}^n}(1))|_{X_i} = g_i^*(\mathcal{O}_{\mathbb{P}^n_A}(1)|_{D_+(x_i)})$, and obtain an $\mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_i)}$-module morphism $G : \mathcal{O}_{\mathbb{P}^n_A}(1)|_{D_+(x_i)} \to (g_i)_*\mathcal{L}|_{X_i}$. Recall in (1.13) the latter is viewed as an $\mathcal{O}_{\mathbb{P}^n_A}|_{D_+(x_i)}$-module via the homomorphism θ_i, so

$$G_{D_+(X_i)}(x_j) = G_{D_+(X_i)}\left(x_i \cdot \frac{x_j}{x_i}\right) = \theta_i\left(\frac{x_j}{x_i}\right) G_{D_+(X_i)}(x_i).$$

Our claim will follow once we prove $G_{D_+(X_i)}(x_j) = f^*x_i \in \mathcal{L}(X_i)$ for any $0 \leqslant j \leqslant n$. This is clear from the definition (1.130). $\square$

1.7 Relative spec

1.134 Let X be a scheme and $\mathcal{R}$ a quasi-coherent sheaf of $\mathcal{O}_X$-algebras. We are going to construct a scheme $\mathbf{Spec}_X\mathcal{R}$, called the **relative spectrum of $\mathcal{R}$ over** X, along with a morphism $\pi : \mathbf{Spec}_X\mathcal{R} \to X$ and an isomorphism $\mathcal{R} \cong \pi_*\mathcal{O}_{\mathbf{Spec}_X\mathcal{R}}$ of $\mathcal{O}_X$-algebras satisfying the following universal property : for any morphism $f : Y \to X$ of schemes and a morphism $\alpha : \mathcal{R} \to f_*\mathcal{O}_Y$ of $\mathcal{O}_X$-algebras, there exists a unique morphism

$$g : Y \to \mathbf{Spec}_X\mathcal{R}$$

of schemes fitting into a commutative triangle

$$\begin{tikzcd} Y \arrow[rr, "g"] \arrow[dr, "f"'] & & \mathbf{Spec}_X\mathcal{R} \arrow[dl, "\pi"] \\ & X & \end{tikzcd}$$

so that α factors as

$$\mathcal{R} \cong \pi_* \mathcal{O}_{\mathbf{Spec}_X \mathcal{R}} \xrightarrow{g_*} f_* \mathcal{O}_Y \ .$$

1.135 As an example, let X be a scheme and consider $\mathcal{O}_X$, which is itself a quasi-coherent $\mathcal{O}_X$-algebra. It is tautological that the morphism $\mathrm{id} : X \to X$ and the equality $\mathcal{O}_X = \mathcal{O}_X$ satisfies the universal property in (1.134). Hence $X = \mathbf{Spec}_X \mathcal{O}_X$ as schemes.

1.136 Let X and $\mathcal{R}$ be as in (1.134). For a scheme Y, define

$$\begin{aligned} F(Y) &= \left\{ (f, \beta) \mid f \in \mathrm{Hom}_{\mathbf{Sch}}(Y, X),\ \beta \in \mathrm{Hom}_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{R}, f_* \mathcal{O}_Y) \right\} \\ &= \mathrm{Hom}_{\mathbf{Alg}_{\mathbf{LRS}}}((Y, \mathcal{O}_Y), (X, \mathcal{R})) \end{aligned}$$

If $Y \to Y'$ is a morphism of schemes, by composition we obtain a map $F(Y') \to F(Y)$. Thus F defines a contravariant functor $F : \mathbf{Sch} \to \mathbf{Set}$. A part of (1.134) says that F is a representable functor that is represented by the scheme $\mathbf{Spec}_X \mathcal{R}$. If we fix some $f \in \mathrm{Hom}_{\mathbf{Sch}}(Y, X)$, we then obtain the isomorphism

$$\mathrm{Hom}_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{R}, f_* \mathcal{O}_Y) \cong \mathrm{Hom}_{\mathbf{Sch}_X}(Y, \mathbf{Spec}_X \mathcal{R}).$$

We will see in (1.137) that this generalizes the isomorphism (1.44). In particular, if we take $Y = \mathbf{Spec}_X \mathcal{T}$ for some quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{T}$ and f to be the canonical morphism $\pi' : Y \to X$, we have

$$\mathrm{Hom}_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{R}, \mathcal{T}) \cong \mathrm{Hom}_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{R}, \pi'_* \mathcal{O}_{\mathbf{Spec}_X \mathcal{T}}) \cong \mathrm{Hom}_{\mathbf{Sch}_X}(\mathbf{Spec}_X \mathcal{T}, \mathbf{Spec}_X \mathcal{R})$$

By restricting to affine opens in X (1.138), this is the same as the anti-equivalence between affine schemes and commutative rings. From this we see $\mathbf{Spec}_X$ defines a fully faithful functor from the category of quasi-coherent $\mathcal{O}_X$-algebras to $\mathbf{Sch}_X$.

1.137 We start our construction of $\mathbf{Spec}_X \mathcal{R}$ by first considering the case X being affine. We claim that

$$\mathbf{Spec}_X \mathcal{R} = \mathrm{Spec}\, \mathcal{R}(X)$$

does the job. In fact, the structure map $\mathcal{O}_X(X) \to \mathcal{R}(X)$ gives, by taking spec, a morphism $\pi : \mathrm{Spec}\, \mathcal{R}(X) \to X$ of schemes. Since $\mathcal{R}$ is quasi-coherent, by (1.57) the canonical map $\alpha : \widetilde{\mathcal{R}(X)} \to \mathcal{R}$, which corresponds to the identity on $\mathcal{R}(X)$, is an isomorphism. Combining with (1.55), we have

$$\mathcal{R} \xrightarrow{\alpha^{-1}} \widetilde{\mathcal{R}(X)} \cong \pi_* \mathcal{O}_{\mathrm{Spec}\, \mathcal{R}(X)}.$$

Suppose $f : Y \to X$ is any morphism of schemes and $\beta : \mathcal{R} \to f_* \mathcal{O}_Y$ is any $\mathcal{O}_X$-algebra morphism. By (1.44), the $\mathcal{O}_X(X)$-algebra homomorphism $\beta_X : \mathcal{R}(X) \to \mathcal{O}_Y(Y)$ gives rises to the morphism $g : Y \to \mathrm{Spec}\, \mathcal{R}(X)$ that satisfies the desired commutative triangle. Using (1.45), the pair (f, β) gives rise to $\widetilde{\mathcal{R}(X)} \to f_* \mathcal{O}_Y$ satisfying

$$\begin{tikzcd} \mathcal{R} \arrow[rr] \arrow[dr] & & f_* \mathcal{O}_Y \\ & \widetilde{\mathcal{R}(X)} \arrow[ur] & \end{tikzcd}$$

It is straightforward to see that once replacing the bottom by $\pi_*\mathcal{O}_{\operatorname{Spec}\mathcal{R}(X)}$, the morphism on the left is given by g_*. The uniqueness follows from the equality $\beta_X = (g_*)_X$, which comes from the factorization for β.

1.138 Now drop the affineness condition and consider X to be an arbitrary scheme. Let $\mathcal{U}$ be an affine open cover of X. To motivate the construction of $\pi : \mathbf{Spec}_X\mathcal{R} \to X$, let $f : Y \to X$ be a morphism with image in some $U \in \mathcal{U}$. We then see in fact the resulting morphism $g : Y \to \mathbf{Spec}_X\mathcal{R}$ goes into $\pi^{-1}(U)$. Moreover, we have $\alpha|_U : \mathcal{R}|_U \to (f_*\mathcal{O}_Y)|_U = (f|^U)_*\mathcal{O}_Y$ and $(\pi_*\mathcal{O}_{\mathbf{Spec}_X\mathcal{R}})|_U = (\pi|_{\pi^{-1}(U)})_*(\mathcal{O}_{\mathbf{Spec}_X\mathcal{R}}|_{\pi^{-1}(U)})$, which means $\pi|_{\pi^{-1}(U)} : \pi^{-1}(U) \to X$ solves the universal property for $\mathbf{Spec}_U\mathcal{R}|_U$. Therefore $\pi^{-1}(U) \cong \mathbf{Spec}_U\mathcal{R}|_U$, and in the notation (1.93) it reads

$$(\mathbf{Spec}_X\mathcal{R})\,|_U \cong \mathbf{Spec}_U\mathcal{R}|_U = \operatorname{Spec}\mathcal{R}.$$

This suggests us to construct $\mathbf{Spec}_X\mathcal{R}$ by patching $\operatorname{Spec}\mathcal{R}(U)$ together.

To this end, for any $U \in \mathcal{U}$, let $\pi_U : \operatorname{Spec}\mathcal{R}(U) \to U$ be the map as in (1.137). For any $U, V \in \mathcal{U}$, we must construct an isomorphism $\theta_{UV} : \pi_U^{-1}(U \cap V) \cong \pi_V^{-1}(U \cap V)$ satisfying the conditions in (1.15). By (1.57) and (1.55), we have

$$\mathcal{R}|_U \cong (\pi_U)_*\mathcal{O}_{\operatorname{Spec}\mathcal{R}(U)}, \tag{$\heartsuit$}$$

and thus $\mathcal{R}(U \cap V) \cong \mathcal{O}_{\operatorname{Spec}\mathcal{R}(U)}((\pi_U)^{-1}(U \cap V))$. Thus from the restriction $\mathcal{R}(V) \to \mathcal{R}(U \cap V)$ we obtain a morphism $\pi_U^{-1}(U \cap V) \to \operatorname{Spec}\mathcal{R}(V)$. This morphism fits into a commutative diagram

$$\begin{tikzcd}
\pi_U^{-1}(U \cap V) \arrow[rr] \arrow[dr, "\pi_U"'] & & \operatorname{Spec}\mathcal{R}(V) \arrow[dl, "\pi_V"] \\
& U \cap V \subseteq V &
\end{tikzcd}$$

(which, as V is affine, follows easily from taking global sections), implying the image lies in $\pi_V^{-1}(U \cap V)$; denote by $\theta_{UV} : \pi_U^{-1}(U \cap V) \to \pi_V^{-1}(U \cap V)$ the resulting morphism. Note that the inclusion $\pi_U^{-1}(U \cap V) \subseteq \operatorname{Spec}\mathcal{R}(U)$ corresponds to the restriction $\mathcal{R}(U) \to \mathcal{R}(U \cap V)$ as well, and from this we easily deduce that θ_{UV} is an isomorphism and $\theta_{UV}^{-1} = \theta_{VU}$. Also, this shows the θ_{UV}'s satisfy the cocycle condition in (1.15), so we obtain a well-defined scheme, which is of course denoted by $\mathbf{Spec}_X\mathcal{R}$. By (1.17), the π_U's patch together to a morphism $\pi : \mathbf{Spec}_X\mathcal{R} \to X$. The way we glue $\operatorname{Spec}\mathcal{R}(U)$ and π also glue the isomorphisms ($\heartsuit$) together to obtain a global isomorphism $\mathcal{R} \cong \pi_*\mathcal{O}_{\mathbf{Spec}_X\mathcal{R}}$.

1.139 Finish of construction. We still need to show that $\pi : \mathbf{Spec}_X\mathcal{R} \to X$ and $\mathcal{R} \cong \pi_*\mathcal{O}_{\mathbf{Spec}_X\mathcal{R}}$ satisfy the universal property in (1.134). Let $f : Y \to X$ be a morphism of schemes and $\alpha : \mathcal{R} \to f_*\mathcal{O}_Y$ a morphism of $\mathcal{O}_X$-algebras. Let $\mathcal{U}$ be an affine open cover of X. Put $f_U = f|^U_{f^{-1}(U)} : f^{-1}(U) \to U$ for any $U \in \mathcal{U}$. Note that $(f_*\mathcal{O}_Y)|_U = (f_U)_*(\mathcal{O}_Y|_{f^{-1}(U)})$. Then as in (1.137), we have a morphism $g_U : f^{-1}(U) \to \operatorname{Spec}\mathcal{R}(U)$, which is, in the level of rings, induced by $\alpha(U) : \mathcal{R}(U) \to \mathcal{O}_Y(f^{-1}(U))$. The g_U's patch together, as for any $U, V \in \mathcal{U}$, we can cover $U \cap V$ by affine open subsets W's, and both $g_U|_{f^{-1}(W)}$ and $g_V|_{f^{-1}(W)}$ are then given by $\mathcal{R}(W) \to \mathcal{O}_Y(f^{-1}(W))$. Thus we obtain a well-defined morphism $g : Y \to \mathbf{Spec}_X\mathcal{R}$. Finally, α

factors as $\mathcal{R} \cong \pi_* \mathcal{O}_{\mathbf{Spec}_X \mathcal{R}} \xrightarrow{g_*} f_* \mathcal{O}_Y$ since it does on every affine open $U \in \mathcal{U}$. The morphism g is unique as the restriction $g|_{f^{-1}(U)}^{\pi^{-1}(U)}$ must equal g_U, which is unique as shown in (1.137).

1.140 Lemma. Let X be a scheme and $\mathcal{R}$ a quasi-coherent $\mathcal{O}_X$-algebra. Let $g : X' \to X$ be a morphism of schemes. By (1.60), $g^*\mathcal{R}$ is a quasi-coherent $\mathcal{O}_{X'}$-algebra. Then there exists a canonical isomorphism

$$\mathbf{Spec}_{X'}(g^*\mathcal{R}) \cong X' \times_X \mathbf{Spec}_X \mathcal{R}$$

Proof. Let F be the functor in (1.136) represented by $\mathbf{Spec}_X \mathcal{R}$, and let G be that of represented by $\mathbf{Spec}_{X'}(g^*\mathcal{R})$. It suffices to construct a natural bijection

$$G(T) \cong h_{X'}(T) \times_{h_X(T)} F(T),$$

where $T \in \mathbf{Sch}$. But this is more than tautological : giving $f' : T \to X'$ with a $\beta' : g^*\mathcal{R} \to (f')_* \mathcal{O}_T$ is the same as giving $f : T \to X$ with $\beta : \mathcal{R} \to f_* \mathcal{O}_T$ and $f = g \circ f'$. Here we use adjunction (1.13) to replace β by $f^*\mathcal{R} \to \mathcal{O}_T$ and replace β' by $(f')^* g^* \mathcal{R} \to \mathcal{O}_T$. □

1.141 Proposition. Let $f : Y \to X$ be a morphism of schemes. TFAE :

(a) f is an affine morphism (1.63).

(b) There exists a quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{R}$ such that $Y \cong \mathbf{Spec}_X \mathcal{R}$ as X-schemes.

Proof. (b)⇒(a) follows from the discussion in (1.138). For (a)⇒(b), note that an affine morphism $f : Y \to X$ satisfies the conditions in Lemma 1.60, so $f_* \mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra. At this stage, it suffices to show that $Y \cong \mathbf{Spec}_X f_* \mathcal{O}_Y$ as X-schemes. By the last isomorphism in (1.136), the identity morphism on $f_* \mathcal{O}_Y$ induces a canonical morphism $g : Y \to \mathbf{Spec}_X f_* \mathcal{O}_Y$. To see this is an isomorphism, let U be an affine open subset of X such that $f^{-1}(U)$ is affine. We only need to show $g|_{f^{-1}(U)}^{\pi^{-1}(U)}$ is an isomorphism, where $\pi : \mathbf{Spec}_X f_* \mathcal{O}_Y \to X$ is the canonical morphism. From construction (1.139) $g|_{f^{-1}(U)}^{\pi^{-1}(U)}$ is built from the isomorphism $\mathrm{id}_{f^{-1}(U)} : f_* \mathcal{O}_Y(U) \to \mathcal{O}_Y(f^{-1}(U))$, so $g|_{f^{-1}(U)}^{\pi^{-1}(U)}$ is an isomorphism as well. □

Note that the proof also gives another way to see the equivalence in (1.63). Further, the claim in (1.62) also follows from the proposition.

1.142 Let X be a scheme. Denote by $\mathbf{QcohAlg}_X$ the subcategory of $\mathbf{Alg}_{\mathcal{O}_X}$ consisting of all quasi-coherent $\mathcal{O}_X$-modules. Recall in (1.136) we see that $\mathbf{Spec}_X$ defines a fully faithful functor :

$$\mathbf{Spec}_X : \mathbf{QcohAlg}_X \longrightarrow \mathbf{Sch}_X$$

(1.141) shows that the essential image of $\mathbf{Spec}_X$ is the full subcategory of $\mathbf{Sch}_X$ consisting of X-schemes with structure morphism being affine ; we say such an X-scheme is **affine over** X. Hence $\mathbf{Spec}_X$ establishes an anti-equivalence from $\mathbf{QcohAlg}_X$ to the category of schemes affine over X.

1.7.1 Vector bundles

1.143 Definition. Let X be a scheme and $n \in \mathbb{Z}_{\geqslant 0}$. An X-scheme $p : Y \to X$ is called a **vector bundle of rank** n **over** X if Y admits an open cover $\mathcal{U}$ such that

(i) for any $U \in \mathcal{U}$, there is an isomorphism $\psi_U : p^{-1}(U) \cong \mathbb{A}^n_U$ of U-schemes, and

(ii) for any $U, V \in \mathcal{U}$ and any affine open subset $W = \operatorname{Spec} A \subseteq U \cap V$, the isomorphism $\psi_V^{-1} \circ \psi_U$ of $\mathbb{A}^n_W = \operatorname{Spec} A[x_1, \ldots, x_n]$ is given by a linear automorphism θ of $A[x_1, \ldots, x_n]$, namely, $\theta|_A = \mathrm{id}_A$ and $\theta(x_i) = \sum_{j=1}^{n} a_{ij} x_j$ for some $a_{ij} \in A$.

We shall call the datum $\{\psi_U\}_{U \in \mathcal{U}}$ satisfying (i) and (ii) a **vector bundle structure over** X on Y. A **morphism between vector bundles** (Y, ψ_U) of rank n and (Y', ψ'_V) of rank n' over X is a morphism $g : Y \to Y'$ of X-schemes such that for any $U \in \mathcal{U}$, $V \in \mathcal{V}$ and $W = \operatorname{Spec} A \subseteq U \cap V$, the restriction $g|_W : Y|_W \to Y'|_W$ comes from a linear homomorphism $A[y_1, \ldots, y_{n'}] \to A[x_1, \ldots, x_n]$.

1.144 Symmetric algebra. Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Put $\operatorname{Sym}^0 \mathcal{F} = \mathcal{O}_X$, $\operatorname{Sym}^1 \mathcal{F} = \mathcal{F}$, and for integers $n \geqslant 2$,

$$\operatorname{Sym}^n \mathcal{F} = \mathcal{F}^{\otimes n} / \mathcal{I}_n$$

where $\mathcal{I}_n$ is the $\mathcal{O}_X$-module generated by the local sections $s_1 \otimes \cdots \otimes s_n - s_{\sigma(1)} \otimes \cdots \otimes s_{\sigma(n)} \in \mathcal{F}^{\otimes n}$ $(\sigma \in S_n)$. This is called the n**-th symmetric power** of $\mathcal{F}$, and it is an $\mathcal{O}_X$-module. Here the tensor product is always over $\mathcal{O}_X$. The direct sum

$$\operatorname{Sym} \mathcal{F} = \bigoplus_{n \geqslant 0} \operatorname{Sym}^n \mathcal{F}$$

is called the **symmetric algebra** of $\mathcal{F}$, which is an $\mathcal{O}_X$-algebra.

1.145 For any integer $n \geqslant 0$, the n-th symmetric power $\operatorname{Sym}^n \mathcal{F}$ is isomorphic to the sheafification of the presheaf $U \mapsto \operatorname{Sym}^n \mathcal{F}(U)$. To see this, for $n \geqslant 2$, the presheaf $U \mapsto \operatorname{Sym}^n \mathcal{F}(U)$ is the quotient of $\mathcal{F} \otimes^p \cdots \otimes^p \mathcal{F}$ by the subsheaf $\mathcal{I}'_n$ generated by the local sections $s_1 \otimes \cdots \otimes s_n - s_{\sigma(1)} \otimes \cdots \otimes s_{\sigma(n)} \in \mathcal{F}^{\otimes n}$ $(\sigma \in S_n)$. By definition, $(\mathcal{F} \otimes^p \cdots \otimes^p \mathcal{F})^\dagger = \mathcal{F}^{\otimes n}$ and $(\mathcal{I}'_n)^\dagger = \mathcal{I}_n$. At this stage the claim follows from (1.21).

1.146 Let $f : Y \to X$ be a morphism of ringed spaces and let $\mathcal{F}$ be a $\mathcal{O}_Y$-module. Then there exists a canonical isomorphism

$$\operatorname{Sym}^n f^* \mathcal{F} \cong f^*(\operatorname{Sym}^n \mathcal{F}).$$

It is clear for $n = 0, 1$. For $n \geqslant 2$, since f^* is right exact, there is an exact sequence

$$f^* \mathcal{I}_n \longrightarrow f^*(\mathcal{F}^{\otimes n}) \longrightarrow f^*(\operatorname{Sym}^n \mathcal{F}) \longrightarrow 0$$

Now it suffices to show that image of $f^*\mathcal{I}_n$ in $f^*(\mathcal{F}^{\otimes n}) \cong (f^*\mathcal{F})^{\otimes n}$ is isomorphic to the ideal defining $\operatorname{Sym}^n f^*\mathcal{F}$. This is clear in the level of presheaves, and by (1.21) the same holds in the level of sheaves as well.

1.147 Lemma. Let A be a ring and M an A-module. Then $\operatorname{Sym}^n \widetilde{M} \cong \widetilde{\operatorname{Sym}^n M}$.

Proof. The cases $n = 0, 1$ are obvious. Let $n \geqslant 2$, and let $f \in A$. Consider the composition

$$\widetilde{\operatorname{Sym}^n M}(D(f)) = (\operatorname{Sym}^n M)_f \cong \operatorname{Sym}^n M_f = \operatorname{Sym}^n \widetilde{M}(D(f)) \to (\operatorname{Sym}^n \widetilde{M})(D(f))$$

where the isomorphism results from the same argument in (1.146), and the last arrow follows from (1.145). Since the arrows involved are natural, by (1.4) this defines a morphism $\alpha : \widetilde{\operatorname{Sym}^n M} \to \operatorname{Sym}^n \widetilde{M}$ of $\mathcal{O}_{\operatorname{Spec} A}$-modules. For any $\mathfrak{p} \in \operatorname{Spec} A$, there is an isomorphism $(\operatorname{Sym}^n M)_{\mathfrak{p}} \cong \operatorname{Sym}^n M_{\mathfrak{p}}$ compatible with the ones above, which coincides with the stalk map $\alpha_{\mathfrak{p}}$. Hence α is an isomorphism by (1.20). □

1.148 Corollary. Let X be a scheme and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module. Then for any affine open subset $U = \operatorname{Spec} A$ of X, we have

$$(U, (\operatorname{Sym}^n \mathcal{F})|_U) \cong (\operatorname{Spec} A, \widetilde{\operatorname{Sym}^n \mathcal{F}}(U)).$$

In particular, $(\operatorname{Sym}^n \mathcal{F})(U) \cong \operatorname{Sym}^n \mathcal{F}(U)$ for any affine open subset U, and

$$(\operatorname{Sym} \mathcal{F})|_U \cong \widetilde{\operatorname{Sym} \mathcal{F}}(U)$$

Proof. By (1.146), (1.147) and (1.57), we have

$$(\operatorname{Sym}^n \mathcal{F})|_U \cong \operatorname{Sym}^n(\mathcal{F}|_U) \cong \operatorname{Sym}^n \widetilde{\mathcal{F}(U)}.$$

For the last assertion,

$$\left(\bigoplus_{n\geqslant 0} \operatorname{Sym}^n \mathcal{F}\right)|_U \cong \bigoplus_{n\geqslant 0} \operatorname{Sym}^n(\mathcal{F}|_U) \cong \bigoplus_{n\geqslant 0} \operatorname{Sym}^n \widetilde{\mathcal{F}(U)} \cong \bigoplus_{n\geqslant 0} \widetilde{\operatorname{Sym}^n} \mathcal{F}(U).$$

The first and the last isomorphisms result from the facts that $(\cdot)|_U$ and $\widetilde{(\cdot)}$ are exact functors (1.20) (1.41). □

1.149 Total space of a locally free sheaf. Let X be a scheme and let $\mathcal{F}$ be a locally free $\mathcal{O}_X$-module of rank $n < \infty$. We can find an affine open cover $\mathcal{U}$ of X consisting such that $\mathcal{F}|_U \cong (\mathcal{O}_X|_U)^{\oplus n}$ for any $U \in \mathcal{U}$. Fix a $U \in \mathcal{U}$ and write $(U, \mathcal{O}_X|_U) \cong (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$. Then

$$(\operatorname{Sym} \mathcal{F})|_U \cong \widetilde{\operatorname{Sym} \mathcal{F}}(U) \cong \widetilde{\operatorname{Sym} A^{\oplus n}} \cong A[x_1, \ldots, x_n].$$

This means $\operatorname{Sym} \mathcal{F}$ is locally free and is locally of finite type. The same argument works for the dual $\mathcal{F}^\vee$; in particular, $\operatorname{Sym}(\mathcal{F}^\vee)$ is a quasi-coherent $\mathcal{O}_X$-algebra, so we can apply the relative spec construction. The X-scheme $\mathbb{V}(\mathcal{F}) := \mathbf{Spec}_X \operatorname{Sym}(\mathcal{F}^\vee)$ is called the **total space** of $\mathcal{F}$.

1.150 Retain the notations in (1.149). Let $U \in \mathcal{U}$. Then

$$(\mathbf{Spec}_X \operatorname{Sym}(\mathcal{F}^\vee))|_U \cong \operatorname{Spec} \operatorname{Sym}(\mathcal{F}^\vee)(U) \cong \operatorname{Spec} \mathcal{O}_X(U)[y_1, \dots, y_n] = \mathbb{A}^n_U.$$

Let $V \in \mathcal{U}$ and pick $W \subseteq U \cap V$ with W affine. Then $\mathcal{F}|_W$ is free. The transition function on W is obtained by applying Sym to the isomorphism

$$(\mathcal{O}_X|_W)^n \cong (\mathcal{F}|_U)|_W = \mathcal{F}|_W = (\mathcal{F}|_V)|_W \cong (\mathcal{O}_X|_W)^n.$$

In particular, this shows the transition function is a linear automorphism (for it is degree-preserving). Hence, we have shown that $\mathbb{V}(\mathcal{F}) \to X$ is a vector bundle of rank n over X.

1.151 If $\mathcal{F} \to \mathcal{G}$ is a morphism of finite locally free $\mathcal{O}_X$-modules, first taking dual and next applying $\mathbf{Spec}_X \operatorname{Sym}$ gives a morphism $g : \mathbb{V}(\mathcal{F}) \to \mathbb{V}(\mathcal{G})$. Locally on an affine open, it is the morphism $g|_U : \operatorname{Spec} \operatorname{Sym} \widetilde{\mathcal{F}^\vee(U)} \to \operatorname{Spec} \operatorname{Sym} \widetilde{\mathcal{G}^\vee(U)}$, and it is induced by the morphism $\mathcal{F}|_U \to \mathcal{G}|_U$. Thus $g|_U$ corresponds to a linear homomorphism in the sense of (1.143), and this shows that g is a vector bundle morphism.

1.152 Sheaf of sections. Let Y be an X-scheme and denote by $f : Y \to X$ the structure morphism. Let U be an open set of X. An element in $\operatorname{Hom}_{\mathbf{Sch}_U}(U, Y|_U)$ is called a **section of** f **over** U. Let us put

$$\Gamma(U, Y/X) = \operatorname{Hom}_{\mathbf{Sch}_U}(U, Y|_U) = \{s \in \operatorname{Hom}_{\mathbf{Sch}}(U, Y) \mid f \circ s = \mathrm{id}_U\}$$

By (1.17), the presheaf $U \mapsto \Gamma(U, Y/X)$ of sections is a sheaf of sets. We denote this sheaf by $\Gamma(\cdot, Y/X)$, and it is a subsheaf of $\operatorname{Hom}_{\mathbf{Sch}}(\cdot, Y)$. It is clear that $\Gamma(\cdot, Y/X)$ defines a functor from $\mathbf{Sch}_X$ to $\mathbf{Set}_X$.

1.153 Let $p : Y \to X$ be a vector bundle of rank n over a scheme X. We show that $\Gamma(\cdot, Y/X)$ has a natural $\mathcal{O}_X$-module structure. First consider the case $X = \operatorname{Spec} A$ being affine and Y is a trivial bundle, i.e., $Y = \mathbb{A}^n_X$. Then $\Gamma(X, Y/X) \cong \operatorname{Hom}_{\mathbf{Alg}_A}(A[x_1, \dots, x_n], A) \cong A^{\oplus n}$, so $\Gamma(X, Y/X)$ has a naturally an A-module (note that the resulting module structure is independent of the last isomorphism). For general case, let $U = \operatorname{Spec} A$ be an affine open set of X trivializing Y and let $W = \operatorname{Spec} B \subseteq U$ be another such affine open in X. We then know $\Gamma(U, Y/X)$ (resp. $\Gamma(W, Y/X)$) is a A- (resp. B-) module. Choose a free basis of $\mathcal{O}_{\mathbb{A}^n_W}(W)$ in a way that $\mathcal{O}_{\mathbb{A}^n_W}(W) = B[x_1, \dots, x_n]$ and $\mathcal{O}_{\mathbb{A}^n_U}(U) = A[x_1, \dots, x_n]$. Then there is a commutative diagram

$$\begin{tikzcd}
\Gamma(U, Y/X) = \operatorname{Hom}_{\mathbf{Sch}_U}(U, \mathbb{A}^n_U) \arrow[r, "\sim"] \arrow[d, "\text{restriction}"] & \operatorname{Hom}_{\mathbf{Alg}_A}(A[x_1, \dots, x_n], A) \arrow[r, "\sim"] \arrow[dd, "\circ\varphi"] & A^{\oplus n} \arrow[dd, "\varphi"] \\
\{s \in \operatorname{Hom}_{\mathbf{Sch}}(W, \mathbb{A}^n_U) \mid p \circ s = \mathrm{id}_W\} \arrow[d, "\wr\ (1.28)"] & & \\
\Gamma(W, Y/X) = \operatorname{Hom}_{\mathbf{Sch}_W}(W, \mathbb{A}^n_W) \arrow[r, "\sim"] & \operatorname{Hom}_{\mathbf{Alg}_B}(B[x_1, \dots, x_n], B) \arrow[r, "\sim"] & B^{\oplus n}
\end{tikzcd}$$

where we denote by $\varphi : A \to B$ the restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$. From this we see the module structures are compatible with the restriction, provided the affine open subsets are concerned. Finally, we extend

the action to all open sets by (1.4). Moreover, the argument above also shows that $\Gamma(\cdot, Y/X)$ is locally free of rank n.

1.154 Let X be a scheme. Denote by $\mathbf{FinLoc}_X$ the full subcategory of $\mathbf{Mod}_{\mathcal{O}_X}$ consisting of finite locally free $\mathcal{O}_X$-modules, and denote by $\mathbf{VB}_X$ the category of vector bundles over X (1.143). So far we have two functors

$$\mathbb{V} : \mathbf{FinLoc}_X \longrightarrow \mathbf{VB}_X \ , \qquad \Gamma(\cdot, \cdot/X) : \mathbf{VB}_X \longrightarrow \mathbf{FinLoc}_X$$

In fact, these are mutually inverse to each other, and hence define an anti-equivalence between $\mathbf{FinLoc}_X$ and $\mathbf{VB}_X$.

1.155 Lemma. Let $\mathcal{G}$ be a locally free $\mathcal{O}_X$-module of rank n, and let $\mathcal{F} = \Gamma(\cdot, \mathbb{V}(\mathcal{G})/X)$ be the sheaf of sections of the vector bundle $\mathbb{V}(\mathcal{G})$. Then $\mathcal{F} \cong \mathcal{G}$.

1.7.2 Relative proj

2 Morphisms of Schemes

3 Varieties

3.1 Classical varieties

3.1.1 Affine varieties

3.1 In this subsection, let k be a field. For any positive integer $n \geqslant 1$, we write $\mathbb{A}^n_k$ for the product space k^n, and called the **n-dimensional affine space over** k. If the field involved is clear from the context, we usually omit the subscript and simply write $\mathbb{A}^n$. For a subset S of the polynomial ring $k[x_1, \ldots, x_n]$, we put $V(S)$ to denote the common zeroes in $\mathbb{A}^n$ of the polynomials in S, i.e.,

$$V(S) := \{p \in \mathbb{A}^n \mid f(p) = 0 \text{ for all } f \in S\}.$$

If $I \trianglelefteq k[x_1, \ldots, x_n]$ is the ideal generated by S, then clearly $V(I) = V(S)$. A set of the form $V(S)$ is called an **(affine) algebraic set**. There are some properties of the assignment $S \mapsto V(S)$ that can be observed directly from the definition :

(i) If $\{I_\alpha\}_\alpha$ is a family of ideals in $k[x_1, \ldots, x_n]$, then $V\left(\bigcup_\alpha I_\alpha\right) = \bigcap_\alpha V(I_\alpha)$.

(ii) If $I_1, I_2 \trianglelefteq k[x_1, \ldots, x_n]$ are two ideals, then $V(I_1) \cup V(I_2) = V(I_1 \cap I_2) = V(I_1 I_2)$.

(iii) $\varnothing = V(1)$ and $\mathbb{A}^n = V(0)$.

(iv) If $I \trianglelefteq k[x_1, \ldots, x_n]$ is an ideal, then $V(I) = V(\sqrt{I})$.

(v) If $S_1 \subseteq S_2 \subseteq k[x_1, \ldots, x_n]$, then $V(S_2) \subseteq V(S_1)$.

By (i), (ii) and (iii), the sets of the form $V(I)$, $I \trianglelefteq k[x_1, \ldots, x_n]$ define a topology (of closed sets) on $\mathbb{A}^n$. The defined topology is called the **Zariski topology** on the affine space $\mathbb{A}^n$.

It is easy to write down an open basis for the Zariski topology. To start with, since $k[x_1, \ldots, x_n]$ is Noetherian, every ideal I is finitely generated ; say $I = \langle f_1, \ldots, f_m \rangle$. Then by (i),

$$V(I) = V((f_1)) \cap \cdots \cap V((f_m)).$$

For $f \in k[x_1, \ldots, x_n]$, put $D(f) = \{p \in \mathbb{A}^n \mid f(p) \neq 0\}$; a set of this form is called a **principal affine open set**. Then taking complement, the above equation becomes

$$\mathbb{A}^n \backslash V(I) = D(f_1) \cup \cdots \cup D(f_m).$$

Every open set in $\mathbb{A}^n$ has the form as the left hand side. For $f, g \in k[x_1, \ldots, x_n]$, by (ii) we have $D(f) \cap D(g) = D(fg)$. Thus the sets of the form $D(f)$ are a basis for the Zariski topology.

3.2 For a subset $S \subseteq \mathbb{A}^n$, put $I(S)$ to denote the set of all polynomials that vanish on S, i.e.,

$$I(S) = \{f \in k[x_1, \ldots, x_n] \mid f(p) = 0 \text{ for all } p \in S\}.$$

This is a radical ideal in $k[x_1, \ldots, x_n]$. There are some formal properties of the assignment $S \mapsto I(S)$:

(i) If $S_1, S_2 \subseteq \mathbb{A}^n$, then $I(S_1 \cup S_2) = I(S_1) \cap I(S_2)$.

(ii) If $S_1 \subseteq S_2 \subseteq \mathbb{A}^n$, then $I(S_2) \subseteq I(S_1)$.

(iii) If $V \subseteq \mathbb{A}^n$ is an algebraic set, then $V(I(V)) = V$.

(iv) For $S \subseteq \mathbb{A}^n$, we have $V(I(S)) = \overline{S}$, the Zariski closure of S in $\mathbb{A}^n$.

(v) $I(\varnothing) = k[x_1, \ldots, x_n]$, and if $\#k = \infty$, $I(\mathbb{A}^n) = 0$.

For an algebraic set $V \subseteq \mathbb{A}^n$, the quotient

$$k[V] := k[x_1, \ldots, x_n]/I(V)$$

is called the **coordinate ring** of V. Since $I(V)$ is radical, the quotient ring $k[V]$ is reduced.

3.3 Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be algebraic sets. A map $\varphi : V \to W$ is called a **morphism** if for any integer $1 \leqslant i \leqslant m$, we have $\mathrm{pr}_i \circ \varphi \in k[x_1, \ldots, x_n]$, where pr_i denotes the projection to the i-th component.

3.4 If $\varphi : V \to W$ is a morphism, then the pullback $f \mapsto f \circ \varphi$ defines a k-algebra homomorphism $\varphi^* : k[W] \to k[V]$. In fact, this defines a bijection

$$\begin{array}{ccc} \{\text{morphisms } V \to W\} & \longrightarrow & \mathrm{Hom}_{\mathbf{Alg}_k}(k[W], k[V]) \\ \varphi & \longmapsto & \varphi^* \end{array}$$

3.5 Lemma. Let $\varphi : V \to W$ be a morphism between algebraic sets. Then

(a) $\ker \varphi^* = I(\varphi(V))$.

(b) $\overline{\varphi(V)} = V(\ker \varphi^*)$.

In particular, $\overline{\varphi(I)}$ is an algebraic set with coordinate ring $k[W]/\ker \varphi^*$.

3.1.2 Nullstellensatz.

3.6 Noether's normalization lemma. Let A be a k-algebra of finite type. Then there exist $y_1, \ldots, y_d \in A$ $(0 \leqslant d \leqslant m)$ such that the y_i are algebraically independent and A is finite over $k[y_1, \ldots, y_d]$.

Proof. Write $A = k[r_1, \ldots, r_m]$. We prove this by induction on m.

1° $m = 1$: Say $A = k[r]$. If r is transcendental over k, pick $y_1 = r$. Otherwise, r is algebraic over k so that A is finite over k.

2° $m > 1$: If the $r_1, \ldots, r_m$ are algebraically independent over k, then done. Otherwise, there's a nonzero $f \in k[x_1, \ldots, x_m]$ such that $f(r_1, \ldots, r_m) = 0$. Renumbering the subscripts, if necessary, we assume $f(x_1, \ldots, x_m)$ is not a constant in the variable x_m. Let $d = \deg f$, the maximum of the total monomial degrees. For $j = 1, \ldots, m-1$, define

$$X_j := x_j - x_m^{(1+d)^j}$$

For each monomial $x_1^{e_1} \cdots x_m^{e_m}$, we have

$$\begin{aligned} x_1^{e_1} \cdots x_m^{e_m} &= (X_1 + x_m^{1+d})^{e_1} \cdots (X_{m-1} + x_m^{(1+d)^{m-1}})^{e_{m-1}} x_m^{e_m} \\ &= x_m^{e_m + e_1(1+d) + \cdots + e_{m-1}(1+d)^{m-1}} + \cdots \end{aligned}$$

Note that different $(e_1, \ldots, e_m)$ give polynomials in $X_1, \ldots, X_{m-1}, x_m$ with the different highest degrees of x_m.

Now write

$$g(X_1, \ldots, X_{m-1}, x_m) = f(X_1 + x_m^{e_1(1+d)^1}, \ldots, x_m^{e_m}) = c x_m^N + \sum_{j=0}^{N-1} h(X_1, \ldots, X_{m-1}) x_m^j$$

for non-zero $c \in k$. For $j = 1, \ldots, m-1$, let $s_j = r_j - r_m^{(1+d)^j}$. Then $\frac{1}{c} g(s_1, \ldots, s_{m-1}, r_m) = \frac{1}{c} f(r_1, \ldots, r_m) = 0$, i.e, r_m is integral over $B := k[s_1, \ldots, s_{m-1}]$. By induction hypothesis, there exists $y_1, \ldots, y_d$, $(0 \leqslant d \leqslant m-1)$ such that $y_1, \ldots, y_d$ are algebraically independent over k and B is finite over $k[y_1, \ldots, y_d]$ and thus A is finite over $k[y_1, \ldots, y_d]$.

□

3.7 Zariski's lemma. Let K/k be a field extension. If K is of finite type over k, it's finite over k.

Proof. By normalization lemma, $k \subseteq k[y_1, \ldots, y_d] \subseteq K$ with K finite over $k[y_1, \ldots, y_d]$ for some algebraically independent elements $y_1, \ldots, y_d$ over k. Since K is a field, it follows that $k[y_1, \ldots, y_d]$ is a field, and thus $d = 0$, i.e, K is algebraic over k. Since K is finitely generated as k-algebra, $[K : k]$ is finite. □

3.8 Hilbert's Nullstellensatz. In what follows, we assume k is an algebraically closed field.

(i) (Weak form) There is a bijection

$$\begin{aligned} \mathbb{A}_k^n &\longrightarrow \operatorname{mSpec} k[x_1, \ldots, x_n] \\ (a_1, \ldots, a_n) &\longmapsto (x_1 - a_1, \ldots, x_n - a_n) \end{aligned}.$$

In particular, if $I \trianglelefteq k[x_1, \ldots, x_n]$ is a proper ideal, then $V(I) \neq \varnothing$ in $\mathbb{A}_k^n$.

(ii) For any ideal I of $k[x_1, \ldots, x_n]$, we have the equality

$$I(V(I)) = \sqrt{I}.$$

Proof.

(i) Clearly, $(x_1 - a_1, \ldots, x_n - a_n)$ is a maximal ideal. Conversely, let $\mathfrak{m} \in \operatorname{mSpec} k[x_1, \ldots, x_n]$. Then $K := k[x_1, \ldots, x_n]/\mathfrak{m}$ is a field of finite type over k, so by (3.7) K is finite over k. In particular, K/k is algebraic. But k is algebraically closed, this implies $K = k$. For each $1 \leqslant i \leqslant n$, we have $x_i - a_i \in \mathfrak{m}$ for some $a_i \in k$, and thus $\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n)$.

(ii) The nontrivial part if $I(V(I)) \subseteq \sqrt{I}$. Assume $g \in I(V(I))$ and $I = (f_1, \ldots, f_m)$. Introduce a new indeterminate x_{m+1}, and consider the ideal

$$I' = (f_1, \ldots, f_m, gx_{n+1} - 1) \trianglelefteq k[x_1, \ldots, x_n, x_{n+1}]$$

Then $V(I') = \varnothing$, so by (i), it must be the case $(f_1, \ldots, f_m, gx_{m+1} - 1) = k[x_1, \ldots, x_n, x_{n+1}]$. But note

$$\frac{k[x_1, \ldots, x_n]_g}{I_g} \cong \frac{k[x_1, \ldots, x_n, x_{n+1}]}{I'},$$

so that $1 \in I_g$. This means

$$1 = \frac{a_1 f_1 + \cdots + a_m f_m}{g^\ell}$$

for some $a_i \in k$ and $\ell \in \mathbb{Z}_{\geqslant 0}$, which exactly means $g \in \sqrt{I}$.

□

3.9 Let A be a k-algebra of finite type. Say $\{y_1, \ldots, y_s\}$ is a generating set of A. Then there is a surjection

$$\begin{aligned} \Phi : k[x_1, \ldots, x_s] &\longrightarrow A = k[y_1, \ldots, y_s] \\ x_i &\longmapsto y_i \end{aligned}$$

Since $k[x_1, \ldots, x_s]$ is Noetherian, $\ker(\Phi) = (f_1, \ldots, f_m)$ for some $f_1, \ldots, f_m \in k[x_1, \ldots, x_s]$, and we can form the algebraic set

$$V = V(\ker(\Phi)) = V((f_1, \ldots, f_m)) \subseteq \mathbb{A}^s_k.$$

The coordinate ring of V is by definition $k[V] = k[x_1, \ldots, x_s]/I(V)$. By Nullstellensatz, we have $I(V) = \sqrt{\ker(\Phi)}$. Hence if A is a reduced ring, we have $k[V] \cong A$ as k-algebras.

Suppose $\{z_1, \ldots, z_r\}$ is another generating set of A. Then there is a similarly defined surjection $\Psi : k[x_1, \ldots, x_r] \to A$ and the algebraic set $V' = V(\ker(\Psi)) \subseteq \mathbb{A}^r_k$. We have the isomorphism

$$k[V'] \cong A \cong k[V]$$

as k-algebras, so $V \cong V'$ as algebraic sets over k.

Let $\mathbf{AffVar}_k$ be the category of affine algebraic sets over k whose morphisms are defined as (3.3), and let $\mathbf{redfgAlg}_k$ be the full subcategory of $\mathbf{Alg}_k$ consisting of reduced k-algebra of finite type. Together with (3.4), we then have shown that there is an anti-equivalence of categories

$$\begin{aligned} \mathbf{AffVar}_k &\longrightarrow \mathbf{redfgAlg}_k \\ V &\longmapsto k[V] \end{aligned}$$

3.10 Let A be a reduced k-algebra of finite type. Let $\{y_1, \ldots, y_n\}$ be a generating set of A and form the surjection $\Phi : k[x_1, \ldots, x_n] \to A$. Let $V = V(\ker(\Phi)) \subseteq \mathbb{A}^n_k$. Let $p = (a_1, \ldots, a_n) \in V$ and consider the

maximal ideal $\mathfrak{m}_p = (x_1 - a_1, \dots, x_n - a_n)$ of $k[x_1, \dots, x_n]$. By (3.2).(ii), we have $\ker(\Phi) \subseteq \mathfrak{m}_p$. Conversely, if $\mathfrak{m}$ is a maximal ideal of $k[V]$, regarding it as a maximal ideal in $k[x_1, \dots, x_n]$ containing $\ker(\Phi)$, by Nullstellensatz we obtain a point $p \in \mathbb{A}^n_k$ with $\mathfrak{m}_p = \mathfrak{m}$. The point lies in V as $\{p\} = V(\mathfrak{m}_p) \subseteq V(\ker(\Phi)) = V$. This establishes a bijection between V and mSpec $k[V]$, and by the isomorphism $k[V] \cong A$, we have a bijection

$$\begin{aligned} V &\longrightarrow \operatorname{mSpec} A \\ p &\longmapsto \Phi(\mathfrak{m}_p). \end{aligned}$$

Now let $a \in A$ and let $f \in k[V]$ be the corresponding element. Then the value of f at a point $p \in V$ is the same as the class of a in the residue field $\kappa(\Phi(\mathfrak{m}_p))$, i.e., the image of A in $A/\Phi(\mathfrak{m}_p)$. Indeed, we have $A/\Phi(\mathfrak{m}_p) \cong k[V]/\mathfrak{m}_p \cong k$ as k-algebras.

3.11 Retain the notations in (3.10). For a ring R, equip mSpec R with the subspace topology from the topology on the affine scheme Spec R. Then clearly mSpec $k[V] \cong$ mSpec A as topological spaces. Also, it follows from the very definition that the bijection

$$\begin{aligned} \mathbb{A}^n_k &\longrightarrow \operatorname{mSpec} k[x_1, \dots, x_n] \\ (a_1, \dots, a_n) &\longmapsto (x_1 - a_1, \dots, x_n - a_n) \end{aligned}.$$

in Nullstellensatz is a homeomorphism. For an ideal I, we have a commutative diagram

$$\begin{tikzcd} \mathbb{A}^n_k \arrow[r, "\sim"] & \operatorname{mSpec} k[x_1, \dots, x_n] \\ V(I) \arrow[u] \arrow[r, "\sim"] & \operatorname{mSpec} k[V] \arrow[u] \end{tikzcd}$$

The vertical maps are all closed embeddings, so the horizontal map on the bottom is a homeomorphism as well. Altogether we see the bijection

$$V \longrightarrow \operatorname{mSpec} A$$

in (3.10) is a homeomorphism.

Let $f : B \to A$ be a k-algebra homomorphism of reduced k-algebra of finite types. Let W be any affine algebraic set constructed as in (3.10) with A replaced by B. Then we have a k-algebra homomorphism $f' : k[W] \cong B \xrightarrow{f} A \cong k[V]$, and by (3.4) it gives a morphism $F : V \to W$ of algebraic sets. We claim the diagram

$$\begin{tikzcd} W \arrow[r, "\sim"] & \operatorname{mSpec} k[W] \\ V \arrow[u, "F"] \arrow[r, "\sim"] & \operatorname{mSpec} k[V] \arrow[u, "f'^{-1}"'] \end{tikzcd}$$

commutes. This is clear from the construction of F (3.4).

3.12 Consider the projective space $\mathbb{P}^n$ over k with homogeneous coordinates $x_0, \dots, x_n$. For a point $p \in \mathbb{P}^n$, it lies in some affine pieces $D_+(x_i)$ which is isomorphic to the affine space $\mathbb{A}^n_k$. By Nullstellensatz it corresponds to a maximal ideal in $k[x_0, \dots, x_n]_{x_i}$, which can be viewed as a maximal ideal in $k[x_0, \dots, x_n]$ not containing x_i; call this maximal ideal $\mathfrak{m}_p$. If p also lies in $D_+(x_j)$, it will then corresponds to a maximal ideal in $k[x_0, \dots, x_n]_{x_j}$. Nevertheless, since $p \in D_+(x_i) \cap D_+(x_j)$, it gives a maximal ideal in $k[x_0, \dots, x_n]_{x_i x_j}$, and it can be obtained by localizing $\mathfrak{m}_p$ at x_j. By symmetry we see $\mathfrak{m}_p \in \mathrm{mSpeck}[x_0, \dots, x_n]$ is well-defined. Thus we obtain a well-defined map

$$\begin{aligned} \mathbb{P}^n_k &\longrightarrow \{\mathfrak{m} \in \mathrm{mSpeck}[x_0, \dots, x_n] \mid x_0 \cdots x_n \notin \mathfrak{m}\} \\ p &\longmapsto \mathfrak{m}_p. \end{aligned}$$

The procedure of obtaining $\mathfrak{m}_p$ from p above also shows that this is a bijection. It is this bijection that motivates the Proj construction in (1.113).

3.13 Projective Hilbert's Nullstellensatz. Let I be a homogeneous ideal of $k[x_0, \dots, x_n]$.

3.1.3 Sheaf of regular functions

Recall that we are assuming k is algebraically closed.

3.14 Let Y be an algebraic set and $V \subseteq Y$ a Zariski open set. A **regular function on** V is a map $g : V \to \mathbb{A}^1$ such that any point $y \in V$ admits an open neighborhood $W \subseteq Y$ and $u, v \in k[Y]$ with u nonvanishing on W such that $g|_W = \dfrac{v|_W}{u|_W}$. A regular function on V is continuous in the Zariski topology.

We put $\mathcal{O}_Y(V)$ to be the set of regular functions on V. The assignment $V \mapsto \mathcal{O}_Y(V)$ is clearly a sheaf of k-algebras on Y, and $(Y, \mathcal{O}_Y)$ is a local-ringed space. Indeed, if a regular function defined near a point $y \in Y$ is nonzero at y, then by continuity it is nonvanishing on an open neighborhood of y, making it an invertible element in the stalk $\mathcal{O}_{Y,y}$.

3.15 Let Y be an algebraic set. There is an inclusion $k[Y] \to \mathcal{O}_Y(Y)$. Then for $f \in k[Y]$, we can form the open set Y_f as in (1.27), and we see there that $f \in \mathcal{O}_Y(Y_f)^\times$. We can compare Y_f with $D(f)$. In fact, $D(f) = Y_f$. Indeed, $y \in Y_f$ if and only if $f_y \in \mathcal{O}^\times_{Y,y}$, if and only if $f(y) \neq 0$, or $y \in D(f)$.

By the universal property of localization, the composition $k[Y] \to \mathcal{O}_Y(Y) \to \mathcal{O}_Y(Y_f)$ induces a homomorphism $\theta : k[Y]_f \to \mathcal{O}_Y(Y_f)$. This is injective, for if $\theta(gf^{-m}) = 0$, then $g|_{Y_f} = 0$, or $Y_f \subseteq V(g)$, which implies $\varnothing = Y_f \cap Y_g = D(fg)$, or $fg = 0 \in k[Y]$. This exactly means $gf^{-m} = 0$ in $k[Y]_f$. Moreover,

Theorem. The canonical map $\theta : k[Y]_f \to \mathcal{O}_Y(Y_f)$ is a k-algebra isomorphism.

Proof. It remains to show the surjectivity. The proof is similar to (1.40), with some extra topological concern. Let $g \in \mathcal{O}_Y(Y_f)$. By definition, each $y \in Y_f$ admits an open neighborhood $W_y \subseteq Y_f$ and

v_y, $u_y \in k[Y]$ with u_y nonvanishing on W_y such that $g|_{W_y} = \dfrac{v_y|_{W_y}}{u_y|_{W_y}}$. Since $\{Y_g \mid g \in k[Y]\}$ is a basis for Y, we can find $u'_y \in k[Y]$ such that $y \in Y_{u'_y} \subseteq W_y \subseteq Y_{u_y}$. Taking complement gives $V(u'_y) \supseteq V(u_y)$, and by Nullstellensatz, $\sqrt{(u'_y)} \subseteq \sqrt{(u_y)}$. Thus $u'^m_y = u_y u''_y$ for some $u''_y \in k[Y]$ and $m \in \mathbb{Z}_{\geqslant 1}$; then

$$\frac{v_y}{u_y} = \frac{v_y u''_y}{u_y u''_y} = \frac{v_y u''_y}{(u'_y)^m}.$$

Since $k[Y]$ is Noetherian, Y is Noetherian, which implies Y_f is Noetherian, and hence compact. Thus we can find a finite set $\{y_i\}_{i\in I} \subseteq Y_f$ such that $\{Y_{u'_{y_i}} = Y_{(u'_{y_i})^m}\}_{i\in I}$ covers Y_f. Replacing v_y by $v_y u''_y$ and u_y by $(u'_y)^m$, we find $u_1, \ldots, u_n$, $v_1, \ldots, v_n \in k[Y]$ such that $g|_{Y_{u_j}} = \dfrac{v_j|_{Y_{u_j}}}{u_j|_{Y_{u_j}}}$ with $Y_f = Y_{u_1} \cup \cdots \cup Y_{u_n}$.

For each $i \neq j$, we have

$$\frac{v_i|_{Y_{u_i}\cap Y_{u_j}}}{u_i|_{Y_{u_i}\cap Y_{u_j}}} = g|_{Y_{u_i}\cap Y_{u_j}} = \frac{v_j|_{Y_{u_i}\cap Y_{u_j}}}{u_j|_{Y_{u_i}\cap Y_{u_j}}}$$

so $(v_i u_j - v_j u_i)|_{Y_{u_i}\cap Y_{u_j}} = 0$. This implies $u_i u_j (v_i u_j - v_j - u_i) = 0$ on Y. Further replacing v_j by $v_j u_j$ and u_j by u_j^2, we may assume $u_i v_j - u_j v_i = 0$ on Y for any i, j. Since $Y_f = \bigcup_{i=1}^n Y_{u_i}$, we have $\sqrt{(f)} = \sqrt{(u_1, \ldots, u_n)}$ so that

$$f^m = a_1 u_1 + \cdots + a_n u_n$$

for some $a_1, \ldots, a_n \in k[Y]$ and $m \in \mathbb{Z}_{\geqslant 1}$. Define $v = a_1 v_1 + \cdots + a_n v_n$. We claim $gf^m = v$, which will imply $\theta(vf^{-m}) = g$, completing the proof. This is easy, as for any $1 \leqslant j \leqslant n$,

$$gf^m|_{Y_{u_j}} = \sum_{i=1}^n a_i|_{Y_{u_j}} (gu_i)|_{Y_{u_j}} = \sum_{i=1}^n a_i|_{Y_{u_j}} (\frac{v_j}{u_j} u_i)|_{Y_{u_j}} = \sum_{i=1}^n a_i|_{Y_{u_j}} v_i|_{Y_{u_j}} = v|_{Y_{u_j}}.$$

□

3.16 Corollary. Let Y be an algebraic set.

(a) $\mathcal{O}_Y(Y) = k[Y]$.

(b) For any $y \in Y$, we have $\mathcal{O}_{Y,y} \cong k[Y]_{I(\{y\})}$.

Note these strike a resemblance with the results in (1.39).

Proof. (a) follows directly from Theorem 3.15. For (b), we compute

$$\mathcal{O}_{Y,y} = \varinjlim_{y\in U} \mathcal{O}_Y(U) \cong \varinjlim_{f\in k[Y],\, f(y)\neq 0} \mathcal{O}_Y(Y_f) \cong \varinjlim_{f\notin I(\{y\})} k[Y]_f \cong k[Y]_{I(\{y\})}.$$

The third isomorphism results from the functoriality of the isomorphism in Theorem 3.15, and the final isomorphism follows from the same reason as in (1.39). □

3.17 Let X be a topological space. The assignment $U \mapsto \mathrm{Hom}_{\mathbf{Set}}(U, k)$ defines a sheaf of k-algebras on X, which we denote by k^X. A local-ringed space $(X, \mathcal{O}_X)$ is called a **basic k-space** if $\mathcal{O}_X$ is a subsheaf of k-algebras of k^X and for any $x \in X$, the unique maximal ideal of $\mathcal{O}_{X,x}$ is

$$\mathfrak{m}_{X,x} = \{g_x \in \mathcal{O}_{X,x} \mid g(x) = 0 \in k\}.$$

A **morphism of basic k-spaces** is a continuous map $f : X \to Y$ such that for any open $V \subseteq Y$ and $g \in \mathcal{O}_Y(V)$, we have $g \circ f \in \mathcal{O}_X(f^{-1}(V))$. In other words, a morphism of basic k-space is a morphism in **LRS** such that the morphism on sheaves is given by function pullback. The category of basic k-spaces is denote by $\mathbf{bSp}_k$.

As an example, if Y is an algebraic set, we see in (3.14) that $(Y, \mathcal{O}_Y)$ is a basic k-space.

3.18 Definition. Let $(X, \mathcal{O}_X)$ be a basic k-space.

1. X is called an **affine variety over** k if there exist an algebraic set Y over k such that $(X, \mathcal{O}_X) \cong (Y, \mathcal{O}_Y)$ as basic k-spaces.
2. X is called an **(algebraic) variety over** k if it admits a finite open cover $\mathcal{U}$ such that $(U, \mathcal{O}_X|_U)$ is an affine variety over k for any $U \in \mathcal{U}$.

Denote by $\mathbf{AffVar}_k$ and $\mathbf{Var}_k$ the full subcategories of $\mathbf{bSp}_k$ whose objects consist of affine varieties and varieties over k, respectively.

3.19 Note that by (3.4), $\mathbf{AffVar}_k$ defined in (3.18) is equivalent to the one defined in (3.9), and by (3.16) the anti-equivalence there now takes the form

$$\begin{aligned} \mathbf{AffVar}_k &\longrightarrow \mathbf{redfgAlg}_k \\ X &\longmapsto \mathcal{O}_X(X) \end{aligned}$$

A coordinate-free description of the inverse is given by $A \mapsto \mathrm{mSpec}\, A$.

3.20 Theorem. Let X be a basic k-space and Y an affine variety over k. There exists a bijection

$$\begin{aligned} \mathrm{Hom}_{\mathbf{bSp}_k}(X, Y) &\longrightarrow \mathrm{Hom}_{\mathbf{Alg}_k}(k[Y], \mathcal{O}_X(X)) \\ f : X \to Y &\longmapsto k[Y] = \mathcal{O}_Y(Y) \ni g \mapsto g \circ f \end{aligned}$$

Proof. The proof is similar to that of Theorem 1.44. In fact, the proof for injectivity is exactly the same. For surjectivity, let $\theta \in \mathrm{Hom}_{\mathbf{Alg}_k}(k[Y], \mathcal{O}_X(X))$ and define $f : X \to \mathrm{Spec}\, k[Y]$ by setting $f(x) = \theta^{-1}(\mathrm{res}^X_x)^{-1}(\mathfrak{m}_{X,x})$. We claim $f(x)$ is in fact a maximal ideal. In fact, from the definition, the composition $k \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathfrak{m}_{X,x}$ is an isomorphism. Then the homomorphism $\mathrm{res}^X_x\, \theta : k[Y] \to \mathcal{O}_{X,x}$ gives rise to an isomorphism $k[Y]/(\mathrm{res}^X_x\, \theta)^{-1}(\mathfrak{m}_{X,x}) \cong \mathcal{O}_{X,x}/\mathfrak{m}_{X,x} \cong k$, which implies $f(x) = (\mathrm{res}^X_x\, \theta)^{-1}(\mathfrak{m}_{X,x})$ is a maximal ideal. Thus f is in fact a map $f : X \to \mathrm{mSpec}\, k[Y]$, which by Nullstellensatz gives a map $f : X \to Y$ in turn. This is a continuous map, as shown in Theorem 1.44. It remains to show

1. for any open $V \subseteq Y$ and $g \in \mathcal{O}_Y(V)$, $g \circ f \in \mathcal{O}_X(f^{-1}(V))$, and
2. for any $g \in k[Y]$, $\theta(g) = g \circ f \in \mathcal{O}_X(X)$.

For 2., if $x \in X$, then the value $\theta(g)(x)$ is the same as the class of $\theta(g)$ in $\mathcal{O}_{X,x}/\mathfrak{m}_{X,x} \cong k$. On the other hand, $g(f(x))$ is the same as the class of g in $\mathcal{O}_{Y,f(x)}/\mathfrak{m}_{Y,f(x)}$, which is isomorphic to $\mathcal{O}_{X,x}/\mathfrak{m}_{X,x}$ as k-algebras. Hence $\theta(g)(x) = g(f(x)) \in k$. For 1., we use the argument in Theorem 1.44 to obtain, for any $h \in k[Y]$, a map $\theta_h : \mathcal{O}_Y(Y_h) \to \mathcal{O}_X(X_{\theta(h)})$. Explicitly,

$$\theta_h\left(\frac{g}{h^m}\right) = \frac{\theta(g)}{\theta(h)^m}.$$

By 2. we have $\theta_h(gh^{-m}) = \dfrac{g \circ f}{(h \circ f)^m} = (gh^{-m}) \circ f$, so 1. holds for open sets of the form $Y_h = D(h)$. The general case follows from covering V with open sets of this form and the sheaf axiom. □

3.21 Remark. A k**-space** is a local-ringed space $(X, \mathcal{O}_X)$ over k (1.26) such that the composition $k \to \mathcal{O}_{X,x} \to \kappa(x)$ is a field isomorphism for any $x \in X$. Denote by $\mathbf{Sp}_k$ the full subcategory of $\mathbf{LRS}_k$ whose objects are all k-spaces.

It is straightforward to check that a basic k-space is a k-space, and there is a faithful functor $\mathbf{bSp}_k \to \mathbf{Sp}_k$. There is also a bijection

$$\mathrm{Hom}_{\mathbf{Sp}_k}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{Alg}_k}(k[Y], \mathcal{O}_X(X))$$

defined as in Theorem 1.44. The proof is the same as (3.20).

3.1.4 Dimension

3.22 Definition. Let X be a topological space.

1. A subset $C \subseteq X$ is called **locally closed** if it is an intersection of an open set and a closed set in X.
2. A subset of X is called **constructible** if it is a finite union of locally closed subsets.

3.23 Theorem (Chevalley). Let $f : X \to Y$ be a morphism of algebraic varieties. If $C \subseteq X$ is constructible, then $f(C) \subseteq Y$ is constructible.

3.1.5 Associated complex analytic spaces

In this subsubsection by a variety we mean an irreducible algebraic variety.

3.24 Now we consider $\mathbb{C}^n$ as the usual euclidean space. Let U be an open subset of $\mathbb{C}^n$ and denote by $\mathcal{O}_U$ the sheaf of holomorphic functions on U. Let $f_1, \ldots, f_m \in \mathcal{O}_U(U)$ be holomorphic functions and let $Y = V(f_1, \ldots, f_m)$ be the common zero locus of these $f_1, \ldots, f_m$. Put $\mathcal{O}_Y = (\mathcal{O}_U/(f_1, \ldots, f_m))|_Y$, where $(f_1, \ldots, f_m) \trianglelefteq \mathcal{O}_U$ is the ideal sheaf generated by $f_1, \ldots, f_m$. It is clear that $(Y, \mathcal{O}_Y)$ is a local-ringed space, and the sheaf $\mathcal{O}_Y$ is a sheaf of $\mathbb{C}$-algebra.

3.25 Definition. A **complex analytic space** is a local-ringed space $(X, \mathcal{O}_X)$ over $\mathbb{C}$ that admits an open cover $\mathcal{U}$ such that for any $U \in \mathcal{U}$, $(U, \mathcal{O}_X|_U) \cong (Y, \mathcal{O}_Y)$ in $\mathbf{LRS}_{\mathbb{C}}$ for some Y defined as in (3.24).

3.26 Let X be a variety over $\mathbb{C}$. We cover X with affine open subsets Y_i. Each Y_i is isomorphic to some algebraic set, so we can write $Y_i = V(f_1, \ldots, f_m) \subseteq \mathbb{A}^n_{\mathbb{C}}$ for some polynomials $f_1, \ldots, f_m \in \mathbb{C}[x_1, \ldots, x_n]$. If we regard the f_i as holomorphic functions defined on $\mathbb{C}^n$, then Y_i is an complex analytic space, which we denote by $(Y_i)^{\mathrm{an}}$. To distinguish the sheaf of holomorphic functions from that of regular functions, we denote the former by $\mathcal{O}^{\mathrm{an}}_{Y_i}$. The gluing data for $(X, \mathcal{O})$ allows us to glue $(Y_i, \mathcal{O}^{\mathrm{an}}_{Y_i})$ together, by (1.15), to a complex analytic space. We denote the resulting space by $(X^{\mathrm{an}}, \mathcal{O}^{\mathrm{an}}_X)$.

The underlying spaces of X and X^{an} are the same; the only difference is the topology. We refer to the topology on X^{an} the **classical topology/analytic topology** on X.

3.27 Let A be a reduced $\mathbb{C}$-algebra of finite type. By picking a finite generating set $\{y_1, \ldots, y_s\}$ of A, we obtain a closed embedding $\mathrm{mSpec}(A) \to \mathbb{A}^s_{\mathbb{C}}$ with image $V(\ker(\Phi))$, where $\Phi : k[x_1, \ldots, x_s] \to A$ is given by $\Phi(x_i) = y_i$ (3.9). In this way the affine variety $\mathrm{mSpec}(A)$ has a analytic topology inherited from $\mathbb{A}^s_{\mathbb{C}}$. If $\{z_1, \ldots, z_r\}$ is another finite generating set of A, via $\Psi : \mathbb{C}[X_1, \ldots, X_r] \to A$ the affine variety $\mathrm{mSpec}(A)$ has another analytic topology from $\mathbb{A}^r_{\mathbb{C}}$. We show these topologies on $\mathrm{mSpec}(A)$ are homeomorphic in analytic topology. But this is rather clear, for a morphism $V(\ker(\Phi)) \to V(\ker(\Psi))$ comes from a polynomial map $\mathbb{A}^s_{\mathbb{C}} \to \mathbb{A}^r_{\mathbb{C}}$, and a polynomial is continuous.

3.28 Lemma. Let X be a variety over $\mathbb{C}$. If $U \subseteq X$ is a nonempty Zariski open set, then U is classically dense.

Proof. Covering X by affine open sets, we may assume X is affine. Since $D(f) \subseteq U$ for some $0 \neq f \in \mathbb{C}[X]$, to prove the lemma we may assume $U = D(f)$. Taking complement, we need to show $V(f) \subseteq X$ is classically nowhere dense, and it suffices to show $V(f)$ has empty classical interior. If $V(f)$ contains a nonempty open set V, then f is trivial on V, which by identity principle $f \equiv 0$ throughout, a contradiction to $f \neq 0$. □

3.29 Lemma. Let X be a variety over $\mathbb{C}$ and Y a construtible subset of X. Then the Zariski closure of Y in X coincides with the classical closure in X.

Proof. We may assume Y is nonempty Zariski locally closed. Denote by $\overline{Y}^Z$ and $\overline{Y}^C$ the Zariski closure of Y and the classical closure of Y in X, respectively; we always have $\overline{Y}^C \subseteq \overline{Y}^Z$. By definition, Y is Zariski open in $\overline{Y}^Z$, so by (3.28) Y is classically dense in $\overline{Y}^Z$, which means $\overline{Y}^Z \subseteq \overline{Y}^C$. □

3.30 Lemma. Let $U \subseteq \mathbb{A}^n_{\mathbb{C}}$ be a Zariski open set. Then U is classically path-connected.

Proof. Let $x \neq y \in U$ and pick any affine line $L \subseteq \mathbb{A}^n_{\mathbb{C}}$ connecting x, y. Since $U \cap L \subseteq L$ is Zariski open and $L \cong \mathbb{A}^1_{\mathbb{C}}$, $\#(L \backslash U) < \infty$. But $L \cong \mathbb{R}^2$ is a plane, so $U \cap L = L \backslash (L \backslash U)$ is still path-connected. In particular, there is a (smooth) path in $U \cap L$ connecting x and y. □

3.31 Theorem. A variety X over $\mathbb{C}$ is separated if and only if the complex analytic space X^{an} is Hausdorff.

Proof. Since the diagonal $\Delta : X \to X \times_{\mathbb{C}} X$ is always an immersion, we only need to show $\Delta(X)$ is Zariski closed if and only if it is classically closed. But $\Delta(X)$ is construtible by Chevalley theorem, so the result follows from Lemma 3.29.

□

3.32 Theorem. A variety X over $\mathbb{C}$ is complete if and only if the complex analytic space X^{an} is compact.

3.33 Theorem. A morphism $X \to Y$ of varieties over $\mathbb{C}$ is proper if and only if the continuous map $X^{an} \to Y^{an}$ is classically proper.

3.2 Schemes and varieties

3.2.1 Generalities on topology

3.34 Definition. A topological space is called **sober** if every irreducible closed subset has a unique generic point.

3.35 Let X be a sober space. For example, a scheme is sober (1.76). In fact, from the proof there we see a topological space with an open cover consisting of sober spaces is again sober. We have a bijection

$$\begin{aligned} X &\longrightarrow \{\text{irreducible closed subset of } X\} \\ x &\longmapsto \overline{\{x\}}. \end{aligned}$$

Not every topological space is sober. Nevertheless, with each topological space we may associate a sober space in a functorial way, which we will discuss in the next paragraph.

3.36 Soberification. Comparing the topology on $\operatorname{Spec} A$ (1.38) and that on an affine variety (3.1), we are naturally led to the following construction. For a topological space, define $\operatorname{Sob}(X)$ to be the set of all irreducible closed subsets in X. For a closed subset C of X, define

$$V(C) = V^X(C) := \{Z \in \operatorname{Sob}(X) \mid Z \subseteq C\}.$$

Then $\left\{V(C) \mid C \underset{\text{closed}}{\subseteq} X\right\}$ forms a topology on $\operatorname{Sob}(X)$. Namely,

(i) $V(C_1) \cup V(C_2) = V(C_1 \cup C_2)$ for any closed subsets C_1, C_2 of X.

(ii) $V\left(\bigcap_\alpha C_\alpha\right) = \bigcap_\alpha V(C_\alpha)$ for any family $\{C_\alpha\}$ of closed subsets of X.

(iii) $V(\varnothing) = \varnothing$, and $V(X) = \operatorname{Sob}(X)$.

These are clear. Recall that by definition an irreducible space is nonempty (1.74). The resulting topological space $\operatorname{Sob}(X)$ is called the **soberification** of X. There is a natural inclusion $\iota : X \to \operatorname{Sob}(X)$ defined by

$x \mapsto \overline{\{x\}}$. This is a continuous map, for

$$\iota^{-1}(V(C)) = \{x \in X \mid \overline{\{x\}} \in C\} = \{x \in X \mid x \in C\} = C$$

for any closed subset C of X. In fact, this shows the inverse ι^{-1} induces a bijection between $\mathrm{Top}(\mathrm{Sob}(X))$ and $\mathrm{Top}(X)$.

As its name indicates, the space $\mathrm{Sob}(X)$ is a sober topological space. Indeed, the aforementioned bijection implies that C is irreducible if and only if $V(C)$ is irreducible. If $Z \in \mathrm{Sob}(X)$ and $Z \in V(C)$ for some closed C', it is obvious that $V(Z) \subseteq V(C)$. This implies that $V(Z) = \overline{\{Z\}}$ in $\mathrm{Sob}(X)$. If $V(Z) = V(Z')$ for two $Z, Z' \in \mathrm{Sob}(X)$, then clearly $Z = Z'$. This finishes the proof that $\mathrm{Sob}(X)$ is sober. We will see in the following paragraph that $\iota = \iota_X : X \to \mathrm{Sob}(X)$ is universal among all the other sober spaces. To give an intuition, we first observe that ι is a homeomorphism if and only if X is sober. Indeed, if X is sober, then x is uniquely determined by $\overline{\{x\}}$, so ι is a bijection. A continuous bijection whose inverse ι^{-1} induces a bijection between topologies is by definition a homeomorphism. In particular, this shows $\iota : X \to \mathrm{Sob}(X)$ is a homeomorphism.

3.37 Universality of soberification. Denote by **Sob** the full subcategory of **Top** consisting of sober topological spaces. Let X be a topological space, and define a functor $\mathbf{Sob} \to \mathbf{Set}$ by $Y \mapsto \mathrm{Hom}_{\mathbf{Top}}(X, Y)$. Then the soberification $\iota : X \to \mathrm{Sob}(X)$ represents this functor.

As a first step, we show Sob actually defines a functor from **Top** to **Sob**. If $f : X \to Y$ is a continuous map, define $\mathrm{Sob}(f) : \mathrm{Sob}(X) \to \mathrm{Sob}(Y)$ by sending Z to $\overline{f(Z)}$. This is continuous, as for any $C' \underset{\text{closed}}{\subseteq} Y$, we have

$$\mathrm{Sob}(f)^{-1}(V(C')) = \{Z \in \mathrm{Sob}(X) \mid \overline{f(Z)} \in V(C')\} = \{Z \in \mathrm{Sob}(X) \mid f(Z) \subseteq C'\} = V(f^{-1}(C')).$$

If f, g are continuous maps with $g \circ f$ being defined, then $\overline{g(\overline{f(Z)})} = \overline{g(f(Z))}$. Also, $\mathrm{Sob}(\mathrm{id}_X) = \mathrm{id}_{\mathrm{Sob}(X)}$. These prove that $\mathrm{Sob} : \mathbf{Top} \to \mathbf{Sob}$ is really a functor. Additionally, for continuous $f : X \to Y$, since $\overline{f(x)} = \overline{f(\overline{\{x\}})}$ for any $x \in X$, there is a commutative square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\iota_X"'] & Y \arrow[d, "\iota_Y"] \\ \mathrm{Sob}(X) \arrow[r, "\mathrm{Sob}(f)"'] & \mathrm{Sob}(Y) \end{tikzcd}$$

In particular, if Y is sober, the map ι_Y is a homeomorphism, and the diagram gives a functorial map

$$\begin{aligned} \mathrm{Hom}_{\mathbf{Top}}(X, Y) &\longrightarrow \mathrm{Hom}_{\mathbf{Sob}}(\mathrm{Sob}(X), Y) \\ f &\longmapsto \iota_Y^{-1} \circ \mathrm{Sob}(f). \end{aligned}$$

In fact, this is a bijection with inverse $f' \mapsto f' \circ \iota_X$. To see this, let $f' \in \mathrm{Hom}_{\mathbf{Sob}}(\mathrm{Sob}(X), Y)$ and put $f = f' \circ \iota_X$. The identity $\mathrm{Sob}(f) = \iota_Y \circ f'$ is a tautology. This exactly tells what we claim in the beginning of this paragraph.

3.38 We can also describe open sets for the soberification of a space X. Let U be an open subspace of X and C its complement in X. Clearly for $Z \in \mathrm{Sob}(X)$, we have $Z \nsubseteq C$ if and only if $Z \cap U \neq \varnothing$. Thus, the set of the form

$$D(U) = D^X(U) = \mathrm{Sob}(X) \backslash V(C) = \{Z \in \mathrm{Sob}(X) \mid Z \cap U \neq \varnothing\}$$

is precisely an open set in $\mathrm{Sob}(X)$.

We record a fact that will be used later. Let $\mathcal{U}$ be an open cover of X. Then $\{D^X(U) \mid U \in \mathcal{U}\}$ also covers $\mathrm{Sob}(X)$. This is tautological. Let $Z \in \mathrm{Sob}(X)$. Then $Z \cap U \neq \varnothing$ for some $U \in \mathcal{U}$, so that $Z \in D(U)$.

Denote by $\alpha : U \to X$ the inclusion. Then we have a map $\mathrm{Sob}(\alpha) : \mathrm{Sob}(U) \to \mathrm{Sob}(X)$. This is in fact an open embedding with image precisely $D(U)$. Indeed, by definition, for $Z \in \mathrm{Sob}(U)$, we have

$$\mathrm{Sob}(\alpha)(Z) = \overline{\alpha(Z)} = \overline{Z} \in \mathrm{Sob}(X).$$

If $Z, Z' \in \mathrm{Sob}(U)$ satisfy $\overline{Z} = \overline{Z}'$ in X, then $Z \cap Z'$ and $Z \cap (\overline{Z} \backslash Z')$ are two closed subsets of Z whose union is Z. Since Z is irreducible and $Z \cap Z' \neq \varnothing$, we must have $Z \cap Z' = Z$, or $Z \subseteq Z'$. By symmetry we then obtain $Z = Z'$, proving that $\mathrm{Sob}(\alpha)$ is injective. For a closed subset $C \subseteq U$, we have

$$\mathrm{Sob}(\alpha)(V_U(C)) = \{\overline{Z} \mid Z \in \mathrm{Sob}(U),\ Z \subseteq C\} = V_X(\overline{C}) \cap \mathrm{Im}\,\mathrm{Sob}(\alpha).$$

The last equality holds, as $\overline{Z} \subseteq \overline{C}$ implies $Z = \overline{Z} \cap U \subseteq \overline{C} \cap U = C$. This shows $\mathrm{Sob}(\alpha)$ is an embedding, and it remains to show $\mathrm{Im}\,\mathrm{Sob}(\alpha) = D(U)$. The containment $\subseteq$ is clear. For $\supseteq$, if $Z \cap U \neq \varnothing$, since Z is irreducible, $Z \cap U$ is dense in Z, showing that $Z = \overline{Z \cap U}$ with $Z \cap U \in \mathrm{Sob}(U)$.

There is counterpart for a closed subspace $C \subseteq X$, stating that $\mathrm{Sob}(C) \to \mathrm{Sob}(X)$ is a closed embedding, and the proof is much easier.

3.39 Let $f : X \to Y$ be a closed continuous map between topological spaces. Then its

3.40 By definition, a topological space X is $\mathbf{T_0}$ / **Kolmogorov** if for any $x \neq y \in X$ there exists a closed set in X containing exactly either x or y. We note here that the image $\iota(X)$ of X under $\iota : X \to \mathrm{Sob}(X)$ is T_0. To see this, let $x, y \in X$ with $\overline{\{x\}} \neq \overline{\{y\}}$. If every closed subset in $\iota(X)$ that contains, say, x also contains y, then $y \in \overline{\{x\}}$. By symmetry, we have $x \in \overline{\{y\}}$, which altogether gives that $\overline{\{x\}} = \overline{\{y\}}$, a contradiction.

Denote by $\mathbf{Top}_{T_0}$ the full subcategory of $\mathbf{Top}$ consisting of T_0 spaces. Let X be a topological space. We show that the map $i : \iota|^{\iota(X)} : X \to \iota(X)$ represents the functor $\mathrm{Hom}_{\mathbf{Top}}(X, -) : \mathbf{Top} \to \mathbf{Top}_{T_0}$. To see this, we only need to show that if X is T_0, then $i : X \to \iota(X)$ is a homeomorphism. Once this is proven, the rest of the proof follows exactly the same as that in (3.37). Since i^{-1} induces a bijection between topologies, just as what ι^{-1} does, it suffices to show $i : X \to \iota(X)$ is bijective, and it remains to show injectivity. This is clear as X is T_0.

3.41 Jacobson space. Let k be an algebraically closed field. By Nullstellensatz, the closed points of the affine scheme $\mathrm{Spec}\, k[x_1, \ldots, x_n]$ are in bijection with the affine variety k^n. Under this bijection, we see k^n is a topological subspace of $\mathrm{Spec}\, k[x_1, \ldots, x_n]$, and for $I \trianglelefteq k[x_1, \ldots, x_n]$ we have

$$V^{\mathrm{var}}(I) = V^{\mathrm{sch}}(I) \cap \{\text{closed points in } \mathrm{Spec}\, A\}$$

In view of this relation, we are led to the following definition.

Let X be a topological space. Denote by X_{cl} the set of closed points in X, and equip X_{cl} with the subspace topology. We say the space X is **Jacobson** if for any closed subset Z of X, the subset Z_{cl} is dense in Z. Note that $Z_{\mathrm{cl}} = Z \cap X_{\mathrm{cl}}$ as Z is closed.

3.42 Example. Let A be a ring. The closed points in $\operatorname{Spec} A$ are exactly the maximal ideals in A; in other words, $(\operatorname{Spec} A)_{\mathrm{cl}} = \operatorname{mSpec} A$. For an ideal $I \trianglelefteq A$, we see that $V(I)_{\mathrm{cl}}$ is the set of maximal ideals containing A. For $V(I)_{\mathrm{cl}}$ to be dense in $V(I)$, it is sufficient and necessary that $\sqrt{I} = \operatorname{Jac}(I)$, where $\operatorname{Jac}(I)$ is the intersection of all maximal ideals in A containing I, and it called the **Jacobson radical** of I. To see this, first note that for $f \in A$, $D(f) \cap V(I) \neq \varnothing$ if and only if $f \notin \sqrt{I}$. For such f, clearly we have $D(f) \cap V(I)_{\mathrm{cl}} \neq \varnothing$ if and only if $f \notin \operatorname{Jac}(I)$. This implies the claim. The ring A with $\operatorname{Spec} A$ being Jacobson is called a **Jacobson ring**. The Jacobson radical of the zero ideal is denoted by $\operatorname{rad}(A)$, and is called the **Jacobson radical of the ring** A**.**

For example, Nullstellensatz implies that every algebra of finite type over an algebraically closed field k is Jacobson. Indeed, if I is an ideal of $k[x_1, \dots, x_n]$, then

$$\begin{aligned} I(V(I)) &= \{f \in k[x_1, \dots, x_n] \mid f(p) = 0 \text{ for all } p \in V(I)\} \\ &= \{f \in k[x_1, \dots, x_n] \mid f \in \mathfrak{m}_p \text{ for all } p \in V(I)\} = \operatorname{Jac}(I) \end{aligned}$$

where the second equality results from (3.10). This shows $k[x_1, \dots, x_n]$ is Jacobson, and clearly this implies what we say.

3.43 Definition. A continuous map $f : X \to Y$ is called a **quasi-homeomorphism** if the inverse f^{-1} establishes a bijection $\operatorname{Top}(Y) \to \operatorname{Top}(X)$ of topologies. Equivalently, it establishes a bijection between closed sets in X and Y.

As an example, we see in (3.36) that for a topological space X, the natural map $\iota_X : X \to \operatorname{Sob}(X)$ into its soberification is a quasi-homeomorphism.

3.44 The inverse f^{-1} of a function f is well-behaved with arbitrary intersection and arbitrary union. If $f : X \to Y$ is a quasi-homeomorphism, many identities happening in $\operatorname{Top}(Y)$ can be pulled back to identities in $\operatorname{Top}(X)$, and vice versa. Thus those topological properties that are defined or can be checked only using intersection and union are preserved under quasi-homeomorphism. We list some in the following lemma.

Lemma. Let $f : X \to Y$ be a quasi-homeomorphism between topological spaces.

(i) Let $\mathcal{U}$ be a collection of open sets in Y. Then $\mathcal{U}$ is a cover (resp. a basis) of Y if and only if $f^{-1}\mathcal{U} := \{f^{-1}(U)\}_{U \in \mathcal{U}}$ is a cover (resp. a basis) of X.

(ii) X is compact if and only if Y is compact.

(iii) X is connected (resp. irreducible) if and only if Y is connected (resp. irreducible).

3.45 If $f : X \to Y$ is a quasi-homeomorphism, then the pushforward f_* of presheaves on X to Y induces an equivalence of categories

$$\begin{aligned} \mathcal{C}_X^{\text{pre}} &\longrightarrow \mathcal{C}_Y^{\text{pre}} \\ \mathcal{F} &\longmapsto f_*\mathcal{F} \end{aligned}$$

where $\mathcal{C} = \mathbf{Set}$, $\mathbf{Ab}$, $\mathbf{Ring}$, $\mathbf{Mod}_R$ etc. This is obviously a fully faithful functor. Now given a presheaf $\mathcal{G}$ on Y, consider the pullback presheaf $f^{\text{pre}}\mathcal{G}$. We claim $f_*f^{\text{pre}}\mathcal{G} \cong \mathcal{G}$ as presheaves canonically. Indeed, for any open set V in Y,

$$f_*f^{\text{pre}}\mathcal{G}(V) = \varinjlim_{V' \supseteq f(f^{-1}(V))} \mathcal{G}(V') = \varinjlim_{V' \supseteq V} \mathcal{G}(V') \cong \mathcal{G}(V)$$

Passing to sheafification, we see $f_* : \mathcal{C}_X \to \mathcal{C}_Y$ is again an equivalence of categories with inverse f^{-1}.

3.46 Lemma. Let X be a topological space. TFAE :

(a) X is Jacobson.

(b) The inclusion $X_{\text{cl}} \to X$ is a quasi-homeomorphism.

(c) Every nonempty locally closed subset of X meets nontrivially with X_{cl}.

Proof. The equivalence (a)⇔(b) is simply a paraphrase of the very definition, and (a)⇒(c) is clear. To see (c)⇒(a), note that for any closed $Z \subseteq X$, if $\overline{Z_{\text{cl}}} \subsetneq Z$, then $Z \backslash \overline{Z_{\text{cl}}}$ is a nonempty locally closed set in X. However, we have $Z \backslash \overline{Z_{\text{cl}}} \subseteq Z \backslash X_{\text{cl}}$, which is a contradiction to (c). □

3.47 Corollary. Let A be a ring. Assume A satisfies the property that for any $f \in A$ and $\mathfrak{p} \in D(f)$, $\mathfrak{p} \in D(f)_{\text{cl}}$ implies $\mathfrak{p} \in \operatorname{mSpec} A$. Then A is Jacobson.

Proof. We use (3.46).(c). Assume that $A \neq 0$. Let $I \trianglelefteq A$ and $f \in A \backslash \sqrt{I}$. It suffices to show that $D(f) \cap V(I)$ contains a closed point in $\operatorname{Spec} A$. The space $D(f) \cap V(I)$ is homeomorphic to the affine scheme $\operatorname{Spec} A_f/I_f$, so it contains a closed point. By hypothesis, it is also a closed point in $\operatorname{Spec} A$. □

Note that the proof is essentially based on the fact that $\operatorname{Spec} A$ has a closed point as long as $A \neq 0$.

3.48 Lemma.

1. Any locally closed subset of a Jacobson space is Jacobson.
2. A topological space that admits an open cover consisting of Jacobson spaces is Jacobson.

Proof.

1. Let X be a Jacobson. It follows from definition that every closed subset of X is Jacobson, so we only need to show every open subset of X is Jacobson. Let $U \subseteq X$ be nonempty open, and let $Z \subseteq U$ be nonempty closed. By (3.46).(c), Z contains a closed point in X which is, a fortiori, a closed point in U. By (3.46).(c) again, U is Jacobson.

2. Let X be a space and $\mathcal{U}$ be an open cover of X with each $U \in \mathcal{U}$ Jacobson. We use (3.46).(c). Let $W \subseteq X$ be a nonempty locally closed subset. Then $W \cap U \neq \varnothing$ for some $U \in \mathcal{U}$, and by (3.46).(c) $W \cap U$ contains a closed point x in U. In particular, $\{x\}$ is locally closed in X. If $x \in V$ for some $V \in \mathcal{U}$, then $x \in V_{cl}$ for $\{x\}$ is locally closed in V and by (3.46).(c). Thus $\{x\}$ is closed in X, for $\{x\} \cap V$ is closed for any $V \in \mathcal{U}$.

□

3.49 Example. Let A be a ring. The universal property of soberification gives rise to a continuous map

$$\mathrm{Sob}(\mathrm{mSpec}\, A) \longrightarrow \mathrm{Spec}\, A.$$

We are going to write down this map explicitly under some additional conditions imposed on A. A closed set of the form $V^{\mathrm{mspec}}(I)$ is irreducible if and only if $\mathrm{Jac}(I)$ is a prime ideal. For such I, the image of $V^{\mathrm{mspec}}(I)$ in $\mathrm{Spec}\, A$ would be the generic point of $\overline{V^{\mathrm{mspec}}(I)} \subseteq \mathrm{Spec}\, A$. For another ideal I, we have $V^{\mathrm{mspec}}(\mathrm{Jac}(I)) \subseteq V(J)$ if and only if $J \subseteq \mathrm{Jac}(I)$, and this implies $V(\mathrm{Jac}(I)) \subseteq V(J)$, which further implies $J \subseteq \sqrt{\mathrm{Jac}(I)} = \sqrt{I}$. If A is Jacobson (3.42), we then see $\overline{V^{\mathrm{mspec}}(I)} = V(\sqrt{I})$ whose generic point is $\sqrt{I} = \mathrm{Jac}(I)$. For a prime $\mathfrak{p}$, if A is Jacobson, we have $\mathrm{Jac}(\mathfrak{p}) = \sqrt{\mathfrak{p}} = \mathfrak{p}$. Hence, if A is a Jacobson ring, every element in $\mathrm{Sob}(\mathrm{mSpec}\, A)$ has the form $V^{\mathrm{mspec}}(\mathfrak{p})$ with $\mathfrak{p}$ prime and it has image $V(\mathfrak{p})$ in $\mathrm{Spec}\, A$.

Assume A is Jacobson. The map is then clearly bijective. A closed set in $\mathrm{Sob}(\mathrm{mSpec}\, A)$ has the form $\{V^{\mathrm{mspec}}(\mathfrak{p}) \mid I \subseteq \mathfrak{p}\}$, where I is an ideal. Its image in $\mathrm{Spec}\, A$ is then $V(I)$, which is closed. Thus the canonical inclusion $\mathrm{mSpec}\, A \to \mathrm{Spec}\, A$ is the soberification when A is Jacobson.

Let $f : A \to B$ be a homomorphism between Jacobson rings with $f^{-1}(\mathrm{mSpec}\, B) \subseteq \mathrm{mSpec}\, A$. Let $\theta = \mathrm{Spec}(f)$ and $\theta' = \theta|^{\mathrm{mSpec}\, A}_{\mathrm{mSpec}\, B}$. Then there is a commutative diagram

$$\begin{tikzcd}
\mathrm{Sob}(\mathrm{mSpec}\, A) \arrow[r, "\sim"] & \mathrm{Spec}\, A \\
\mathrm{Sob}(\mathrm{mSpec}\, B) \arrow[r, "\sim"] \arrow[u, "\mathrm{Sob}(\theta')"] & \mathrm{Spec}\, B \arrow[u, "\theta"']
\end{tikzcd}$$

Indeed, this follows from (1.49).(iii).

3.50 By definition, every point in a T_1 **topological space** is a closed point. Denote by $\mathbf{Top}_{T_1}$ the full subcategory of **Top** consisting of T_1 spaces. Let X be a T_1 topological space and consider the soberification $\iota : X \to \mathrm{Sob}(X)$. Since every point $x \in X$ is a closed point, we have $\iota(x) = \{x\} \in \mathrm{Sob}(X)$; in particular, ι is injective, and $X \subseteq \mathrm{Sob}(X)_{cl}$. A point $Z \in \mathrm{Sob}(X)$ is closed if and only if $\{Z\} = \overline{\{Z\}} = V(Z)$ (3.36), which means that Z is a minimal closed irreducible subset in X with respect to inclusion. But $X = X_{cl}$, we see $Z \subseteq X$ is a singleton. This proves $X = \mathrm{Sob}(X)_{cl}$. Since ι is a quasi-homeomorphism (3.36), by (3.46) we see $\mathrm{Sob}(X)$ is a Jacobson space.

Denote by **JacSob** the subcategory of **Top** whose objects consist of Jacobson sober spaces and whose morphisms consist of continuous maps $f : X \to Y$ with $f(X_{cl}) \to Y_{cl}$. Then the functor $\mathbf{Sob} : \mathbf{Top} \to \mathbf{Sob}$

restricts to a functor $\mathrm{Sob} : \mathbf{Top}_{T_1} \to \mathbf{JacSob}$. Furthermore, this is an equivalence of categories. We state it as a theorem in the next paragraph.

3.51 Theorem. The functor

$$\mathrm{Sob} : \mathbf{Top}_{T_1} \to \mathbf{JacSob}$$

is an equivalence of categories with inverse $Y \mapsto Y_{cl}$.

Proof. We already show in (3.51) that for a T_1 space X, the map $\iota : X \to \mathrm{Sob}(X)$ is an embedding with $\iota(X) = \mathrm{Sob}(X)_{cl}$. Conversely, let Y be a Jacobson sober space. By definition Y_{cl} is T_1. By the universal property of soberification (3.37), there exists a continuous map $f : \mathrm{Sob}(Y_{cl}) \to Y$ with $f \circ \iota_{Y_{cl}}$ equal the inclusion $Y_{cl} \to Y$. For any closed irreducible $Z \subseteq Y_{cl}$, we have $\overline{Z} \cap Y_{cl} = Z$, where the closure is taken in Y. In particular, this implies f is injective. Conversely, if $y \in Y$, since Y is Jacobson, $Z_y := \overline{\{y\}} \cap Y_{cl} \neq \varnothing$ is a closed irreducible subset of Y_{cl}. Since $\overline{Z_y} \cap Y_{cl} = Z_y$, we have $f(Z_y) = y$. To show f is a homeomorphism, at this stage it suffices to show f is a quasi-homeomorphism. This follows as $Y_{cl} \to Y$ and $Y_{cl} \to \mathrm{Sob}(Y_{cl})$ are quasi-homeomorphisms.

The way we define the morphisms in **JacSob** is a sufficient condition for $Y \mapsto Y_{cl}$ to be functor $\mathbf{JacSob} \to \mathbf{Top}_{T_1}$. If $f : X \to Y$ is a morphism of T_1 spaces, by (3.37) we have a commuting square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\iota_X"'] & Y \arrow[d, "\iota_Y"] \\ \mathrm{Sob}(X) \arrow[r, "\mathrm{Sob}(f)"'] & \mathrm{Sob}(Y). \end{tikzcd}$$

Since $X = \mathrm{Sob}(X)_{cl}$ and similar for Y, we see $\mathrm{Sob}(f) \in \mathrm{Hom}_{\mathbf{JacSob}}(\mathrm{Sob}(X), \mathrm{Sob}(Y))$. Also from this diagram we conclude that Sob is an equivalence of categories with inverse $Y \mapsto Y_{cl}$. □

3.2.2 Equivalence of categories

3.52 We paraphrase the results in (3.11) in terms of the languages in the preceding subsubsection. For an affine variety V, its soberification $\mathrm{Sob}(V)$ is homeomorphic to that of $\mathrm{mSpec}\, k[V]$, which is homeomorphic to $\mathrm{Spec}\, k[V]$ by (3.42) and (3.49). Precisely, we have a commutative diagram with horizontal maps being isomorphisms

$$\begin{tikzcd} \mathrm{Sob}(V) \arrow[r, "\sim"] & \mathrm{Spec}\, k[V] \\ V \arrow[r, "\sim"] \arrow[u, "\iota_V"] & \mathrm{mSpec}\, k[V]. \arrow[u, "\text{inclusion}"'] \end{tikzcd}$$

Denote by α the bottom horizontal isomorphism and by ι the inclusion on the right. By Theorem 3.15, the sheaf $\iota_*\alpha_*\mathcal{O}_V$ on $\operatorname{Spec} k[V]$ is naturally isomorphic to the structure sheaf $\mathcal{O}_{\operatorname{Spec} k[V]}$. In this way, we see $(\operatorname{Sob}(V), (\iota_V)_*\mathcal{O}_V)$ is isomorphic to $\operatorname{Spec} k[V]$ in $\mathbf{LRS}_k$, and we obtain a functor

$$\begin{aligned} \mathbf{AffVar}_k &\longrightarrow \mathbf{Sch}_k \\ V &\longmapsto (\operatorname{Sob}(V), (\iota_V)_*\mathcal{O}_V), \end{aligned}$$

where for a morphism $f : V \to W$, we define $(\operatorname{Sob}(V), (\iota_V)_*\mathcal{O}_V) \to (\operatorname{Sob}(W), (\iota_W)_*\mathcal{O}_W)$ as follows. The map on topological spaces is certainly $\operatorname{Sob}(f)$. To define a sheaf map $\theta : (\iota_W)_*\mathcal{O}_W \to \operatorname{Sob}(f)_*(\iota_V)_*\mathcal{O}_V = (\iota_W)_*f_*\mathcal{O}_V$, we only need to choose a map $\mathcal{O}_W \to f_*\mathcal{O}_V$. We already have one : the function pullback. In fact, under the isomorphism $(\operatorname{Sob}(V), (\iota_V)_*\mathcal{O}_V) \cong \operatorname{Spec} k[V]$, we see the morphism $(\operatorname{Sob}(V), (\iota_V)_*\mathcal{O}_V) \to (\operatorname{Sob}(W), (\iota_W)_*\mathcal{O}_W)$ coincides $\operatorname{Spec} f^* : \operatorname{Spec} k[V] \to \operatorname{Spec} k[W]$. The maps on spaces are the same by (3.49). The maps on sheaves are the same as well, for both are induced by $f^* : k[W] \to k[V]$.

Similarly, we can define a functor

$$\begin{aligned} \mathbf{Var}_k &\longrightarrow \mathbf{Sch}_k \\ X &\longmapsto (\operatorname{Sob}(X), (\iota_X)_*\mathcal{O}_X). \end{aligned}$$

The only issue is to show $(\operatorname{Sob}(X), (\iota_X)_*\mathcal{O}_X)$ is really a scheme. Let V be an affine open subset of X. By (3.38) we can regard $U = \operatorname{Sob}(V)$ as an open subset of $\operatorname{Sob}(X)$. We show that

$$(\operatorname{Sob}(X), (\iota_X)_*\mathcal{O}_X)|_U \cong (U, (\iota_V)_*(\mathcal{O}_X|_V)).$$

For this we only need to notice that $((\iota_X)_*\mathcal{O}_X)|_U = (\iota_V)_*(\mathcal{O}_X|_V)$.

3.53 Definition. A morphism $f : X \to Y$ of schemes is called **locally of finite type** if Y admits an affine open cover $\mathcal{V}$ such that for any $V = \operatorname{Spec} A \in \mathcal{V}$, the open set $f^{-1}(V)$ $(V \in \mathcal{V})$ admits an affine open cover $\mathcal{U}_V$ such that for any $U = \operatorname{Spec} B \in \mathcal{U}_V$, the corresponding ring homomorphism $A \to B$ is of finite type.

If each $\mathcal{U}_V$ can be chosen to be a finite set, $f : X \to Y$ is called **of finite type**.

3.54 Let $A \to B$ be a ring homomorphism of finite type. For any $g \in B$, the composition $A \to B \to B_g$ is also of finite type, as $B_g \cong B[x]/(xg - 1)$ is a B-algebra of finite type. If $A \to B$ factors through some localization $A \to A_f$ at $f \in A$, then $A_f \to B$ is still of finite type.

These imply that for a morphism $f : X \to Y$ of schemes, being locally of finite type is really local on both X and Y. Precisely, let $f : X \to Y$ be locally of finite type. Then

(i) For any open $U \subseteq X$, $f|_U : U \to Y$ is locally of finite type.

(ii) For any open $V \subseteq Y$ and open $U \subseteq f^{-1}(V)$, $f|_U^V : U \to V$ is locally of finite type.

Also, let $f : X \to Y$ be a morphism of schemes.

(iii) If Y admits an open cover $\mathcal{V}$ such that $f|_{f^{-1}(V)}^V : f^{-1}(V) \to V$ is locally of finite type for any $V \in \mathcal{V}$, then f is locally of finite type.

(iv) If X admits an open cover $\mathcal{U}$ such that $f|_U : U \to Y$ is locally of finite type for any $U \in \mathcal{U}$, then f is locally of finite type.

3.55 Lemma. Consider morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ of schemes.

(i) If f and g are locally of finite type, then so is $g \circ f$.

(ii) If $g \circ f$ is locally of finite type, then so is f.

Proof. Only (ii) deserves a proof. Let $y \in Y$ and $x \in f^{-1}(y)$. Pick an affine open neighborhood $W = \operatorname{Spec} A$ of $g(y)$ in Z and an affine open neighborhood $U = \operatorname{Spec} C$ of x in X such that $A \to C$ induced by $(g \circ f)|_U^W$ is of finite type. Choose an affine neighborhood $V = \operatorname{Spec} B$ of y such that $g(V) \subseteq W$. Pick $h \in C$ with $f(U_h) \subseteq V$. Then $U_h \to V \to W$ gives $A \to B \to C_h$. Since $A \to C$ is of finite type, so is $A \to C_h$. In particular, $B \to C_h$ is of finite type. □

3.56 Lemma. A morphism $f : X \to Y$ is locally of finite type if and only if for every affine open $V = \operatorname{Spec} A$ in Y and every affine open $U = \operatorname{Spec} B$ in $f^{-1}(V)$, the corresponding ring homomorphism $A \to B$ is of finite type.

In particular, a morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ of affine schemes is locally of finite type if and only $A \to B$ is of finite type.

Proof. We need to prove the only if part. We see in (3.54) that $f|_U^V$ is locally of finite type. By definition, for any $h \in A$ we can find $g \in B$ such that $f(U_g) \subseteq V_h$ with $A_h \to B_g$ of finite type. In particular, B_g is of finite type over A. All such U_g cover U, and since U is compact, we can find $g_1, \dots, g_n \in B$ with $B = (g_1, \dots, g_n)$ and each B_{g_i} of finite type over A.

Let $b_i \in B$ be such that $1 = b_1 g_1 + \cdots + b_n g_n$. Raising to arbitrary power, for any $N \in \mathbb{Z}_{\geqslant 1}$, we can find $b_{iN} \in B$ with $1 = \sum_{i=1}^{n} b_{iN} g_i^N$. Say $B_{g_i} = A[x_{ij}]_{j=1}^{a_i}$ for some $x_{i1}, \dots, x_{ia_i} \in B_{g_i}$. Let $N_i \in \mathbb{Z}_{\geqslant 1}$ be such that $x_{ij} g_i^{N_i} \in B$ for any i, j. Let $N = \max_{1 \leqslant i \leqslant n} N_i$. Then $B = A[x_{ij} g_i^N]_{\substack{1 \leqslant i \leqslant n \\ 1 \leqslant j \leqslant a_i}}$. To see this, for $b \in B$, let $f_i \in A[X_{ij}]_{j=1}^{a_i}$ be such that $b = f_i(x_{ij})$. Let $M = \max_{1 \leqslant i \leqslant n} \deg f_i$ and put $L = MN$. Then

$$b = b1 = \sum_{i=1}^{n} b_{iL} f_i(x_{ij}) g_i^{MN} \in A[x_{ij} g_i^N]_{\substack{1 \leqslant i \leqslant n \\ 1 \leqslant j \leqslant a_i}}$$

□

3.57 Our goal is to show the functor

$$\begin{aligned} \mathbf{Var}_k &\longrightarrow \mathbf{Sch}_k \\ X &\longmapsto (\operatorname{Sob}(X), (\iota_X)_* \mathcal{O}_X). \end{aligned}$$

is fully faithful and describe its essential image. If V is an affine k-variety, by construction we see $\operatorname{Sob}(V)$ is a reduced affine k-scheme locally of finite type over $\operatorname{Spec} k$. Hence for a general k-variety X, we see $\operatorname{Sob}(X)$ is a reduced k-scheme locally of finite type over $\operatorname{Spec} k$.

To proceed, first note that an affine k-variety is a T_1 topological space. Next, suppose X is a topological space that admits an open cover $\mathcal{U}$ consisting of T_1 space. Then X is itself T_1. Indeed, for $x \in X$, if $\overline{\{x\}} \cap U \neq \varnothing$ for some $U \in \mathcal{U}$, then $x \in U$, as U is open. Since each $U \in \mathcal{U}$ is T_1, the intersection $\overline{\{x\}} \cap U$ is either $\varnothing$ or $\{x\}$, proving that $x \in X$ is a closed point. As a result, we see that a k-variety is T_1. By (3.50) and (1.76), the scheme $\mathrm{Sob}(X)$ is a Jacobson sober space.

If $f : V \to W$ is a morphism of affine k-varieties, in (3.52) we also see that the morphism

$$(\mathrm{Sob}(V), (\iota_V)_* \mathcal{O}_V) \to (\mathrm{Sob}(W), (\iota_W)_* \mathcal{O}_W)$$

is the same as $\operatorname{Spec} f^* : \operatorname{Spec} k[V] \to \operatorname{Spec} k[W]$. Since $f^* : k[W] \to k[V]$ is automatically of finite type, we see $(\mathrm{Sob}(V), (\iota_V)_* \mathcal{O}_V) \to (\mathrm{Sob}(W), (\iota_W)_* \mathcal{O}_W)$ is locally of finite type. Thus, the image of a morphism between k-varieties is a morphism of k-scheme locally of finite type.

Now let $f : X \to Y$ be a morphism of k-varieties. We see $\mathrm{Sob}(X)$ and $\mathrm{Sob}(Y)$ are Jacobson sober spaces. A natural question is whether for any morphism

$$(\mathrm{Sob}(X), (\iota_X)_* \mathcal{O}_X) \to (\mathrm{Sob}(Y), (\iota_Y)_* \mathcal{O}_Y)$$

of schemes, the underlying continuous map sends closed points to closed points, i.e., a morphism of Jacobson sober spaces. Such a morphism is necessarily locally of finite type by (3.55), so we are now in the situation stated in the following, which is essentially the Nullstellensatz.

3.58 Lemma. Let A, B be k-algebras of finite type and $\varphi : A \to B$ a k-algebra homomorphism.

(i) A prime ideal $\mathfrak{p} \in \operatorname{Spec} A$ is maximal if and only if $k = A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}$.

(ii) If $\mathfrak{m}$ is a maximal ideal of B, then $\varphi^{-1}(\mathfrak{m})$ is a maximal ideal in A.

Proof.

(i) The if part is clear, for it forces that $k = A/\mathfrak{p}$. For the only if part, if $\mathfrak{p}$ is maximal, then $k \to A/\mathfrak{p}$ is a finitely generated field extension, and Zariski's lemma implies that it is a finite extension. Since k is algebraically closed, it forces $k = A/\mathfrak{p}$. Also, since $\mathfrak{p}$ is maximal, we have $A/\mathfrak{p} = A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}$, so $k = A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}$.

(ii) The homomorphism induces a chain $k \to A/\varphi^{-1}(\mathfrak{m}) \to B/\mathfrak{m}$. By Zariski's lemma again we see $k = B/\mathfrak{m}$, so $k = A/\varphi^{-1}(\mathfrak{m})$. In particular, $\varphi^{-1}(\mathfrak{m})$ is maximal.

□

3.59 Corollary. Let $f : X \to Y$ be a morphism of k-schemes locally of finite type over $\operatorname{Spec} k$. Then

(i) X is Jacobson, and $x \in X$ is a closed point if and only if $k = \kappa(x)$.

(ii) $f(X_{\mathrm{cl}}) \subseteq Y_{\mathrm{cl}}$.

Proof.

(i) These follow from (3.48), (3.42) and (3.58).(i).

(ii) Let $x \in X_{\mathrm{cl}}$. Choose an affine open neighborhood $V = \operatorname{Spec} A$ of $f(x)$ in Y and an affine open neighborhood $U = \operatorname{Spec} B$ of x in X such that $A \to B$ is of finite type. By (3.58).(ii), $f(x)$ is a closed point in V, so $k = \kappa(f(x))$ by (i), which implies $f(x)$ is closed in Y, or $f(x) \in Y_{\mathrm{cl}}$.

□

3.60 Let us continue the discussion in (3.57). Let $(f, \theta) : (\mathrm{Sob}(V), (\iota_V)_* \mathcal{O}_V) \to (\mathrm{Sob}(W), (\iota_W)_* \mathcal{O}_W)$ be a k-morphism. Since $\mathrm{Sob}(V)$ is locally of finite type over k, by (3.55).(ii) we see (f, θ) is locally of finite type, and (3.59).(ii) implies that $f(\mathrm{Sob}(V)_{\mathrm{cl}}) \subseteq \mathrm{Sob}(W)_{\mathrm{cl}}$, i.e., f is a morphism in **JacSob** (3.50). Put $g = f_{\mathrm{cl}} : V \to W$ be the restriction of f to the closed points (c.f. (3.51)). Then we have a commutative diagram in **Top**

$$\begin{tikzcd} V \arrow[r, "g"] \arrow[d, "\iota_V"'] & W \arrow[d, "\iota_W"] \\ \mathrm{Sob}(V) \arrow[r, "f"'] & \mathrm{Sob}(W). \end{tikzcd}$$

If W is an affine k-variety, by (1.44) and (3.20), we have functorial bijections

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Sob}(V), \mathrm{Sob}(W)) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathbf{Alg}_k}(\mathcal{O}_W(W), \mathcal{O}_V(V)) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathbf{Var}_k}(V, W) \\ (f, \theta) & \longmapsto & \theta_{\mathrm{Sob}(W)} & & \\ & & g^* & \longleftarrow\!\shortmid & g \end{array}$$

From their proofs (and as ι_V is an embedding and (1.11)), we see $\theta_{\mathrm{Sob}(W)} = g^*$ if and only if $g = f_{\mathrm{cl}}$. In particular, this shows θ is given by pullback of functions by g. The general case follows by covering W by affine opens. These altogether prove that

$$\begin{array}{ccc} \mathbf{Var}_k & \longrightarrow & \mathbf{Sch}_k \\ X & \longmapsto & (\mathrm{Sob}(X), (\iota_X)_* \mathcal{O}_X). \end{array}$$

is fully faithful.

3.61 Inspired by (3.57), we say a k-scheme is **algebraic** if it is locally of finite type over $\operatorname{Spec} k$. Thus, if X is a k-variety, its soberification $\mathrm{Sob}(X)$ is a reduced algebraic k-scheme. We are now going to show any such a scheme comes from a k-variety, and then prove that the essential image of the functor

$$\begin{array}{ccc} \mathbf{Var}_k & \longrightarrow & \mathbf{Sch}_k \\ X & \longmapsto & (\mathrm{Sob}(X), (\iota_X)_* \mathcal{O}_X). \end{array}$$

is the full subcategory of $\mathbf{Sch}_k$ consisting of reduced algebraic k-scheme.

3.62 For a starter, we show if A is a reduced k-algebra of finite type, then the local-ringed space $(\mathrm{mSpec}\, A, \mathcal{O}_{\mathrm{Spec}\, A}|_{\mathrm{mSpec}\, A})$ is isomorphism to an affine variety in $\mathbf{LRS}_k$. To see this, we are back to the situation in the first paragraph of (3.52), and we need to show there is an isomorphism

$$(V, \mathcal{O}_V) \cong (\mathrm{mSpec}\, k[V], \mathcal{O}_{\mathrm{Spec}\, k[V]}|_{\mathrm{mSpec}\, k[V]})$$

The map on topological space is clearly given by $\alpha : V \to \mathrm{mSpec}\, k[V]$ there. For the sheaf map, put $\iota : \mathrm{mSpec}\, k[V] \to \mathrm{Spec}\, k[V]$ to be the inclusion. For $f \in k[V]$, since ι is a quasi-homeomorphism (3.49), we have

$$\begin{aligned}
\iota^{\mathrm{pre}}\mathcal{O}_{\mathrm{Spec}\, k[V]}(D^{\mathrm{mspec}}(f)) &\cong \varinjlim_{D^{\mathrm{spec}}(g) \supseteq \iota(D^{\mathrm{mspec}}(f))} \mathcal{O}_{\mathrm{Spec}\, k[V]}(D^{\mathrm{spec}}(g)) \\
&= \varinjlim_{D^{\mathrm{spec}}(g) \supseteq D^{\mathrm{spec}}(f)} k[V]_g \\
&\cong k[V]_f \\
&\overset{3.15}{\cong} \mathcal{O}_V(D^{\mathrm{Var}}(f)) = \alpha_* \mathcal{O}_V(D^{\mathrm{mspec}}(f)).
\end{aligned}$$

Every isomorphism is functorial, so this defines an isomorphism $\iota^{\mathrm{pre}}\mathcal{O}_{\mathrm{Spec}\, k[V]} \to \alpha_*\mathcal{O}_V$ of presheaves on the principal open sets. But $\alpha_*\mathcal{O}_V$ is a sheaf, this means $\iota^{\mathrm{pre}}\mathcal{O}_{\mathrm{Spec}\, k[V]}$ is a sheaf on principal open sets. By (1.4) this extends to an isomorphism $\mathcal{O}_{\mathrm{Spec}\, k[V]}|_{\mathrm{mSpec}\, k[V]} \to \alpha_*\mathcal{O}_V$ of sheaves on $\mathrm{mSpec}\, k[V]$.

3.63 Let X be an algebraic k-scheme. Put $V = X_{\mathrm{cl}}$ and denote by $\iota = \iota_V : V \to X$ the inclusion. By (3.59), X is Jacobson and $V = \{x \in X \mid k = \kappa(x)\}$. For a section $f \in \mathcal{O}_X(U)$ on some open set U of X, we can regard it as a function on $U \cap V$ by means of

$$\begin{aligned}
\iota_* f : U \cap V &\longrightarrow k \\
x &\longmapsto \text{class of } f \text{ in } k = \kappa(x)
\end{aligned}$$

That $k = \kappa(x)$ follows from (3.59).(ii). In this way we have defined a morphism $\alpha = \alpha_V : \mathcal{O}_X \to \iota_* k^V$ of sheaves, where k^V denotes the sheaves $U \mapsto \mathrm{Hom}_{\mathbf{Set}}(U, k)$. In other words, we obtain a morphism $(\iota, \alpha) : (V, k^V) \to (X, \mathcal{O}_X)$ in $\mathbf{RS}_k$. But it follows from definition that this is a morphism in $\mathbf{LRS}_k$. By adjunction (1.12) α gives a morphism $\iota^{-1}\mathcal{O}_X \to k^V$. Denote by $\mathcal{O}_V$ the image sheaf, which is a subsheaf of k^V. Since X is Jacobson, ι is a quasi-homeomorphism (3.46), which implies that the sheaf $\mathcal{O}_V$ is the unique sheaf of k^V such that $\iota_*\mathcal{O}_V$ is the image of α (3.45). By definition we have a chain of morphisms $\iota^{-1}\mathcal{O}_X \twoheadrightarrow \mathcal{O}_V \hookrightarrow k^V$. Computing stalks (c.f. (1.24)) reads that $(V, \mathcal{O}_V)$ is a basic k-space (3.17).

Let $f : X \to Y$ be a morphism between algebraic k-schemes. Let $V = X_{\mathrm{cl}}$, $U = Y_{\mathrm{cl}}$ and ι_V, ι_U be the respective inclusions. By (3.59).(ii), we have $f(V) \subseteq U$. Let us denote by $g = f^U_V : V \to U$. There is a

commutative diagram in $\mathbf{LRS}_k$

$$\begin{tikzcd}
(V, k^V) \arrow[r, "(\iota_V, \alpha_V)"] \arrow[d, "(g, g^*)"'] & (X, \mathcal{O}_X) \arrow[d, "(f, f^\sharp)"] \\
(U, k^U) \arrow[r, "(\iota_U, \alpha_U)"'] & (Y, \mathcal{O}_Y).
\end{tikzcd}$$

To see this, for open $W \subseteq Y$, $g \in \mathcal{O}_Y(W)$ and $x \in f^{-1}(W) \cap V$, we must show the class of $f^\sharp_W(g)$ in $k = \kappa(x)$ is the same as that of g in $k = \kappa(f(x))$. This is clear (c.f. (3.20)). This implies that $f : X \to Y$ induces a morphism $(g, g^*) : (V, \mathcal{O}_V) \to (U, \mathcal{O}_U)$ in $\mathbf{bSp}_k$. The construction is entirely functorial, and this shows the assignment $X \mapsto (V, \mathcal{O}_V)$ defines a functor from the category of algebraic k-schemes to $\mathbf{bSp}_k$.

We shall expect $\iota_* \mathcal{O}_V$ is isomorphic to the structure sheaf $\mathcal{O}_X$ of X. The morphism α induces a surjective morphism $\alpha : \mathcal{O}_X \to \iota_* \mathcal{O}_V$, so it would be nice if α is injective. Let $U = \operatorname{Spec} A$ be an affine open set of X. By (3.59) we have $U \cap V = \operatorname{mSpec} A$. For $\alpha_V(U)$ to be injective, it is the same as saying that if $f \in A$ is such that $f \in \mathfrak{m}_\mathfrak{m} \subseteq A_\mathfrak{m}$ for all maximal $\mathfrak{m}$, then $f = 0$ in A, i.e., $\operatorname{rad}(A) = 0$. Since A is Jacobson by (3.48) and (3.42), this is equivalent to saying that $0 = \sqrt{0}$, i.e., A is reduced. Hence, if X is a reduced algebraic k-scheme, X can be recovered from the local-ringed space $(V, \mathcal{O}_V)$, in the sense that $\iota_* \mathcal{O}_V \cong \mathcal{O}_X$ canonically.

Therefore, we assume X is reduced in the following. We are going to show $(V, \mathcal{O}_V)$ is a k-variety. Let U be an affine open set in X; we see $U_{\mathrm{cl}} = U \cap V$ in the previous paragraph. Put $U' = U \cap V$ and let $j : U' \to U$ be the inclusion. Since U is also an algebraic k-scheme, we may construct $(U', \mathcal{O}_{U'})$. By functoriality we have a morphism $(U', \mathcal{O}_{U'}) \to (V, \mathcal{O}_V)$ in $\mathbf{bSp}_k$. By (1.28) this gives $(U', \mathcal{O}_{U'}) \to (U', \mathcal{O}_V|_{U'})$ in $\mathbf{LRS}_k$, which is necessarily in $\mathbf{bSp}_k$. This morphism fits into a similarly obtained commutative diagram

$$\begin{tikzcd}
(\mathcal{O}_X|_U)|_{U'} \arrow[r, two heads] & \mathcal{O}_{U'} \arrow[r, hook] & k^{U'} \\
(\mathcal{O}_X|_V)|_{U'} \arrow[r, two heads] \arrow[u] & \mathcal{O}_V|_{U'} \arrow[r, hook] \arrow[u] & k^V|_{U'} \arrow[u]
\end{tikzcd}$$

The vertical arrows except the middle one are clearly isomorphisms, and thus so is the middle one. Hence $(U', \mathcal{O}_{U'}) \to (U', \mathcal{O}_V|_{U'})$ in $\mathbf{bSp}_k$.

We are now reduced to the case X being affine. Let A be a reduced k-algebra of finite type. Let $\iota : \operatorname{mSpec} A \to \operatorname{Spec} A$ be the inclusion. The third paragraph shows that $\iota_* \mathcal{O}_{\operatorname{mSpec} A} \cong \mathcal{O}_{\operatorname{Spec} A}$ in $\mathbf{LRS}_k$. Since ι is a quasi-isomorphism (3.49), it follows from (3.45) that we have an isomorphism $\mathcal{O}_{\operatorname{mSpec} A} \cong \mathcal{O}_{\operatorname{Spec} A}|_{\operatorname{mSpec} A}$. By (3.62), if W is an affine k-variety constructed from A by means of (3.9), we then have an isomorphism

$$(W, \mathcal{O}_W) \cong (\operatorname{mSpec} A, \mathcal{O}_{\operatorname{mSpec} A})$$

in $\mathbf{LRS}_k$, where the map on topological spaces is given by $\varphi : W \cong \operatorname{mSpec} k[W] \cong \operatorname{mSpec} A$. It remains to show the map on sheaves are given by pullback of functions by φ. This is easily checked on every

principal open set. Therefore we see that the isomorphism is in fact an isomorphism in $\mathbf{bSp}_k$, proving that $(\mathrm{mSpec}\, A, \mathcal{O}_{\mathrm{mSpec}\, A})$ is an affine k-variety.

3.64 We summarize what have been done so far. Denote by $\mathbf{AlgSch}_k$ (resp. $\mathbf{redAlgSch}_k$) the full subcategory of $\mathbf{Sch}_k$ consisting of algebraic (resp. reduced algebraic) k-schemes. Categorically speaking, (3.59) implies that $\mathbf{AlgSch}_k$ admits a forgetful functor $\mathbf{AlgSch}_k \to \mathbf{JacSob}$, given by sending schemes to its underlying topological spaces.

In (3.52) we constructed a functor

$$\begin{aligned} \mathrm{Sob} : \mathbf{Var}_k &\longrightarrow \mathbf{redAlgSch}_k \\ X &\longmapsto (\mathrm{Sob}(X), (\iota_X)_* \mathcal{O}_X). \end{aligned}$$

which is proved to be fully faithful (3.60). This functor fits into a commutative diagram

$$\begin{tikzcd} \mathbf{Var}_k \arrow[r] \arrow[d] & \mathbf{AlgSch}_k \arrow[d] \\ \mathbf{Top}_{T_1} \arrow[r] & \mathbf{JacSob} \end{tikzcd}$$

with the bottom horizontal arrow $X \mapsto \mathrm{Sob}(X)$ being the equivalence (3.51) with inverse $Y \mapsto Y_{\mathrm{cl}}$.

In (3.63) we constructed a functor

$$\begin{aligned} \mathrm{cl} : \mathbf{AlgSch}_k &\longrightarrow \mathbf{bSp}_k \\ X &\longmapsto (X_{\mathrm{cl}}, \mathcal{O}_{X_{\mathrm{cl}}}) \end{aligned}$$

and showed that it restricts to a functor $\mathbf{redAlgSch}_k \longrightarrow \mathbf{Var}_k$ that is injective and inverse to the functor Sob on objects. From the discussion there, we easily see that it is also inverse to Sob on morphisms. Thus they are really inverse to each other. Pictorially,

$$\begin{tikzcd} \mathbf{Var}_k \arrow[r, bend left, "\mathrm{Sob}"] & \mathbf{redAlgSch}_k \arrow[l, bend left, "\mathrm{cl}"] \end{tikzcd}$$

4 Sheaf Cohomology

4.1 Definition

Recall that if R is a unital ring (not necessarily commutative), then the abelian category ${}_R\mathrm{Mod}$ of left R-modules has enough injectives. Now let $(X, \mathcal{O}_X)$ be a ringed space. Denote by $\mathrm{Mod}_{\mathcal{O}_X}$ the abelian category of $\mathcal{O}_X$-modules.

Lemma 4.1. $\mathrm{Mod}_{\mathcal{O}_X}$ has enough injectives.

Proof. Let $\mathcal{F} \in \mathrm{Mod}_{\mathcal{O}_X}$. For each $x \in X$, the stalk $\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$-module, so there exists an injective $\varphi_x : \mathcal{F}_x \to I_x$ for some injective $\mathcal{O}_{X,x}$-module I_x. For each $x \in X$, denote by $\iota_x : \{x\} \to X$ the inclusion and consider the sheaf

$$\mathcal{I} := \prod_{x \in X} (\iota_x)_*(I_x)$$

Here we consider I_x as a sheaf on the one point space $\{x\}$, and view $(\{x\}, \mathcal{O}_{X,x})$ as a ringed space. Explicitly, for each open $U \subseteq X$,

$$\mathcal{I}(U) = \prod_{x \in U} I_x.$$

Then $\mathcal{I} \in \mathrm{Mod}_{\mathcal{O}_X}$, and there is a natural morphism $\mathcal{F} \to \mathcal{I}$ given as follows.

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} I_x = \mathcal{I}(U) \\ f & \longmapsto & (f_x)_{x \in U} & \longmapsto & (\varphi_x(f_x))_{x \in U} \end{array}$$

This is injective, for $\mathcal{F}$ is a sheaf (so the first arrow is injective) and each φ_x is injective.

It remains to show $\mathcal{I}$ is an injective $\mathcal{O}_X$-module. Let $\mathcal{G} \in \mathrm{Mod}_{\mathcal{O}_X}$. The universal property of direct products gives an isomorphism

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{I}) \xrightarrow[\sim]{} \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, (\iota_x)_*(I_x)).$$

As the direct product preserves exactness, it suffices to show each factor on the right is exact. But the adjunction (or argue directly) gives

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, (\iota_x)_*(I_x)) \xrightarrow[\sim]{} \mathrm{Hom}_{\mathcal{O}_{X,x}}((\iota_x)^*\mathcal{G}, I_x) = \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{G}_x, I_x).$$

Since the stalk functor $\mathcal{G} \mapsto \mathcal{G}_x$ is exact, we conclude that $\mathrm{Hom}_{\mathcal{O}_X}(\cdot, \mathcal{I})$ is an exact functor, i.e., $\mathcal{I}$ is injective. $\square$

Corollary 4.1.1. The abelian category of sheaves of abelian groups on a topological space X has enough injectives.

Proof. Regard X with the ringed space $(X, \underline{\mathbb{Z}}_X)$ where $\underline{\mathbb{Z}}_X$ is the constant sheaf with values in $\mathbb{Z}$. $\square$

Let X be a topological space. There is a global section functor $\Gamma(X, \cdot) : \mathrm{Mod}_{\mathbb{Z}_X} \to \mathrm{Mod}_{\mathbb{Z}}$, which is left exact but not right exact in general. We then define the **sheaf cohomology functors** $H^p(X, \cdot)$ to be the right derived functors of $\Gamma(X, \cdot)$.

It should be noted that even if X or $\mathcal{F}$ has some additional structure (e.g. X a manifold or scheme, and $\mathcal{F}$ a coherent sheaf), we always take cohomology by regarding X merely as a topological space and $\mathcal{F}$ as a sheaf of abelian groups over X.

Definition. A sheaf $\mathcal{F}$ on a space X is **flasque** if for every open U, the restriction $\mathcal{F}(X) \to \mathcal{F}(U)$ is surjective.

— Equivalently, $\mathcal{F}$ is flasque if and only if for all opens $V \subseteq U$, the restriction $\mathcal{F}(U) \to \mathcal{F}(V)$ is surjective.

Before continuing our discussion on cohomology, we talk about some operation on a sheaf. Let Z be a closed subset of X and $U := X \backslash Z$. Denote by $i : Z \to X$ and $j : U \to X$ the canonical inclusions.

1. If $\mathcal{F}$ is a sheaf on Z, the direct image $i_*\mathcal{F}$ is a sheaf on X. Then

$$(i_*\mathcal{F})_x = \begin{cases} 0 & \text{, if } x \notin Z \\ \mathcal{F}_x & \text{, if } x \in Z \end{cases}$$

   This follows from the definition of stalks, and the fact that the topology on Z is subspace topology. Because of this, we speak of $i_*\mathcal{F}$ the sheaf obtained by **extending $\mathcal{F}$ by zero outside Z.**

2. If $\mathcal{F}$ is a sheaf on U, denote by $j_!\mathcal{F}$ the sheafification of the presheaf $\mathcal{F}'$ defined by $V \mapsto \mathcal{F}(V)$ if $V \subseteq U$ and $\mapsto 0$ otherwise. Then

$$(j_!\mathcal{F})_x = \begin{cases} 0 & \text{, if } x \notin U \\ \mathcal{F}_x & \text{, if } x \in U \end{cases}$$

   This follows from the description of $\mathcal{F}'$, and the fact that stalk remain unchanged after sheafification. Moreover, $j_!\mathcal{F}$ is the unique sheaf on X such that the above identities hold and whose restriction to U to $\mathcal{F}$. Indeed, we can easily construct a morphism $\mathcal{F}' \to \mathcal{G}$ of presheaves by extending the isomorphism $\mathcal{G}|_U \cong \mathcal{F}$ by "zero outside U", so we obtain a morphism $j_!\mathcal{F} \to \mathcal{G}$ of sheaves. Taking stalks shows this is an isomorphism. We call $j_!\mathcal{F}$ is the sheaf obtained by **extending $\mathcal{F}$ by zero outside U.**

3. Let $\mathcal{F}$ be a sheaf on X. Then there exists an exact sequence

$$0 \longrightarrow j_!(\mathcal{F}|_U) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_Z) \longrightarrow 0$$

   The first arrow is easily constructed, and the second is from the adjunction. Exactness is easily checked on the stalks. Another important property is that $j_!$ is left adjoint to j^{-1}, that is, we have a functorial isomorphism

$$\mathrm{Hom}_{\mathcal{O}_X}(j_!\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}, j^{-1}\mathcal{G})$$

Lemma 4.2. Let $(X, \mathcal{O}_X)$ be a ringed space.

1. Any injective $\mathcal{O}_X$-module is flasque.
2. If $0 \to \mathcal{F}_1 \xrightarrow{\alpha} \mathcal{F}_2 \xrightarrow{\beta} \mathcal{F}_3 \to 0$ is an exact sequence of sheaves with $\mathcal{F}_1$ flasque, then

$$0 \longrightarrow \mathcal{F}_1(X) \xrightarrow{\alpha_X} \mathcal{F}_2(X) \xrightarrow{\beta_X} \mathcal{F}_3(X) \longrightarrow 0$$

   is an exact sequence of abelian groups.
3. If $0 \to \mathcal{F}_1 \xrightarrow{\alpha} \mathcal{F}_2 \xrightarrow{\beta} \mathcal{F}_3 \to 0$ is an exact sequence of sheaves with $\mathcal{F}_1$, $\mathcal{F}_2$ flasque, then $\mathcal{F}_3$ is also flasque.
4. Any flasque sheaf $\mathcal{F}$ on X is **acyclic**, i.e, $H^i(X, \mathcal{F}) = 0$ for $i \geqslant 1$.

Proof.

1. Let $\mathcal{J}$ be an injective $\mathcal{O}_X$-module. Let $U \subseteq X$ be an open set and denote by $j : U \to X$ the inclusion. Applying $\mathrm{Hom}_{\mathcal{O}_X}(\cdot, \mathcal{J})$ to the inclusion $0 \to j_!(\mathcal{O}_X|_U) \to \mathcal{O}_X$, we obtain a surjection $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{J}) \to \mathrm{Hom}_{\mathcal{O}_X}(j_!(\mathcal{O}_X|_U), \mathcal{J}) \to 0$, which is equal to the map $\mathcal{J}(X) \to \mathcal{J}(U)$.
2. It suffices to show $\beta_X : \mathcal{F}_2(X) \to \mathcal{F}_3(X)$ is surjective. Fix an element in $h \in \mathcal{F}_3(X)$ and consider the set

$$S = \left\{ (U, g) \mid g \in \mathcal{F}_2(U), U \underset{\text{open}}{\subseteq} X,\ \beta_U(g) = h|_U \right\}.$$

   Partially ordered S by the rule : $(U_1, g_1) \leqslant (U_2, g_2)$ if and only if $U_1 \subseteq U_2$ and $g_2|_{U_1} = g_1$. By exactness S is nonempty. By Zorn's lemma S admits a maximal element (U_0, g_0).

   We claim $U_0 = X$. Suppose otherwise and pick $x \in X \backslash U_0$. By looking at the stalk at x we can find a neighborhood W of X and $g_W \in \mathcal{F}_2(W)$ with $\beta_W(g_W) = h|_W$. Then $g_0|_{U_0 \cap W} = g_W|_{U_0 \cap W}$, so by exactness there exists an $f' \in \mathcal{F}(U_0 \cap W)$ such that

$$\alpha_{U_0 \cap W}(f') = g_0|_{U_0 \cap W} - g_W|_{U_0 \cap W}$$

   Since $\mathcal{F}_1$ is flasque, $f' = f|_{U_0 \cap W}$ for some $f \in \mathcal{F}_1(X)$. Then $\alpha_X(f)|_{U_0 \cap W} = g_0|_{U_0 \cap W} - g_W|_{U_0 \cap W}$, or

$$g_0|_{U_0 \cap W} = \alpha_X(f)|_{U_0 \cap W} + g_W|_{U_0 \cap W} = \big(\alpha_X(f)|_W + g_W\big)|_{U_0 \cap W}.$$

   This means g_0 and $\alpha_X(f)|_W + g_W$ glue to a section $g \in \mathcal{F}_2(U_0 \cup W)$, and $\beta_{U_0 \cup W}(g) = h|_{U_0 \cup W}$. This contradicts the maximality.
3. This follows from 2. and snake lemma.
4. Choose an embedding $\mathcal{F} \to \mathcal{J}$ into an injective object $\mathcal{J}$ in $\mathrm{Mod}_{\mathbb{Z}_X}$. Put $\mathcal{G} = \mathcal{J}/\mathcal{F}$; then we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{J} \longrightarrow \mathcal{G} \longrightarrow 0$$

of sheaves. By 2. we have an exact sequence

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{J}(X) \longrightarrow \mathcal{G}(X) \longrightarrow 0$$

Since $\mathcal{J}$ is injective, it is acyclic, so in view of the long exact sequence of sheaf cohomology, we obtain $H^1(X, \mathcal{F}) = 0$ and $H^i(X, \mathcal{F}) = H^{i-1}(X, \mathcal{G})$ for each $i \geqslant 2$. But by 3. $\mathcal{G}$ is also flasque, so 4. follows from induction. on i.

□

Corollary 4.2.1. Let $(X, \mathcal{O}_X)$ be a ringed space. The derived functors of $\Gamma(X, \cdot) : \mathrm{Mod}_{\mathcal{O}_X} \to \mathrm{Mod}_{\mathbb{Z}}$ coincide with the sheaf cohomology functors $H^i(X, \cdot)$.

Proof. This follows from the previous lemma and the fact that we can compute $H^\bullet(X, \mathcal{F})$ by taking a acyclic resolution of $\mathcal{F}$. □

We conclude this subsection by introducing a functorial flasque resolution of an abelian sheaf. Let $\mathcal{F}$ be a sheaf of abelian groups on X. Define a sheaf $\mathcal{C}^0(\mathcal{F})$ on X by (c.f. (4.1))

$$\mathcal{C}^0(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x.$$

This is clearly a flasque sheaf on X, and there is a natural morphism $\iota_{\mathcal{F}} : \mathcal{F} \to \mathcal{C}^0(\mathcal{F})$ given by $s \mapsto (s_x)_{x \in U}$ (c.f. (1.5)), which is injective as $\mathcal{F}$ is a sheaf. If we denote by $Z^1(\mathcal{F})$ the cokernel of ι_F, we can obtain a morphism $\delta^0_{\mathcal{F}} \mathcal{C}^0(\mathcal{F}) \to \mathcal{C}^1(\mathcal{F}) := \mathcal{C}^0(Z^1(\mathcal{F}))$ by injecting $Z^1(\mathcal{F})$ into $\mathcal{C}^1(\mathcal{F})$. Continuing this process, we obtain a flasque resolution of $\mathcal{F}$

$$0 \longrightarrow \mathcal{F} \xrightarrow{\iota_{\mathcal{F}}} \mathcal{C}^0(\mathcal{F}) \xrightarrow{\delta^0_{\mathcal{F}}} \mathcal{C}^1(\mathcal{F}) \xrightarrow{\delta^1_{\mathcal{F}}} \cdots .$$

This is called the **Godement resolution** of $\mathcal{F}$. It is clear from the construction that the resolution is functorial in $\mathcal{F}$. Additionally, if $\mathcal{F}$ is an $\mathcal{O}_X$-module for some sheaf of rings $\mathcal{O}_X$ on X, then each $\mathcal{C}^i(\mathcal{F})$ has a natural structure of $\mathcal{O}_X$-modules. Particularly, we see

Lemma 4.3. Let $(X, \mathcal{O}_X)$ be a ringed space. Every $\mathcal{O}_X$-module admits a flasque resolution in $\mathbf{Mod}_{\mathcal{O}_X}$.

4.2 Fine sheaves

The main reference of this section is [Voi02].

Definition. Let X be a topological space. A **fine sheaf** $\mathcal{F}$ on X is a $\mathcal{A}$-module, where $\mathcal{A}$ is a sheaf of rings satisfying the following property : For every open cover $\mathcal{U}$ of X, there exists a partition of unity $(f_U)_{U \in \mathcal{U}}$ subordinate to $\mathcal{U}$.

By a partition of unity we mean the family $(f_U)_{U \in \mathcal{U}}$ satisfies

(i) $f_U \in \mathcal{A}(X)$ with $\operatorname{supp} f_U \subseteq U$.

(ii) $\{\operatorname{supp} f_U\}_{U\in\mathcal{U}}$ is a locally finite family of subsets of X, and $\sum_{U\in\mathcal{U}} f_U = 1$.

Here $\operatorname{supp} f_U := \{x \in X \mid (f_U)_x \neq 0 \in \mathcal{A}_x\}$, and it is a closed set in X.

Proposition 4.4. A fine sheaf $\mathcal{F}$ on X is acyclic.

Proof. Say $\mathcal{A}$ is the sheaf of ring on X making $\mathcal{F}$ a fine sheaf. Choose any acyclic resolution $\mathcal{I}^\bullet$ of $\mathcal{F}$ in $\mathbf{Mod}_{\mathcal{A}}$. Then

$$H^k(X,\mathcal{F}) \cong \frac{\ker(\mathcal{I}^k(X) \to \mathcal{I}^{k+1}(X))}{\operatorname{image}(\mathcal{I}^{k-1}(X) \to \mathcal{I}^k(X))}.$$

Let $k \geqslant 1$ and $\alpha \in \mathcal{I}^k(X)$ be in the kernel. By exactness we can find an open cover $\mathcal{U}$ such that $\alpha|_U$ comes from some $\beta_U \in \mathcal{I}^{k-1}(U)$ for every $U \in \mathcal{U}$. Let $(f_U)_{U\in\mathcal{U}}$ be a partition of unity subordinate to $\mathcal{U}$, and put

$$\beta = \sum_{U\in\mathcal{U}} f_U \beta_U.$$

Note that the sum is locally finite and we view $f_U\beta_U \in \mathcal{I}^{k-1}(X)$ by setting $(f_U\beta_U)_x = 0$ for $x \in X\backslash U$ and $(f_U\beta_U)_x = (f_U)_x(\beta_U)_x$ for $x \in U$ (c.f. (1.6)). Thus $\beta \in \mathcal{I}^{k-1}(X)$ is well-defined and is mapped to α as $\alpha = \sum_{U\in\mathcal{U}} f_U\alpha|_U$. Then $\alpha = 0$ in $H^k(X,\mathcal{F})$, and since α is arbitrary, this proves $H^k(X,\mathcal{F}) = 0$. □

Corollary 4.4.1. Let M be a (paracompact) smooth manifold and $\mathcal{C}^\infty_M$ the sheaf of (real-valued) smooth functions on M. Then every $\mathcal{C}^\infty_M$-module is fine, and hence acyclic by (4.4).

4.2.1 de Rham cohomology

Let M be a smooth manifold and $p \geqslant 0$. Let $\mathcal{A}^p_{M,\mathbb{R}}$ denote sheaf of smooth sections of the p-th cotangent bundle $\bigwedge^p(TM)^\vee \to M$; note that $\mathcal{A}^0_{M,\mathbb{R}} = \mathcal{C}^\infty_M$. Together with the exterior derivatives d, we obtain the **de Rham complex**

$$0 \longrightarrow \mathcal{A}^0_{M,\mathbb{R}} \xrightarrow{d} \mathcal{A}^1_{M,\mathbb{R}} \xrightarrow{d} \mathcal{A}^2_{M,\mathbb{R}} \longrightarrow \cdots$$

Note that $\ker(\mathbb{A}^0_{M,\mathbb{R}} \xrightarrow{d} \mathbb{A}^1_{M,\mathbb{R}}) = \underline{\mathbb{R}}_M$ consists of locally constant functions on M. We define the **p-th de Rham cohomology group** $H^p_{dR}(M,\mathbb{R})$ as

$$H^p_{dR}(M,\mathbb{R}) := H^p(\mathcal{A}^\bullet_{M,\mathbb{R}}(M), d).$$

A fundamental result is that every closed form on M is locally exact. Precisely,

Lemma 4.5 (Poincaré lemma)**.** $H^p_{dR}(\mathbb{R}^n,\mathbb{R}) = 0$ for $p \geqslant 1$.

Proof. We are going the construct an integration operator $L_p : \mathcal{A}^p_{\mathbb{R}^n,\mathbb{R}}(\mathbb{R}^n) \to \mathcal{A}^{p-1}_{\mathbb{R}^n,\mathbb{R}}(\mathbb{R}^n)$ such that $L_{p+1}d + dL_k = \mathrm{id}$.

□

It follows from Poincaré lemma and (4.4.1) that

$$0 \longrightarrow \underline{\mathbb{R}}_M \longrightarrow \mathcal{A}^0_{M,\mathbb{R}} \xrightarrow{d} \mathcal{A}^1_{M,\mathbb{R}} \xrightarrow{d} \mathcal{A}^2_{M,\mathbb{R}} \longrightarrow \cdots$$

is an acyclic resolution of $\underline{R}_M$.

Theorem 4.6. For any smooth manifold M, there is a natural isomorphism

$$H^\bullet(M, \underline{\mathbb{R}}_M) \cong H^\bullet_{dR}(M, \mathbb{R}).$$

Assume M is a complex smooth manifold. Similarly we define $\mathcal{A}^p_{M,\mathbb{C}}$ to be the sheaf of smooth sections of the p-th complexified cotangent bundle $\bigwedge^p(TM \otimes_{\mathbb{R}} \mathbb{C})^\vee \to M$, and there is a de Rham complex over $\mathbb{C}$

$$0 \longrightarrow \mathcal{A}^0_{M,\mathbb{C}} \xrightarrow{d} \mathcal{A}^1_{M,\mathbb{C}} \xrightarrow{d} \mathcal{A}^2_{M,\mathbb{C}} \longrightarrow \cdots$$

which resolves the locally constant sheaf $\underline{\mathbb{C}}_M$. If we put

$$H^p_{dR}(M, \mathbb{C}) := H^p(\mathcal{A}^\bullet_{M,\mathbb{C}}(M), d),$$

the same reason then shows that there is a natural isomorphism

$$H^\bullet(M, \underline{\mathbb{C}}_M) \cong H^\bullet_{dR}(M, \mathbb{C}).$$

4.2.2 Dolbeault cohomology

Let M be a complex (smooth) manifold. Multiplication by $\sqrt{-1}$ on local coordinates of TM gives rise to a smooth vector bundle isomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}_{TM}$. This is called the **complex structure** on M. Consider the complexified tangent bundle $TM \otimes_{\mathbb{R}} \mathbb{C}$. On each fibre, the linear map J has eigenvalues $\pm\sqrt{-1}$, and this gives a global decomposition

$$TM \otimes_{\mathbb{R}} \mathbb{C} = T^{1,0}_M \oplus T^{0,1}_M$$

of $TM \otimes_{\mathbb{R}} \mathbb{C}$, where $T^{1,0}_M$ and $T^{0,1}_M$ are complex subbundles corresponding to the eigenvalue $\sqrt{-1}$ and $-\sqrt{-1}$ respectively. Both $T^{1,0}_M$ and $T^{0,1}_M$ are isomorphic to TM as real vector bundles. This decomposition induces a decomposition

$$\mathcal{A}^k_{M,\mathbb{C}} = \bigoplus_{p+q=k} \mathcal{A}^{p,q}_M \tag{$\clubsuit$}$$

on differential forms. Explicitly, in a local chart U of M with local coordinates $z_1, \ldots, z_n$, $\mathcal{A}^{p,q}_M(U)$ is the subspace of $\mathcal{A}^k_M(U)$ generated by the k-forms of the form

$$f dz_{i_1} \wedge \cdots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$$

with $f : U \to \mathbb{C}$ a smooth function. An element in $\mathcal{A}^{p,q}_M(U)$ is called a (p,q)**-form**.

Define two differentials ∂, $\overline{\partial}$ by

$$\partial : \mathcal{A}_M^{p,q} \hookrightarrow \mathcal{A}_{M,\mathbb{C}}^{p+q} \xrightarrow{d} \mathcal{A}_{M,\mathbb{C}}^{p+q+1} \twoheadrightarrow \mathcal{A}_M^{p+1,q}$$

$$\overline{\partial} : \mathcal{A}_M^{p,q} \hookrightarrow \mathcal{A}_{M,\mathbb{C}}^{p+q} \xrightarrow{d} \mathcal{A}_{M,\mathbb{C}}^{p+q+1} \twoheadrightarrow \mathcal{A}_M^{p,q+1}$$

where the last arrows are the projections from (♣). It is straightforward to see that

(i) a smooth function $f : U \to \mathbb{C}$ is holomorphic if and only if $\overline{\partial} f = 0$,

(ii) $d = \partial + \overline{\partial}$ and $\partial\overline{\partial} + \overline{\partial}\partial = 0$,

(iii) $(\mathcal{A}^{\bullet,q}, \partial)$ and $(\mathcal{A}^{p,\bullet}, \overline{\partial})$ are complexes,

(iv) for any k-form α and k'-form β, we have

$$\partial(\alpha \wedge \beta) = (\partial\alpha) \wedge \beta + (-1)^k \alpha \wedge (\partial\beta)$$

and the same for $\overline{\partial}$.

In what follows we put $\mathcal{C}_M^\infty$ to be the sheaf of complex-valued smooth functions on M. Let E be a holomorphic vector bundle over M and let $\mathcal{E}$ be the sheaf of *holomorphic* sections of $E \to M$. Define

$$\mathcal{A}_M^{p,q}(E) := \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{A}_M^{p,q}.$$

Explicitly, if $U \subseteq M$ is a local chart with holomorphic coordinates $z_1, \ldots, z_n$ that trivializes the bundle E, and if $\sigma_1, \ldots, \sigma_r \in \mathcal{E}(U)$ such that $\sigma_1(x), \ldots, \sigma_r(x)$ form a basis of the fibre E_x for each $x \in U$, then

$$\mathcal{A}_M^{p,q}(E)(U) = \left\{ \sum_{i=1}^{r} \sum_{\#I=p, \#J=q} f_{I,J,i} \cdot \sigma_i \otimes dz_I \wedge d\overline{z}_J \mid f_{I,J,i} \in \mathcal{C}_M^\infty(U) \right\}$$

where I, J run over all ordered multi indices in $[n]$ of length p and q, respectively. With this notation, define

$$\overline{\partial}_{E,U}(f_{I,J,i} \cdot \sigma_i \otimes dz_I \wedge d\overline{z}_J) = \sigma_i \otimes (\overline{\partial} f_{I,J,i} \wedge dz_I \wedge d\overline{z}_J).$$

This extends by linearity to a map $\overline{\partial}_{E,U} : \mathcal{A}_M^{p,q}(E)(U) \to \mathcal{A}_M^{p,q+1}(E)(U)$.

Lemma 4.7. Let $V \subseteq M$ be another local trivialization of E. Then $\overline{\partial}_{E,U}$ and $\overline{\partial}_{E,V}$ agree on $\mathcal{A}_M^{p,q}(E)(U \cap V)$.

This means we can glue $\overline{\partial}_{E,U}$ to a global morphism

$$\overline{\partial}_E : \mathcal{A}_M^{p,q}(E) \to \mathcal{A}_M^{p,q+1}(E).$$

As above, we can show that

(a) a smooth section σ of $E \to M$ is holomorphic if and only if $\overline{\partial}_E \sigma = 0$,

(b) $(\mathcal{A}_M^{p,\bullet}(E), \overline{\partial}_E)$ is a complex,

(c) for any k-form α and any local section σ of $\mathcal{A}_M^{p,q}(E)$, we have

$$\overline{\partial}_E(\alpha \wedge \sigma) = (\overline{\partial}\alpha) \wedge \sigma + (-1)^k \alpha \wedge (\overline{\partial}_E \sigma).$$

Definition. Let $E \to M$ be a holomorphic vector bundle on a complex manifold M and $\mathcal{E}$ the sheaf of holomorphic sections of $E \to M$. For $p, q \in \mathbb{Z}_{\geqslant 0}$, the (p, q)**-th Dolbeault cohomology group** of E is defined as

$$H_{\overline{\partial}}^{p,q}(X, E) = \frac{\ker(\overline{\partial}_E : \mathcal{A}_M^{p,q}(E)(X) \to \mathcal{A}_M^{p,q+1}(E)(X))}{\operatorname{image}(\overline{\partial}_E : \mathcal{A}_M^{p,q-1}(E)(X) \to \mathcal{A}_M^{p,q}(E)(X))}$$

Note that each $H^{p,q}(X, E)$ is a $\mathbb{C}$-vector space, and $H^{0,0}(X, E) = \mathcal{E}(M)$ by (a).

To relate it with the sheaf cohomology, we need a following $\overline{\partial}$-analogue of Poincaré lemma.

Lemma 4.8 (Dolbeault-Grothendieck lemma)**.** The complex

$$\mathcal{A}_M^{p,q-1} \xrightarrow{\overline{\partial}} \mathcal{A}_M^{p,q} \xrightarrow{\overline{\partial}} \mathcal{A}_M^{p,q+1}$$

is exact for $p \geqslant 0,\ q \geqslant 1$.

It follows that the complex

$$\mathcal{A}_M^{p,q-1}(E) \xrightarrow{\overline{\partial}_E} \mathcal{A}_M^{p,q}(E) \xrightarrow{\overline{\partial}_E} \mathcal{A}_M^{p,q+1}(E)$$

is also exact for $q \geqslant 1$. By (i) we see the kernel of $\overline{\partial} : \mathcal{A}_M^{p,0} \to \mathcal{A}_M^{p,q}$ is the sheaf Ω_M^p of holomorphic p-forms on M. Hence by (4.4.1)

$$0 \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_M} \Omega_M^p \longrightarrow \mathcal{A}_M^{p,0}(E) \xrightarrow{\overline{\partial}_E} \mathcal{A}_M^{p,1}(E) \xrightarrow{\overline{\partial}_E} \cdots$$

is an acyclic resolution of $\mathcal{E} \otimes_{\mathcal{O}_M} \Omega_M^p$, where $\mathcal{O}_M$ is the sheaf of holomorphic functions on M. This proves

Theorem 4.9 (Dolbeault theorem)**.** There is a natural isomorphism

$$H^q(M, \mathcal{E} \otimes_{\mathcal{O}_M} \Omega_M^p) \cong H^{p,q}(X, E).$$

4.3 Grothendieck vanishing theorem

Theorem 4.10. Let X be a noetherian topological space of dimension n. Then for all $i > n$ and all sheaves of abelian groups $\mathcal{F}$ on X, we have $H^i(X, \mathcal{F}) = 0$.

4.4 Serre's theorem on affineness

Let X be a topological space and $\mathcal{F}$ a sheaf of abelian groups on X. For an open U, if we denote by $\iota : U \to X$ the natural inclusion, then we have a canonical map $\mathcal{F} \to \iota_*(\mathcal{F}|_U)$, and hence a map $H^i(X, \mathcal{F}) \to H^i(X, \iota_*(\mathcal{F}|_U))$ on sheaf cohomology groups . The image of an element $x \in H^i(X, \mathcal{F})$ in the latter group will be denoted by $x|_U$.

We start with a general topological lemma.

Lemma 4.11. Let X be a compact topological space and $\mathcal{B}$ a basis of the topology of X. For a sheaf $\mathcal{F}$ of abelian groups and $i \geqslant 0$, we say $\mathcal{F}$ satisfies (P_i) if

(P_i) for all $\gamma \in H^i(X, \mathcal{F})$ there exists a finite cover $U_1, \ldots, U_r$ of X by elements in $\mathcal{B}$ such that $\alpha|_{U_i} = 0$ for any $1 \leqslant i \leqslant r$.

Then we have the following :

(i) (P_1) holds for all sheaves of abelian groups on X.

(ii) Suppose in addition $\mathcal{B}$ is closed under finite intersection. Let $\mathcal{F}$ be a sheaf and $i > 1$. If $H^p(U, \mathcal{F}|_U) = 0$ for all $0 < p < i$ and $U \in \mathcal{B}$, then (P_i) holds for $\mathcal{F}$.

Proof. Let $\mathcal{F}$ be a sheaf of abelian groups on X. Let $0 \to \mathcal{F} \to \mathcal{I}$ be an embedding into a flasque sheaf $\mathcal{I}$. If we write $\mathcal{G}$ to be the quotient sheaf $\mathcal{I}/\mathcal{F}$, we get an exact sequence of sheaves :

$$0 \longrightarrow \mathcal{F} \xrightarrow{\varphi} \mathcal{I} \xrightarrow{\psi} \mathcal{G} \longrightarrow 0$$

Taking cohomology, we have a long (but quite short) exact sequence

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{I}(X) \longrightarrow \mathcal{G}(X) \longrightarrow H^1(X, \mathcal{F}) \longrightarrow 0$$

Let $\gamma \in H^1(X, \mathcal{F})$ and lift it to $\alpha \in \mathcal{G}(X)$. Since $\mathcal{I} \to \mathcal{G}$ is surjective (as sheaves) and since X is compact, we can find a finite cover $U_1, \ldots, U_r$ of X by $\mathcal{B}$ such that $\alpha|_{U_i} = \psi_{U_i}(s_i)$ for some $s_i \in \mathcal{I}(U_i)$.

Let ι_i denote the canonical inclusion $U_i \to X$. Then we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{I} \arrow[r, "\psi"] \arrow[d] & \mathcal{G} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & (\iota_i)_*(\mathcal{F}|_{U_i}) \arrow[r] & (\iota_i)_*(\mathcal{I}|_{U_i}) \arrow[r] & (\iota_i)_*(\mathcal{G}|_{U_i}) &
\end{tikzcd}$$

Replacing $(\iota_i)_*(\mathcal{G}|_{U_i})$ by $\mathcal{G}_i := \operatorname{coker}\big((\iota_i)_*(\mathcal{F}|_{U_i}) \to (\iota_i)_*(\mathcal{I}|_{U_i})\big)$, we obtain a commutative diagram with exact rows (note that $(\cdot)|_{U_i}$ is exact!)

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{I} \arrow[r, "\psi"] \arrow[d] & \mathcal{G} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & (\iota_i)_*(\mathcal{F}|_{U_i}) \arrow[r] & (\iota_i)_*(\mathcal{I}|_{U_i}) \arrow[r] & \mathcal{G}_i \arrow[r] & 0.
\end{tikzcd}$$

Taking cohomology, by functoriality we have (note that $(\iota_i)_*(\mathcal{I}|_{U_i})$ is flasque)

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(X) \arrow[r, "\varphi"] \arrow[d] & \mathcal{I}(X) \arrow[r, "\psi"] \arrow[d] & \mathcal{G}(X) \arrow[r] \arrow[d] & H^1(X,\mathcal{F}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{F}(U_i) \arrow[r] & \mathcal{I}(U_i) \arrow[r] & \mathcal{G}_i(X) \arrow[r] & H^1(X,(\iota_i)_*(\mathcal{F}|_{U_i})) \arrow[r] & 0
\end{tikzcd}$$

The image of α in $\mathcal{G}_i(X) \subseteq (\iota_i)_*(\mathcal{G}|_{U_i})(X) = \mathcal{G}(U_i)$ is $\alpha|_{U_i}$, and it comes from $s_j \in \mathcal{I}(U_i)$ via ψ_{U_j}. Thus $\gamma|_{U_i}$ is the image of s_j in $H^1(X,(\iota_i)_*(\mathcal{F}|_{U_i}))$, which is zero. This proves (i).

Next we prove (ii). Proceed by induction on $i > 1$. Let $U_1, \ldots, U_r$ be an arbitrary finite open cover of X by $\mathcal{B}$, and put $\iota_j : U_j \to X$ to be the natural inclusion. If $U \in \mathcal{B}$, then $U_j \cap U \in \mathcal{B}$ and hence $H^1(U_j \cap U, \mathcal{F}|_{U_j \cap U})$ by our assumption. The sequence $0 \to \mathcal{F}|_{U_j\cap U} \to \mathcal{I}_{U_j \cap U} \to \mathcal{G}|_{U_j \cap U} \to 0$ is still exact, so taking cohomology gives $0 \longrightarrow \mathcal{F}(U_j \cap U) \longrightarrow \mathcal{I}(U_j \cap U) \longrightarrow \mathcal{G}(U_j \cap U) \longrightarrow 0$, or equivalently

$$0 \longrightarrow (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)(U) \longrightarrow (\iota_j)_*\left(\mathcal{I}|_{U_j}\right)(U) \longrightarrow (\iota_j)_*\left(\mathcal{G}|_{U_j}\right)(U) \longrightarrow 0$$

On the other hand, we have an exact sequence $0 \to (\iota_j)_*\left(\mathcal{F}|_{U_j}\right) \to (\iota_j)_*\left(\mathcal{I}|_{U_j}\right) \to \mathcal{G}_j \to 0$ (with $\mathcal{G}_j$ being the cokernel of the former map). Restricting to U and taking cohomology gives

$$0 \longrightarrow (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)(U) \longrightarrow (\iota_j)_*\left(\mathcal{I}|_{U_j}\right)(U) \longrightarrow \mathcal{G}_j(U) \longrightarrow H^1(U,(\iota_j)_*\left(\mathcal{F}|_{U_j}\right)|_U) \longrightarrow 0.$$

(Note that $(\iota_j)_*\left(\mathcal{I}|_{U_j}\right)$ is flasque, and so is its restriction to U.) If $0 \to \mathcal{F} \to \mathcal{C}^\bullet$ is a flasque resolution of $\mathcal{F}$, then $\mathcal{C}^\bullet|_{U_j\cap U}$ is also a flasque resolution of $\mathcal{F}|_{U_j\cap U}$, and $(\iota_j)_*(\mathcal{C}^\bullet|_{U_j})|_U$ is also a flasque resolution of $(\iota_j)_*(\mathcal{F}|_{U_j})|_U$. Since

$$\Gamma(U,(\iota_j)_*(\mathcal{C}^n|_{U_j})|_U) = \Gamma(U_j \cap U, \mathcal{C}^n|_{U_j\cap U})$$

we see particularly that $H^1(U,(\iota_j)_*(\mathcal{F}|_{U_j})|_U) = H^1(U_j \cap U, \mathcal{F}|_{U_j\cap U}) = 0$, and hence

$$0 \longrightarrow (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)(U) \longrightarrow (\iota_j)_*\left(\mathcal{I}|_{U_j}\right)(U) \longrightarrow \mathcal{G}_j(U) \longrightarrow 0$$

is exact. Regarding $\mathcal{G}_j$ naturally as a subsheaf of $(\iota_j)_*\left(\mathcal{G}|_{U_j}\right)$, since we see that these two sheaves agrees on every basis element $U \in \mathcal{B}$, they are the same. In particular, we have a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{I} \arrow[r, "\psi"] \arrow[d] & \mathcal{G} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & (\iota_j)_*\left(\mathcal{F}|_{U_j}\right) \arrow[r] & (\iota_j)_*\left(\mathcal{I}|_{U_j}\right) \arrow[r] & (\iota_j)_*\left(\mathcal{G}|_{U_j}\right) \arrow[r] & 0
\end{tikzcd}$$

Taking cohomology and using the fact that $\mathcal{I}$ and $(\iota_j)_*\left(\mathcal{I}|_{U_j}\right)$ are flasque, for each $i \geqslant 2$ we have a commuting square

$$\begin{tikzcd} H^{i-1}(X, \mathcal{G}) \arrow[r] \arrow[d] & H^i(X, \mathcal{F}) \arrow[d] \\ H^{i-1}(X, (\iota_j)_*\left(\mathcal{G}|_{U_j}\right)) \arrow[r] & H^i(X, (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)) \end{tikzcd}$$

with two horizontal arrows being isomorphisms. By (i) $\mathcal{G}$ satisfies (P_1), so if we choose those $U_1, \ldots, U_r$ to be as in (P_1) for $\mathcal{G}$, this says that $\mathcal{F}$ satisfies (P_2). Now suppose (ii) holds for all i with $2 \leqslant i < n$. To show (ii) in n, it suffices to show $\mathcal{G}$ satisfies (P_{n-1}). By induction hypothesis it is sufficient to show $H^p(U, \mathcal{G}|_U) = 0$ for all $U \in \mathcal{B}$ and $0 < p < n-1$. This of course holds, as by the exact sequence

$$0 \longrightarrow \mathcal{F}|_U \longrightarrow \mathcal{I}|_U \longrightarrow \mathcal{G}|_U \longrightarrow 0$$

and as $H^p(U, \mathcal{I}|_U) = 0$ and $H^{p+1}(U, \mathcal{F}|_U) = 0$ (this is true because $p+1 < n$). □

Theorem 4.12. Let A be a ring and $\mathcal{F}$ a quasi-coherent sheaf on $X = \operatorname{Spec} A$. Then $H^i(X, \mathcal{F}) = 0$ for all $i > 0$.

Proof. We use Lemma 4.11 with $\mathcal{B}$ the collection of all affine open subsets of X, which is a basis for the Zariski topology on X. Note also that X is (quasi-)compact. We prove this by induction on $i \geqslant 1$.

Assume $i = 1$. Let $\gamma \in H^1(X, \mathcal{F})$ By Lemma 4.11.(i) we can find affine opens $U_1, \ldots, U_r$ such that γ maps to 0 in all $H^1(X, (\iota_j)_*\left(\mathcal{F}|_{U_j}\right))$, where $\iota_j : U_j \to X$ denotes the inclusion. Now each $(\iota_j)_*\left(\mathcal{F}|_{U_j}\right)$ is quasi-coherent, and so is the cokernel $\mathcal{G} := \operatorname{coker}\left(\mathcal{F} \to \prod_{j=1}^{r} (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)\right)$. Since X is affine, we obtain a short exact sequence

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \prod_{j=1}^{r} (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)(X) \longrightarrow \mathcal{G}(X) \longrightarrow 0. \tag{$*$}$$

In view of long exact sequence of cohomology, we see $H^1(X, \mathcal{F}) \to \prod_{j=1}^{r} H^1\left(X, (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)\right)$ is injective. This shows $\gamma = 0$.

For $i > 1$, by induction hypothesis, $H^p(U, \mathcal{F}|_U) = 0$ for each $0 < p < i$ and $U \in \mathcal{B}$. Now let $\gamma \in H^i(X, \mathcal{F})$ be given. By Lemma 4.11.(ii) we can find affine opens $U_1, \ldots, U_r$ that covers X such that γ maps to zero in each $H^i(X, (\iota_j)_*\left(\mathcal{F}|_{U_j}\right))$. By induction hypothesis, $H^{i-1}(X, \mathcal{G}) = 0$, so $(*)$ implies $H^i(X, \mathcal{F}) \to \prod_{j=1}^{r} H^i\left(X, (\iota_j)_*\left(\mathcal{F}|_{U_j}\right)\right)$ is injective. This again shows $\gamma = 0$. □

4.5 Base change

4.5.1 Higher direct image

Let X, Y be spaces and $f : X \to Y$ be a continuous map. The direct image of f defines a functor $f_* : \mathbf{Ab}_X \to \mathbf{Ab}_Y$. This is a left exact functor, as it is a right adjoint functor (1.13). Since $\mathbf{Ab}_X$ has enough injective, we then can consider the right derived functors $R^i f_*$ of f_*.

Definition. The right derived functors $R^i f_* : \mathbf{Ab}_X \to \mathbf{Ab}_Y$ of the direct image f_* are called the **higher direct image** functors of f.

Lemma 4.13. For $i \geqslant 0$ and any sheaf $\mathcal{F}$ of abelian groups on X, the sheaf $R^i f_* \mathcal{F}$ is isomorphic to the sheafification of the presheaf $V \mapsto H^i(f^{-1}(V), \mathcal{F}|_{f^{-1}(V)})$ on Y.

Proof.

□

4.5.2 cohomology of base change

Theorem 4.14. Let $f : X \to Y$ be a proper morphism of Noetherian schemes with $Y = \operatorname{Spec} A$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module that is flat over Y, i.e., each stalk $\mathcal{F}_x$ is flat over $\mathcal{O}_{Y,f(x)}$. Then there exist a finite complex $K^\bullet : 0 \to K^0 \to K^1 \to \cdots \to K^n \to 0$ of finitely generated projective A-modules and a natural isomorphism

$$H^\bullet(X \times_Y \operatorname{Spec} B, \mathcal{F} \otimes_A B) \cong H^\bullet(K^\bullet \otimes_A B)$$

on the category of A-algebras B.

Proof. First we make a reduction so that this becomes a purely homological problem. Choose a finite affine open cover $\mathcal{U}$ of X. By Corollary 5.13.1 (and since X is separated) and Theorem 5.6, we can compute the cohomology by the ordered Čech complex $C^\bullet = (C^\bullet_{\mathrm{ord}}(\mathcal{U}, \mathcal{F}), d^\bullet)$, which is a finite complex as $\#\mathcal{U} < \infty$, and consists of A-flat modules.

Also, for any A-algebra B, $\{U \times_Y \operatorname{Spec} B \mid U \in \mathcal{U}\}$ is an finite affine cover of $X \times_Y \operatorname{Spec} B$, and $C^\bullet \otimes_A B$ is the Čech complex of $\mathcal{F} \otimes_A B$ for this cover. Therefore,

$$H^\bullet(X \times_Y \operatorname{Spec} B, \mathcal{F} \otimes_A B) \cong H^\bullet(C^\bullet \otimes_A B)$$

for all B, which is in fact functorial in B.

□

5 Čech Cohomology

5.1 Definitions

Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ an open cover of X and $\mathcal{F}$ a *presheaf* of abelian groups on X. For $p \geqslant 0$, define a complex $(C^\bullet(\mathcal{U}, \mathcal{F}), d^\bullet)$ as follows. For each $p \geqslant 0$, define the **Čech p-th cochain group with values in $\mathcal{F}$** by

$$C^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \ldots, i_p \in I} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p}).$$

For a p-th cochain $s \in C^p(\mathcal{U}, \mathcal{F})$, we write $s(i_0, \ldots, i_p)$ to denote its component in $\mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p})$.

For $p \geqslant 0$, define the p-th coboundary map $d^p : C^p(\mathcal{U}, \mathcal{F}) \to C^{p+1}(\mathcal{U}, \mathcal{F})$ by the formula

$$(ds)(i_0, \ldots, i_{p+1}) = \sum_{j=0}^{p+1} (-1)^j s(i_0, \ldots, \widehat{i_j}, \ldots, i_{p+1})\big|_{U_{i_0} \cap \cdots \cap U_{i_{p+1}}}.$$

Lemma 5.1. For $p \geqslant 0$, the composition

$$C^p(\mathcal{U}, \mathcal{F}) \xrightarrow{d^p} C^{p+1}(\mathcal{U}, \mathcal{F}) \xrightarrow{d^{p+1}} C^{p+2}(\mathcal{U}, \mathcal{F})$$

is zero.

Proof. Let $s \in C^p(\mathcal{U}, \mathcal{F})$. Then

$$\begin{aligned} d(ds)(i_0, \ldots, i_{p+2}) &= \sum_{j=0}^{p+2} (-1)^j (ds)(i_0, \ldots, \widehat{i_j}, \ldots, i_{p+2})\big|_{U_{i_0} \cap \cdots \cap U_{i_{p+2}}} \\ &= \sum_{j=0}^{p+2} \Bigg(\sum_{0 \leqslant k < j} (-1)^{j+k} s(i_0, \ldots, \widehat{i_k}, \ldots, \widehat{i_j}, \ldots, i_{p+2})\big|_{U_{i_0} \cap \cdots \cap U_{i_{p+2}}} \\ &\qquad + \sum_{j < k \leqslant p+2} (-1)^{j+k-1} s(i_0, \ldots, \widehat{i_j}, \ldots, \widehat{i_k}, \ldots, i_{p+2})\big|_{U_{i_0} \cap \cdots \cap U_{i_{p+2}}} \Bigg) = 0. \end{aligned}$$

□

In this way we define a cochain complex $(C^\bullet(\mathcal{U}, \mathcal{F}), d)$ of abelian groups, called the **Čech complex**. The **p-th Čech-cohomology group $\check{H}^p(\mathcal{U}, \mathcal{F})$ with respect to the cover $\mathcal{U}$** is then defined to be the p-th cohomology group of the Čech complex :

$$\check{H}^p(\mathcal{U}, \mathcal{F}) := H^p(C^\bullet(\mathcal{U}, \mathcal{F}), d^\bullet).$$

Suppose $\mathcal{V} = \{V_j\}_{j \in J}$ is another open cover X that refines $\mathcal{U}$. Pick any map $\sigma : J \to I$ such that $V_j \subseteq U_{\sigma(j)}$ for all $j \in J$. The restriction induces a map on cochain group

$$\mathrm{ref}^{\mathcal{U},p}_{\mathcal{V},\sigma} : C^p(\mathcal{U}, \mathcal{F}) \longrightarrow C^p(\mathcal{V}, \mathcal{F})$$

given by

$$\mathrm{ref}^{\mathcal{U},p}_{\mathcal{V},\sigma}(s)(j_0,\dots,j_p) = s(\sigma(j_0),\dots,\sigma(j_p))\big|_{V_{j_0}\cap\cdots\cap V_{j_p}}.$$

Lemma 5.2. One has

$$d^{p+1}\circ \mathrm{ref}^{\mathcal{U},p}_{\mathcal{V},\sigma} = \mathrm{ref}^{\mathcal{U},p+1}_{\mathcal{V},\sigma}\circ d^p.$$

This shows $\mathrm{refs}^{\mathcal{U},\bullet}_{\mathcal{V},\sigma}$ induces a map on the Čech cohomology group, still denoted by

$$\mathrm{ref}^{\mathcal{U},\bullet}_{\mathcal{V},\sigma} : \check{H}^\bullet(\mathcal{U},\mathcal{F}) \longrightarrow \check{H}^\bullet(\mathcal{V},\mathcal{F}).$$

Proof.

□

The construction of $\mathrm{ref}^{\mathcal{U},\bullet}_{\mathcal{V},\sigma}$ depends on the map σ between index sets, but it turns out that the resulting map on cohomology levels is independent of σ. Suppose $\tau : J \to I$ is another map with $V_j \subseteq U_{\tau(j)}$. In particular, we have

$$V_j \subseteq U_{\sigma(j)} \cap U_{\tau(j)}.$$

Suppose s is a 1-cocycle (i.e. $ds = 0$) for the cover $\mathcal{U}$. Then

$$\begin{aligned}
\mathrm{ref}^{\mathcal{U},1}_{\mathcal{V},\sigma} s(j_0,j_1) &= s(\sigma(j_0),\sigma(j_1))\big|_{V_{j_0}\cap V_{j_1}} \\
\text{(omit the restriction)} &= s(\sigma(j_0),\tau(j_1)) - s(\sigma(j_1),\tau(j_1)) \\
&= \big(s(\sigma(j_0),\tau(j_0)) + s(\tau(j_0),\tau(j_1))\big) - s(\sigma(j_1),\tau(j_1)) \\
&= \mathrm{ref}^{\mathcal{U},1}_{\mathcal{V},\tau} s(j_0,j_1) + s(\sigma(j_0),\tau(j_0)) - s(\sigma(j_1),\tau(j_1)).
\end{aligned}$$

If we define $t \in C^1(\mathcal{V},\mathcal{F})$ by $t(j) := s(\sigma(j),\tau(j)) \in \mathcal{F}(V_j)$, the above computation shows

$$\mathrm{ref}^{\mathcal{U},1}_{\mathcal{V},\sigma} s - \mathrm{ref}^{\mathcal{U},1}_{\mathcal{V},\tau} s = d^0 t$$

i.e., two different ref differ only from a 1-coboundary. In general

Lemma 5.3. If s is a p-cocycle for the cover $\mathcal{U}$, then

$$\mathrm{ref}^{\mathcal{U},p}_{\mathcal{V},\sigma} s - \mathrm{ref}^{\mathcal{U},p}_{\mathcal{V},\tau} s = d^{p-1} t$$

where $t \in C^{p-1}(\mathcal{V},\mathcal{F})$ is given by

$$t(j_0,\dots,j_p) = \sum_{k=0}^{p-1}(-1)^k s(\sigma(j_0),\dots,\sigma(j_k),\tau(j_k),\dots,\tau(j_{p-1}))\big|_{V_{j_0}\cap\cdots\cap V_{j_p}}.$$

Proof.

□

This shows $\mathrm{ref}^{\mathcal{U},\bullet}_{\mathcal{V},\sigma}$ is in fact independent of the choice of σ in the cohomology level, and hence we can drop the subscript when mentioning the cohomology map :

$$\mathrm{ref}^{\mathcal{U},\bullet}_{\mathcal{V}} : \check{H}^\bullet(\mathcal{U},\mathcal{F}) \longrightarrow \check{H}^\bullet(\mathcal{V},\mathcal{F}).$$

This also shows that all $\check{H}^\bullet(\mathcal{U},\mathcal{F})$, with $\mathcal{U}$ ranging over all open covers of X, forms a directed system under the partial order given by the refinement. Now we can define the p**-th Čech cohomology group** by

$$\check{H}^\bullet(X,\mathcal{F}) = \varinjlim_{\mathcal{U}} \check{H}^\bullet(\mathcal{U},\mathcal{F}).$$

Introduce the **alternating Čech cochain group** :

$$C^p_{\mathrm{alt}}(\mathcal{U},\mathcal{F}) = \left\{ s \in C^p(\mathcal{U},\mathcal{F}) \mid \begin{array}{l} s(i_0,\dots,i_p) = 0 \text{ if } i_n = i_m \text{ for some } n \neq m \\ s(i_{\sigma(0)},\dots,s_{\sigma(p)}) = \mathrm{sgn}(\sigma)s(i_0,\dots,i_p) \text{ for all } \sigma \in \mathfrak{S}_{p+1}. \end{array} \right\}$$

When $p = 1$, every 1-cocycle is automatically alternating : if $s \in C^1(\mathcal{U},\mathcal{F})$ with $ds = 0$, then $s(i_0,i_1) = s(i_0,i_2) - s(i_1,i_2)$, and in particular, if $i_0 = i_1 = i$, choosing $i_2 = i$ shows $s(i,i) = 0$; if we choose $i_2 = i_0$, then $s(i_0,i_1) = s(i_0,i_0) - s(i_1,i_0) = -s(i_1,s_0)$. This is no longer holds for $p > 1$.

Lemma 5.4. For $p \geqslant 0$, one has $d^p C^p_{\mathrm{alt}}(\mathcal{U},\mathcal{F}) \subseteq C^{p+1}_{\mathrm{alt}}(\mathcal{U},\mathcal{F})$.

Proof. Let $s \in C^p_{\mathrm{alt}}(\mathcal{U},\mathcal{F})$. Then

$$(ds)(i_0,\dots,i_{p+1}) = \sum_{j=0}^{p+1} (-1)^j s(i_0,\dots,\widehat{i_j},\dots,i_{p+1})\big|_{U_{i_0}\cap\cdots\cap U_{i_{p+1}}}.$$

If $i_n = i_m$ for some $n \neq m$, this formula readily implies that $(ds)(i_0,\dots,i_{p+1})$. To show ds is alternating, it suffices to check

$$ds(i_0,\dots,i_n,\dots,i_m,\dots,i_{p+1}) = -ds(i_0,\dots,i_m,\dots,i_n,\dots,i_{p+1}).$$

for all $n < m$. □

Thus $(C^\bullet_{\mathrm{alt}}(\mathcal{U},\mathcal{F}), d^\bullet)$ forms a subcomplex of $(C^\bullet(\mathcal{U},\mathcal{F}), d^\bullet)$. A significant result is the following

Theorem 5.5. The cohomology of $(C^\bullet_{\mathrm{alt}}(\mathcal{U},\mathcal{F}), d^\bullet)$ is isomorphic to that of $(C^\bullet(\mathcal{U},\mathcal{F}), d^\bullet)$.

Proof. Let $<$ be a total order on I. For each $k \geqslant 0$, define $D^p_{(k)} \leqslant C^p(\mathcal{U},\mathcal{F})$ by

$$D^p_{(k)} = \{ s \in C^p(\mathcal{U},\mathcal{F}) \mid s(i_0,\dots,i_p) = 0 \text{ if } i_0 < \cdots < i_{p-k} \}.$$

Clearly we have $0 = D^p_{(p)} \subseteq D^p_{(1)} \subseteq \cdots \subseteq D^p_{(0)}$ and $d^p D^p_{(k)} \subseteq D^{(p+1)}_{(k)}$. Moreover, the inclusion induces an isomorphism

$$D^p_{(0)} \oplus C^p_{\mathrm{alt}}(\mathcal{U},\mathcal{F}) \cong C^p(\mathcal{U},\mathcal{F}).$$

To see this, let $s \in C^p(\mathcal{U}, \mathcal{F})$, and define $s_a \in C^p_{\text{alt}}(\mathcal{U}, \mathcal{F})$ by

$$s_a(i_0, \ldots, i_p) = \begin{cases} 0 & \text{, if } i_n = i_m \text{ for some } n \neq m \\ \operatorname{sgn}(\pi) s(i_{\pi(0)}, \ldots, i_{\pi(p)}) & \text{, if } \pi \in \mathfrak{S}_{p+1} \text{ is the unique element such that } i_{\pi(0)} < \cdots < i_{\pi(p)}. \end{cases}$$

It is clear from the construction that s_a is alternating and $s - s_a \in D^p_{(0)}$. This proves the above map is an isomorphism. In fact, we have $D^\bullet_{(0)} \oplus C^\bullet_{\text{alt}}(\mathcal{U}, \mathcal{F}) \cong C^\bullet(\mathcal{U}, \mathcal{F})$, and taking cohomology gives

$$\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \cong H^\bullet(C^\bullet_{\text{alt}}(\mathcal{U}, \mathcal{F}), d^\bullet) \oplus H^\bullet(D^\bullet_{(0)}, d^\bullet).$$

We claim $H^\bullet(D^\bullet_{(0)}, d^\bullet) = 0$, i.e., $(D^\bullet_{(0)}, d^\bullet)$ is exact. It suffices to show the quotient complex $(D^\bullet_{(k)}/D^\bullet_{(k+1)}, d^\bullet)$ is exact for each k.

There is an isomorphism induced by the projection

$$D^p_{(k)}/D^p_{(k+1)} \cong \left\{ s \in \prod_{\substack{i_0, \ldots, i_p \in I \\ i_0 < \cdots < i_{p-k-1}}} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p}) \mid s(i_0, \ldots, i_p) = 0 \text{ if } i_0 < \cdots < i_{p-k} \right\}$$

The right-handed side also defines a complex with coboundary map defined as before, and this isomorphism in fact an isomorphism of these two complexes. From now on we identify $(D^\bullet_{(k)}/D^\bullet_{(k+1)}, d^\bullet)$ with the right-handed side. Define

$$h^p : D^p_{(k)}/D^p_{(k+1)} \longrightarrow D^{p-1}_{(k)}/D^{p-1}_{(k+1)}$$

as follows. For $i_0, \ldots, i_{p-1} \in I$ with $i_0 < \cdots < i_{p-k-2}$, set

$$h^p s(i_0, \ldots, i_{p-1}) = \begin{cases} 0 & \text{, if } i_{p-k-1} = i_n \text{ for some } 0 \leqslant n \leqslant p-k-2 \\ (-1)^{n+1} s(i_0, \ldots, i_n, i_{p-k-1}, i_{n+1}, \ldots, i_{p-1}) & \text{, if } i_n < i_{p-k-1} < i_{n+1} \text{ for some } 0 \leqslant n \leqslant p-k-3 \\ s(i_{p-k-1}, i_0, \ldots, i_{p-1}) & \text{, if } i_{p-k-1} < i_0 \\ 0 & \text{, if } i_{p-k-1} > i_{p-k-2} \end{cases}$$

We show $d^{p-1} \circ h^p + h^{p+1} \circ d^p = \text{id}$, proving the exactness of $(D^\bullet_{(k)}/D^\bullet_{(k+1)}, d^\bullet)$. Let $i_0, \ldots, i_p \in I$ with $i_0 < \ldots < i_{p-k-1}$. We have

$$\begin{aligned} d^{p-1} \circ h^p(s)(i_0, \ldots, i_p) &= \sum_{j=0}^{p} (-1)^j h^p s(i_0, \ldots, \widehat{i_j}, \ldots, i_p) \\ &= \sum_{0 \leqslant j \leqslant p-k-1} (-1)^j h^p s(i_0, \ldots, \widehat{i_j}, \ldots, i_{p-k-1}, \ldots, i_p) \\ &\quad + \sum_{p-k-1 < j \leqslant p} (-1)^j h^p s(i_0, \ldots, i_{p-k-1}, \ldots, \widehat{i_j}, \ldots, i_p) \\ &= \sum_{0 \leqslant j \leqslant p-k-1} (-1)^j h^p s(i_0, \ldots, \widehat{i_j}, \ldots, i_{p-k-1}, \ldots, i_p). \end{aligned}$$

There are four cases.

— $i_{p-k} = i_n$ for some $0 \leqslant n \leqslant p-k-1$. Then

$$h^{p+1} \circ d^p(s)(i_0, \dots, i_p) = 0$$

and

$$d^{p-1} \circ h^p(s)(i_0, \dots, i_p) = (-1)^n (-1)^n s(i_0, \dots, i_{n-1}, i_{p-k}, i_{n+1}, \dots, i_p) = s(i_0, \dots, i_p).$$

So in this case the homotopy relation holds.

— $i_n < i_{p-k} < i_{n+1}$ for some $0 \leqslant n \leqslant p-k-2$. Then

$$\begin{aligned} d^{p-1} \circ h^p(s)(i_0, \dots, i_p) &= \sum_{0 \leqslant j \leqslant p-k-1} (-1)^j h^p s(i_0, \dots, \widehat{i_j}, \dots, i_{p-k-1}, \dots, i_p) \\ &= \sum_{0 \leqslant j \leqslant n} (-1)^j (-1)^n s(i_0, \dots, \widehat{i_j}, \dots, i_n, i_{p-k}, i_{n+1}, \dots,)+ \\ &\quad + \sum_{n+1 \leqslant j \leqslant p-k-1} (-1)^j (-1)^{n+1} s(i_0, \dots, i_n, i_{p-k}, i_{n+1}, \dots, \widehat{i_j}, \dots, i_{p-k-1}, \dots) \end{aligned}$$

$$\begin{aligned} h^{p+1} \circ d^p(s)(i_0, \dots, i_p) &= (-1)^{n+1} d^p s(i_0, \dots, i_n, i_{p-k}, i_{n+1}, \dots, i_{p-k-1}, \dots, i_p) \\ &= (-1)^{n+1} \Bigg(\sum_{0 \leqslant j \leqslant n} (-1)^j s\left(i_0, \dots \widehat{i_j}, \dots, i_n, i_{p-k}, i_{n+1}, \dots\right) + (-1)^{n+1} s(i_0, \dots, i_p) \\ &\quad + \sum_{n+1 \leqslant j \leqslant p} (-1)^{j+1} s(i_0, \dots, i_n, i_{p-k}, i_{n+1}, \dots, \widehat{i_j}, \dots) \Bigg) \\ &= s(i_0, \dots, i_p) + (-1)^{n+1} \sum_{0 \leqslant j \leqslant n} (-1)^j s\left(i_0, \dots \widehat{i_j}, \dots, i_n, i_{p-k}, i_{n+1}, \dots\right) \\ &\quad + (-1)^{n+1} \sum_{n+1 \leqslant j \leqslant p-k-1} (-1)^{j+1} s(i_0, \dots, i_n, i_{p-k}, i_{n+1}, \dots, \widehat{i_j}, \dots, i_{p-k-1}, \dots, i_p) \end{aligned}$$

Hence the homotopy relation holds as well.

— $i_{p-k} > i_{p-k-1}$. Each term is zero.

— $i_{p-k} < i_0$. Then

$$d^{p-1} \circ h^p(s)(i_0, \dots, i_p) = \sum_{0 \leqslant j \leqslant p-k-1} (-1)^j s(i_{p-k}, i_0, \dots, \widehat{i_j}, \dots, i_p)$$

$$\begin{aligned} h^{p+1} \circ d^p(s)(i_0, \dots, i_p) &= d^p(s)(i_{p-k}, i_0, \dots, i_p) \\ &= s(i_0, \dots, i_p) - \sum_{0 \leqslant j \leqslant p-k-1} (-1)^j s(i_{p-k}, i_0, \dots, \widehat{i_j}, \dots, i_{p-k-1}, \dots, i_p). \end{aligned}$$

Thus the homotopy relation holds.

In any case we see $d^{p-1} \circ h^p + h^{p+1} \circ d^p = \mathrm{id}$ holds, and this completes the proof. $\square$

There is an equivalent formulation of the alternating Čech complex. Let $<$ be a total order on I. Define the **ordered Čech complex** $(C^\bullet_{\mathrm{ord}}(\mathcal{U}, \mathcal{F}), d^\bullet)$ as follows : set

$$C^p_{\mathrm{ord}}(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \cdots < i_p} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p})$$

and set d^p as before. There is a natural projection

$$\pi^p : C^p(\mathcal{U}, \mathcal{F}) \longrightarrow C^p_{\mathrm{ord}}(\mathcal{U}, \mathcal{F}).$$

Its restriction to $C^p_{\mathrm{alt}}(\mathcal{U}, \mathcal{F})$ is an isomorphism of complexes with inverse

$$\psi^p : C^p_{\mathrm{ord}}(\mathcal{U}, \mathcal{F}) \longrightarrow C^p_{\mathrm{alt}}(\mathcal{U}, \mathcal{F})$$

given by

$$\psi^p(s)(i_0, \ldots, i_p) = \begin{cases} 0 & \text{, if } i_n = i_m \text{ for some } n \neq m \\ \mathrm{sgn}(\pi) s(i_{\pi(0)}, \ldots, i_{\pi(p)}) & \text{, if } \pi \in \mathfrak{S}_{p+1} \text{ is the unique element such that } i_{\pi(0)} < \cdots < i_{\pi(p)}. \end{cases}$$

The last theorem together with the above discussion leads to

Theorem 5.6. Let X be a topological space, $\mathcal{U}$ an open cover of X and $\mathcal{F}$ a presheaf on X. Then all arrows below are quasi-isomorphism.

$$C^p_{\mathrm{alt}}(\mathcal{U}, \mathcal{F}) \xrightarrow{\text{inclusion}} C^p(\mathcal{U}, \mathcal{F}) \xrightarrow{\text{projection}} C^p_{\mathrm{ord}}(\mathcal{U}, \mathcal{F}).$$

5.2 Basic properties

5.2.1 Long exact sequence for cohomology

An easy argument shows that Čech cohomology is a δ-functor from the category of *presheaves* of abelian groups over X to that of abelian groups. Explicitly, for every short exact sequence

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

of presheaves, and for every open cover $\mathcal{U}$ of X, there exists a long exact sequence on the cohomology groups :

$$0 \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{F}_1) \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{F}_2) \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{F}_3) \xrightarrow{\delta} \check{H}^1(\mathcal{U}, \mathcal{F}_1) \longrightarrow \check{H}^1(\mathcal{U}, \mathcal{F}_2) \longrightarrow \cdots$$

Moreover, by passing to refinement we obtain (recall that $\varinjlim$ is exact in the category of $\mathbb{Z}$-modules)

$$0 \longrightarrow \check{H}^0(X, \mathcal{F}_1) \longrightarrow \check{H}^0(X, \mathcal{F}_2) \longrightarrow \check{H}^0(X, \mathcal{F}_3) \xrightarrow{\delta} \check{H}^1(X, \mathcal{F}_1) \longrightarrow \check{H}^1(X, \mathcal{F}_2) \longrightarrow \cdots$$

But in applications, we are interested in the cohomology of *sheaves*, and a short exact sequence of sheaves is barely exact as a sequence of presheaves. But the long exact cohomology sequence continues to hold in reasonable cases. Say

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is an exact sequence of *sheaves*. Define a subpresheaf $\mathcal{F}_3^*$ of $\mathcal{F}_3$ by

$$\mathcal{F}_3^*(U) := \operatorname{Im}\left(\mathcal{F}_2(U) \to \mathcal{F}_3(U)\right).$$

Then we obtain a short exact sequence of presheaves

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3^* \longrightarrow 0$$

whence a long exact sequence

$$\cdots \longrightarrow \check{H}^i(X, \mathcal{F}_1) \longrightarrow \check{H}^i(X, \mathcal{F}_2) \longrightarrow \check{H}^i(X, \mathcal{F}_3) \xrightarrow{\delta} \check{H}^{i+1}(X, \mathcal{F}_1) \longrightarrow \cdots$$

Now $\mathcal{F}_3$ is the sheafification of $\mathcal{F}_3^*$, so we can replace $\mathcal{F}_3^*$ by $\mathcal{F}_3$ in the above sequence if we can prove the following assertion

- $\diamond$ for all presheaves $\mathcal{F}$ over X, the canonical maps

$$\check{H}^i(X, \mathcal{F}) \longrightarrow \check{H}^i(X, \mathcal{F}^\dagger)$$

  are isomorphisms.

The statement ($\diamond$) will follow if the following assertion holds :

- $\diamond\diamond$ If $\mathcal{F}$ is a presheaf over X such that $\mathcal{F}^\dagger = 0$, then $\check{H}^i(X, \mathcal{F}) = 0$.

Proposition 5.7. ($\diamond\diamond$) holds when X is a Hausdorff paracompact space.

Proof. Since X is paracompact, locally finite covers of X form a cofinal system among all open covers of X, and hence it suffices to compute Čech cohomology group with respect to every locally finite cover. So let $\mathcal{U} = \{U_i\}_i$ be such a cover. Each point $x \in X$ is contained in some U_i, and by regularity (a paracompact space is normal, and hence regular) we can find an open $W_x \ni x$ with $W_x \subseteq \overline{W_x} \subseteq U_i$. Pick a locally finite open refinement $\mathcal{W}$ of $\{W_x\}$. Define $f : \mathcal{W} \to I$ by requiring $W \subseteq U_{f(W)}$ for each $W \in \mathcal{W}$, and set

$$V_i := \bigcup_{f(W)=i} W$$

Then $V_i \subseteq \overline{V_i} \subseteq U_i$. We claim $\mathcal{V} = \{V_i\}_{i \in I}$ is locally finite. Let $x \in X$ and pick a neighborhood N of x such that $\#\{W \in \mathcal{W} \mid W \cap N \neq \varnothing\} < \infty$. If $N \cap V_i$ is nonempty, then N intersects nontrivially with some W with $f(W) = i$, and vice versa. Hence

$$\#\{i \in I \mid V_i \cap N = \varnothing\} \leqslant \#\{W \in \mathcal{W} \mid W \cap N \neq \varnothing\} < \infty$$

and this proves the locally finiteness of $\mathcal{V}$.

Now we prove ($\diamond\diamond$). Let $s \in C^p(\mathcal{U}, \mathcal{F})$. Let $x \in X$; by locally finiteness we can find a neighborhood N_x such that

(i) $x \in V_i$ implies $N_x \subseteq V_i$,

(ii) $N_x \cap V_i \neq \varnothing$ implies $N_x \subseteq V_i$, and

(iii) $x \in U_{i_0} \cap \cdots \cap U_{i_p}$ implies $N_x \subseteq U_{i_0} \cap \cdots \cap U_{i_p}$ and $s(i_0, \ldots, i_p)|_{N_x} = 0$.

Indeed, (i) and (iii) follow from locally finiteness and assumption. For (ii), we can first pick N'_x so that it intersects only with finitely many V_i and then choose

$$N_x \subseteq N'_x \cap \bigcup_{x \in V_i} V_i \cap \bigcup_{x \notin V_i} (N'_x \backslash \overline{V}_i).$$

Now take $\mathcal{W} = \{N_x\}_{x \in X}$. Then $\mathcal{W}$ refines $\mathcal{V}$ and $\mathrm{ref}^{\mathcal{U},p}_{\mathcal{W}}(s) = 0$ as a cochain. Indeed, suppose $N_{x_0} \cap \cdots \cap N_{x_p} \neq \varnothing$ and pick $i_0, \ldots, i_p \in I$ with $N_{x_n} \subseteq V_{i_n}$. We must show $s(i_0, \ldots, i_p)|_{N_{x_0} \cap \cdots \cap N_{x_p}} = 0$. But $N_{x_0} \cap \cdots \cap N_{x_p} \subseteq N_{x_n} \subseteq V_{i_n}$ implies $N_{x_0} \cap V_{i_n} \neq \varnothing$, so by (ii) $N_{x_0} \subseteq V_{i_n}$ for each $0 \leqslant n \leqslant p$. Hence

$$s(i_0, \ldots, i_p)|_{N_{x_0} \cap \cdots \cap N_{x_p}} = \big(s(i_0, \ldots, i_p)|_{N_{x_0}}\big)\big|_{N_{x_0} \cap \cdots \cap N_{x_p}} = 0$$

by (iii). □

It is sad that we do not have a long exact sequence for Čech cohomology for sheaves in general. Nevertheless, this is true in low dimension.

Proposition 5.8. Let X be a topological space and

5.2.2 Comparison between refinement

Proposition 5.9. Let X be a topological space, $\mathcal{F}$ a *sheaf* of abelian groups on X, and $\mathcal{U} = \{U_i\}_{i \in I}$ and $\mathcal{V} = \{V_j\}_{j \in I}$ two open covers of X. Suppose $\mathcal{V}$ refines $\mathcal{U}$. For every finite subset $I_0 \subseteq I$, put $U_{I_0} = \bigcap_{i \in I_0} U_i$, and denote by $\mathcal{V}|_{U_{S_0}}$ the cover of U_{S_0} induced by $\mathcal{V}$. Assume

$$\check{H}^p(\mathcal{V}|_{U_{S_0}}, \mathcal{F}|_{U_{S_0}}) = 0 \text{ for all } S_0 \text{ and } p > 0.$$

Then $\mathrm{ref}^{\mathcal{U}^\bullet}_{\mathcal{V}} : \check{H}^\bullet(\mathcal{U}, \mathcal{F}) \to \check{H}^\bullet(\mathcal{V}, \mathcal{F})$ is an isomorphism.

Proof. We compare two cohomologies by double complexes. For $p, q \geqslant 0$, define

$$C^{p,q} = \prod_{i_0, \ldots, i_p \in I} \prod_{j_0, \ldots, j_q \in J} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p} \cap V_{j_0} \cap \cdots \cap V_{j_q}).$$

Define two differentials $d_1 : C^{p,q} \to C^{p+1,q}$ and $d_2 : C^{p,q} \to C^{p,q+1}$ by the usual differentials but ignoring either $i_0, \ldots, i_p$ or $j_0, \ldots, j_q$. Then $d_i^2 = 0$, $i = 1, 2$ and $d_1 d_2 = d_2 d_1$, and we obtain a commutative first quadrant double complex

$$
\begin{tikzcd}
\vdots & \vdots & \vdots & \\
C^{0,2} \arrow[u, "d_2"] \arrow[r, "d_1"] & C^{1,2} \arrow[u, "d_2"] \arrow[r, "d_1"] & C^{2,2} \arrow[u, "d_2"] \arrow[r] & \cdots \\
C^{0,1} \arrow[u, "d_2"] \arrow[r, "d_1"] & C^{1,1} \arrow[u, "d_2"] \arrow[r, "d_1"] & C^{2,1} \arrow[u, "d_2"] \arrow[r] & \cdots \\
C^{0,0} \arrow[u, "d_2"] \arrow[r, "d_1"] & C^{1,0} \arrow[u, "d_2"] \arrow[r, "d_1"] & C^{2,0} \arrow[u, "d_2"] \arrow[r] & \cdots
\end{tikzcd}
$$

Put

$$C^{(n)} = \mathrm{tot}(C^{\bullet,\bullet})^n = \sum_{\substack{p+q=n \\ p,q\geqslant 0}} C^{p,q}$$

to be the total complex of this double complex, and put

$$d := \sum_{\substack{p+q=n \\ p,q\geqslant 0}} (d_1 + (-1)^p d_2) : C^{(n)} \longrightarrow C^{(n+1)}$$

to be the differential of $C^{(n)}$. By our assumption, the d_2-cohomology of columns vanish for all $p \geqslant 0$ and $q > 0$. On the other hand, the d_1-cohomology of rows also vanish for all $p > 0$ and $q \geqslant 0$. This is because the cover used in that Čech complex contains the whole space, and the following lemma witnesses the vanishing.

Lemma 5.10. Let X be a topological space, $\mathcal{F}$ a sheaf of abelian groups, and $\mathcal{U}$ an open cover of X with $X \in \mathcal{U}$. Then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.

Proof. Say $X = U_\zeta$ for some $\zeta \in I$. If s is a p-cocycle, define $t \in C^{p-1}(\mathcal{U}, \mathcal{F})$ by

$$t(i_0, \ldots, i_{p-1}) = s(\zeta, i_0, \ldots, i_{p-1}).$$

Then $dt = s$. Indeed,

$$dt(i_0, \ldots, i_p) = \sum_{j=0}^{p} (-1)^j s(\zeta, i_0, \ldots, \widehat{i_j}, \ldots, i_p).$$

On the other hand,

$$0 = ds(\zeta, i_0, \ldots, i_p) = s(i_0, \ldots, i_p) - \sum_{j=0}^{p} (-1)^j s(\zeta, i_0, \ldots, \widehat{i_j}, \ldots, i_p).$$

so summing these two up gives $dt = s$. $\square$

To finish the proof, we use the following homological lemma whose proof is omitted.

Lemma 5.11. Consider the following double complex

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \vdots & \vdots & \\
0 \arrow[r] & F^3 \arrow[r] \arrow[u] & K^{0,3} \arrow[r] \arrow[u] & K^{1,3} \arrow[r] \arrow[u] & K^{2,3} \arrow[r] \arrow[u] & K^{3,3} \arrow[r] \arrow[u] & \cdots \\
0 \arrow[r] & F^2 \arrow[r] \arrow[u] & K^{0,2} \arrow[r] \arrow[u] & K^{1,2} \arrow[r] \arrow[u] & K^{2,2} \arrow[r] \arrow[u] & K^{3,2} \arrow[r] \arrow[u] & \cdots \\
0 \arrow[r] & F^1 \arrow[r] \arrow[u] & K^{0,1} \arrow[r] \arrow[u] & K^{1,1} \arrow[r] \arrow[u] & K^{2,1} \arrow[r] \arrow[u] & K^{3,1} \arrow[r] \arrow[u] & \cdots \\
0 \arrow[r] & F^0 \arrow[r] \arrow[u] & K^{0,0} \arrow[r] \arrow[u] & K^{1,0} \arrow[r] \arrow[u] & K^{2,0} \arrow[r] \arrow[u] & K^{3,0} \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

with all rows exact. Then the natural map $F \to \mathrm{tot}(K)$ is a quasi-isomorphism.

Since $\mathcal{F}$ is a sheaf, the kernel $d_1 : C^{0,q} \to C^{1,q}$ is exactly $C^q(\mathcal{V}, \mathcal{F})$, and similarly $\ker(d_2 : C^{p,0} \to C^{p,1}) = C^p(\mathcal{U}, \mathcal{F})$. These natural inclusions give rise to the isomorphisms

$$\check{H}^\bullet(\mathcal{V}, \mathcal{F}) \xrightarrow{\sim} H^\bullet(C^{(\bullet)}, d)$$

and

$$\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \xrightarrow{\sim} H^\bullet(C^{(\bullet)}, d).$$

It remains to show the composition

$$\check{H}^n(\mathcal{U}, \mathcal{F}) \longrightarrow H^n(C^{(\bullet)}, d) \longrightarrow \check{H}^n(\mathcal{U}, \mathcal{F})$$

coincides with $\mathrm{ref}_{\mathcal{V}}^{\mathcal{U},n}$. Let s be an n-cocycle for $\mathcal{U}$. Then $s \in C^{n,0}$ and $\mathrm{ref}_{\mathcal{V}}^{\mathcal{U},n}(s) \in C^{0,n}$. It is enough to show they are cohomologous in the total complex. To be precise, we have $\mathrm{ref}_{\mathcal{V}}^{\mathcal{U},n}(s) - s \in C^{(n)}$, and we show there exists $t \in C^{(n-1)}$ such that $dt = \mathrm{ref}_{\mathcal{V}}^{\mathcal{U},n}(s) - s$. Define $t \in C^{(n-1)}$ by setting its $(\ell, n-1-\ell)$-th component t_ℓ to be

$$t_\ell(i_0, \ldots, i_\ell, j_0, \ldots, j_{n-1-\ell}) = (-1)^\ell s(i_0, \ldots, i_\ell, \sigma(j_0), \ldots, \sigma(j_{n-1-\ell}))\big|_{U_{\{i_0,\ldots,i_\ell\}} \cap V_{\{j_0,\ldots,j_{n-1-\ell}\}}}$$

where $\sigma : J \to I$ is a map such that $V_j \subseteq U_{\sigma(j)}$. Then

$$\begin{aligned}
d_1 t_{n-1}(i_0, \ldots, i_n, j_0) &= \sum_{k=0}^{n} (-1)^k t_{n-1}(i_0, \ldots, \widehat{i_k}, \ldots, i_n, j_0) \\
&= \sum_{k=0}^{n} (-1)^k (-1)^{n-1} s(i_0, \ldots, \widehat{i_k}, \ldots, i_n, \sigma(j_0)) \\
\text{(cocycle condition)} &= -s(i_0, \ldots, i_n)
\end{aligned}$$

for $0 \leqslant \ell \leqslant n-2$

$$\begin{aligned}
&d_1 t_\ell(i_0,\dots,i_\ell,i_{\ell+1},j_0,\dots,j_{n-1-\ell}) + (-1)^{\ell+1} d_2 t_{\ell+1}(i_0,\dots,i_\ell,i_{\ell+1},j_0,\dots,j_{n-1-\ell}) \\
&\quad = \sum_{k=0}^{\ell+1} (-1)^k (-1)^\ell s(i_0,\dots,\widehat{i_k},\dots,i_{\ell+1},\sigma(j_0),\dots,\sigma(j_{n-1-\ell})) \\
&\qquad + (-1)^{\ell+1} \sum_{k=0}^{n-1-\ell} (-1)^k (-1)^{\ell+1} s(i_0\dots,i_{\ell+1},\sigma(j_0),\dots,\widehat{\sigma(j_k)},\dots,\sigma(j_{n-1-\ell})) \\
&\quad = (-1)^\ell ds(i_0,\dots,i_{\ell+1},\sigma(j_0),\dots,\sigma(j_{n-1-\ell})) = 0
\end{aligned}$$

and

$$\begin{aligned}
d_2 t_0(i_0,j_0,\dots,j_n) &= \sum_{k=0}^{n} (-1)^k s(i_0,\sigma(j_0),\dots,\widehat{\sigma(j_k)},\dots,\sigma(j_n)) \\
\text{(cocycle condition)} &= s(\sigma(j_0),\dots,\sigma(j_n)) = \mathrm{ref}^{\mathcal{U},n}_{\mathcal{V},\sigma}(s)(j_0,\dots,j_n)
\end{aligned}$$

The above computation shows $dt = \mathrm{ref}^{\mathcal{U},n}_{\mathcal{V},\sigma}(s) - s$, and the proof is completed. □

5.2.3 Sheafified Čech complex

Let X be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ be an open cover and $\mathcal{F}$ a *presheaf* of abelian groups. For $p \geqslant 0$, we define a presheaf $\mathcal{C}^p(\mathcal{U},\mathcal{F})$ by

$$\mathcal{C}^p(\mathcal{U},\mathcal{F})(V) = \prod_{i_0,\dots,i_p\in I} \mathcal{F}(V \cap U_{i_0} \cap \cdots \cap U_{i_p}).$$

and for two open $V_1 \subseteq V_2 \subseteq X$, two groups are connected with restriction :

$$\begin{tikzcd}
\mathcal{C}^p(\mathcal{U},\mathcal{F})(V_2) \arrow[rr] \arrow[d, equal] \arrow[dr, phantom, "\circlearrowleft"] & & \mathcal{C}^p(\mathcal{U},\mathcal{F})(V_1) \arrow[d, equal] \\
\prod_{i_0,\dots,i_p\in I} \mathcal{F}(V_1 \cap U_{i_0} \cap \cdots \cap U_{i_p}) \arrow[rr, "\text{restriction}"] & & \prod_{i_0,\dots,i_p\in I} \mathcal{F}(V_2 \cap U_{i_0} \cap \cdots \cap U_{i_p})
\end{tikzcd}$$

Equivalently, if for each tuple $(i_0,\dots,i_p) \in I^{p+1}$ we put $\iota_{i_0,\dots,i_p} : U_{i_0} \cap \cdots \cap U_{i_p} \to X$ to be the natural inclusion, then

$$\mathcal{C}^p(\mathcal{U},\mathcal{F}) = \prod_{i_0,\dots,i_p\in I} (\iota_{i_0,\dots,i_p})_* \left(\mathcal{F}|_{U_{i_0}\cap\cdots\cap U_{i_p}}\right)$$

The Čech differential d on each $\mathcal{C}^p(\mathcal{U},\mathcal{F})(V)$ is packed together to a morphism

$$d^p : \mathcal{C}^p(\mathcal{U},\mathcal{F}) \longrightarrow \mathcal{C}^{p+1}(\mathcal{U},\mathcal{F})$$

and we obtain a complex of presheaves :

$$0 \longrightarrow \mathcal{F} \xrightarrow{\epsilon} \mathcal{C}^0(\mathcal{U},\mathcal{F}) \xrightarrow{d^0} \mathcal{C}^1(\mathcal{U},\mathcal{F}) \xrightarrow{d^1} \mathcal{C}^2(\mathcal{U},\mathcal{F}) \longrightarrow \cdots$$

where $\epsilon : \mathcal{F} \to \mathcal{C}^0(\mathcal{U},\mathcal{F})$ is the product of maps $\mathcal{F} \to (\iota_i)_*(\mathcal{F}|_{U_i})$ $(i \in I)$. Observe the following properties.

— If $\mathcal{F}$ is a sheaf (resp. flasque sheaf), so is each $\mathcal{C}^p(\mathcal{U},\mathcal{F})$.

— $\mathcal{F}$ is a sheaf if and only if the above complex is exact at the first two places.

— $\Gamma(X,\mathcal{C}^p(\mathcal{U},\mathcal{F})) = C^p(\mathcal{U},\mathcal{F})$.

In fact, more is true.

Lemma 5.12. For any sheaf $\mathcal{F}$ of abelian groups on X, the sequence

$$0 \longrightarrow \mathcal{F} \xrightarrow{\epsilon} \mathcal{C}^0(\mathcal{U},\mathcal{F}) \xrightarrow{d^0} \mathcal{C}^1(\mathcal{U},\mathcal{F}) \xrightarrow{d^1} \mathcal{C}^2(\mathcal{U},\mathcal{F}) \longrightarrow \cdots$$

is exact as sheaves.

Proof. The first two places are already exact. To show the exactness at remaining places, we check them on the stalks. To this end, we construct $k^0 : \mathcal{C}^0(\mathcal{U},\mathcal{F})_x \to \mathcal{F}_x$ and for each $p \geqslant 1$

$$k^p : \mathcal{C}^p(\mathcal{U},\mathcal{F})_x \longrightarrow \mathcal{C}^{p-1}(\mathcal{U},\mathcal{F})_x$$

and prove $\epsilon \circ k^0 + k^1 \circ d^0 = \mathrm{id}$ and $d^{p-1} \circ k^p + k^{p+1} \circ d^p = \mathrm{id}$.

Let $x \in X$ be given and an index $j \in I$ such that $x \in U_j$. Given $s_x \in \mathcal{C}^p(\mathcal{U},\mathcal{F})_x$, it is represented by some $s \in \Gamma(V,\mathcal{C}^p(\mathcal{U},\mathcal{F})_x)$ for some neighborhood V of x; choose V so small that $V \subseteq U_j$. Now for $i_0,\ldots,i_{p-1} \in I$, put

$$k^p s(i_0,\ldots,i_{p-1}) = s(j,i_0,\ldots,i_{p-1})\big|_x.$$

This makes sense as $V \cap U_{i_0} \cap \cdots U_{i_{p-1}} = V \cap U_j \cap U_{i_0} \cap \cdots U_{i_{p-1}}$, and this map is independent of the choice of $V \subseteq U_j$. Now compute

$$k^{p+1} \circ d^p s(i_0,\ldots,i_p) = d^p s(j,i_0,\ldots,i_p)|_x$$

$$\begin{aligned} d^{p-1} \circ k^p s(i_0,\ldots,i_p) &= \sum_{k=0}^{p} (-1)^k k^p s(i_0,\ldots,\widehat{i_k},\ldots,i_p) \\ &= \sum_{k=0}^{p} (-1)^k s(j,i_0,\ldots,\widehat{i_k},\ldots,i_p) = s(i_0,\ldots,i_p) - d^p s(j,i_0,\ldots,i_p) \end{aligned}$$

and

$$\epsilon \circ k^0 s(i) = k^0 s(i) = s(j,i).$$

These proves the homotopy relations, and the proof is completed. □

Corollary 5.12.1. Let X denote a topological space, $\mathcal{U}$ an open cover of X and $\mathcal{F}$ a sheaf of abelian groups on X. Then for any $p \geqslant 0$ there exists a natural map $\check{H}^p(\mathcal{U}, \mathcal{F}) \to H^p(X, \mathcal{F})$ functorial in $\mathcal{F}$.

Proof. Let $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ denote an injective resolution of $\mathcal{F}$ in $\mathbf{Mod}_{\underline{\mathbb{Z}}_X}$. Consider the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d, "\mathrm{id}"] & \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \arrow[r] & \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \arrow[r] & \cdots \\
0 \arrow[r] & \mathcal{F} \arrow[r] & \mathcal{I}^0 \arrow[r] & \mathcal{I}^1 \arrow[r] & \cdots
\end{tikzcd}$$

A general result says that there exists a morphism of complexes $C^\bullet(\mathcal{U}, \mathcal{F}) \to \mathcal{I}^\bullet$ making the above diagram commute, and the morphism is unique up to homotopy. Now the result follows once we taking global sections and cohomology. □

Corollary 5.12.2. Let X denote a topological space and $\mathcal{U}$ an open cover of X. If $\mathcal{F}$ is a flasque sheaf, then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p \geqslant 1$.

Proof. Since $\mathcal{F}$ is flasque, the sheafified Čech complex is a flasque, hence acyclic, resolution of $\mathcal{F}$, and thus it computes the sheaf cohomology. But since $\mathcal{F}$ is flasque, by Lemma 4.2 $\mathcal{F}$ is acyclic, and the result follows. □

5.2.4 Comparison to sheaf cohomology

Theorem 5.13. Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ an open cover of X and $\mathcal{F}$ a sheaf of abelian groups on X. Suppose that for any $p > 0$ and any finite intersection $V = U_{i_0} \cap \cdots \cap U_{i_k}$ of elements in $\mathcal{U}$, we have $H^p(V, \mathcal{F}|_V) = 0$. Then for any $p \geqslant 0$, the natural map in Corollary 5.12.1 is an isomorphism.

Proof. When $p = 0$, the natural map $\check{H}^p(\mathcal{U}, \mathcal{F}) \to H^0(X, \mathcal{F})$ is induced by the identity map $\mathrm{id} : \mathcal{F}(X) \to \mathcal{F}(X)$. It follows from definition that $\check{H}^p(\mathcal{U}, \mathcal{F}) = \mathcal{F}(X) = H^0(X, \mathcal{F})$, so the theorem holds for $p = 0$.

For $p > 0$, embed $\mathcal{F}$ into a flasque sheaf $\mathcal{I}$ and consider the resulting short exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0.$$

By assumption for any $k \geqslant 0$ and $i_0, \ldots, i_k \in I$, the sequence

$$0 \longrightarrow \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p}) \longrightarrow \mathcal{I}(U_{i_0} \cap \cdots \cap U_{i_p}) \longrightarrow \mathcal{G}(U_{i_0} \cap \cdots \cap U_{i_p}) \longrightarrow 0$$

is exact, and taking product over all $(k+1)$-tuples gives an exact sequence of Čech complexes :

$$0 \longrightarrow C^\bullet(\mathcal{U}, \mathcal{F}) \longrightarrow C^\bullet(\mathcal{U}, \mathcal{I}) \longrightarrow C^\bullet(\mathcal{U}, \mathcal{G}) \longrightarrow 0$$

This gives a long exact sequence of Čech cohomology groups; in view of Corollary 5.12.2, we have an exact sequence

$$0 \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{I}) \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{G}) \longrightarrow \check{H}^1(\mathcal{U}, \mathcal{F}) \longrightarrow 0 \qquad (\star)$$

and isomorphisms

$$\check{H}^p(\mathcal{U}, \mathcal{G}) \cong \check{H}^{p+1}(\mathcal{U}, \mathcal{F}) \tag{$\star\star$}$$

for $p \geqslant 1$ (given by connecting homomorphisms). Since the maps $\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \to H^\bullet(X, \mathcal{F})$ are functorial, by the case $p = 0$, the sequence $(\star)$ implies $\check{H}^1(\mathcal{U}, \mathcal{F}) \to H^1(X, \mathcal{F})$ is an isomorphism.

Let V be any finite intersection of elements in $\mathcal{U}$. The restriction $\mathcal{F} \mapsto \mathcal{F}|_V$ being exact, we have an exact sequence

$$0 \longrightarrow \mathcal{F}|_V \longrightarrow \mathcal{I}|_V \longrightarrow \mathcal{G}|_V \longrightarrow 0$$

with $\mathcal{I}|_V$ still flasque. The resulting long exact sequence on cohomology shows that $H^k(V, \mathcal{G}|_V) = 0$ for any $k > 0$. Now the result follows from induction on p and the isomorphisms $(\star\star)$. □

Corollary 5.13.1. Let X be a separated scheme, $\mathcal{U} = \{U_i\}_{i \in I}$ an open affine cover of X, and let $\mathcal{F}$ a quasi-coherent sheaf on X. Then for $p \geqslant 0$, the natural map in Corollary 5.12.1 is an isomorphism.

Proof. It suffices to take $\mathcal{U}$ to be an affine cover of X. Then the assumption in Theorem 5.13 holds by Theorem 4.12 (that X is separated is use here!). □

In general, Čech cohomology does not coincide with sheaf cohomology. But in the degree 1, two cohomology groups are always isomorphic. Precisely,

Proposition 5.14. Let X be a topological space and $\mathcal{F}$ a sheaf of abelian groups on X.

1. For open covers $\mathcal{U}$ and $\mathcal{V}$ of X, if $\mathcal{V}$ refines $\mathcal{U}$, we have a commutative triangle

$$\begin{tikzcd}
\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \arrow[dd, "\mathrm{ref}_{\mathcal{V}}^{\mathcal{U},\bullet}"'] \arrow[dr] & \\
& H^\bullet(X, \mathcal{F}) \\
\check{H}^\bullet(\mathcal{V}, \mathcal{F}) \arrow[ur] &
\end{tikzcd}$$

   where the slanted arrows are natural maps in Corollary 5.12.1.

By 1. we can pass to limit, obtaining a map $\check{H}^\bullet(X, \mathcal{F}) \to H^\bullet(X, \mathcal{F})$.

2. The map $\check{H}^1(X, \mathcal{F}) \to H^1(X, \mathcal{F})$ is an isomorphism.

Proof. Using the same map on index sets for induced covers, various refinement maps are packed to a morphism of sheafified Čech complexes $\mathrm{ref} : \mathcal{C}^\bullet(\mathcal{U}, \mathcal{F}) \to \mathcal{C}^\bullet(\mathcal{V}, \mathcal{F})$. We then have a commutative

diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d, "\mathrm{id}"] & \mathcal{C}^0(\mathcal{U},\mathcal{F}) \arrow[r] \arrow[d, "\mathrm{ref}"] & \mathcal{C}^1(\mathcal{U},\mathcal{F}) \arrow[r] \arrow[d, "\mathrm{ref}"] & \cdots \\
0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d, "\mathrm{id}"] & \mathcal{C}^0(\mathcal{V},\mathcal{F}) \arrow[r] & \mathcal{C}^1(\mathcal{V},\mathcal{F}) \arrow[r] & \cdots \\
0 \arrow[r] & \mathcal{F} \arrow[r] & \mathcal{I}^0 \arrow[r] & \mathcal{I}^1 \arrow[r] & \cdots
\end{tikzcd}$$

Now 1. follows from the algebraic fact used in Corollary 5.12.1.

For 2., embed $\mathcal{F}$ into a flasque sheaf $\mathcal{I}$ and take the quotient $\mathcal{G} = \mathcal{I}/\mathcal{F}$. Define a complex $D^\bullet(\mathcal{U})$ by the exact sequence

$$0 \longrightarrow C^\bullet(\mathcal{U},\mathcal{F}) \longrightarrow C^\bullet(\mathcal{U},\mathcal{G}) \longrightarrow D^\bullet(\mathcal{U}) \longrightarrow 0$$

The natural map $D^\bullet(\mathcal{U}) \to C^\bullet(\mathcal{U},\mathcal{G})$ and this exact sequence are compatible with refinement. Now taking cohomology, together with Corollary 5.12.2, gives

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(X) \arrow[r] \arrow[d, equal] & \mathcal{I}(X) \arrow[r] \arrow[d, equal] & H^0(D^\bullet(\mathcal{U})) \arrow[r] \arrow[d] & \check{H}^1(\mathcal{U},\mathcal{F}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{F}(X) \arrow[r] & \mathcal{I}(X) \arrow[r] & \mathcal{G}(X) \arrow[r] & H^1(X,\mathcal{F}) \arrow[r] & 0
\end{tikzcd}$$

Each map is natural as in Corollary 5.12.1. Moreover, the third map is the composition $H^0(D^\bullet(\mathcal{U})) \to \check{H}^0(\mathcal{U},\mathcal{G}) \to H^0(X,\mathcal{G})$. Being compatible with refinement, we can pass the first sequence to the limit, obtaining

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(X) \arrow[r] \arrow[d, equal] & \mathcal{I}(X) \arrow[r] \arrow[d, equal] & \varinjlim_{\mathcal{U}} H^0(D^\bullet(\mathcal{U})) \arrow[r] \arrow[d] & \varinjlim_{\mathcal{U}} \check{H}^1(\mathcal{U},\mathcal{F}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{F}(X) \arrow[r] & \mathcal{I}(X) \arrow[r] & \mathcal{G}(X) \arrow[r] & H^1(X,\mathcal{F}) \arrow[r] & 0
\end{tikzcd}$$

To show that last vertical arrow is an isomorphism, it suffices to show the third vertical one is an isomorphism. The injectivity is clear ; in fact, it is already injective before taking limit. To see surjectivity, for $x \in \mathcal{G}(X)$, by surjectivity of stalks we can find an open cover $\mathcal{U} = \{U_i\}_{i\in I}$ such that $x|_{U_i} = s_i$ for some $s_i \in \mathcal{I}(U_i)$. This means x lies in $\mathrm{Im}(C^0(\mathcal{U},\mathcal{I}) \to C^0(\mathcal{U},\mathcal{G})) \cap \mathcal{G}(X) = H^0(D^\bullet(\mathcal{U}))$, demonstrating the surjectivity.

$\square$

6 Singular (co)homology

6.1 Definitions

Let $\mathbb{R}^\infty$ have the standard basis $e_0, e_1, \ldots$. For $n \in \mathbb{Z}_{\geqslant 0}$, the **standard** p**-simplex** is the set

$$\Delta_n := \left\{ \sum_{i=0}^{p} a_i e_i \mid 0 \leqslant a_i \leqslant 1, \sum_{i=0}^{p} a_i = 1 \right\}.$$

For $v_0, \ldots, v_p \in \mathbb{R}^N$, define

$$\begin{aligned} [v_0, \ldots, v_p] : \Delta_n &\longrightarrow \mathbb{R}^N \\ \sum_{i=0}^{p} a_i e_i &\longmapsto \sum_{i=0}^{p} a_i v_i. \end{aligned}$$

Let X be a topological space and $p \in \mathbb{Z}_{\geqslant 0}$. By definition, a **singular** p**-simplex** in X is a continuous map $\Delta_p \to X$. For $p \geqslant 0$, define the **singular** p**-chain group** $S_p(X)$ to be the free abelian group based on all the singular p-simplices in X. For convenience, we put $S_p(X) = 0$ for $p < 0$.

For a singular p-simplex $\sigma : \Delta_p \to X$ and $0 \leqslant i \leqslant p$, the i-th face of σ is defined as 0 if $p \leqslant 0$, and

$$\sigma|_{[e_0, \ldots, \widehat{e_i}, \ldots, e_n]} := \sigma \circ [e_0, \ldots, e_{i-1}, e_{i+1}, \ldots, e_n] : \Delta_{n-1} \to X$$

if $p \geqslant 1$. Define the **boundary** map $\partial_p : S_p(X) \to S_{p-1}(X)$ by the formula

$$\partial_p \sigma = \sum_{i=0}^{p} (-1)^i \sigma|_{[e_0, \ldots, \widehat{e_i}, \ldots, e_p]}.$$

In particular, $\partial_p = 0$ for $p \leqslant 0$.

Lemma 6.1. For $n \in \mathbb{Z}_{\geqslant 0}$, one has $\partial_n \circ \partial_{n+1} = 0$. Thus there is a chain complex

$$\cdots \longrightarrow S_{p+1}(X) \xrightarrow{\partial_{p+1}} S_p(X) \xrightarrow{\partial_p} S_{p-1}(X) \xrightarrow{\partial_{p-1}} \cdots$$

Definition. Let G be an abelian group. Tensoring the chain complex in Lemma 6.1 with G, we obtain

$$\cdots \longrightarrow S_{p+1}(X) \otimes_{\mathbb{Z}} G \xrightarrow{\partial_{p+1} \otimes \mathrm{id}_G} S_p(X) \otimes_{\mathbb{Z}} G \xrightarrow{\partial_p \otimes \mathrm{id}_G} S_{p-1}(X) \otimes_{\mathbb{Z}} G \xrightarrow{\partial_{p-1} \otimes \mathrm{id}_G} \cdots$$

The p**-th singular homology group** $H_p(X; G)$ **of the space** X **with coefficient in** G is defined as

$$H_p^{\mathrm{sing}}(X; G) := H_p(S_\bullet(X) \otimes_{\mathbb{Z}} G, \partial_\bullet \otimes \mathrm{id}_G).$$

For simplicity, we write $S_p(X; G)$ for the group $S_p(X) \otimes_{\mathbb{Z}} G$ and write $\partial_\bullet$ for the boundary map $\partial_\bullet \otimes \mathrm{id}_G$.

Definition. Let G be an abelian group. Applying $\mathrm{Hom}_{\mathbb{Z}}(\cdot, G)$ to the complex in Lemma 6.1, we obtain a cochain complex

$$\cdots \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(S_{p-1}(X), G) \xrightarrow{\delta_p} \mathrm{Hom}_{\mathbb{Z}}(S_p(X), G) \xrightarrow{\delta_{p+1}} \mathrm{Hom}_{\mathbb{Z}}(S_{p+1}(X), G) \longrightarrow \cdots$$

The p**-th singular cohomology group** $H^p_{\mathbf{sing}}(X; G)$ **of the space** X **with coefficient in** G is defined as

$$H^p_{\mathrm{sing}}(X; G) := H^p(\mathrm{Hom}_{\mathbb{Z}}(S_\bullet(X), G), \delta_\bullet).$$

For simplicity, we write $S^p(X; G)$ for the group $\mathrm{Hom}_{\mathbb{Z}}(S_p(X), G)$.

The relation between singular homology and singular cohomology is established by the universal coefficient theorem, which is purely algebraic. A definition : a ring R is **left** (resp. **right**) **hereditary** if any R-submodule of a projective left (resp. ring) R-module is projective.

Theorem 6.2 (Universal coefficient theorem for cohomology)**.** Let R be a left hereditary ring, $\mathcal{C}^\bullet$ a complex of projective left R-modules, and M any left R-module. For every $n \in \mathbb{Z}$ there is an exact sequence

$$0 \longrightarrow \mathrm{Ext}^1_R(H_{n-1}(\mathcal{C}^\bullet), M) \longrightarrow H^n(\mathrm{Hom}_R(\mathcal{C}^\bullet, M)) \longrightarrow \mathrm{Hom}_R(H_n(\mathcal{C}^\bullet), M) \longrightarrow 0$$

natural in $\mathcal{C}^\bullet$ and M. Moreover, it splits by a homomorphism that is natural in M (but not in $\mathcal{C}^\bullet$).

There is also a universal coefficient theorem for homology, which states that the singular homology with coefficient in G is determined by that with coefficient in $\mathbb{Z}$.

Theorem 6.3 (Universal coefficient theorem for homology)**.** Let R be a right hereditary ring, $\mathcal{C}^\bullet$ a complex of projective right R-modules and M any left R-module. For every $n \in \mathbb{Z}$ there is an exact sequence

$$0 \longrightarrow H_n(\mathcal{C}^\bullet) \otimes_R M \longrightarrow H_n(\mathcal{C}^\bullet \otimes_R M) \longrightarrow \mathrm{Tor}^R_1(H_{n-1}(\mathcal{C}^\bullet), M) \longrightarrow 0$$

natural in $\mathcal{C}^\bullet$ and M. Moreover, it splits by a homomorphism that is natural in M (but not in $\mathcal{C}^\bullet$).

6.2 Relation to sheaf cohomology

Definition. Let X, Y be topological spaces.

1. Two continuous maps $f, g : X \to Y$ are called **homotopic** if there exists a continuous map $F : X \times [0,1] \to Y$ such that $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for any $x \in X$. Such an F is called a **homotopy from** f **to** g, and we write $f \cong g$.
2. Let A be a subspace of X. A continuous map $F : X \times [0,1] \to Y$ is called a **homotopy rel(ative to)** A if $F(a, t) = a$ for any $a \in A$, $t \in [0,1]$.
3. A continuous map $f : X \to Y$ is a **homotopy equivalence** if there exists another continuous map $g : Y \to X$ such that $f \circ g \cong \mathrm{id}_Y$ and $g \circ f \cong \mathrm{id}_X$. If such f exists, we say X and Y are **homotopic equivalent**, and write $X \cong Y$.

4. X is called **contractible** if $\mathrm{id}_X : X \to X$ is homotopic to a constant map $X \to X$. Equivalently, $X \cong \{*\}$ for some $* \in X$.
5. X is called **locally contractible** if every point in X admits a neighborhood basis consisting of contractible open sets.

Let $f : X \to Y$ be a continuous map. For a singular p-simplex $\sigma : \Delta_p \to X$, the composition $f_*\sigma = f \circ \sigma : \Delta_p \to Y$ is a singular p-simplex in Y. Extending by linearity, we obtain a homomorphism $f_* : S_p(X) \to S_p(Y)$. The map f_* is a chain map, i.e., $f_*\partial = \partial f_*$, so it induces maps on homology groups $f_* : H_p^{\mathrm{sing}}(X; G) \to H_p^{\mathrm{sing}}(X; G)$ and cohomology groups $f^* : H^p_{\mathrm{sing}}(Y; G) \to H^p_{\mathrm{sing}}(X; G)$.

Lemma 6.4. If $f, g : X \to Y$ are two continuous map homotopic to each other, then

$$f_* = g_* : H_p^{\mathrm{sing}}(X; G) \to H_p^{\mathrm{sing}}(Y; G).$$

In particular, if $f : X \to Y$ is an homotopy equivalence, $f_* : H_p^{\mathrm{sing}}(X; G) \to H_p^{\mathrm{sing}}(Y; G)$ is an isomorphism. Similarly, $f^* = g^* : H^p_{\mathrm{sing}}(Y; G) \to H^p_{\mathrm{sing}}(X; G)$.

Proof. Let $F : X \times [0, 1] \to Y$ be a homotopy from f to g. We are going to utilize F to show f_* is chain homotopic to g_* as maps $S_p(X) \to S_p(Y)$. Precisely, define $P : S_p(X) \to S_{p+1}(Y)$ by

$$P(\sigma) = \sum_{i=0}^{p} (-1)^i F \circ (\sigma \times \mathrm{id}_{[0,1]})|_{[v_0,\ldots,v_i,w_i,\ldots,w_p]}.$$

Here $v_0, \ldots, v_p$ (resp. $w_0, \ldots, w_p$) are the vertices of $\Delta^p \times \{0\}$ (resp. $\Delta^p \times \{1\}$) such that v_i and w_i have the same image under the projection $\Delta^p \times [0, 1] \to \Delta^p$. We claim

$$g_* - f_* = \partial P + P\partial.$$

□

The goal of this subsection is to prove the following theorem.

Theorem 6.5. Let X be a locally contractible topological space and R a ring. Then there is a canonical isomorphism

$$H^p_{\mathrm{sing}}(X; R) \cong H^p(X, \underline{R}_X)$$

6.3 Relative homology groups

Let G be an abelian group, X be a topological space and A be its subspace. For $n \in \mathbb{Z}$, define

$$S_n(X, A; G) := S_n(X; G)/S_n(A; G).$$

The boundary map $\partial_n : S_n(X; G) \to S_{n-1}(X; G)$ takes $S_n(A; G)$ and $S_{n-1}(A; G)$, so it induces a boundary map

$$\partial_n : S_n(X, A; G) \longrightarrow S_{n-1}(X, A; G)$$

on the quotient groups. The homology groups of this chain complex are called the **relative homology groups**, and are denoted as

$$H_\bullet^{\text{sing}}(X, A; G) := H_\bullet(S_\bullet(X, A; G), \partial_\bullet)$$

Similarly, put $S^n(X, A; G) = \text{Hom}_{\mathbb{Z}}(S_n(X, A), G)$ and define the **relative cohomology group**

$$H^\bullet_{\text{sing}}(X, A; G) := H^\bullet(S^\bullet(X, A; G), \delta^\bullet)$$

where $\delta^\bullet = \text{Hom}_{\mathbb{Z}}(\partial_\bullet, G)$.

By definition, we have an exact sequence of chain complexes

$$0 \to S_\bullet(A; G) \to S_\bullet(X; A) \to S_\bullet(X, A; G) \to 0$$

so it induces a long exact sequence on homology groups :

$$\cdots \longrightarrow H_n^{\text{sing}}(A; G) \longrightarrow H_n^{\text{sing}}(X; G) \longrightarrow H_n^{\text{sing}}(X, A; G) \longrightarrow H_{n-1}^{\text{sing}}(A, G) \longrightarrow \cdots.$$

For the cohomology, note that the exact sequence

$$0 \to S_n(A) \to S_n(X) \to S_n(X, A) \to 0$$

splits : $S_n(X, A)$ is naturally identified as the subgroup of $S_n(X)$ free over the singular n-simplices $\Delta_n \to X$ with image not contained in A. It follows that the dual sequence

$$0 \to S^n(X, A; G) \to S^n(X; G) \to S^n(A; G) \to 0$$

is exact, so it induces a long exact sequence of cohomology groups

$$\cdots \longrightarrow H^n_{\text{sing}}(X, A; G) \longrightarrow H^n_{\text{sing}}(X; G) \longrightarrow H^n_{\text{sing}}(A; G) \longrightarrow H^{n+1}_{\text{sing}}(X, A; G) \longrightarrow \cdots.$$

Definition.

1. A pair (X, A) consists of a space X and a subspace $A \subseteq X$.
2. An arrow $f : (X, A) \to (Y, B)$ is a continuous map $f : X \to Y$ such that $f(A) \subseteq B$.

A map $f : (X, A) \to (Y, B)$, as usual, induces maps on homology groups $f_* : H_n^{\text{sing}}(X, A; G) \to H_n^{\text{sing}}(Y, B; G)$ and cohomology groups $f^* : H^n_{\text{sing}}(Y, B; G) \to H^n_{\text{sing}}(X, A; G)$. The map f_* is natural in the sense that there exists a commutative diagram

$$\begin{tikzcd}
\cdots \arrow[r] & H_n^{\text{sing}}(A; G) \arrow[r] \arrow[d, "f_*"] & H_n^{\text{sing}}(X; G) \arrow[r] \arrow[d, "f_*"] & H_n^{\text{sing}}(X, A; G) \arrow[r] \arrow[d, "f_*"] & H_{n-1}^{\text{sing}}(A, G) \arrow[r] \arrow[d, "f_*"] & \cdots \\
\cdots \arrow[r] & H_n^{\text{sing}}(B; G) \arrow[r] & H_n^{\text{sing}}(Y; G) \arrow[r] & H_n^{\text{sing}}(Y, B; G) \arrow[r] & H_{n-1}^{\text{sing}}(B, G) \arrow[r] & \cdots.
\end{tikzcd}$$

The similar statement holds for f^*. Also, a careful look of the proof of Lemma 6.4 gives

Lemma 6.6. If $f, g : (X, A) \to (Y, B)$ are continuous map homotopic through maps of pairs $(X, A) \to (Y, B)$, then $f_* = g_* : H_n^{\text{sing}}(X, A; G) \to H_n^{\text{sing}}(Y, B; G)$ and $f^* = g^* : H^n_{\text{sing}}(Y, B; G) \to H^n_{\text{sing}}(X, A; G)$.

Theorem 6.7 (Excision)**.** Given a chain $Z \subseteq A \subseteq X$ of subspaces with $\overline{Z} \subseteq \operatorname{int} A$, the inclusion $(X \backslash Z, A \backslash Z) \to (X, A)$ induces an isomorphisms

$$H_n(X \backslash Z, A \backslash Z) \xrightarrow{\sim} H_n(X, A)$$

for all $n \in \mathbb{Z}$. Equivalently, for subspaces $A, B \subseteq X$ with $X = \operatorname{int} A \cup \operatorname{int} B$, the inclusion $(B, A \cap B) \to (X, A)$ induces isomorphisms $H_n(B, A \cap B) \to H_n(X, A)$ for all $n \in \mathbb{Z}$.

6.4 Cup product

Let X be a topological space and R be a ring. Here a ring is not necessarily commutative nor unital. Define the **cup product**

$$\begin{aligned} S^p(X; R) \times S^q(X; R) &\longrightarrow S^{p+q}(X; R) \\ (\varphi, \psi) &\longmapsto \varphi \smile \psi \end{aligned}$$

by

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma|_{[e_0, \ldots, e_p]})\psi(\sigma|_{[e_p, e_{p+1}, \ldots, e_{p+q}]}) \in R$$

It is associative in the obvious sense, and it is distributive. If R is unital, then the 0-cocycle defined by $\sigma \in S^0(X) \mapsto 1 \in R$ is the identity element for $\smile$.

Lemma 6.8. $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^p \varphi \smile \delta\psi$ for $\varphi \in S^p(X, R)$ and $\psi \in S^q(X, R)$.

Proof. This is a direct computation. For $\sigma \in S_{p+q+1}(X)$,

$$\begin{aligned}
\delta(\varphi \smile \psi))(\sigma) &= \sum_{i=0}^{p+q+1} (-1)^i (\varphi \smile \psi)(\sigma|_{[e_0, \ldots, \widehat{e_i}, \ldots, e_{p+q+1}]}) \\
&= \sum_{i=0}^{p} (-1)^i \varphi(\sigma|_{[e_0, \ldots, \widehat{e_i}, \ldots, e_{p+1}]})\psi(\sigma|_{[e_{p+1}, e_{p+2}, \ldots, e_{p+q+1}]}) \\
&\quad + \sum_{i=p+1}^{p+q+1} (-1)^i \varphi(\sigma|_{[e_0, \ldots, e_p]})\psi(\sigma|_{[e_p, e_{p+1}, \ldots, \widehat{e_i}, \ldots, e_{p+q+1}]}) \\
&= \sum_{i=0}^{p} (-1)^i \varphi(\sigma|_{[e_0, \ldots, \widehat{e_i}, \ldots, e_{p+1}]})\psi(\sigma|_{[e_{p+1}, e_{p+2}, \ldots, e_{p+q+1}]}) + (-1)^{p+1} \varphi(\sigma|_{[e_0, \ldots, e_p]})\psi(\sigma|_{[e_{p+1}, e_{p+2}, \ldots, e_{p+q+1}]}) \\
&\quad + (-1)^p \varphi(\sigma|_{[e_0, \ldots, e_p]})\psi(\sigma|_{[e_{p+1}, \ldots, e_{p+q+1}]}) + \sum_{i=p+1}^{p+q+1} (-1)^i \varphi(\sigma|_{[e_0, \ldots, e_p]})\psi(\sigma|_{[e_p, e_{p+1}, \ldots, \widehat{e_i}, \ldots, e_{p+q+1}]}) \\
&= (\delta\varphi \smile \psi)(\sigma) + (-1)^p (\varphi \smile \delta\psi)(\sigma)
\end{aligned}$$

□

From this lemma we see the cup product descends the level of cohomology groups :

$$H^p_{\text{sing}}(X;R) \times H^q_{\text{sing}}(X;R) \xrightarrow{\smile} H^{p+q}_{\text{sing}}(X;R).$$

If we form the direct sum

$$H^*_{\text{sing}}(X;R) := \bigoplus_{p \geqslant 0} H^p_{\text{sing}}(X;R),$$

the cup product then makes $H^*_{\text{sing}}(X;R)$ into an (associative and distributive) $\mathbb{Z}_{\geqslant 0}$-graded R-algebras (there is a canonical map $R \to S^0(X;R) \to H^0_{\text{sing}}(X;R)$). This is called the **singular cohomology ring**. It is unital as long as R is unital.

If $f : X \to Y$ is a continuous map, the induced map $f^* : H^*(Y;R) \to H^*(X;R)$ is a ring homomorphism. Indeed, for $\varphi \in S^p(Y;R)$ and $\psi \in S^q(Y;R)$,

$$\begin{aligned}(f^*\varphi \smile f^*\psi)(\sigma) &= f^*\varphi(\sigma|_{[e_0,\dots,e_p]})f^*\psi(\sigma|_{[e_p,\dots,e_{p+q}]}) \\ &= \varphi(f \circ \sigma|_{[e_0,\dots,e_p]})\psi(f \circ \sigma_{[e_p,\dots,e_{p+q}]}) = (\varphi \smile \psi)(f \circ \sigma) = f^*(\varphi \smile \psi)(\sigma).\end{aligned}$$

Theorem 6.9. If R is commutative, then $\alpha \smile \beta = (-1)^{pq}\beta \smile \alpha$ for all $\alpha \in H^p_{\text{sing}}(X;R)$ and $\beta \in H^q_{\text{sing}}(X;R)$.

Proof. For a singular n-simplex $\sigma : \Delta_n \to X$, define

$$\begin{aligned}\overline{\sigma} : \Delta_n &\xrightarrow{\text{linear}} \Delta_n \xrightarrow{\sigma} X \\ e_i &\longmapsto e_{n-i}\end{aligned}$$

and define

$$\begin{aligned}\rho : S_n(X) &\longrightarrow S_n(X) \\ \sigma &\longmapsto \varepsilon_n \overline{\sigma}\end{aligned}$$

where $\varepsilon_n := (-1)^{\frac{n(n+1)}{2}}$. We are going to show ρ is a chain map homotopic to the identity. This, together with the discussion preceding the theorem, at once shows the desired identity.

That ρ is a chain map follows from a direct computation and a trivial fact that $\varepsilon_n = (-1)^n \varepsilon_{n-1}$. To construct a chain homotopy from ρ to id, we use the notation in Lemma 6.4 and define $Q : S_n(X) \to S_{n+1}(X)$ by

$$Q(\sigma) = \sum_{i=0}^{n} (-1)^i \varepsilon_{n-i} (\sigma \circ \pi)|_{[v_0,\dots,v_i,w_n,\dots,w_i]}$$

where $\pi = \text{pr}_1 : \Delta_n \times [0,1] \to \Delta_n$ is the projection to the first component. We claim $\partial Q + Q\partial = \rho - \text{id}$. $\square$

6.5 smooth singular theory and deRham theorem

7 Curves

Let k be a field. Conceptually, a **curve** over k is an integral scheme of dimension 1 separated and of finite type over k. There are various definitions in the literature. We will note specify a definition for curves; instead, we will qualify our schemes appropriately in different context.

7.1 Compact Riemann surface

7.1 Definition. A **Riemann surface** X is a complex manifold of dimension 1. We denote by $\mathcal{O}_X$ (resp. C^∞_X) the sheaf of holomorphic (resp. smooth) functions on X. Note that C^∞_X is a flasque sheaf.

7.2 Various tangent bundles. Let M be a complex manifold. As a real manifold, we can form the **(real) tangent bundle** $T_{M,\mathbb{R}} \to M$. As M is a complex manifold, $T_{M,\mathbb{R}}$ admits a holomorphic structure. With this structure, we call it the **holomorphic tangent bundle** of M, and denote it by $T_M \to M$. Multiplication by $\sqrt{-1}$ on local coordinates of $T_{M,\mathbb{R}}$ gives a smooth bundle isomorphism $J : T_{M,\mathbb{R}} \to T_{M,\mathbb{R}}$ that satisfies $J^2 = -\operatorname{id}$. The function J is called the **complex structure** of M.

Put $T_{M,\mathbb{C}} = T_{M,\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to M$ to be the complexified tangent bundle, which also has a holomorphic structure inherited from T_M. The complex structure J induces a complex bundle isomorphism $J_{\mathbb{C}} : T_{M,\mathbb{C}} \to T_{M,\mathbb{C}}$, and on each fibre $J_{\mathbb{C}}$ has eigenvalues $\pm\sqrt{-1}$. This gives a global decomposition

$$T_{M,\mathbb{C}} = T^{1,0}_M \oplus T^{0,1}_M$$

The bundle homomorphism $\operatorname{Re} : T_{M,\mathbb{C}} \to T_{M,\mathbb{R}}$ defined by taking real part maps $T^{1,0}_M$ and $T^{0,1}_M$ isomorphically onto $T_{M,\mathbb{R}}$ as smooth vector bundles.

7.3 Differential form. Let M be a complex manifold, U an open subset of M and $k \geqslant 0$. The sheaf $\mathcal{A}^k_{M,\mathbb{C}}$ of smooth sections of $\bigwedge^k T^\vee_{M,\mathbb{C}} \to M$ is called the **sheaf of smooth k-forms on M with coefficient $\mathbb{C}$.** The decomposition in (7.2) induces a decomposition

$$\mathcal{A}^k_{M,\mathbb{C}} = \bigoplus_{p+q=k} \mathcal{A}^{p,q}_M$$

Explicitly, in a local chart U of M with local coordinates $z_1, \ldots, z_n$, $\mathcal{A}^{p,q}_M(U)$ is the subspace of $\mathcal{A}^k_M(U)$ generated by the k-forms of the form

$$f dz_{i_1} \wedge \cdots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$$

with $f : U \to \mathbb{C}$ a smooth function. An element in $\mathcal{A}^{p,q}_M(U)$ is called a (p,q)**-form**.

The sheaf Ω^k_M of holomorphic sections of $\bigwedge^k T_M \to M$ is called the **sheaf of holomorphic k-forms on** M. By definition, we see a holomorphic k-form can be viewed as a smooth $(k,0)$-form.

7.2 Serre duality

7.2.1 Traces and residues

In this subsubsection, we do not suppose an algebra is commutative. Still, we assume an algebra is unital.

7.4 Traces. Let k be a field and V a k-vector space. We say $\theta \in \operatorname{End}_k V$ is **finite potent** if $\dim_k \theta^n V < \infty$ for some integer $n \geqslant 0$. For a finite potent $\theta \in \operatorname{End}_k V$, a **trace** $\operatorname{tr}_V \theta \in k$ can be defined, having the following properties :

(i) If $\dim_k V < \infty$, then $\operatorname{tr}_V \theta$ is the usual trace.

(ii) If W is a subspace of V with $\theta W \subseteq W$, then

$$\operatorname{tr}_V \theta = \operatorname{tr}_W \theta + \operatorname{tr}_{V/W} \theta.$$

(iii) If θ is nilpotent, then $\operatorname{tr}_V \theta = 0$.

These properties also characterizes traces. In fact, if $\theta \in \operatorname{End}_k V$ is finite potent, there exists $n \gg 0$ such that the subspace $W := \theta^n V$ satisfies $\theta W \subseteq W$ and $\dim_k W < \infty$. Then

$$\operatorname{tr}_V \theta = \operatorname{tr}_W \theta + \operatorname{tr}_{V/W} \theta = \operatorname{tr}_W \theta$$

for θ in nilpotent on the quotient V/W. This also gives a well-defined definition of traces. There are some additional properties of traces.

(iv) If $F \subseteq \operatorname{End}_k V$ is a finite potent subspace (i.e. there exists some $n \geqslant 0$ such that $\dim_k \theta_1 \cdots \theta_n V < \infty$ for any $\theta_1, \ldots, \theta_n \in F$), then $\operatorname{tr}_V : F \to k$ is k-linear.

Indeed, we may assume F is finite dimensional, and compute the traces of all elements of F on the finite dimensional subspace

$$W = F^n V = \sum_{\sigma \in S_n} \theta_{\sigma(1)} \cdots \theta_{\sigma(n)} V$$

where $\{\theta_1, \ldots, \theta_n\}$ is a k-basis for F.

(v) If $\varphi : V' \to V$ and $\psi : V \to V'$ are k-linear with $\varphi\psi$ and $\psi\varphi$ finite potent, then

$$\operatorname{tr}_V(\varphi\psi) = \operatorname{tr}_{V'}(\psi\varphi).$$

Indeed, for $n \gg 0$ the maps φ and ψ induce mutually inverse isomorphisms between the subspace $W' = (\psi\varphi)^n V'$ and $W = (\varphi\psi)^n V$, under which the endomorphisms $(\psi\varphi)|_{W'}$ and $(\varphi\psi)|_W$ correspond.

7.5 Definition. Let V be a k-vector space and A, B be two k-subspaces of V.

1. A is **not much bigger** than B if $\dim_k (A + B)/B < \infty$, or equivalently, $A \subseteq B + W$ for some finite dimensional subspace $W \subseteq V$. In this case we write $A < B$.

2. A is **about the same size** as B if $A < B$ and $B < A$. In this case we write $A \sim B$.

It follows from definition at once that the relation $<$ is transitive and $\sim$ is an equivalence relation. Note that $<$ is preserved under $\operatorname{End}_k V$. Namely, if $A < B$, then $\varphi A < \varphi B$ for any $\varphi \in \operatorname{End}_k V$. Also, we have

$$\sum_{i=1}^{m} A_i < \bigcap_{j=1}^{n} B_j \text{ if and only if } A_i < B_j \text{ for all } i, j.$$

7.6 Let V be a k-vector space and fix a k-subspace A of V. Define

$$\begin{aligned} E &= \{\theta \in \operatorname{End}_k V \mid \theta A < A\} \\ E_1 &= \{\theta \in \operatorname{End}_k V \mid \theta V < A\} \\ E_2 &= \{\theta \in \operatorname{End}_k V \mid \theta A < 0\} = \{\theta \in \operatorname{End}_k V \mid \dim_k \operatorname{Im} \theta < \infty\} \\ E_0 &= E_1 \cap E_2 = \{\theta \in \operatorname{End}_k V \mid \theta V < A,\ \theta A < 0\}. \end{aligned}$$

Proposition.

(a) E is a k-subalgebra of $\operatorname{End}_k V$.

(b) The E_i's are two sided ideals in E, $E_1 + E_2 = E$ and E_0 is finite potent.

(c) E and the E_i's depend only on the equivalence class of A under $\sim$.

(d) If $(\varphi, \psi) \in E_0 \times E \cup E_1 \times E_2$, the commutator $[\varphi, \psi] = \varphi\psi - \psi\varphi$ lies in E_0 and has zero trace.

Proof. (a) is clear, and the first statement in (b) follows from the facts that $\psi V < V$ for any $\psi \in \operatorname{End}_k V$ and $<$ is transitive. That E_0 if finite potent is clear. Write $V = A \oplus A'$ for some k-subspace A' of V. If we denote by π and π' the projections to A and A', respectively, then $\mathrm{id}_V = \pi + \pi'$ with $\pi \in E_1$, $\pi' \in E_2$. This implies $E = E_1 + E_2$, so (b) is proved. (c) is clear. (d) follows from definition and (7.4).(v).

□

7.7 Definition-Theorem. Let K be a commutative k-algebra, V a K-module, and $A \subseteq V$ a k-subspace such that $fA < A$ for all $f \in K$. Let E and E_i be as in (7.6). The condition on A implies that the image $K \to \operatorname{End}_k V$ lies in E; for an element $f \in K$, we shall denote its image in E by the same letter f.

Then, there exists a unique k-linear map, the **residue map**

$$\operatorname{res}_A^V : \Omega_{K/k} \to k$$

such that for any pair $f, g \in K$, we have

$$\operatorname{res}_A^V(f\,dg) = \operatorname{tr}_V[f_1, g_1]$$

for $f_1, g_1 \in E$ such that

(a) $f \equiv f_1 \pmod{E_2}$, $g \equiv g_1 \pmod{E_2}$, and

(b) either $f_1 \in E_1$ or $g_1 \in E_1$.

Proof. Let $f, g \in K$. Since $E = E_1 + E_2$ (7.6), f_1 and g_1 that satisfy (a) and (b) always exist. Then $[f_1, g_1] \in E_1$ by (b), and $[f_1, g_1] \equiv [f, g] = 0 \pmod{E_2}$ by (b) and the commutativity of K. Hence $[f_1, g_1] \in E_1 \cap E_2 = E_0$, so the trace $\mathrm{tr}_V[f_1, g_1] \in k$ is defined. By (7.6).(d), the value of $\mathrm{tr}_V[f_1, g_1]$ is unaltered if f_1 or g_1 is changed by an element in E_2, provided that the other is in E_1. Moreover, by (7.4).(iv), $(f, g) \mapsto \mathrm{tr}_V[f_1, g_1]$ is a k-bilinear map on K, so it gives rise to a k-linear map

$$\begin{aligned} r : K \otimes_k K &\longrightarrow k \\ f \otimes g &\longmapsto \mathrm{tr}_V[f_1, g_1]. \end{aligned}$$

Recall that by the very definition of $\Omega_{K/k}$ there is a surjective k-linear map

$$\begin{aligned} c : K \otimes_k K &\longrightarrow \Omega_{K/k} \\ f \otimes g &\longmapsto f dg \end{aligned}$$

such that $\ker c$ is generated over k by elements of the form $f \otimes gh - fg \otimes h - fh \otimes g$. By surjectivity of c, the residue res_A^V, if it exists, is the only map r' such that $r' \circ c = r$. Such r' exists if and only if r vanishes on $\ker c$. This is the case, for if $f, g, h \in K$, we can choose suitable $f_1, g_1, h_1 \in E_1$ and let $(fg)_1 = f_1 g_1$, $(gh)_1 = g_1 h_1$, $(fh)_1 = f_1 h_1$. Now the identity

$$[f_1, g_1 h_1] = [f_1 g_1, h_1] + [f_1 h_1, g_1]$$

implies that r vanishes on $\ker c$, proving the existence and the uniqueness res_A^V. □

7.8 Retain the notation in (7.7). For $f, g \in K$, let

$$\begin{aligned} B &= A + gA \\ C &= B \cap f^{-1}(A) \cap (fg)^{-1}(A) = \{v \in B \mid fv \in A,\ fgv \in A\}. \end{aligned}$$

Then $\dim_k B/C < \infty$, and

$$\mathrm{res}_A^V(f dg) = \mathrm{tr}_{B/C}[\pi f, g],$$

where $\pi : V \to A$ is a k-linear projection.

7.9 Retain the notation in (7.7). We list some properties for the residue map res_A^V.

(R1) If V' is a K-submodule of V with $V \supseteq V' \supseteq A$, then $\mathrm{res}_A^V = \mathrm{res}_A^{V'}$. For this, we usually suppress the superscript V and simply write res_A.

(R2) If $fA + fgA + fg^2A \subseteq A$, then $\mathrm{res}_A(f dg) = 0$. In particular, this is so if $fA,\ gA \subseteq A$. Thus res_A is identically zero if A is a K-submodule of V.

(R3) For $g \in K$, $\mathrm{res}_A(g^n dg) = 0$ for all integers $n \geqslant 0$, and if g is invertible in K, then the same holds for $n \leqslant -2$. In particular, $\mathrm{res}_A(dg) = 0$ for all $g \in K$.

(R4) If $g \in K^\times$ and $h \in K$ such that $hA \subseteq A$, then

$$\mathrm{res}_A(hg^{-1}dg) = \mathrm{tr}_{A/(A\cap gA)}(h) - \mathrm{tr}_{gA/(A\cap gA)}(h).$$

In particular, $\mathrm{res}_A(g^{-1}dg) = \dim_k A/gA$.

(R5) Suppose $B \subseteq V$ is another k-subspace such that $fB < B$ for all $f \in K$. Then

$$f(A+B) < A+B \text{ and } f(A\cap B) < A\cap B \text{ for all } f \in K$$

and $\mathrm{res}_A + \mathrm{res}_B = \mathrm{res}_{A+B} + \mathrm{res}_{A\cap B}$.

(R6) Let K' be a commutative K-algebra which is a free K-module of finite rank. Let $V' = K' \otimes_K V$ and let $A' = \sum_i x_i \otimes A \subseteq V'$, where $\{x_i\}$ is a K-basis for K'. Then $f'A' < A'$ for all $f' \in K'$, the $\sim$-equivalence class of A' depends only on that of A, not on the choice of basis $\{x_i\}$, and we have

$$\mathrm{res}_{A'}(f'dg) = \mathrm{res}_A((\mathrm{tr}_{K'/K} f)dg)$$

for $f' \in K'$ and $g \in K$.

7.2.2 Residues on algebraic curves

8 Affine Group Schemes

8.1 Group schemes. Let S be a scheme. An S-scheme $f : G \to S$ is called a **group scheme** over S if there are three S-morphisms

$$\mathrm{mult} = \mathrm{mult}_G : G \times_S G \to G, \qquad \mathrm{inv} = \mathrm{inv}_G : G \to G, \qquad e = e_G : S \to G$$

satisfying the following commutative diagrams :

(i) Associativity :

$$\begin{tikzcd}
G \times_S (G \times_S G) \arrow[r, "\mathrm{id}_G \times \mathrm{mult}"] \arrow[d, "\mathrm{canonical}"', "\wr"] & G \times_S G \arrow[dr, "\mathrm{mult}"] & \\
 & & G \\
(G \times_S G) \times_S G \arrow[r, "\mathrm{mult} \times \mathrm{id}_G"'] & G \times_S G \arrow[ur, "\mathrm{mult}"'] &
\end{tikzcd}$$

(ii) Left and right identity :

$$\begin{tikzcd}
G \times_S S \arrow[r, "\mathrm{id}_G \times e"] & G \times_S G \arrow[dr, "\mathrm{mult}"] & \\
G \arrow[u, "\wr"] \arrow[d, "\wr"'] \arrow[rr, "\mathrm{id}_G"] & & G \\
S \times_S G \arrow[r, "e \times \mathrm{id}_G"'] & G \times_S G \arrow[ur, "\mathrm{mult}"'] &
\end{tikzcd}$$

(iii) Left and right inverse :

$$\begin{tikzcd}
 & G \times_S G \arrow[r, "\mathrm{id}_G \times \mathrm{inv}"] & G \times_S G \arrow[dr, "\mathrm{mult}"] & \\
G \arrow[ur, "\Delta"] \arrow[rr, "f"] \arrow[dr, "\Delta"'] & & S \arrow[r, "e"] & G \\
 & G \times_S G \arrow[r, "\mathrm{inv} \times \mathrm{id}_G"] & G \times_S G \arrow[ur, "\mathrm{mult}"'] &
\end{tikzcd}$$

A morphism between two group scheme over S is an S-morphism $\phi : G \to G'$ such that

$$\phi \circ \mathrm{mult}_G = \mathrm{mult}_{G'} \circ (\phi \times \phi).$$

Denote by $\mathbf{GpSch}_S$ the category of group schemes over S whose morphisms are the one defined above. If $S = \operatorname{Spec} A$ is affine, we simply write $\mathbf{GpSch}_A = \mathbf{GpSch}_{\operatorname{Spec} A}$, and $\mathbf{GpSch} = \mathbf{GpSch}_{\mathbb{Z}}$.

8.2 Functor of points. By Yoneda's lemma, the three S-morphisms $\mathrm{mult} : G \times_S G \to G$, $\mathrm{inv} : G \to G$, $e : S \to G$ and their compatibility are completely determined by the functorial maps

$$\begin{aligned} \mathrm{mult}(T) &: G(T) \times_{S(T)} G(T) = G(T) \times G(T) \to G(T) \\ \mathrm{inv}(T) &: G(T) \to G(T) \\ e(T) &: S(T) = \{g\} \to G(T) \end{aligned}$$

and the corresponding compatibility, where $T \in \mathbf{Sch}_S$ with structure map g. From the compatibility diagrams we see these define a group structure on the set $G(T)$ of T-points of $G_{/S}$. This suggests that a group scheme $G_{/S}$ is exactly the S-scheme such that its functor of points $h_G : \mathbf{Sch}_S \to \mathbf{Set}$ factors through the category of groups : we have a commutative triangle

$$\begin{tikzcd} \mathbf{Sch}_S \arrow[r, "h_G"] \arrow[dr, "h_G"'] & \mathbf{Set} \\ & \mathbf{Gp} \arrow[u, "\text{forgetful functor}"'] \end{tikzcd}$$

8.3 Let k be a ring. Recall in (1.92) we have an anti-equivalence

$$\begin{aligned} \mathrm{Spec} : \mathbf{Alg}_k &\longrightarrow \mathbf{AffSch}_k \\ A &\longmapsto \mathrm{Spec}\, A \end{aligned} .$$

If $G = \mathrm{Spec}\, A$ is a group ring, the multiplication, identity and inversion are transformed to k-homomorphisms :

$$\Delta : A \to A \otimes_k A, \qquad \iota : A \to A, \qquad \varepsilon : A \to k$$

These are called the **comultiplication**, **antipode**, and **counit** on A, respectively.

9 Convex Geometry

9.1 Convex polyhedral cones

In this subsection, let V be a finite dimensional real vector space with dual space $V^\vee$. Then the evaluation $V^\vee \times V \xrightarrow{\langle , \rangle} \mathbb{R}$ is a perfect pairing. We identify $(V^\vee)^\vee$ with V naturally.

Definition.

1. A **convex polyhedral cone** in V is a set of the form

$$\sigma = \mathrm{cone}(S) := \left\{ \sum_{u \in S} a_u u \mid a_u \geqslant 0 \right\} \subseteq V$$

for some finite subset $S \subseteq V$. We also say σ is the **cone generated by** S. By convention, $\mathrm{cone}(\varnothing) = \{0\}$.

2. The **dimension** of a convex polyhedral cone σ, denoted by $\dim \sigma$, is the dimension of the linear span of σ over $\mathbb{R}$; in other words, $\dim \sigma = \dim_{\mathbb{R}}(\sigma + (-\sigma))$.
3. The **dual** $\sigma^\vee$ of some set $\sigma \subseteq V$ is

$$\sigma^\vee := \{u \in V^\vee \mid \langle u, v \rangle \geqslant 0 \text{ for all } v \in \sigma\}.$$

Also, we put

$$\sigma^\perp = \{u \in V^\vee \mid \langle u, v \rangle = 0 \text{ for all } v \in \sigma\}.$$

- If $\sigma = \mathrm{cone}(S)$ for some finite subset $S \subseteq V$, then

$$\sigma^\vee = \bigcap_{v \in S} \{u \in V^\vee \mid \langle u, v \rangle \geqslant 0\}.$$

Lemma 9.1 (Farkas'). If σ is a convex polyhedral cone and $v_0 \notin \sigma$, then there exists $u_0 \in \sigma^\vee$ such that $\langle u_0, v_0 \rangle < 0$. In particular, this implies $(\sigma^\vee)^\vee = \sigma$.

Proof. By choosing a basis for V, we do not distinguish V, $V^\vee$ and $\mathbb{R}^n$, and under this assumption, $\langle , \rangle$ becomes the standard inner product. Let B be the closed ball centered at 0 with radius $\|v_0\|$. Then $\sigma \cap B$ is nonempty and compact, so the function $w \mapsto \|w - v_0\|$ attains its minimal at some $w_0 \in \sigma \cap B \subseteq \sigma$. Put $u_0 = w_0 - v_0$. We claim $u_0 \in \sigma^\vee$ and $\langle u_0, v_0 \rangle < 0$.

- Let $w \in \sigma$. Then $w_0 + tw \in \sigma$ for all $t \geqslant 0$, and

$$\|u_0\|^2 = \|w_0 - v_0\|^2 \leqslant \|w_0 + tw - v_0\|^2 = \|u_0 + tw\|^2 = \|u_0\|^2 + 2t\langle u_0, w \rangle + t^2 \|w\|^2.$$

This implies $\langle u_0, w \rangle \geqslant 0$, and since $w \in \sigma$ is arbitrary, we see $u_0 \in \sigma^\vee$.

- Since $\langle u_0, v_0\rangle = \langle w_0, v_0\rangle - \|v_0\|^2$, if $w_0 = 0$, then we are done. Suppose now $w_0 \neq 0$; then $u_0 \neq 0$ as well. Consider

$$\|tw_0 - v_0\|^2 = t^2 \|w_0\| - 2t\langle w_0, v_0\rangle + \|v_0\|^2 .$$

  By our choice of w_0, the quadratic function on the right minimizes at $t = 1$. This implies $\|w_0\|^2 = \langle w_0, v_0\rangle$, and hence

$$0 < \|u_0\|^2 = \|w_0\|^2 - 2\langle w_0, v_0\rangle + \|v_0\|^2 = -\langle w_0, v_0\rangle + \|v_0\|^2 = -\langle u_0, v_0\rangle.$$

□

Definition. Let σ be a convex polyhedral cone in V.

1. A **supporting hyperplane** of σ is the hyperplane in V determined by some vector in $\sigma^\vee$.
2. A **face** of σ is the intersection of σ with some support hyperplane. In particular, σ is a face of itself. The faces of σ other than itself are called **proper faces**.

— For $u \in V^\vee$, we put

$$u^\perp := \{v \in V \in \langle u, v\rangle = 0\}.$$

  Then a supporting hyperplane is of the form $u^\perp$ for some $u \in \sigma^\vee$.

— If τ is a face of σ, we write $\tau \preceq \sigma$. We call $\preceq$ the **face relation**.

Lemma 9.2. Let σ be a convex polyhedral cone in V.

1. Any linear subspace of σ is contained in every face of σ.
2. Every face of σ is a convex polyhedral cone.
3. An intersection of two faces of σ is a face of σ.
4. A face of a face of σ is a face of σ.

Proof.

1. If $v \in \sigma$ is such that $-v \in \sigma$, then for any $u \in \sigma^\vee$, we have $\langle u, v\rangle \geqslant 0 \leqslant \langle u, -v\rangle$, proving that $\langle u, v\rangle = 0$.
2. In fact, if $\sigma = \mathrm{cone}(S)$ for some finite subset $S \subseteq V$ and $\tau = \sigma \cap u^\perp$ for some $u \in \sigma^\vee$, then $\tau = \mathrm{cone}(S \cap u^\perp)$. Clearly, we have $\mathrm{cone}(S \cap u^\perp) \subseteq \tau$. Conversely, if $v \in \tau$, write $v = \sum_{s \in S} a_s s$ with $a_s \geqslant 0$. By the following lemma, we see $a_s s \in \tau$ for each s, so $s \in S \cap \tau = S \cap u^\perp$, implying $v \in \mathrm{cone}(S \cap u^\perp)$.

Lemma 9.3. Let $\tau \preceq \sigma$. If $v, w \in \sigma$ are such that $v + w \in \tau$, then $v, w \in \tau$.

Proof. Say τ is cut off by the hyperplane determined by $u \in \sigma^\vee$. Then $\langle u, v\rangle \geqslant 0 \leqslant \langle u, w\rangle$. By assumption we have $\langle u, v + w\rangle = 0$, so this forces that $\langle u, v\rangle = \langle u, w\rangle = 0$. This shows $v, w \in \tau$. □

3. Let $u_1, u_2 \in \sigma^\vee$. Then

$$(\sigma \cap u_1^\perp) \cap (\sigma \cap u_2^\perp) = \sigma \cap (u_1 + u_2)^\perp.$$

This identity is proved as before, and clearly implies 3.

4. Let $\tau = \sigma \cap u^\perp$ for some $u \in \sigma^\vee$ and $\gamma = \tau \cap v^\perp$ for some $v \in \tau^\vee$. Consider the vector $v + tu$; since σ has a finite set of generators, we see $v + tu \in \sigma^\vee$ for $t \gg 0$. Indeed, if $v \in S$, we have either $\langle u, v\rangle > 0$, or $\langle u, v\rangle = 0$ and $\langle u', v\rangle \geqslant 0$. Thus

$$\tau \cap (v + tu)^\perp = (\tau \cap v^\perp) \cap (\tau \cap u^\perp) = \gamma \cap \tau = \gamma.$$

□

The converse of Lemma 9.3 holds in the following sense. Note that this lemma is not used in this section.

Lemma 9.4. Let σ be a convex polyhedral cone in V and $\tau \subseteq \sigma$ a convex cone such that if $v, w \in \sigma$ and $v + w \in \tau$, then $v, w \in \tau$. Then τ is a face of σ.

Proof. Replacing V by $\operatorname{span}_{\mathbb{R}} \sigma$, we may assume σ has maximal dimension. Further, by replacing σ by a face of which, we may assume τ is not contained in any proper face of σ. Under these assumptions and the condition of the lemma, we prove $\tau = \sigma$.

By Corollary 9.8, $\sigma^\vee$ is a convex polyhedral cone, so we may write $\sigma^\vee = \mathrm{cone}(u_1, \ldots, u_s)$ for some minimal generating set $\{u_1, \ldots, u_s\}$ of $\sigma^\vee$. By Lemma 9.1, $\sigma = \mathrm{cone}(u_1, \ldots, u_s)^\vee$. Fix an integer $1 \leqslant i \leqslant s$. If $\langle u_i, \tau\rangle = 0$, then $\tau \subseteq \sigma \cap (u_i)^\perp$, contradicting to our assumption that τ is not contained in any proper face. Thus we can find $v_i \in \tau$ such that $\langle u_i, v_i\rangle > 0$. If we take $v_0 = v_1 + \cdots + v_s$, we see $\langle u_i, v_0\rangle > 0$ for any i, so that $v_0 \in \operatorname{int}(\sigma) \cap \tau$.

If $\tau \subsetneq \sigma$, then for $v \in \sigma \backslash \tau$, consider the vector $tv_0 - v$; if $t \gg 0$, then $\langle u_i, tv_0 - v\rangle \geqslant 0$ for any $1 \leqslant i \leqslant s$, so that $tv_0 - v \in (\sigma^\vee)^\vee = \sigma$. This implies $tv_0 = (tv_0 - v) + v \in \tau$ with $tv_0 - v$, $v \in \sigma$, and our condition implies that $tv_0 - v$, $v \in \sigma$, a contradiction.

□

Definition. Let σ be a convex polyhedral cone in V.

1. A **facet** of σ is a face τ of codimension one, namely, $\dim \tau = \dim \sigma - 1$.
2. An **edge** of σ is a face of dimension one.

Lemma 9.5. Any proper face of a convex polyhedral cone σ in V is contained in some facet.

Proof. Put $\sigma = \mathrm{cone}(S)$ for some finite subset S of V. Let $u \in \sigma^\vee$ be such that $\tau = \sigma \cap u^\perp$ has codimension > 1 in σ. We saw that $\tau = \mathrm{cone}(S \cap u^\perp)$. By replacing V with $\sigma + (-\sigma)$, we may assume V is spanned by σ. Also put $W = \tau + (-\tau)$. The image of S in V/W is contained in the closed half-space determined by u. Since $\dim_{\mathbb{R}} V/W \geqslant 2$, if we rotate (in V/W) this half-space about the origin, we can obtain a closed half space containing $0 + W$ and at least one nonzero image of S in V/W. In other words, we can find $u_0 \in \sigma^\vee$ such that $W \subsetneq \sigma \cap u_0^\perp$. □

Corollary 9.5.1.

1. Any face of codimension 2 is an intersection of two facets.
2. Any proper face is the intersection of all facets containing it.

Proof. Let σ be a convex polyhedral cone.

1. Retain the notation in the proof of the last lemma. Under our assumption, we have $\dim_{\mathbb{R}} V/W = 2$, i.e., V/W is a plane. This means the image of σ in W has two supporting lines, and the result is clear.
2. Let τ be a proper face of σ. If the codimension of τ in σ is > 1, then by the last lemma, τ is contained in the facet γ of σ. Since $\sigma^\vee \subseteq \gamma^\vee$, we may regard τ as a face of γ, so by induction we see τ is the intersection of all facets of γ containing τ. But each facet of γ, by 1., is an intersection of two facets in σ, so τ is an intersection of facets.

□

Lemma 9.6. Let σ be a convex polyhedral cone in V such that $V = \sigma + (-\sigma)$. Then $\partial\sigma$ is the union of all facets of σ.

Proof. A facet is an intersection of σ with some hyperplane determined by some vector in $\sigma^\vee$, so any neighborhood of a point p in a facet intersects nontrivially with the complement of σ. Since σ has nonempty interior and is convex, the segment connecting p and any interior point entirely lies in σ. This shows any neighborhood of p meets the interior of σ. This shows $p \in \partial\sigma$.

Conversely, let $p \in \partial\sigma$ and let (w_n) be a sequence outside σ such that $w_n \to p$. By Lemma 9.1, we can find $u_n \in \sigma^\vee$ such that $\langle u_n, w_n\rangle < 0$. Rescaling, we assume $\|u_n\| = 1$. Since V is finite dimensional, the unit sphere is compact; by passing the convergent subsequence, we may assume u_n converges to some point $q \in \sigma^\vee$. Then $0 \leqslant \langle q, p\rangle \leqslant 0$, so $p \in \sigma \cap q^\perp$. □

Lemma 9.7. If σ is a convex polyhedral cone spanning the whole vector space V with $\sigma \neq V$, then

$$\sigma = \bigcap_\tau \{v \in V \mid \langle u_\tau, v\rangle \geqslant 0\}$$

where τ runs over all facets of σ and u_τ is the unique vector (up to multiplication by a positive scalar) such that $\tau = \sigma \cap u_\tau^\perp$.

Proof. By definition we always have the containment $\subseteq$. If v were in the intersection but not in σ, take any vector v' in the interior of σ and denote by w the last point in σ on the segment from v' to v. Then $w \in \partial\sigma$, so w lies in some facet τ of σ by Lemma 9.6. We have $\langle u_\tau, v'\rangle > 0$ and $\langle u_\tau, w\rangle = 0$, so $\langle u_\tau, v\rangle = 0$, a contradiction. □

Lemma 9.8. The dual of a convex polyhedral cone σ is a convex polyhedral cone.

Proof. If σ spans V, then $\sigma^\vee$ is generated by the u_τ in Lemma 9.8. Indeed, if $u \in \sigma^\vee \backslash \mathrm{cone}\{u_\tau\}_\tau$, by Lemma 9.1, we can find $v \in V$ with $\langle u_\tau, v\rangle \geqslant 0$ for all facets τ and $\langle u, v\rangle < 0$; the former implies $v \in \sigma$ by the last lemma, and the latter leads to a contradiction.

If σ spans a smaller space W, then $\sigma^\vee$ is generated by lifts of generators of its image in $V^\vee/W^\perp$ along with vectors u and $-u$, where u runs over a basis of $W^\perp$.

□

Corollary 9.8.1. A subset σ in V is a convex polyhedral cone if and only if it is a finite intersection of closed half spaces in V.

Proof. The only if part follows from the last lemma and the fact $\sigma^\vee = \bigcap_{v \in S} \{u \in V^\vee \mid \langle u, v\rangle \geqslant 0\}$. For the if part, say $\sigma = u_1^\vee \cap \cdots \cap u_\ell^\vee$ with $u_i \in V^\vee$. Put $\gamma = \mathrm{cone}\{u_1, \ldots, u_\ell\} \subseteq V^\vee$; then $\gamma^\vee = \sigma$. If $\gamma \subsetneq \sigma^\vee$, say $u \in \sigma^\vee \backslash \gamma$, then by Lemma 9.1 we can find $v \in \gamma^\vee = \sigma$ such that $\langle u, v\rangle < 0$, a contradiction to the facts $u \in \sigma^\vee$ and $v \in \sigma$. Hence $\sigma^\vee = \gamma$ is a convex polyhedral cone, so $\sigma = \gamma^\vee$ is also a convex polyhedral cone by the last lemma.

□

Definition.

1. A **lattice** in V is a $\mathbb{Z}$-submodule of V of rank $\dim_{\mathbb{R}} V$.
2. A convex polyhedral cone σ in V is N**-rational** if N is a lattice in V and $\sigma = \mathrm{cone}(S)$ for some finite subset $S \subseteq N$.

- Suppose N and M are lattices of V and $V^\vee$, respectively, so that N and M are dual to each other under the pairing $\langle\, ,\, \rangle$. Then
  (i) Any face of σ is N-rational. For if $\tau = \sigma \cap u^\perp$ and $\sigma = \mathrm{cone}(S)$ for some $S \subseteq N$, then $\tau = \mathrm{cone}(S \cap u^\perp)$.
  (ii) If σ is N-rational, then $\sigma^\vee$ is M-rational. This can be seen from the 1. and the proof of Lemma 9.8.

Lemma 9.9 (Gordan's)**.** Suppose N and M are lattices of V and $V^\vee$, respectively, so that N and M are dual to each other under the pairing $\langle\, ,\, \rangle$. If σ is a N-rational convex polyhedral cone, then

$$S_\sigma := \sigma^\vee \cap M$$

is a finitely generated semigroup.

Proof. Say $\sigma^\vee = \mathrm{cone}(S)$ for some $S \subseteq M$. Let $K = \{\sum_{u \in S} a_u u \mid 0 \leqslant a_u \leqslant 1 \text{ for all } u \in S\}$ be the fundamental parallelotope. Since K is compact and M is discrete, $K \cap M$ is finite. Then $K \cap M$ generates $\sigma^\vee \cap M$. Indeed, if $v \in \sigma^\vee \cap M$, write

$$v = \sum_{u \in S} a_u u = \sum_{u \in S} \lfloor a_u \rfloor u + \sum_{u \in S} \{a_u\} u$$

One sees that the latter two terms lie in $K \cap M$. □

Definition. The **relative interior** $\mathrm{relint}(\sigma)$ of a convex polyhedral cone σ is the interior of σ in the topological space $\mathrm{span}_{\mathbb{R}}\, \sigma$.

- Say $\sigma = \mathrm{cone}(S)$, and let $T \subseteq S$ be the independent subset of size $\dim \sigma$. Then every positive linear combination of vectors in T lies in $\mathrm{relint}(\sigma)$. In particular, if $S \neq \varnothing$, then $\mathrm{relint}(\sigma) \neq \varnothing$. Also, if σ is N-rational, then $\mathrm{relint}(\sigma) \cap N \neq \varnothing$.
- We have $v \in \mathrm{relint}(\sigma)$ if and only if $\langle u, v\rangle > 0$ for all $u \in \sigma^\vee \backslash \sigma^\perp$. To see this, first consider the case $V = \mathrm{span}_{\mathbb{R}}\sigma$. Then $\mathrm{relint}(\sigma)$ is just the interior of σ in V. By Lemma 9.6 we have $v \in \mathrm{relint}(\sigma)$ if and only if $\langle u_\tau, v\rangle > 0$ for all facets τ of σ. Each element in $\sigma^\vee \backslash \sigma^\perp$ gives a proper face of σ, and each proper face is contained in some facet. Now the result follows.

  If σ spans a smaller space W, then the preceding paragraph shows that

  $$\mathrm{relint}(\sigma) = \bigcap_{u \in W^\vee} \{v \in \sigma \mid \langle u, v\rangle > 0\}$$

  But $W^\vee \cong V^\vee / W^\perp = V^\vee/\sigma^\perp$ in a way that $\langle\, ,\, \rangle$ is preserved, the result follows.
- Every vector in σ is contained in the relative interior of some face of σ. This can be seen by induction on dimensions of cones, by virtue of Lemma 9.6.

Lemma 9.10. Let σ be a convex polyhedral cone in V.

1. If τ is a face of σ, then $\tau^* := \sigma^\vee \cap \tau^\perp$ is a face of $\sigma^\vee$ with $\dim \tau + \dim \tau^* = \dim_{\mathbb{R}} V$.
2. There is an inclusion-reversing bijection

   $$\begin{aligned} \{\text{faces of } \sigma\} &\longrightarrow \{\text{faces of } \sigma^\vee\} \\ \tau &\longmapsto \tau^* \end{aligned}$$

3. The smallest face of σ is $\sigma \cap (-\sigma)$.

Proof. First note that every face of $\sigma^\vee$ has the form $\sigma^\vee \cap v^\perp$ for some $v \in (\sigma^\vee)^\vee = \sigma$. If v is contained in the relative interior of some face τ, then

$$\sigma^\vee \cap v^\perp = \sigma^\vee \cap \tau^\vee \cap v^\perp = \sigma^\vee \cap \tau^\perp.$$

The first equality results from $\sigma^\vee \subseteq \tau^\vee$, and the second follows from $\langle u, v\rangle > 0$ for any $u \in \tau^\vee \backslash \tau^\perp$. This shows the map 2. is surjective. The map is clearly inclusion-reversing, and it follows formally from $\tau \subseteq (\tau^*)^*$ that $\tau^* = ((\tau^*)^*)^*$, so the map is bijective. In particular, the smallest face is

$$(\sigma^\vee)^\vee \cap (\sigma^\vee)^\perp = (\sigma^\vee)^\perp = \sigma \cap (-\sigma)$$

and $\dim \sigma \cap (-\sigma) + \dim \sigma^\vee = \dim V$ (for $\langle\, ,\, \rangle$ is nondegenerate). In general, if τ is a proper face, find a sequence

$$\sigma \cap (-\sigma) = \tau_0 \subsetneq \tau_1 \subsetneq \cdots \subsetneq \tau_\ell = \tau \subsetneq \cdots \subsetneq \tau_m \subsetneq \sigma$$

of faces of σ such that τ_{i-1} has codimension one in τ_i and τ_m is a facet. Taking * and computing dimensions gives $\dim \tau + \dim \tau^* = \dim_{\mathbb{R}} V$. $\square$

Lemma 9.11. Let σ be a convex polyhedral cone in V. TFAE :

(a) $\{0\}$ is a face of σ.

(b) σ contains no positive dimensional subspace of V.

(c) $\sigma \cap (-\sigma) = \{0\}$.

(d) $\dim \sigma^\vee = \dim_{\mathbb{R}} V$.

If either condition holds, we say σ is **strongly convex**.

Proof. We saw in Lemma 9.2.1. that any linear subspace of σ is contained in every face of σ. Since $\sigma \cap (-\sigma)$ is the smallest face of σ, we see (b)$\Leftrightarrow$(c). Also, Lemma 9.10 shows (c)$\Leftrightarrow$(d). If $\{0\}$ is a face, it must be the smallest one, so (a)$\Leftrightarrow$(c). $\square$

Corollary 9.11.1. If σ is a strongly convex polyhedral cone, then σ is generated by its edges.

Proof. For an edge ρ of σ, by Lemma 9.11.(b), we see ρ is a ray, i.e., $\rho = \mathbb{R}_{\geqslant 0} e_\rho$ for some vector $e_\rho \in \sigma$. By saying σ is generated by its edges we actually mean that $\sigma = \mathrm{cone}\{e_\rho\}$, where ρ runs over all edges of σ.

Say $\sigma = \mathrm{cone}(S)$ for some finite subset $S \subseteq V$. We assume that S is minimal among all generating sets of σ. We claim that $\{\mathbb{R}_{\geqslant 0} v \mid v \in S\}$ is the collection of all edges of σ.

— Let $v \in S$. Applying Lemma 9.1 to $\mathrm{cone}(S \backslash \{v\})$, we obtain a vector $u \in \mathrm{cone}(S \backslash \{v\})^\vee$ with $\langle u, v \rangle < 0$. Take $u' \in \mathrm{relint}(\sigma^\vee)$; then $\langle u', v' \rangle > 0$ for any $v' \in (\sigma^\vee)^\vee \backslash (\sigma^\vee)^\perp = \sigma \backslash \{0\}$. Let $\lambda > 0$ be the unique number such that $\langle u + \lambda u', v \rangle = 0$. Then $u + \lambda u' \in \sigma^\perp$, and

$$\sigma \cap (u + \lambda u')^\perp = \mathbb{R}_{\geqslant 0} v$$

for $\langle u + \lambda u', v' \rangle > 0$ for any $v' \in S \backslash \{v\}$. This shows $\mathbb{R}_{\geqslant 0} v$ is an edge.

— Let $\tau = \sigma \cap u^\perp$ be an edge of σ. Then $\tau = \mathrm{cone}(S \cap u^\perp)$ is one dimensional, so $S \cap u^\perp$ must be a singleton $\{v\}$, and $\tau = \mathbb{R}_{\geqslant 0} v$ by Lemma 9.11.(b).

$\square$

Remark. Let σ be a strongly convex N-rational polyhedral cone. If ρ is an edge of σ, then ρ is a ray. Since ρ is also N-rational (and N is discrete), the semigroup $\rho \cap N$ is generated by a unique element $u_\rho \in \rho \cap N$. This vector u_ρ is called the **ray generator** of ρ. Thus Corollary 9.11.1 implies that σ is generated by the ray generators of its edges.

Lemma 9.12. Let σ be a convex polyhedral cone. If $\tau = \sigma \cap u^\perp$ for some $u \in \sigma^\vee$, then $\tau^\vee = \sigma^\vee + \mathbb{R}_{\geqslant 0}(-u)$.

Proof. Both sides of the identity being convex polyhedral cones, we only need to show their duals are equal. But

$$(\sigma^\vee + \mathbb{R}_{\geqslant 0}(-u))^\vee = (\sigma^\vee)^\vee \cap (-u)^\vee = \sigma \cap (-u)^\vee = \sigma \cap u^\perp = \tau,$$

we are done. $\square$

Lemma 9.13 (Separation lemma)**.** If σ and σ' are convex polyhedral cones whose intersection is a face τ of each, then $\tau = \sigma \cap u^\perp = \sigma' \cap u^\perp$ for any $u \in \operatorname{relint}(\sigma^\vee \cap (-\sigma')^\vee)$.

In the following we fix a lattice N in V and denote by $M \subseteq V^\vee$ the dual lattice of N. By a rational convex polyhedral cone in V (resp. in $V^\vee$) we always mean a convex polyhedral cone that is N-rational (resp. M-rational).

9.1.1 Hilbert bases

Definition. Let σ be a rational convex polyhedral cone in V and $S_\sigma := \sigma^\vee \cap M$. An element $x \in S_\sigma$ is called **irreducible** if $x = x' + x''$ for some $x', x'' \in S_\sigma$ implies $x' = 0$ or $x'' = 0$.

Lemma 9.14. Let σ be a strongly convex rational polyhedral cone of maximal dimension in V and $S_\sigma := \sigma^\vee \cap M$. Put

$$\mathcal{H} := \{x \in S_\sigma \mid x \text{ is irreducible}\}.$$

Then

(a) $\mathcal{H}$ is finite and generates S_σ.

(b) $\mathcal{H}$ contains the ray generators of the edges of $\sigma^\vee$.

(c) $\mathcal{H}$ is the minimal generating set of S_σ with respect to inclusion.

The generating set $\mathcal{H}$ is called the **Hilbert basis** of the semigroup S_σ.

Proof. Since $\sigma = (\sigma^\vee)^\vee$, Lemma 9.11.(d) implies that $\sigma^\vee$ is strongly convex, so $\{0\}$ is a face of $\sigma^\vee$ by Lemma 9.11.(a), i.e., there exists $v \in \sigma \cap N \backslash \{0\}$ such that $\langle u, v\rangle \in \mathbb{Z}_{\geqslant 0}$ for any $u \in S_\sigma$, and for $u \in S_\sigma$, $\langle u, v\rangle = 0$ if and only if $u = 0$.

Suppose $u \in S_\sigma$ is not irreducible. Then $u = u' + u''$ for some $u', u'' \in S_\sigma \backslash \{0\}$. It follows that

$$\langle u, v\rangle = \langle u', v\rangle + \langle u'', v\rangle$$

with $\langle u', v\rangle, \langle u'', v\rangle \in \mathbb{Z}_{>0}$, so both of them are strictly smaller than $\langle u, v\rangle$. By induction on $\langle u, v\rangle$ we conclude that every element in S_σ is a finite sum of irreducible elements, meaning that $\mathcal{H}$ generates S_σ. By Lemma 9.9, $\mathcal{H}$ is finite. This proves (a).

For (b), let ρ be an edge of $\sigma^\vee$ and $u_\rho \in \rho \cap M$ be its ray generator. We must show that $u_\rho \in S_\sigma$ is irreducible. Write $\rho = \sigma^\vee \cap v^\perp$ for some $v \in \sigma \cap N \backslash \{0\}$. If $u_\rho = u_1 + u_2$ for some $u_i \in S_\sigma \subseteq \sigma^\vee$, then

$$0 = \langle u_\rho, v\rangle = \langle u_1, v\rangle + \langle u_2, v\rangle$$

with each $\langle u_i, v\rangle$ non-negative. This forces that $\langle u_1, v\rangle = 0 = \langle u_2, v\rangle$, so u_1, $u_2 \in \rho \cap M$. But u_ρ is the unique generator of $\rho \cap M$, so either $u_1 = 0$ or $u_2 = 0$, demonstrating the irreducibility of the ray generator u_ρ. Finally, (c) results from the fact that every element in $\mathcal{H}$ is irreducible.

□

9.2 Convex polytopes

We retain the notation in the last subsection.

Definition.

1. For any subset $S \subseteq V$, the **convex hull** of S is the smallest convex set $\mathrm{conv}(S)$ in V containing S.
2. A **convex polytope** in V is the convex hull of a finite subset of V
3. An **affine subspace** of V is a subset of the form $v + W$, where $v \in V$ and $W \subseteq V$ is a subspace. The **dimension** of this affine subspace is the dimension of W.
4. For any subset $S \subseteq V$, the **affine span** of S is the smallest affine subspace $\mathrm{aff}(S)$ of V containing S. In fact, we have

$$\mathrm{aff}(S) = \left\{ \sum_{v \in S} a_v v \mid \sum_{v \in S} a_v = 1,\ a_v = 0 \text{ for all but finitely many } v \in S \right\}$$

5. The **dimension** $\dim P$ of a polytope P is $\dim \mathrm{aff}(P)$.

- For a finite subset S of V, we have

$$\mathrm{conv}(V) = \left\{ \sum_{v \in S} a_v v \mid 0 \leqslant a_v,\ \sum_{v \in S} a_v = 1 \right\}$$

- Let $S \subseteq V$ be a finite subset. Consider $V' = V \oplus \mathbb{R}$. If we identify V as $V \times \{1\} \subseteq V'$, then

$$\mathrm{conv}(S) = \mathrm{cone}(S \times \{1\}) \cap V.$$

  Clearly, $\mathrm{conv}(S) \subseteq \mathrm{cone}(S \times \{1\}) \cap V$. Conversely, if $v \in \mathrm{cone}(S \times \{1\}) \cap V$, write $v = \sum_{s \in S} a_s(s, 1)$ with $a_s \geqslant 0$. Since $v \in V \times \{1\}$, $\sum_{s \in S} a_s = 1$, so $v \in \mathrm{conv}(S)$ by the last property.
- If S is a finite subset, then $\mathrm{cone}(S \times \{1\})$ is a strongly convex polyhedral cone. To see this, let $u' \in V^\vee$ be such that $\langle u', S \rangle > 0$. Let $u = (u', 0) \in V^\vee \oplus \mathbb{R}$; then $u \in \mathrm{cone}(S \times \{1\})^\vee$ and $\mathrm{cone}(S \times \{1\}) \cap u^\perp = \{0\}$, implying that $\{0\}$ is a face of $\mathrm{cone}(S \times \{1\})$.

Definition. Let P be a convex polytope in V. A **face** of P is a subset of the form

$$P \cap \{v \in V \mid \langle u, v \rangle = b\}$$

for some $u \in V^\vee$ and $b \in \mathbb{R}$ with $P \subseteq H^+_{u,b} := \{v \in V \mid \langle u, v \rangle \geqslant b\}$. In this situation, $H_{u,b} := \{v \in V \mid \langle u, v \rangle = b\}$ is called the **supporting affine hyperplane**.

- A **facet**, **edge**, and **vertex** of P is a face of dimension $\dim P - 1$, 1 and 0, respectively.

Lemma 9.15. If $P = \mathrm{conv}(S)$ for some finite subset $S \subseteq V$ and $Q = H_{u,b} \cap P$ is a face, then $Q = \mathrm{conv}(S \cap H_{u,b})$. In particular, any face of a convex polytope is a convex polytope.

Proof. Clearly we have $\subseteq$. Conversely, if $q \in Q$, write $q = \sum_{v \in S} a_v v$ with $0 \leqslant a_v$ and $\sum_{v \in S} a_v = 1$. Since $q \in H_{u,b}$, we have $\sum_{v \in S} a_v \langle u, v \rangle = b$. But $P \subseteq H^+_{u,b}$, we have

$$b = \sum_{v \in S} a_v \langle u, v \rangle \geqslant \sum_{v \in S} a_v b = b$$

and hence the equality $\langle u, v \rangle = b$ for any $v \in S$ with $a_v \neq 0$. This proves $q \in \mathrm{conv}(S \cap H_{u,b})$. □

Lemma 9.16. There is a bijection

$$\begin{aligned} \{\text{faces of } P\} &\longrightarrow \{\text{faces of } \mathrm{cone}(P \times \{1\})\} \\ Q &\longmapsto \mathrm{cone}(Q \times \{1\}) \end{aligned}$$

with empty face of P corresponding to $\{0\}$.

Proof. Write $P = \mathrm{conv}(S)$ for some finite subset $S \subseteq V$; then $\mathrm{cone}(P \times \{1\}) = \mathrm{cone}(S)$, where we identify S with $S \times \{1\}$. Let $u \in \mathrm{cone}(S)^\vee$ and write $u = (u', r)$ for some $u' \in V^\vee$ and $r \in \mathbb{R}$. Then

$$\mathrm{cone}(S) \cap u^\perp \cap V = P \cap H_{u',-r}$$

and $P \subseteq H^+_{u',-r}$. Indeed, $v = (v, 1) \in$ LHS if and only if $\langle u', v \rangle + r = 0$, if and only if $v \in$ RHS. Since $u \in \mathrm{cone}(S)^\vee$, for any $v \in P$, we have $\langle u', v \rangle + r \geqslant 0$, i.e., $v \in H^+_{u',-r}$.

If $u \in V^\vee$ and $b \in \mathbb{R}$, then $(u, -b) \in V^\vee \oplus \mathbb{R}$, and the condition $P \subseteq H^+_{u,-b}$ implies $(u, -b) \in \mathrm{cone}(S)^\vee$. Thus, the map in the lemma is well-defined, and it is clear that it is bijective. □

Corollary 9.16.1. Let $P = \mathrm{conv}(S)$ be a polytope in V.

1. Every vertex of P lies in S.
2. P is the convex hull of its vertices.
3. If Q is a face of P, the faces of Q are exactly the faces of P contained in Q.
4. Every proper face Q of P is the intersection of facets of F containing Q.

Proof.

1. Say $H_{u,b} \cap P = \mathrm{conv}(S \cap H_{u,b})$ is a vertex of P. This means $\mathrm{conv}(S \cap H_{u,b})$ is a singleton, so $\mathrm{conv}(S \cap H_{u,b}) = S \cap H_{u,b} \subseteq S$.
2. A vertex of P corresponds to an edge of $\mathrm{cone}(P \times \{1\})$. Now the result follows from Corollary 9.11.1.
3. Lemma 9.2.4.

4. Corollary 9.5.1.2.

□

Lemma 9.17. A subset P of V is a polytope if and only if it is bounded and is a finite intersection of affine half-spaces.

Proof. The only if part follows from Corollary 9.8.1. For the if part, write $P = \bigcap_{i=1}^{\ell} H^+_{u_i,b_i}$ for some $u_i \in V^\vee$ and $b_i \in \mathbb{R}$. Then

$$C(P) = \bigcap_{i=1}^{\ell} \{(v,\lambda) \in V \oplus \mathbb{R}_{\geqslant 0} \mid \langle u_i, v\rangle \geqslant \lambda b_i\} = \bigcap_{i=1}^{\ell} (u_i, -b_i)^\vee$$

is a convex polyhedral cone, so $C(P) = \mathrm{cone}(S)$ for some finite subset $S \subseteq V \oplus \mathbb{R}_{\geqslant 0}$. We assume $S = \{v_i\}_{i=1}^a \cup \{w_j\}_{j=1}^b$ with, by abuse of notation, $v_i = (v_i, 1) \in V \times \{1\}$ and $w_j = (w_j, 0) \in V \times \{0\}$. Clearly $C(P) \cap V = P$, so $P = \mathrm{conv}\{v_i\}_{i=1}^a + \mathrm{cone}\{w_j\}_{k=1}^b$. By boundedness of P we must have $b = 0$, i.e., $P = \mathrm{conv}(S)$. □

Remark. A set of the form $\mathrm{conv}(S) + \mathrm{cone}(T)$ with $S, T \subseteq V$ finite is called a **polyhedron**. By definition, a convex polytope is exactly a bounded polyhedron. One can show that a subset of V is a polyhedron if and only if it is a finite intersection of affine half-spaces.

When $\dim P = \dim_{\mathbb{R}} V$, by Lemma 9.7 we can write P as the intersection of the half spaces determined its facets, and each facet F has a unique supporting affine hyperplane

$$H_F := \{v \in V \mid \langle u_F, v\rangle = -a_F\}$$

for some pair $(u_F, a_F) \in V^\vee \oplus \mathbb{R}$ unique up to a multiplication of a positive scalar. The vector u_F is called an **inward-pointing facet normal** of F. If we put $H_F^+ = H^+_{u_F,-a_F}$, we have

$$P = \bigcap_F H_F^+ = \{v \in V \mid \langle u_F, v\rangle \geqslant -a_F \text{ for all facets } F\},$$

where F runs over all facets of P. This is called the **facet presentation** of P.

Definition. Convex polytopes P_1 and P_2 are **combinatorially equivalent** if there is a bijection

$$\{\text{faces of } P_1\} \longrightarrow \{\text{faces of } P_2\}$$

that preserves dimensions, intersections, and the face relation (this means if Q' is a face of Q with Q, Q' being faces of P_1, then under this bijection, the image of Q' is still a face of the image of Q).

Definition. Let P be a convex polytope of dimension d in V.

1. P is a **simplex**/d**-simplex** if it has $d+1$ vertices.

2. P is **simplicial** if every facet of P is a simplex.
3. P is **simple** if every vertex is the intersection of precisely d facets.

Definition. Let $P = \mathrm{conv}(S)$ be a convex polytope in V.

1. For $r \geqslant 0$, put $rP = \mathrm{conv}(rS) = \mathrm{conv}\{rv \mid v \in S\}$, which is again a convex polytope.
2. If $P' = \mathrm{conv}(S')$ is another convex polytope, the Minkowski sum $P + P' = \mathrm{conv}(S + S')$ is still a convex polytope.
3. When $\dim P = \dim_{\mathbb{R}} V$ and $0 \in \mathrm{int}(P)$, define the **polar/polar set** of P to be

$$P^\circ = \{u \in V^\vee \mid \langle u, v\rangle \geqslant -1 \text{ for all } v \in P\} \subseteq V^\vee.$$

- Note that the distribution law holds : for $r, t \geqslant 0$, we have

$$rP + tP = (r + t)P.$$

— The equality $P + P' = \mathrm{conv}(S + S')$ needs a verification. The containment $\mathrm{conv}(S + S') \subseteq P + P'$ is relatively clear. The other can be dealt with greedy.

Lemma 9.18. Let P be a convex polytope in V with $\dim P = \dim_{\mathbb{R}} V$ and $0 \in \mathrm{int}(P)$.

1. P° is convex polytope, and $(P^\circ)^\circ = P$.
2. If Q is a face of P,

$$Q^* = \{u \in P^\circ \mid \langle u, v\rangle = -1 \text{ for all } v \in Q\}$$

is a face of P°, and this establishes an inclusion-reversing bijection

$$\begin{array}{ccc} \{\text{faces of } P\} & \longrightarrow & \{\text{faces of } P^\circ\} \\ Q & \longmapsto & Q^* \end{array}$$

with $\dim Q + \dim Q^* = \dim V - 1$.
3. If $P = \{v \in V \mid \langle u_F, v\rangle \geqslant -a_F \text{ for all facets } F\}$, then $P^\circ = \mathrm{conv}\{a_F^{-1} u_F\}_{F \text{ facets}}$. Note that $a_F > 0$ since $0 \in \mathrm{int}(P)$.

Proof. Let $\sigma = \mathrm{cone}(P \times \{1\})$. Then $\sigma^\vee = \mathrm{cone}(P^\circ \times \{1\})$, so P° is a convex polytope. Now $(P^\circ)^\circ = P$ follows from $(\sigma^\vee)^\vee = \sigma$. For a face Q of P, $\tau = \mathrm{cone}(Q \times \{1\})$ is a face of σ, and $\tau^* = \sigma^\vee \cap \tau^\perp = \mathrm{cone}(Q^* \times \{1\})$. Now 2. follows from Lemma 9.10, and 3. follows from the first paragraph of the proof of Lemma 9.8. $\square$

9.2.1 Normal lattice polytopes

Definition. Let M be a lattice in V. An M**-lattice**/M**-rational polytope** is a convex polytope $\mathrm{conv}(S)$ with $S \subseteq M$.

- A convex polytope is M-rational if and only if its vertices all lie in M.
- Faces of an M-rational polytope is also M-rational. Also, every M-rational polytope is an intersection of affine half-spaces $H^+_{u,b}$ defined over M, i.e., $u \in N$ and $b \in \mathbb{Z}$.
- Let $N \subseteq V^\vee$ be the dual lattice of M. If P is an M-rational polytope with $\dim P = \dim_{\mathbb{R}} V$ and $0 \in \mathrm{int}(P)$, then P° is an N-rational polytope in $V^\vee$.
- Let N be as above and P be M-rational and $\dim P = \dim_{\mathbb{R}} V$. For any facet F of P, the inward-pointing facet normal lies on a rational ray in $V^\vee$; let $u_F \in N$ be the unique ray generator. If v is a vertex contained in F, then $\langle u_F, v\rangle = -a_F \in \mathbb{Z}$ is integral.

In the following, fix an lattice M in V and let N be the dual lattice in $V^\vee$. By a lattice polytope in V (resp. in $V^\vee$) we always mean an M-rational (resp. N-rational) convex polytope. Notice that in the last subsection N denoted a lattice in V, while here we use M to denote a lattice in V.

Definition. A lattice polytope P in V is called **normal** if

$$(kP) \cap M + (\ell P) \cap M = ((k+\ell)P) \cap M$$

for all $k, \ell \in \mathbb{Z}_{\geqslant 0}$.

- The inclusion $\subseteq$ is automatic, so the normality actually means that all lattice points of $(k+\ell)P$ come from lattice points of kP and ℓP.
- A lattice polytope P is normal if and only if

$$\underbrace{P \cap M + \cdots + P \cap M}_{\text{k-copies}} = (kP) \cap M$$

  for all $k \in \mathbb{Z}_{\geqslant 0}$.
- Lattices polytopes of dimension one are normal.

Definition. A lattice polytope P of V is a **basic simplex**/**unimodular simplex** if P is a simplex and has a vertex m_0 such that $\{m - m_0 \mid m \neq m_0 : \text{vertices of } P\}$ forms a subset of a $\mathbb{Z}$-basis of M.

- The definition is independent of the choice of the vertex $m_0 \in P$.
- The standard simplex Δ_d $(0 \leqslant d \leqslant n)$ in $\mathbb{R}^n$ is basic.
- Any basic simplex is normal.

Theorem 9.19. Let $P \subseteq V$ be a full dimensional lattice polytope of dimension $n \geqslant 2$. Then kP is normal for all $k \geqslant n-1$.

Proof. We will prove that

$$(kP) \cap M + P \cap M = ((k+1)P) \cap M$$

for $k \geqslant n-1$. This implies kP is normal for all $k \geqslant n-1$.

First consider the case where P is a simplex with no interior lattice point. Let $v_0, \ldots, v_n$ be vertices of P and take $k \geqslant n-1$. Then $(k+1)P$ has vertices $(k+1)v_0, \ldots, (k+1)v_n$, so that any point $v \in ((k+1)P) \cap M$ has the form

$$v = \sum_{i=0}^{n} \mu_i (k+1) v_i$$

with $\mu_i \geqslant 0$ and $\sum_{i=0}^{n} \mu_i = 1$. Put $\lambda_i = (k+1)\mu_i$; we have $\sum_{i=0}^{n} \lambda_i = k+1$.

(i) $\lambda_i \geqslant 1$ for some $i = 0, \ldots, n$. Then $v - v_i \in (kP) \cap M$, so $v = (v - v_i) + v_i \in (kP) \cap M + P \cap M$, as we desire.

(ii) $\lambda_i < 1$ for all $i = 0, \ldots, n$. Then

$$n = (n-1) + 1 \leqslant k + 1 = \sum_{i=0}^{n} \lambda_i < n + 1$$

so $k = n-1$ and $\sum_{i=0}^{n} \lambda_i = n$. Consider the point

$$\tilde{v} := (v_0 + \cdots + v_n) - v = \sum_{i=0}^{n} (1 - \lambda_i) v_i$$

The coefficients are positive, and their sum is $\sum_{i=0}^{n} (1 - \lambda_i) = (n+1) - n = 1$. Thus $\tilde{v}$ is a lattice point, and since $1 - \lambda_i > 0$ for each i, $\tilde{v}$ lies in the interior of P. This is a contradiction.

For general P, it suffices to show P is a finite union of n-dimensional lattice simplices with no interior lattice points. For this, we invoke

Lemma 9.20 (Carathéodory's)**.** For a finite set $S \subseteq V$, we have

$$\mathrm{conv}(S) = \bigcup_{T} \mathrm{conv}(T)$$

where T runs over all subsets of S consisting of $\dim \mathrm{conv}(S) + 1$ affinely independent vectors.

Proof. Let $x \in \mathrm{conv}(S)$ and write $x = \sum_{v \in S} a_v v$ with $0 \leqslant a_v$ and $\sum_{v \in S} a_v = 1$. Among all such presentations of x, we choose a minimal one, i.e., the one so that

$$T := \{v \in S \mid a_v \neq 0\}$$

is minimal. We claim $\#T \leqslant \dim \mathrm{conv}(S) + 1$. Suppose for contradiction that $\#T > \dim \mathrm{conv}(S) + 1$. Pick any $t_0 \in T$. Then $\{t - t_0 \mid t \neq t_0, t \in T\}$ is linear dependent, so $0 = \sum_{T \ni t \neq t_0} b_t(t - t_0)$ for some nonzero sequence $(b_t)_{t \neq t_0}$. If we define $b_{t_0} = -\sum_{t \neq t_0} b_t$, we see $0 = \sum_{t \in T} b_t t$ and $0 = \sum_{t \in T} b_t$. Since $(b_t)_t$ is nonzero, $\#\{t \in T | b_t > 0\} \geqslant 1$. For any $r \in \mathbb{R}$, we may write

$$x = \sum_{v \in T} a_v v - r \sum_{v \in T} b_v v = \sum_{v \in T} (a_v - r b_v) v.$$

If we choose $r = \min \left\{ \frac{a_v}{b_v} \mid v \in T,\ b_v > 0 \right\} > 0$, then $a_v - r b_v \geqslant 0$ for any $v \in T$ and $\sum_{v \in T} (a_v - r b_v) = 1$. But $r = \frac{a_{v'}}{b_{v'}}$ for some $v' \in T$, so $a_{v'} - r b_{v'} = 0$, contradicting the minimality of T. In addition, we also proved that T is affinely independent. The proof is completed once we notice that every affinely independent subset of S can be extended to an affinely independent subset of S of size $\dim \mathrm{conv}(S) + 1$. □

Thus we can write P as a finite union of n-dimensional lattice simplices. If $Q = \mathrm{conv}\{w_0, \ldots, w_n\}$ is an n-dimensional lattice simplex with an interior lattice point v, then

$$Q = \bigcup_{i=0}^{n} \mathrm{conv}\{w_0, \ldots, \widehat{w_i}, \ldots, w_n, v\}$$

with each $\mathrm{conv}\{w_0, \ldots, \widehat{w_i}, \ldots, w_n, v\}$ an n-dimensional lattice simplex having fewer interior lattice points. We repeat this process so that Q, and hence P, is written as a finite union of n-dimensional lattice simplices with no interior lattice points. □

Corollary 9.20.1. Every lattice polygon in $\mathbb{R}^2$ is normal.

Lemma 9.21. Let P be a lattice polytope in V. Then P is normal if and only if $(P \cap M) \times \{1\}$ generates the semigroup $C(P) \cap (M \times \mathbb{Z})$, where $C(P) = \mathrm{cone}(P \times \{1\})$.

Proof. Note that

$$C(P) \cap (M \times \mathbb{Z}) = \{0\} \cup \bigcup_{k \geqslant 1} ((kP) \cap M) \times \{k\}$$

Thus $(P \cap M) \times \{1\}$ generates $C(P) \cap (M \times \mathbb{Z})$ if and only if for any $k \geqslant 1$ and $v \in (kP) \cap M$, we can find $v_1, \ldots, v_\ell \in P \cap M$ such that $v = v_1 + \cdots + v_\ell$ and $k = \ell$, which is the same as saying that P is normal. □

Remark. Elements in $(P \cap M) \times \{1\}$ are clearly irreducible, so $(P \cap M) \times \{1\}$ is contained in the Hilbert basis of the semigroup $C(P) \cap (M \times \mathbb{Z})$. This lemma simply says that P is normal if and only if $(P \cap M) \times \{1\}$ is exactly the Hilbert basis of $C(P) \cap (M \times \mathbb{Z})$.

Lemma 9.22. Let P be a full dimensional lattice polytope in V of dimension $n \geqslant 2$ and let k_0 be the maximal height of an element in the Hilbert basis of $C(P)$. Then $k_0 \leqslant n-1$.

Proof. This follows from the identity

$$(kP) \cap M + P \cap M = ((k+1)P) \cap M \qquad (k \geqslant n-1)$$

proved in Theorem 9.19. □

9.2.2 Very ample polytopes

Definition. A lattice polytope P in V is called **very ample** if for every vertex v of P, the semigroup

$$S_{P,v} := \mathbb{Z}_{\geqslant 0}(P \cap M - v)$$

generated by the set $\{v' - v \mid v' \in P \cap M\}$ is saturated in M.

- Recall that an affine semigroup $S \subseteq M$ is **saturated** if for all $k \in \mathbb{Z}_{>0}$ and $v \in M$, $kv \in S$ implies $v \in S$.

Lemma 9.23. A normal lattice polytope P is very ample.

Proof. Fix a vertex v_0 of P and take $w \in M$ such that $kw \in S_{P,v_0}$ for some $k \in \mathbb{Z}_{>0}$. Write

$$kw = \sum_{v' \in P \cap M} a_{v'}(v' - v_0)$$

for some $a_{v'} \in \mathbb{Z}_{\geqslant 0}$. Take $d \in \mathbb{Z}_{\geqslant 0}$ so that $kd \geqslant \sum_{v' \in P \cap N} a_{v'}$. Then

$$kw + kdv_0 = \sum_{v' \in P \cap M} a_{v'} v' + \left(kd - \sum_{v' \in P \cap M} a_{v'} \right) v_0 \in kdP$$

Dividing out k, we see $w + dv_0 \in dP$. By normality we see $w + dv_0 = \sum_{m \in T} m$ for some $T \subseteq P \cap M$ with $\#T = d$. Thus

$$w = \sum_{m \in T} (m - v_0) \in S_{P,v_0}$$

as desired. □

Corollary 9.23.1. Let $P \subseteq V$ be a full dimensional lattice polytope. If $\dim P \geqslant 2$, then kP is very ample for all $k \geqslant n-1$. In particular, P is very ample if $\dim P = 2$.

Example. There exists non-normal very ample polytope. Given $1 \leqslant i < j < k \leqslant 6$, let $[ijk]$ denote the vector in $\mathbb{Z}^6$ with 1 in positions i, j, k and 0 elsewhere. Let

$$A = \{[123], [124], [135], [146], [156], [236], [245], [256], [345], [346]\} \subseteq \mathbb{Z}^6.$$

The lattice polytope $P = \mathrm{conv}(A)$ lies in the affine hyperplane $\sum_{i=1}^{6} x_i = 3$ in $\mathbb{R}^6$. It is straightforward to see that $A = P \cap \mathbb{Z}^6$, and A is the set of vertices of P. Label the points in A as $m_1, \ldots, m_{10}$. Then

$$(1,1,1,1,1,1) = \frac{1}{5}\sum_{i=1}^{10} m_i = \sum_{i=1}^{10} \frac{1}{10}(2m_i)$$

which shows that $v = (1,1,1,1,1,1) \in 2P$. Since v is not a sum of lattices point of P (when $[ijk] \in A$, the vector $v - [ijk] \notin A$), we conclude that P is not a normal polytope.

To show P is very ample, firstly, by computer we see $A \times \{1\} \cup \{(v,2)\} \subseteq \mathbb{R}^6 \times \mathbb{R}$ is a Hilbert basis of the semigroup $C(P) \cap \mathbb{Z}^7$, where $C(P) = \mathrm{cone}(P \times \{1\}) \subseteq \mathbb{R}^6 \times \mathbb{R}$. Next fix i and let S_{P,m_i} be the semigroup generated by the $m_j - m_i$. Take $m \in \mathbb{Z}^6$ such that $km \in S_{P,m_i}$. As in the proof of the last lemma, this implies $m + dm_i \in dP$ for some $d \in \mathbb{Z}_{\geqslant 0}$, and thus $(m + dm_i, d) \in C(P) \cap \mathbb{Z}^7$. Expressing this in terms of the above Hilbert basis easily implies that

$$m = a(v - 2m_i) + \sum_{j=1}^{10} a_j(m_j - m_i)$$

for some $a, a_j \in \mathbb{Z}_{\geqslant 0}$. If we can show $v - 2m_i \in S_{P,m_i}$, then $m \in S_{P,m_i}$ follows and this proves S_{P,m_i} is saturated. When $i = 1$, one checks that

$$v + [123] = [124] + [135] + [236]$$

which implies that

$$v - 2m_1 = (m_2 - m_1) + (m_3 - m_1) + (m_6 - m_1) \in S_{P,m_1}.$$

One computes this for $i = 2, \ldots, 10$ similarly, and this proves that P is very ample.

9.2.3 Normal fans

Let P be a full dimensional lattice polytope. We have the facet presentation of P

$$P = \{x \in V \mid \langle u_F, x\rangle \geqslant -a_F \text{ for all facets } F\}.$$

A vertex v of P gives cones

$$C_v = \mathrm{cone}(P \cap M - v) \subseteq V \text{ and } \sigma_v = C_v^\vee \subseteq V^\vee.$$

where $P \cap M - v = \{w - v \mid w \in P \cap M\}$. There is a bijection

$$\begin{array}{ccc} \{\text{faces of } P \text{ containing } v\} & \longrightarrow & \{\text{faces of } C_v\} \\ Q & \longmapsto & Q_v = \mathrm{cone}(Q \cap M - v) \\ Q = (Q_v + v) \cap P & \longleftarrow\!\shortmid & Q_v \end{array}$$

that preserves dimensions, inclusions and intersections. In particular, all facets of C_v come from facets of P containing v, so that Lemma 9.7 gives

$$C_v = \{x \in V \mid \langle u_F, x\rangle \geqslant 0 \text{ for all facets } F \text{ of } P \text{ containing } v\}.$$

It follows that the proof of Lemma 9.8 that

$$\sigma_v = \mathrm{cone}\,\{u_F \mid F : \text{facets of } P \text{ containing } v\}.$$

For any face Q of P, we set

$$\sigma_Q = \mathrm{cone}\,\{u_F \mid F : \text{facets of } P \text{ containing } Q\}.$$

In particular, $\sigma_F = \mathbb{R}_{\geqslant 0} u_F$ and $\sigma_P = \mathrm{cone}\,\varnothing = \{0\}$.

Theorem 9.24. Let P be a full dimensional lattice polytope and set $\Sigma_P = \{\sigma_Q \mid Q \text{ is a face of } P\}$. Then :

1. For all $\sigma_Q \in \Sigma_P$, each face of σ_Q also lies in Σ_P.
2. The intersection $\sigma_Q \cap \sigma_{Q'}$ of any two cones in Σ_P is a face of each.

We call Σ_P the **(inner) normal fan** of P.

To avoid cumbersome sentences, we will always write Q, F, v to denote faces, facets, vertices of P.

Lemma 9.25. Let Q be a face of P and let $H_{u,b}$ be a supporting affine hyperplane of P. Then $u \in \sigma_Q$ if and only if $Q \subseteq H_{u,b} \cap P$.

Proof. Suppose $u \in \sigma_Q$ and write $u = \sum_{Q \subseteq F} \lambda_F u_F$, $\lambda_F \geqslant 0$. By setting $b_0 = -\sum_{Q \subseteq F} \lambda_F a_F$, we see

$$\langle u, x\rangle = \sum_{Q \subseteq F} \lambda_F \langle u_F, x\rangle \geqslant -\sum_{Q \subseteq F} \lambda_F a_F = b_0$$

for any $x \in P$, and the middle inequality is an equality for $x \in Q$. In other words, $P \subseteq H^+_{u,b_0}$ and $Q \subseteq H_{u,b_0} \cap P$, so H_{u,b_0} is a supporting hyperplane of P. By assumption $H_{u,b}$ is a support hyperplane of P, so we must have $b = b_0$, and $Q \subseteq H_{u,b} \cap P$ in turn.

Conversely, suppose that $Q \subseteq H_{u,b} \cap P$. Take a vertex $v \in Q$. Then $P \subseteq H^+_{u,b}$ and $v \in H_{u,b}$ imply that $C_v \subseteq H^+_{u,0}$. Hence $u \in C_v^\vee = \sigma_v$, so that

$$u = \sum_{v \in F} \lambda_F u_F$$

for some $\lambda_F \geqslant 0$. Let F_0 be a facets of P containing v but not Q, and pick $p \in Q$ with $p \notin F_0$. Then $p, v \in Q \subseteq H_{u,b}$ imply that

$$b = \langle u, p\rangle = \sum_{v \in F} \lambda_F \langle u_F, p\rangle$$
$$b = \langle u, v\rangle = \sum_{v \in F} \lambda_F \langle u_F, v\rangle = -\sum_{v \in F} \lambda_F a_F,$$

so that $\sum_{v \in F} \lambda_F \langle u_F, p\rangle = -\sum_{v \in F} \lambda_F a_F$. But $p \notin F_0$ gives $\langle u_{F_0}, p\rangle > -a_F$, and since $\langle u_F, p\rangle \geqslant -a_F$ for all F, it forces that $\lambda_{F_0} = 0$ whenever $Q \nsubseteq F_0$. Thus $u \in \sigma_Q$, and the proof is completed. $\square$

Corollary 9.25.1. If Q is a face of P and F is a facet of P, then $u_F \in \sigma_Q$ if and only if $Q \subseteq F$.

Proof. The if part follows from definition, and the only if part follows from the last lemma with $u = u_F$ and $b = -a_F$, where $F = P \cap H_{u_F,-a_F}$. □

Theorem 9.24 is an immediate consequence of the following

Lemma 9.26. Let Q and Q' be faces of a full dimensional lattice polytope P in V. Then

(a) $Q \subseteq Q'$ if and only if $\sigma_{Q'} \subseteq \sigma_Q$.

(b) If $Q \subseteq Q'$, then $\sigma_{Q'}$ is a face of σ_Q, and all faces of σ_Q have this form.

(c) $\sigma_Q \cap \sigma_{Q'} = \sigma_{Q''}$, where Q'' is the smallest face of P containing Q and Q'.

Proof.

(a) The only if part is obvious, and the if part follows from Corollary 9.25.1 and Corollary 9.5.1.2.

(b) Recall that a vertex $v \in Q$ of P determines a face Q_v of C_v. By Lemma 9.10.2,

$$Q_v^* := C_v^\vee \cap Q_v^\perp = \sigma_v \cap Q_v^\perp$$

is a face of σ_v. Since $Q_v \subseteq C_v = \sigma_v^\vee$, for $u \in Q_v^*$, if we write $u = \sum_{v \in F} \lambda_F u_F$ with $\lambda_F \geqslant 0$, then for any $x \in Q_v$, we have

$$0 = \langle u, x \rangle = \sum_{v \in F} \lambda_F \langle u_F, x \rangle$$

with each $\langle u_F, x \rangle \geqslant 0$. Thus $Q_v \subseteq H_{u_F,0}$ if $\lambda_F > 0$, proving that $u \in \mathrm{cone}\{u_F \mid v \in F,\ Q_v \subseteq H_{u_F,0}\}$. Thus $Q_v^* \subseteq \mathrm{cone}\{u_F \mid v \in F,\ Q_v \subseteq H_{u_F,0}\}$. The reversed inclusion is clear, so we obtain

$$Q_v^* = \mathrm{cone}\{u_F \mid v \in F,\ Q_v \subseteq H_{u_F,0}\}.$$

Since $v \in Q$, the inclusion $Q_v \subseteq H_{u_F,0}$ is the same as $Q \subseteq H_{u_F,-a_F}$, which is equivalent to $Q \subseteq F$. It follows that

$$Q_v^* = \mathrm{cone}\{u_F \mid Q \subseteq F\} = \sigma_Q,$$

so σ_Q is a face of σ_v, and all faces of σ_v arise in this way (Lemma 9.10.2).

In particular, $Q \subseteq Q'$ implies that $\sigma_{Q'}$ is a face of σ_v, and since $\sigma_{Q'} \subseteq \sigma_Q$ by (a), $\sigma_{Q'}$ is thus a face of σ_Q. Furthermore, every face of σ_Q is a face of σ_v by Lemma 9.2.4, and hence has the form $\sigma_{Q'}$ for some face Q'; by (a) again we see $Q' \subseteq Q$, and (b) follows.

(c) Let Q'' be the smallest face of P containing Q and Q' (whose existence is assured by Lemma 9.2.3). By (b) $\sigma_{Q''}$ is a face of both σ_Q and $\sigma_{Q'}$, so $\sigma_{Q''} \subseteq \sigma_Q \cap \sigma_{Q'}$.

It remains the prove the reversed inclusion. If $\sigma_Q \cap \sigma_{Q'} = \{0\} = \sigma_P$, then $Q'' = P$ and we are done. If $\sigma_Q \cap \sigma_{Q'} \neq \{0\}$, pick a nonzero vector u in the intersection. If we take $b = \min_{v \in P} \langle u, v \rangle$, then $P \subseteq H_{u,b}^+$ and $v \in H_{u,b}$ for at least one vertex v of P, implying that $H_{u,b}$ is a supporting hyperplane. By Lemma 9.25, $u \in \sigma_Q \cap \sigma_{Q'}$ implies $Q \subseteq H_{u,b} \cap P \supseteq Q'$, so $Q'' \subseteq H_{u,b} \cap P$ by minimality of Q''. Using the same lemma again, we obtain $u \in \sigma_{Q''}$, as desired.

□

Lemma 9.27. Let P be a full dimensional lattice polytope in V of dimension n. Then

1. $\dim Q + \dim \sigma_Q = n$ for all faces Q of P.
2. $V^\vee = \bigcup_{v \in P} \sigma_v = \bigcup_{\sigma_Q \in \Sigma_P} \sigma_Q$.

We will use the sentence "the normal fan Σ_P is **complete**" to refer to the statement 2.

Proof.

1. This follows from the proof of Lemma 9.26.(b) and Lemma 9.10.1.
2. In the proof of Lemma 9.26.(c) we saw for every nonzero element $u \in V^\vee$, we can find $b \in \mathbb{R}$ making $H_{u,b}$ a supporting hyperplane of P. If $v \in H_{u,b}$, then Lemma 9.25 gives $u \in \sigma_v$. The last equality follows from Lemma 9.26.(a).

□

Lemma 9.28. Let P be a full dimensional lattice polytope in V. For any lattice point $m \in M$ and any integer $k \geqslant 1$, the translation $P + m$ and the dilation kP have the same normal fan as P.

10 Toric Varieties

In this section, a variety over k is defined as in (3.18).

Definition. A **toric variety** is an irreducible variety X over $\mathbb{C}$ containing a torus $T_N \cong (\mathbb{C}^\times)^n$ as a Zariski open subset such that the action of T_N on itself extends to an algebraic action of T_N on X.

10.1 Affine toric varieties

10.1 We recall some properties about tori. Denote by **AlgGp** the category of algebraic groups over $\mathbb{C}$. A **torus** (over $\mathbb{C}$) is an affine algebraic group T isomorphic to $(\mathbb{C}^\times)^n$ in **AlgGp**. The set $\mathrm{Hom}_{\mathbf{AlgGp}}(T, \mathbb{C}^\times)$, which is an abstract group, is called the **character group** of T, and its element is called a **character** of T. From the theory of tori, the character group is isomorphic to $\mathbb{Z}^n$. Explicitly, for $m = (m_1, \ldots, m_n) \in \mathbb{Z}^n$, define $\chi^m : T \to \mathbb{C}^\times$ by

$$\chi^m(x_1, \ldots, x_n) = x_1^{m_1} \cdots x_n^{m_n}$$

where $(x_1, \ldots, x_n) \in (\mathbb{C}^\times)^n \cong T$. The assignment $m \mapsto \chi^m$ is an group isomorphism between $\mathbb{Z}^n$ and $\mathrm{Hom}_{\mathbf{AlgGp}}(T, \mathbb{C}^\times)$. Also, if we put $M = \mathrm{Hom}_{\mathbf{AlgGp}}(T, \mathbb{C}^\times)$, then the coordinate ring $\mathbb{C}[T]$ of T is the same as the group algebra $\mathbb{C}[M]$.

On the other hand, the set $\mathrm{Hom}_{\mathbf{AlgGp}}(\mathbb{C}^\times, T)$, which is also an abstract group, is called the **cocharacter group** of T, and its element is called a **cocharacter**. It is also isomorphic to $\mathbb{Z}^n$. Explicitly, for $u = (u_1, \ldots, u_n) \in \mathbb{Z}^n$, define $\lambda^u : \mathbb{C}^\times \to T$ by

$$\lambda^u(t) = (t^{u_1}, \ldots, t^{u_n}) \in (\mathbb{C}^\times)^n \cong T.$$

The assignment $u \mapsto \lambda^u$ is an group isomorphism between $\mathbb{Z}^n$ and $\mathrm{Hom}_{\mathbf{AlgGp}}(\mathbb{C}^\times, T)$.

There is a pairing defined as follows.

$$\begin{aligned} \mathrm{Hom}_{\mathbf{AlgGp}}(T, \mathbb{C}^\times) \times \mathrm{Hom}_{\mathbf{AlgGp}}(\mathbb{C}^\times, T) &\longrightarrow \mathrm{Hom}_{\mathbf{AlgGp}}(\mathbb{C}^\times, \mathbb{C}^\times) \\ (\chi, \lambda) &\longmapsto \chi \circ \lambda \end{aligned}$$

The last group is isomorphic to $\mathbb{Z}$, so $\chi \circ \lambda(x) = x^\ell$ for some unique $\ell \in \mathbb{Z}$. We set $\ell = \langle \chi, \lambda \rangle$. If we identify both the character group and the cocharacter group with $\mathbb{Z}^n$, this becomes the inner product on $\mathbb{Z}^n$:

$$\begin{aligned} \mathbb{Z}^n \times \mathbb{Z}^n &\longrightarrow \mathbb{Z} \\ (m = (m_i), u = (u_i)) &\longmapsto m \cdot n = \sum_{i=1}^n m_i u_i. \end{aligned}$$

In particular, this shows the pairing $\langle\, , \rangle$ is perfect.

10.2 Let T be a torus with character group M. Every point $t \in T$ induces an evaluation $\phi_t : M \to \mathbb{C}^\times$ given by $\phi_t(\chi) = \chi(t)$. The map

$$\begin{aligned} T &\longrightarrow \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \\ t &\longmapsto \phi_t \end{aligned}$$

is an group isomorphism. Indeed, if we identify T with $(\mathbb{C}^\times)^n$, then

$$\begin{aligned} \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{C}^\times) &\longrightarrow (\mathbb{C}^\times)^n \\ \phi &\longmapsto (\phi(e_1), \ldots, \phi(e_n)) \end{aligned}$$

is an group isomorphism whose inverse is exactly the map given above. Thus

$$T \cong \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \cong \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C}^\times \cong N \otimes_{\mathbb{Z}} \mathbb{C}^\times$$

The map is explicitly given by $N \otimes_{\mathbb{Z}} \mathbb{C}^\times \ni u \otimes t \mapsto \lambda^u(t) \in T$.

10.3 We quote, without proof, some facts about tori that we will use in the sequel.

(i) If $\varphi : T_1 \to T_2$ is a morphism of algebraic groups between tori, then the image of φ is closed in T_2, and is a torus.

(ii) If T is a torus and $H \subseteq T$ is an irreducible subvariety that is also an abstract subgroup, then H is a torus.

Suppose that a torus T acts linearly on a finite dimensional $\mathbb{C}$-vector space V. For each $m \in M$, set

$$V_m := \{v \in V \mid t.v = \chi^m(t)v \text{ for all } t \in T\}$$

This is the eigenspace of T with eigencharacter χ^m. Then we always have

(iii) $V = \bigoplus_{m \in M} V_m$.

10.4 Affine toric variety from lattice points. a Let T_N be a torus with character group M and cocharacter group N. Let $A = \{m_1, \ldots, m_s\} \subseteq M$ be a finite subset of M. Using A, we construct a morphism $\Phi_A : T_N \to (\mathbb{C}^\times)^s$ by

$$\Phi_A(t) = (m_1(t), \ldots, m_s(t)) \in (\mathbb{C}^\times)^s$$

Denote by Y_A the Zariski closure in $\mathbb{A}^s_{\mathbb{C}} = \mathbb{C}^s$ of the image of Φ_A. Then Y_A is an affine toric variety whose torus has character group $\mathbb{Z}A \leqslant M$, the abelian group generated by A.

To see this, by 10.3.(i), the image $T = \Phi_A(T)$ of Φ_A in $(\mathbb{C}^\times)^s$ is closed and is a torus. So by definition we have $Y_A \cap (\mathbb{C}^\times)^s = T$, which means that T is open in Y_A. This shows Y_A contains the torus T as a Zariski open subset. To show the action of T on itself extends to an action of T on Y_A, note that for $t \in T$, we have $T = t.T \subseteq t.Y_A$, so taking closure gives $Y_A \subseteq t.Y_A$. Replacing t by t^{-1} shows $Y_A = t.Y_A$, so the action of T on itself really induces an action on Y_A. This shows Y_A is an affine toric variety.

We are left to show the character group of T is $\mathbb{Z}A$. By definition of T, we have a commutative diagram

$$\begin{tikzcd} T_N \arrow[rr] \arrow[dr, two heads] & & (\mathbb{C}^\times)^s \\ & T \arrow[ur, hook, "\text{closed}"'] & \end{tikzcd}$$

Taking coordinate rings gives

$$\begin{tikzcd} \mathbb{C}[M] & & \mathbb{C}[x_1^{\pm}, \dots, x_s^{\pm}] \arrow[ll] \arrow[dl, two heads] \\ & \mathbb{C}[T] \arrow[ul, hook] & \end{tikzcd}$$

The upper horizontal map is given by $x_i \mapsto m_i$, so $\mathbb{C}[T] \cong \mathbb{C}[m_1^{\pm}, \dots, m_s^{\pm}] = \mathbb{C}[\mathbb{Z}A]$. Note that the m_i are viewed as characters of T in a way that for $t = (t_1, \dots, t_s) \in T \subseteq (\mathbb{C}^\times)^s$,

$$m_i(t) = x_i(t_1, \dots, t_s) = t_i.$$

Let us put M' to be the character group of T temporarily. Then $\mathbb{Z}A \leqslant M'$ (written additively). But $\mathbb{C}[\mathbb{Z}A] = \mathbb{C}[M']$, we must have $\mathbb{Z}A = M'$, as we desire.

10.5 Toric ideals Let $A = \{m_1, \dots, m_s\} \subseteq M$ and Y_A be the affine toric variety constructed in 10.4. The map Φ_A there gives a homomorphism on character groups :

$$\begin{aligned} \widehat{\Phi}_A : \mathbb{Z}^s &\longrightarrow M \\ e_i &\longmapsto m_i \end{aligned}$$

Let L be the kernel of this map, so that we have an exact sequence of abelian groups

$$0 \longrightarrow L \longrightarrow \mathbb{Z}^s \xrightarrow{\widehat{\Phi}_A} M$$

For an element $\ell = (\ell_1, \dots, \ell_s) \in L$, write

$$\ell_+ = \sum_{i:\ell_i>0} \ell_i e_i \text{ and } \ell_- = -\sum_{i:\ell_i<0} \ell_i e_i.$$

Then $\ell = \ell_+ - \ell_-$ with $\ell_\pm \in (\mathbb{Z}_{\geqslant 0})^s$. The binomial

$$X^{\ell_+} - X^{\ell_-} = \prod_{i:\ell_i>0} x_i^{\ell_i} - \prod_{i:\ell_i<0} x_i^{-\ell_i}$$

vanishes on the image of Φ_A, and hence on Y_A. Indeed, by definition of L we have $\sum_{i=1}^{s} \ell_i m_i = 0$, so $\sum_{i:\ell_i>0} \ell_i m_i = -\sum_{i:\ell_i<0} \ell_i m_i$, which implies our assertion. Hence

$$\langle X^{\ell_+} - X^{\ell_-} \mid \ell \in L \rangle \subseteq I(Y_A).$$

In fact, we have the equalities :

$$I(Y_A) = \langle X^{\ell_+} - X^{\ell_-} \mid \ell \in L \rangle = \langle X^a - X^b \mid a, b \in (\mathbb{Z}_{\geqslant 0})^s,\ a - b \in L \rangle.$$

The last equality is clear, so it remains to show the first is contained in the second. For this, fix an isomorphism $T_N \cong (\mathbb{C}^\times)^n$ and identify $M = \mathbb{Z}^n$; then the map $\Phi_A : (\mathbb{C}^\times)^n \to (\mathbb{C}^\times)^s$ has the form

$$\Phi_A(t_1, \ldots, t_n) = (t^{m_1}, \ldots, t^{m_s})$$

where for $a = (a_1, \ldots, a_n) \in \mathbb{Z}^n$, we write $t^a := t_1^{a_1} \cdots t_n^{a_n}$. Fix a monomial order $>$ on $\mathbb{C}[x_1, \ldots, x_s]$. If $I_L := \langle X^{\ell_+} - X^{\ell_-} \mid \ell \in L \rangle \subsetneq I(Y_A)$, pick $f \in I(Y_A) \backslash I_L$ with minimal leading monomial $x^\alpha = \prod_{i=1}^{s} x_i^{\alpha_i}$, where $\alpha = (\alpha_1, \ldots, \alpha_s) \in (\mathbb{Z}_{\geqslant 0})^s$. Rescaling, if necessary, we may assume x^α is the leading term of f (that is, its coefficient is one).

Since $f(t^{m_1}, \ldots, t^{m_s}) = 0$ identically, there must be some cancellation involving the term x^α. In other words, there must exist $\beta = (\beta_1, \ldots, \beta_s) \in (\mathbb{Z}_{\geqslant 0})^s$ such that

$$\prod_{i=1}^{s} t^{\alpha_i m_i} = \prod_{i=1}^{s} t^{\beta_i m_i}.$$

which implies $\sum_{i=1}^{s} \alpha_i m_i = \sum_{i=1}^{s} \beta_i m_i$. It follows that $\alpha - \beta \in L$ and $X^\alpha - X^\beta \in I_L$, so $f - (X^\alpha - X^\beta) \in I(Y_A) \backslash I_L$ has smaller leading monomial, a contradiction.

The ideal $I(Y_A)$ is then an example of *toric ideals*, a notion defined in the next paragraph.

10.6 Definition. Let $L \subseteq \mathbb{Z}^s$ be a lattice.

1. The ideal $I_L := \langle X^a - X^b \mid a, b \in (\mathbb{Z}_{\geqslant 0})^s,\ a - b \in L \rangle$ is called a **lattice ideal**.
2. A prime lattice ideal is called a **toric ideal**.

10.7 Proposition. An ideal $I \trianglelefteq \mathbb{C}[x_1, \ldots, x_s]$ is toric if and only if it is prime and is generated by some binomials of the form $X^\alpha - X^\beta$ $(\alpha, \beta \in (\mathbb{Z}_{\geqslant 0})^s)$.

Proof. The only if part is shown in 10.5. For the if part, say I is a prime ideal generated by $X^{\alpha_j} - X^{\beta_j}$ for some $\alpha_j,\ \beta_j \in (\mathbb{Z}_{\geqslant 0})^s$. Then $(1, \ldots, 1) \in V(I) \cap (\mathbb{C}^\times)^s$ and $V(I) \cap (\mathbb{C}^\times)^s$ is a subgroup of $(\mathbb{C}^\times)^s$. Indeed, if $t = (t_1, \ldots, t_s)$, $r = (r_1, \ldots, r_s)$ lies in the intersection, then

$$\prod_{i=1}^{s} t_i^{\alpha_{ji}} = \prod_{i=1}^{s} t_i^{\beta_{ji}} \text{ and } \prod_{i=1}^{s} r_i^{\alpha_{ji}} = \prod_{i=1}^{s} r_i^{\beta_{ji}}$$

so $\prod_{i=1}^{s} (r_i t_i)^{\alpha_{ji}} = \prod_{i=1}^{s} (r_i t_i)^{\beta_{ji}}$, implying $ts \in V(I) \cap (\mathbb{C}^\times)^s$. By 10.3.(ii), the intersection $T := V(I) \cap (\mathbb{C}^\times)^s$ is a torus, so $V(I)$ is an affine toric variety.

Denote by m_i the induced character on T from the projection to the i-th component of $(\mathbb{C}^\times)^s$. Then clearly $V(I) = Y_A$ with $A = \{m_1, \dots, m_s\} \subseteq M$ (here we take $T_N = T$, so M is the character group of T). Since I is a prime, by Hilbert's Nullstellensatz we have $I = I(Y_A)$, and hence I is toric by 10.5. $\square$

10.8 Affine monoids. By definition, a **monoid** is a set S equipped with an associative unital binary operation. We say a monoid S is an **affine monoid** if

(i) S is commutative and finitely generated, and

(ii) S is embedded in some lattice M of a finite dimensional vector space.

For an affine monoid S, the **monoid algebra** $\mathbb{C}[S]$ is a $\mathbb{C}$-algebra with a $\mathbb{C}$-basis S and with multiplication induced from the binary operation on S. If, say, S is embedded in M and M is the character group of a torus T_N, then

$$\mathbb{C}[S] = \left\{ \sum_{m \in S} a_m \chi^m \mid a_m \in \mathbb{C},\ a_m = 0 \text{ for all but finitely many } m \in S \right\}$$

with $\chi^m \cdot \chi^{m'} = \chi^{m+m'}$ for any $m, m' \in S$. If S is generated by $A = \{m_1, \dots, m_s\}$, then

$$\mathbb{C}[S] = \mathbb{C}[\chi^{m_1}, \dots, \chi^{m_s}].$$

10.9 Proposition. Let S be an affine monoid embedded in the lattice M.

1. The monoid algebra $\mathbb{C}[S]$ is an integral domain and is of finite type over $\mathbb{C}$.
2. The affine variety V corresponding to $\mathbb{C}[S]$ is an affine toric variety whose torus has character group $\mathbb{Z}S$, and if S is generated by some finite set $A \subseteq M$, then $V \cong Y_A$.

Proof. Say S is generated by $A = \{m_1, \dots, m_s\} \subseteq M$. Then $\mathbb{C}[S] = \mathbb{C}[\chi^{m_1}, \dots, \chi^{m_s}]$, so $\mathbb{C}[S]$ is finitely generated. Since $\mathbb{C}[S] \subseteq \mathbb{C}[M]$ and $\mathbb{C}[M] \cong \mathbb{C}[\mathbb{Z}^n] \cong \mathbb{C}[t_1^{\pm}, \dots, t_n^{\pm}]$ is an integral domain, so is the monoid algebra $\mathbb{C}[S]$.

There is a $\mathbb{C}$-algebra homomorphism

$$\begin{aligned} \pi : \mathbb{C}[x_1, \dots, x_s] &\longrightarrow \mathbb{C}[M] \\ x_i &\longmapsto \chi^{m_i} \end{aligned}$$

with image $\mathbb{C}[S]$. If M is the character group of the torus T_N, then π corresponds to the morphism $\Phi_A : T_N \to \mathbb{C}^s$ constructed in 10.4. The image of π corresponds to the closure of the image of Φ in $\mathbb{C}^s$, so the coordinate ring of Y_A is $\operatorname{Im} \pi = \mathbb{C}[S]$. The assertion on the character group is clear. $\square$

10.10 Let T_N be a torus with character group M. Then $\mathbb{C}[M]$ is the coordinate ring of T_N. We let T_N acts on $\mathbb{C}[M]$ by left translation : for $t \in T_N$ and $f : T_N \to \mathbb{C}$ in $\mathbb{C}[M]$, define $t.f \in \mathbb{C}[M]$ to be

$$t.f(p) = f(t^{-1}.p)$$

for any $p \in T_N$. We have the following result.

Lemma. Let $W \subseteq \mathbb{C}[M]$ be a subspace stable under the action of T_N. Then

$$W = \bigoplus_{\substack{m \in M \\ \chi^m \in W}} \mathbb{C}\chi^m$$

Proof. The right hand side is of course contained in W, so it suffices to prove the opposite inclusion. For $0 \neq f \in W \subseteq \mathbb{C}[M]$, write $f = \sum_{m \in M} a_m \chi^m$. Let $B = \mathrm{span}_{\mathbb{C}}\{\chi^m \mid m \in M \text{ with } a_m \neq 0\}$, which is a finite dimensional subspace of $\mathbb{C}[M]$. Since $t.\chi^m = \chi^m(t^{-1})\chi^m$, it follows that B is invariant under T_N-action, and hence so is $B \cap W$. By 10.3.(iii), $B \cap W$ is spanned by simultaneous eigenvectors of T_N. An easy computation shows that any simultaneous vector is a character of T_N, so $B \cap W$ is spanned by characters. It follows that $\chi^m \in W$ for those m with $a_m \neq 0$, and hence f lies in the right hand side. □

10.11 Theorem. For an affine variety V, TFAE :

(i) V is an affine toric variety.

(ii) $V = Y_A$ for a finite set A in some lattice (of a finite dimensional vector space).

(iii) $I(V)$ is a toric ideal.

(iv) The coordinate ring of V is $\mathbb{C}[S]$ for some affine monoid S.

Proof. (ii)⇔(iii) follows from 10.5 and (the proof of) Proposition 10.7, and (ii)⇔(iv)⇒(i) follows from 10.9.2. It remains to show (i)⇒(iv). Let V be an affine toric variety containing the torus T_N with character group M. The inclusion $T_N \subseteq V$ induces a homomorphism $\mathbb{C}[V] \to \mathbb{C}[M]$, which is injective for T_N is Zariski dense in V. Thus we can think of $\mathbb{C}[V]$ as a subalgebra of $\mathbb{C}[M]$. The T_N-action on V being given by a morphism $T_N \times V \to V$, it follows that $\mathbb{C}[V]$ is a T_N-invariant subspace of $\mathbb{C}[V]$. By Lemma 10.10

$$\mathbb{C}[V] = \bigoplus_{\substack{m \in M \\ \chi^m \in \mathbb{C}[V]}} \mathbb{C}\chi^m$$

so $\mathbb{C}[V] = \mathbb{C}[S]$ for the monoid $S = \{m \in M \mid \chi^m \in \mathbb{C}[V]\}$. Finally, since $\mathbb{C}[V]$ is finitely generated, it follows that S is a finitely generated monoid, and hence an affine monoid. □

10.12 Affine toric varieties from rational convex polyhedral cones. Let N be a free $\mathbb{Z}$-module of rank n. Let $\sigma \subseteq N_{\mathbb{R}}$ be an N-rational convex polyhedral cone and set $S_\sigma = \sigma^\vee \cap M$, where M is the lattice in $(N_{\mathbb{R}})^\vee$ dual to N with respect to the evaluation pairing $(N_{\mathbb{R}})^\vee \times N_{\mathbb{R}} \to \mathbb{R}$. By Gordan's lemma, the set S_σ is an affine monoid so by 10.11 it gives an affine toric variety, which we denote by U_σ.

10.13 Theorem. Retain the notation in 10.12. TFAE :

(i) $\dim U_\sigma = n$.

(ii) The torus of U_σ is $T_N = N \otimes_{\mathbb{Z}} \mathbb{C}^\times$.

(iii) σ is strongly convex.

Proof. We see in 10.9 that the character lattice of the torus of U_σ is $\mathbb{Z}S_\sigma \subseteq M$. To proceed, we first show the quotient group $M/\mathbb{Z}S_\sigma$ is torsion-free. Let $m \in M$ and suppose $km \in \mathbb{Z}S_\sigma$ for some $k \in \mathbb{Z}_{\geqslant 1}$. Then $km = m_1 - m_2$ for some $m_1, m_2 \in S_\sigma$. Since $\sigma^\vee$ is convex,

$$m + m_2 = \frac{1}{k}m_1 + \frac{k-1}{k}m_2 \in \sigma^\vee$$

and thus $m = (m + m_2) - m_2 \in \mathbb{Z}S_\sigma$.

Since $M/\mathbb{Z}S_\sigma$ is finitely generated and torsion-free, it follows that the torus of U_σ is T_N if and only if $\mathbb{Z}S_\sigma = M$, if and only if $\operatorname{rank}\mathbb{Z}S_\sigma = n$, or $\dim U_\sigma = n$, proving (i)$\Leftrightarrow$(ii). Since σ is strongly convex if and only if $\dim \sigma^\vee = n$ by Lemma 9.11.(d), it remains to show $\operatorname{rank}\mathbb{Z}S_\sigma = n$ if and only if $\dim \sigma^\vee = n$. But $\sigma^\vee$ is rational, we see $\sigma^\vee = \operatorname{cone}(\sigma^\vee \cap M) = \operatorname{cone}(S_\sigma)$, and consequently

$$\operatorname{rank}\mathbb{Z}S_\sigma = \dim \mathbb{R}S_\sigma = \dim \operatorname{cone}(S_\sigma) = \dim \sigma^\vee.$$

□

10.14 Definition. Let A, B be two monoids. A map $\varphi : A \to B$ is called a **monoid homomorphism** if $\varphi(1) = 1$ and $\varphi(xy) = \varphi(x)\varphi(y)$ for any $x, y \in A$.

10.15 Points of affine toric varieties. Let S be an affine monoid and let V be the affine toric variety with coordinate ring $\mathbb{C}[S]$ (10.9). By Nullstellensatz, there is a bijection $V \cong \operatorname{mSpec}\mathbb{C}[S]$. In the case of an affine toric variety V, there is a more useful bijection

$$\begin{aligned} V &\longrightarrow \operatorname{Hom}_{\mathbf{Monoid}}(S, \mathbb{C}) \\ p &\longmapsto \gamma_p : m \mapsto \chi^m(p) \end{aligned}$$

We describe its inverse as follows. Let $\gamma \in \operatorname{Hom}_{\mathbf{Monoid}}(S, \mathbb{C})$. It extends to a nontrivial $\mathbb{C}$-algebra homomorphism $\gamma : \mathbb{C}[S] \to \mathbb{C}$, so $\ker\gamma \in \operatorname{mSpec}\mathbb{C}[S]$, which corresponds to a point $p_\gamma \in V$ in turn.

We can write down p_γ more explicitly in terms of the embedding $V = Y_A \subseteq \mathbb{A}^s_{\mathbb{C}}$, where $A = \{m_1, \ldots, m_s\}$ is a finite generating set of S. Put $q = (\gamma(m_1), \ldots, \gamma(m_s)) \in \mathbb{C}^s$. We claim $q = p_\gamma$. First we must show $q \in Y_A$, and by (10.5) it amounts to showing that all the binomials $X^\alpha - X^\beta$ with $\alpha, \beta \in (\mathbb{Z}_{\geqslant 0})^s$ and $\sum_{i=1}^s \alpha_i m_i = \sum_{i=1}^s \beta_i m_i$ vanishes at q. This is clear, as

$$\prod_{i=1}^s \gamma(m_i)^{\alpha_i} = \gamma\left(\sum_{i=1}^s \alpha_i m_i\right) = \gamma\left(\sum_{i=1}^s \beta_i m_i\right) = \prod_{i=1}^s \gamma(m_i)^{\beta_i}$$

Next, we prove $q = p_\gamma$ by showing $\ker\gamma = \ker\gamma_q$. Let $f \in \mathbb{C}[S] = \mathbb{C}[\chi^{m_1}, \ldots, \chi^{m_s}]$. Write $f = \sum_{\alpha \in (\mathbb{Z}_{\geqslant 0})^s} c_\alpha X^\alpha$; then

$$\gamma(f) = \sum_{\alpha \in (\mathbb{Z}_{\geqslant 0})^s} c_\alpha \gamma(X^\alpha) = \sum_{\alpha \in (\mathbb{Z}_{\geqslant 0})^s} c_\alpha \prod_{i=1}^s \gamma(m_i)^{\alpha_i} = f(q)$$

This implies our contention that $\ker\gamma = \ker\gamma_q$. In fact, it tells more; for any $m \in S$, one has

$$\gamma(m) = \chi^m(p_\gamma),$$

and from this we easily see that $\gamma \mapsto p_\gamma$ is inverse to $p \mapsto \gamma_p$.

Suppose T is the torus of V with character group M. In (10.2) we see there is a bijection

$$\begin{aligned} T &\longrightarrow \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \\ t &\longmapsto [m \mapsto m(t) = \chi^m(t)] \end{aligned}$$

Observe that $\mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) = \mathrm{Hom}_{\mathbf{Monoid}}(M, \mathbb{C})$, so this isomorphism is compatible with the bijection above, i.e., it fits into the commutative diagram

$$\begin{tikzcd} V \arrow[r] & \mathrm{Hom}_{\mathbf{Monoid}}(S, \mathbb{C}) \\ T \arrow[u, "\text{inclusion}"] \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \arrow[u, "\text{restriction}"'] \end{tikzcd}$$

10.16 Let T_N be a torus with character group M. If $A = \{m_1, \ldots, m_s\}$ is a finite set in M, then we have an affine toric variety $V = Y_A \subseteq \mathbb{A}^s_{\mathbb{C}}$. If $p = (p_1, \ldots, p_s) \in Y_A \subseteq \mathbb{A}^s_{\mathbb{C}}$, from (10.4) we see the action of $t \in T_N$ on p is

$$t.p = (m_1(t), \ldots, m_s(t))(p_1, \ldots, p_s) = (m_1(t)p_1, \ldots, m_s(t)p_s).$$

We can also write this action in terms of monoid homomorphism (10.15). The action maps

$$\begin{tikzcd} T_N \times V \arrow[r] & V \\ T_N \times T_N \arrow[u] \arrow[r] & T_N \arrow[u] \end{tikzcd}$$

corresponds to the diagram (here $S = \mathbb{Z}_{\geqslant 0}A \subseteq M$)

$$\begin{tikzcd} \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \times \mathrm{Hom}_{\mathbf{Monoid}}(S_\sigma, \mathbb{C}) \arrow[r] & \mathrm{Hom}_{\mathbf{Monoid}}(S, \mathbb{C}) \\ \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \times \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \arrow[u] \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{C}^\times) \arrow[u] \end{tikzcd}$$

where the horizontal maps are given by multiplication on $\mathbb{C}$.

10.17 Proposition. Let V be an affine toric variety with coordinate ring $\mathbb{C}[S]$ for some affine monoid.

(a) The torus action on V has a fixed point if and only if S is **pointed**, i.e., $S \cap (-S) = \{0\}$, in which case the unique fixed point is given by the monoid homomorphism $\gamma_0 : S \to \mathbb{C}$ defined by

$$m \mapsto \begin{cases} 1 & \text{, if } m = 0 \\ 0 & \text{, if } m \neq 0 \end{cases}$$

(b) If $V = Y_A \subseteq \mathbb{A}^s_{\mathbb{C}}$ for some $A \subseteq S \backslash \{0\}$, then the torus action has a fixed point if and only if $0 \in Y_A$, in which case the unique fixed point is 0.

Proof. Let $p \in V$ be a point and let $\gamma : S \to \mathbb{C}$ be the corresponding monoid homomorphism. Then by (10.16), p is a fixed point if and only if $\gamma(m) = \chi^m(t)\gamma(m)$ for every $t \in T_N$ and $m \in S$. This equation is satisfied for $m = 0$, for $\gamma(0) = 1$. If $m \neq 0$, we can find $t \in T_N$ such that $\chi^m(t) \neq 1$ so that $\gamma(m) = 0$. Thus if a fixed point exists, it is unique and is given by the homomorphism γ_0 described in the proposition. To finish the proof of (a), it suffices to note that γ_0 is a monoid homomorphism if and only if S is pointed. Indeed, if γ_0 is a homomorphism, then for $m, n \in S$ such that $m = -n \in S \cap (-S)$, we have $m + n = 0$ so that $\gamma(m)\gamma(n) = 1$, and thus $\gamma(m) \neq 0 \neq \gamma(n)$. By definition we have $m = n = 0$. Conversely, if $S \cap (-S) \neq \{0\}$, say $m = -n \in (S \cap (-S)) \backslash \{0\}$, then $\gamma(m + n) = 1 \neq 0 = \gamma(m)\gamma(n)$.

For (b), 0 is clearly a fixed point under the $(\mathbb{C}^\times)^s$-action, so if $0 \in Y_A$, 0 is a fixed point under the T_N-action. Conversely, if the torus action has a fixed point, by (a) the fixed point is given by γ_0, and from the proof of (10.16) we see γ_0 corresponds to the origin 0 of $\mathbb{A}^s_{\mathbb{C}}$. Thus $0 \in Y_A$; indeed, the corresponding point is $(\gamma_0(m_1), \ldots, \gamma_0(m_s))$ if we write $A = \{m_i\}_{i=1}^s$, and all components vanish since $0 \notin A$. □

10.18 Corollary. Let U_σ be the affine toric variety of a strongly convex rational polyhedral cone $\sigma \subseteq N_{\mathbb{R}}$. Then the torus action on U_σ has a fixed point if and only if $\dim \sigma = \dim_{\mathbb{R}} N_{\mathbb{R}}$, in which case the fixed point is unique and is given by the maximal ideal

$$\langle \chi^m \mid m \in S_\sigma \backslash \{0\} \rangle \subseteq \mathbb{C}[S_\sigma]$$

where as usual $S_\sigma = \sigma^\vee \cap M$.

Proof. By Proposition 10.17, The torus action on U_σ has a fixed point if and only if S_σ is pointed. But

$$S_\sigma \cap (-S_\sigma) = \sigma^\vee \cap (-\sigma)^\vee \cap M$$

and since $\sigma^\vee \cap (-\sigma)^\cap$ is rational, we see S_σ is pointed if and only if $\sigma^\vee \cap (-\sigma)^\vee = \{0\}$ (if $\sigma^\vee \cap (-\sigma)^\vee \neq \{0\}$, it would contain a rational ray). By Lemma 9.11, this is equivalent of saying that $\dim \sigma = \dim_{\mathbb{R}} N_{\mathbb{R}}$. Finally, it is straightforward to verify that maximal ideal is the kernel of $\gamma_0 : \mathbb{C}[S] \to \mathbb{C}$, where γ_0 is as in Proposition 10.17.(a). □

10.19 Theorem. Let V be an affine toric variety with torus T_N. TFAE :

(i) V is normal.

(ii) $V = \operatorname{mSpec} \mathbb{C}[S]$ for some saturated affine monoid S.

(iii) $V = \operatorname{mSpec} \mathbb{C}[S_\sigma]$, where $S_\sigma = \sigma^\vee \cap M$ and $\sigma \subseteq N_\mathbb{R}$ is a strongly convex rational polyhedral cone.

10.20 Definition. Let V be a finite dimensional real vector space with a lattice N and let σ be a strongly convex N-rational polyhedral cone.

1. σ is called **smooth/regular** if its ray generators of the edges forms a subset of some $\mathbb{Z}$-basis of N.
2. σ is called **simplicial** if its ray generators of the edges are linearly independent over $\mathbb{R}$.

10.21 Let M be the dual lattice of N in $V^\vee$. Let σ be a strongly convex N-rational polyhedral cone of maximal dimension. Then $\sigma^\vee$ is also strongly convex.

Lemma. The cone σ is smooth if and only if its dual $\sigma^\vee$ is smooth.

Proof. In the proof of Corollary 9.11.1, we saw that the ray generators of σ is exactly the minimal generating set of the cone σ; denote by $\{e_1, \dots, e_n\}$ the minimal generating set. If σ is smooth, then $\{e_1, \dots, e_n\}$ is a $\mathbb{Z}$-basis of N. We have $\sigma^\vee = \operatorname{cone}\{e_1^*, \dots, e_n^*\}$, where the e_j^* are the dual basis of the e_i, and $\{e_j^*\}$ is a $\mathbb{Z}$-basis of M. This must be a minimal generating set of $\sigma^\vee$, so $\{e_j^*\}$ consists of ray generators of $\sigma^\vee$. This proves $\sigma^\vee$ is smooth. The other implication follows from $\sigma = (\sigma^\vee)^\vee$. $\square$

10.22 Smooth affine toric varieties. Let the notation be as in 10.12. Then the variety U_σ is smooth if and only if σ is smooth. Furthermore, all smooth affine toric varieties are of this form.

10.23 Faces and affine open subsets. Let N be a free $\mathbb{Z}$-module of rank n and M be the lattice in $(N_\mathbb{R})^\vee$ dual to N with respect to the evaluation pairing. Let $\sigma \subseteq N_\mathbb{R}$ be a strongly convex rational polyhedral cone. Let τ be a face of σ and write $\tau = \sigma \cap m^\perp$ for some $m \in M \cap \sigma^\vee$. Since $\tau \subseteq \sigma$, we have $S_\sigma \subseteq S_\tau = \tau^\vee \cap M$. In fact, we have

$$S_\tau = S_\sigma + \mathbb{Z}_{\geqslant 0}(-m)$$

To see this, since $\langle m, u\rangle = 0$ for any $u \in \tau$, we have $\pm m \in \tau^\vee$, and thus the containment $\supseteq$. For the opposite inclusion, take a finite set $S \subseteq N$ with $\operatorname{cone}(S) = \sigma$ and pick $m' \in S_\tau$. If we set

$$C = \max_{u \in S} |\langle m', u\rangle| \in \mathbb{Z}_{\geqslant 0},$$

then $m' + Cm \in S_\sigma$. Indeed, if $u \in \tau$, then $\langle m', u\rangle \geqslant 0 \leqslant \langle m, u\rangle$, and if $u \in \sigma \backslash \tau$, then $\langle m, u\rangle \geqslant 1$, so $\langle m', u\rangle + \langle Cm, u\rangle \geqslant 0$. This finishes the proof.

From this equality, we obtain $\mathbb{C}[S_\tau] = \mathbb{C}[S_\sigma, \chi^{\pm m}] = \mathbb{C}[S_\sigma]_{\chi^m}$. This implies we may view U_τ as the affine open set $D(\chi^m)$ of U_σ.

10.2 Projective toric varieties

10.24 Affine pieces of projective toric varieties.

10.25 Projective toric variety from a very ample polytope. Let T_N be a torus with character group M and cocharacter group N, and let $P \subseteq M_{\mathbb{R}}$ be a full dimensional very ample M-rational convex polytope. Put $n = \dim P$. If we write $P \cap M = \{m_1, \dots, m_s\}$, then we have a projective toric variety $X_{P\cap M}$, which is by definition the Zariski closure of the image of the map $T_N \to \mathbb{P}^{s-1}$ given by

$$t \longmapsto [m_1(t) : \cdots : m_s(t)] \in \mathbb{P}^{s-1}.$$

For any $m_i \in P \cap M$, let $S_i = \mathbb{Z}_{\geqslant 0}(P \cap M - m_i)$ be the monoid generated by $\{m - m_i \mid m \in P \cap M\}$. Fix a homogeneous coordinates $x_1, \dots, x_s$ of $\mathbb{P}^{s-1}$, and let $U_i = \mathbb{P}^{s-1} \backslash V(x_i) \cong \mathbb{C}^{s-1}$. In 10.24, we saw that intersection $X_{P\cap M} \cap U_i$ is an affine subvariety of U_i, with coordinate ring

$$\mathbb{C}[X_{P\cap M} \cap U_i] \cong \mathbb{C}[S_i]$$

and

$$X_{P\cap M} = \bigcup_{m_i \text{: vertex of } P} X_{P\cap M} \cap U_i$$

For such a projective toric variety from a very ample polytope, we have the following results.

(i) For each vertex m_i of P, the affine piece $X_{P\cap M} \cap U_i$ is the affine toric variety with coordinate ring $\mathbb{C}[\sigma_i^\vee \cap M]$, where

$$\sigma_i = \mathrm{cone}(P \cap M - m_i)^\vee \subseteq N_{\mathbb{R}}$$

is a strongly convex rational polyhedral cone with dimension $n = \dim P$.

(ii) The torus of $X_{P\cap M}$ has character group M, and hence is T_N.

Proof.

(i) Since m_i is a vertex, $\{0\}$ is a face of $\mathrm{cone}(P \cap M - m_i)$, so $\mathrm{cone}(P \cap M - m_i)$ is strongly convex by Lemma 9.11.(a). Clearly, $\mathrm{cone}(P\cap M - m_i)$ has the same dimension as P, so $\sigma_i = \mathrm{cone}(P\cap M - m_i)^\vee$ is strongly convex by Lemma 9.11.(d). Thus both $\mathrm{cone}(P \cap M - m_i)$ and σ_i are strongly convex rational polyhedral cones of dimension n.

We have $S_i \subseteq \mathrm{cone}(P \cap M - m_i) \cap M = \sigma_i^\vee \cap M$. Since P is very ample, by definition S_i is saturated (in M). In the proof of Theorem 10.19), we see this implies $S_i = \sigma_i^\vee \cap M$, so $\mathbb{C}[S_i] = \mathbb{C}[\sigma_i^\vee \cap M]$

(ii) From Theorem 10.12 (applicable as σ_i is strongly convex) we see the torus of $U_{\sigma_i} = X_{P\cap M} \cap U_i$ is T_N. Since $T_N \subseteq U_{\sigma_i} \subseteq X_{P\cap M}$, it follows that T_N is also the torus of $X_{P\cap M}$.

□

10.26 Intersection of affine pieces of $X_{P\cap M}$. Let T_N, N, M, P be as in 10.25, and put $s = \#(P \cap M)$. We can construct the normal fan Σ_P of the polytope P as in Theorem 9.24. If $X_{P\cap M} \cap U_v$ is the affine piece of $X_{P\cap M}$ corresponding to a vertex v of P, then 10.25 shows that $X_{P\cap M} \cap U_v$ is the affine toric variety of the cone σ_v in the normal fan Σ_P.

Next we study the intersection of affine pieces. Suppose $v \neq w$ are two vertices of P. We want to describe the coordinate ring of $X_{P\cap M} \cap U_v \cap U_w$. To this end, let Q be the smallest face in P containing v and w. The result is that

$$X_{P\cap M} \cap U_v \cap U_w = U_{\sigma_Q}$$

where $\sigma_Q \in \Sigma_P$, i.e., the coordinate ring of the intersection of $\mathbb{C}[\sigma_Q^\vee \cap M]$.

Proof. From 10.24 we see the coordinate ring of $X_{P\cap M} \cap U_v \cap U_w$ is $\mathbb{C}[\sigma_v^\vee \cap M]_{\chi^{w-v}} = \mathbb{C}[\sigma_w^\vee \cap M]_{\chi^{v-w}}$, so we only need to show

$$\mathbb{C}[\sigma_v^\vee \cap M]_{\chi^{w-v}} = \mathbb{C}[\sigma_Q^\vee \cap M]$$

But $w - v \in \mathrm{cone}(P \cap M - v) = \sigma_v^\vee$, from 10.23 we see

$$\mathbb{C}[\sigma_v^\vee \cap M]_{\chi^{w-v}} = \mathbb{C}[U_\tau] = \mathbb{C}[\tau^\vee \cap M]$$

where $\tau = (w - v)^\perp \cap \sigma_v$ is a face of σ_v. Thus we only need to show $\tau = \sigma_Q$, or

$$(w - v)^\perp \cap \sigma_v = \sigma_Q = \sigma_v \cap \sigma_w$$

where the last equality is by Lemma 9.24.(c). Let $u \in (w - v)^\perp \cap \sigma_v$. If $u \neq 0$, there exists $b \in \mathbb{R}$ such that $H_{u,b}$ is a supporting hyperplane of P, so by Lemma 9.25, $u \in \sigma_v$ implies $v \in H_{u,b} \cap P$. As $u \in (w - v)^\perp$, we obtain $w \in H_{u,b}$, and using Lemma 9.25 again we see $u \in \sigma_w$. Conversely, let $u \in \sigma_v \cap \sigma_w$. If $u \neq 0$, pick b as above; then by Lemma 9.25, $u \in \sigma_v \cap \sigma_w$ implies $v, w \in H_{u,b} \cap P$, which implies $u \in (w - v)^\perp$ in turn. □

In this proof, we see

$$\mathbb{C}[U_v]_{\chi^{w-v}} = \mathbb{C}[U_\tau] = \mathbb{C}[U_w]_{\chi^{v-w}}$$

for any vertices $v \neq w$. Thus the toric variety $X_{P\cap M}$ can be glued from the affine toric varieties given by $\sigma_v \in \Sigma_P$ along those given by $\sigma_\tau \in \Sigma_P$. Concisely, the normal fan Σ_P completely determines the toric variety $X_{P\cap M}$.

10.27 Projective toric variety from a polytope. Let N and M be as in 10.12, and let $P \subseteq M_\mathbb{R}$ be a full dimensional lattice polytope. By Corollary 9.23.1, for $k \gg 0$, the multiple kP of P becomes very ample, so from 10.25 we can construct the projective variety $X_{kP\cap M}$. In 10.26 we see $X_{kP\cap M}$ is completely determined by the normal fan Σ_{kP}. By Lemma 9.28, $\Sigma_{kP} = \Sigma_P$ for any $k \geqslant 1$, so the resulting toric variety $X_{kP\cap M}$ are all isomorphic as $\mathbb{C}$-varieties. This suggests us to put

$$X_P := X_{kP\cap M}$$

for any $k \gg 0$ such that kP is very ample. This is the toric variety determined by the polytope P.

10.28 Example; toric variety from the standard simplex Δ_n. By definition, $\Delta_n = \mathrm{cone}\{0, e_1, \dots, e_n\} \subseteq \mathbb{R}^n$. Clearly, $k\Delta_n$ is a normal, and hence very ample by Lemma 9.23, lattice polytope for any $k \in \mathbb{Z}_{\geqslant 1}$. We then can construct X_{Δ_n} using $k\Delta_n$ for any integer $k \geqslant 1$. The lattice points of $k\Delta_n$ are all the integer points $(x_1, \dots, x_n) \in (\mathbb{Z}_{\geqslant 0})^n$ with $x_1 + \cdots + x_n \leqslant k$, which corresponds to the monomials in $\mathbb{C}[x_1, \dots, x_n]$ with total degree $\leqslant k$. These monomials are in number $s_k = \binom{n+k}{k}$, so each kP gives a projective embedding $X_{\Delta_n} \subseteq \mathbb{P}^{s_k - 1}$. When $k = 1$, $\Delta_n \cap \mathbb{Z}^n = \{0, e_1, \dots, e_n\}$, which implies

$$X_{\Delta_n} = \mathbb{P}^n.$$

From the definition of $X_{k\Delta_n \cap \mathbb{Z}^n}$, for any integer $k \geqslant 1$, the embedding $X_{\Delta_n} \subseteq \mathbb{P}^{s_k - 1}$ is given by the morphism

$$\nu_k : \mathbb{P}^n \longrightarrow \mathbb{P}^{s_k - 1}$$

defined using all the monomial of total degree $k \leqslant 1$ in $\mathbb{C}[x_0, x_1, \dots, x_n]$. The morphism ν_k is really an embedding, usually called the **Veronese embedding**.

10.29 Example; rational normal scrolls. For $a, b \in \mathbb{Z}$ with $1 \leqslant a \leqslant b$, define the polygon

$$P_{a,b} = \mathrm{conv}\{0, ae_1, e_2, be_1 + e_2\}.$$

The toric variety of $P_{a,b}$ is denoted by $S_{a,b}$, and is called the **rational normal scroll**. By Corollary 9.23.1, $P_{a,b}$ is very ample, so $S_{a,b} = X_{P_{a,b} \cap \mathbb{Z}^2}$ is the closure of the morphism

$$\begin{aligned} (\mathbb{C}^\times)^2 &\longrightarrow \mathbb{P}^{a+b+1} \\ (s, t) &\longmapsto [1 : s : \cdots : s^a : t : st : \cdots : s^b t] \end{aligned}$$

To describe the image, we write this morphism as

$$\begin{aligned} \mathbb{C} \times \mathbb{P}^1 &\longrightarrow \mathbb{P}^{a+b+1} \\ (s, [\lambda : \mu]) &\longmapsto [\lambda : \lambda s : \cdots : \lambda s^a : \mu : s\mu : \cdots : s^b \mu] \end{aligned}$$

When $[\lambda : \mu] = [1 : 0]$, it maps to a rational normal curve C_a, and when $[\lambda : \mu] = [0 : 1]$, it maps to another rational normal curve C_b. One thinks C_a and C_b as edges of $S_{a,b}$. By fixing an $s \in \mathbb{C}$ we obtain points on C_a and C_b respectively, and varying $[\lambda : \mu] \in \mathbb{P}^1$ gives a segment connecting the two points.

Note that the normal fan of $P_{a,b}$ only depends on the difference $b - a$. Thus for any integer $r \geqslant 1$, we have

$$X_{P_{1,r}} = X_{P_{2,r+1}} = X_{P_{3,r+2}} = \cdots$$

as $\mathbb{C}$-varieties. If we consider $S_{a,b}$ as a projective variety embedded in $\mathbb{P}^{a+b+1}$, the roles of integers $a \leqslant b$ can be seen from the defining equations of $S_{a,b}$: if we let $x_0, \dots, x_a, y_0, \dots, y_b$ be the homogeneous

coordinates of $\mathbb{P}^{a+b+1}$, then $I(S_{a,b})$ is generated by the 2×2 minors of the matrix

$$\begin{pmatrix} x_0 & x_1 & \cdots & x_{a-1} & y_0 & y_1 & \cdots & y_{b-1} \\ x_1 & x_2 & \cdots & x_a & y_1 & y_2 & \cdots & y_b \end{pmatrix}$$

10.30 Normality. Let $P \subseteq M_{\mathbb{R}}$ be a full dimensional lattice polytope. Then

(i) X_P is normal.

(ii) X_P is projectively normal under the embedding given by kP if and only if kP is normal.

Proof.

1. This follows from Theorem 10.19.(ii) and the definition of very ample.
2. The affine cone of $X_{kP \cap M}$ is $Y_{((kP)\cap M)\times\{1\}}$, and by Theorem 10.19, $Y_{kP\cap M\times\{1\}}$ is normal if and only if the monoid $S := \mathbb{Z}_{\geqslant 0}(((kP)\cap M)\times\{1\})$ is saturated in $M \times \mathbb{Z}$. But $((kP)\cap M)\times\{1\}$ generates the cone $C(kP)$, so it is equivalent to saying that $((kP) \cap M) \times \{1\}$ generates the monoid $C(kP) \cap (M \times \mathbb{Z})$: if $(m, \ell) \in M \times \mathbb{Z}$ and $(nm, n\ell) \in S$, then $nm \in n\ell((kP) \cap M)$ so that $m \in \ell((kP) \cap M)$, or $(m, \ell) \in S$. Conversely, for $x \in C(kP) \cap (M \times \mathbb{Z})$, let $kP \cap M = \{m_1, \ldots, m_s\}$ and write $x = \left(\sum_{i=1}^{s} a_i m_i, \sum_{i=1}^{s} a_i\right)$ for some $a_i \geqslant 0$. Since $x \in M \times \mathbb{Z}$, every a_s is rational, so there exists $r \in \mathbb{Z}_{\geqslant 1}$ such that $ra_s \in \mathbb{Z}_{\geqslant 0}$. But this implies $rx \in S$, and thus $x \in S$ as S is saturated. Finally, by Lemma 9.21, the last equivalent statement is the same as saying kP is normal.

□

10.31 Definition. Let $P \subseteq M_{\mathbb{R}}$ be a lattice polytope.

1. Given a vertex v of P and an edge E containing v, denote by w_E the unique lattice point on E such that $w_E - v$ is the ray generator of the ray $\mathrm{cone}(E - v)$.
2. P is called **smooth/unimodular** if for every vertex v, the vectors $w_E - v$, where E runs over all edges containing v, form a subset of some $\mathbb{Z}$-basis of M. In particular, if $\dim P = \dim M_{\mathbb{R}}$, then $\{w_E - v\}_E$ forms a $\mathbb{Z}$-basis of M.

10.32 Smoothness. Let $P \subseteq M_{\mathbb{R}}$ be a full dimensional lattice polytope. TFAE :

(i) X_P is a smooth projective variety.

(ii) Σ_P is a smooth fan, in the sense that every cone in Σ_P is smooth (10.20).

(iii) P is a smooth polytope.

Proof. Being smooth is a local property, so X_P is smooth if and only if for any vertex v of P, the affine toric variety U_{σ_v} is smooth. By 10.22, this is true if and only if for any vertex v of P, the cone $\sigma_v \in \Sigma_P$ is smooth. Clearly faces of a smooth cone are again smooth, so by Lemma 9.26.(a) this is the same as saying that every cone in Σ_P is smooth. Thus (i)$\Leftrightarrow$(ii).

For (ii)$\Leftrightarrow$(iii), since σ_v is of maximal dimension, in 10.21 we see σ_v is smooth if and only if $C_v = \sigma_v^\vee$ is smooth. Recall that $C_v = \mathrm{cone}(P \cap M - v)$, so the ray generators of C_v are exactly $w_E - v$, where E runs over all edges of P containing v. Thus P is smooth if and only if C_v, and hence σ_v, is smooth for every vertex v. $\square$

10.33 Lemma. Every smooth full dimensional lattice polytope $P \subseteq M_{\mathbb{R}}$ is very ample.

Proof. This follows from the fact

$$\mathbb{Z}_{\geqslant 0}\{w_E - v\}_{v \in E} = \mathbb{Z}_{\geqslant 0}(P \cap M - v)$$

for any vertex v. $\square$

10.34 Let V_1, V_2 be two finite dimensional real vector spaces and let $P_i \subseteq V_i$ $(i = 1, 2)$ be a convex polytope. Consider the cartesian product $P_1 \times P_2 \subseteq V_1 \times V_2$. If $P_i = \mathrm{conv}(S_i)$ for some finite subset $S_i \subseteq V_i$, then

$$P_1 \times P_2 = \mathrm{conv}(S_1 \times S_2) \subseteq V_1 \times V_2.$$

In particular, this shows $P_1 \times P_2$ is also a convex polytope.

Now let S_i be the set of vertices of P_i. The above equality shows that the vertex of $P_1 \times P_2$ is contained in $S_1 \times S_2$. Conversely, if $v_i \in S_i$ $(i = 1, 2)$, then $v_i = P_i \cap H_{u_i, b_i}$ for some supporting hyperplane H_{u_i, b_i} of P_i. Then $(v_1, v_2) \in H_{u_1, b_2} \times H_{u_2, b_2} \subseteq H_{(u_1, u_2), b_1 + b_2}$ and $P_1 \times P_2 \subseteq H^+_{u_1, b_2} \times H^+_{u_2, b_2} \subseteq H^+_{(u_1, u_2), b_1 + b_2}$, so $H_{(u_1, u_2), b_1 + b_2}$ is a supporting hyperplane of $P_1 \times P_2$ containing (v_1, v_2). If $(x, y) \in (P_1 \times P_2) \cap H_{(u_1, u_2), b_1 + b_2}$, then $b_1 + b_2 = \langle u_1, x \rangle + \langle u_2, y \rangle$. But $P_i \subseteq H^+_{u_i, b_i}$ $(i = 1, 2)$, so $\langle u_1, x \rangle \geqslant b_1$ and $\langle u_2, y \rangle \geqslant b_2$, which forces these two to be equalities in turn. Thus $(x, y) = (v_1, v_2)$, so that $\{(v_1, v_2)\} = (P_1 \times P_2) \cap H_{(u_1, u_2), b_1 + b_2}$, i.e., (v_1, v_2) is a vertex of $P_1 \times P_2$.

10.35 Let $i = 1, 2$ and N_i be a free $\mathbb{Z}$-module of rank n_i. Let M_i be the dual lattice of N_i in $(N_i)_{\mathbb{R}}$ with respect to the evaluation pairing. Let $P_i \subseteq (M_i)_{\mathbb{R}}$ be a full dimensional lattice polytope. The product $P_1 \times P_2$ is then a full dimensional lattice polytope in $(M_1 \times M_2)_{\mathbb{R}} = (M_1)_{\mathbb{R}} \times (M_2)_{\mathbb{R}}$, so we may consider the normal fan $\Sigma_{P_1 \times P_2}$. A result is that

$$\Sigma_{P_1 \times P_2} = \Sigma_{P_1} \times \Sigma_{P_2}.$$

To see this, by virtue of Lemma 9.26, we only need to show :

(i) For any vertices $v_i \in P_i$ $(i = 1, 2)$, we have $\sigma_{(v_1, v_2)} = \sigma_{v_1} \times \sigma_{v_2}$.

(ii) Suppose $\sigma_i \subseteq (N_i)_{\mathbb{R}}$ are convex polyhedral cones and $\tau_i \subseteq \sigma_i$ $(i = 1, 2)$ are faces. Then $\tau_1 \times \tau_2$ is a face of $\sigma_1 \times \sigma_2$, and all faces of $\sigma_1 \times \sigma_2$ arise in this way.

(ii) is clear, and to show (i), we prove $C_{(v_1, v_2)} = C_{v_1} \times C_{v_2}$. This follows from $\mathrm{cone}(S_1) \times \mathrm{cone}(S_2) = \mathrm{cone}(S_1 \times S_2)$ for finite subsets $S_i \subseteq (M_i)_{\mathbb{R}}$.

10.36 Let $P_i \subseteq (M_i)_{\mathbb{R}}$ be full dimensional lattice polytopes of dimension n_i $(i = 1, 2)$. Put $s_i = \#(P_i \cap M_i)$. Replacing P_i by their multiples, we assume the P_i are very ample, so there are projective embeddings $X_{P_i} \to \mathbb{P}^{s_i - 1}$. We obtain a projective embedding by the composition

$$X_{P_1} \times X_{P_2} \longrightarrow \mathbb{P}^{s_1-1} \times \mathbb{P}^{s_2-1} \xrightarrow{\text{Segre embedding}} \mathbb{P}^{s_1 s_2 - 1}$$

Theorem.

(i) $P_1 \times P_2 \subseteq (M_1 \times M_2)_{\mathbb{R}}$ is very ample with lattice points

$$(P_1 \times P_2) \cap (M_1 \times M_2) = (P_1 \cap M_1) \times (P_2 \cap M_2),$$

which has cardinality $s = s_1 s_2$.

(ii) The image of the embedding $X_{P_1 \times P_2} \subseteq \mathbb{P}^{s-1}$ determined by $P_1 \times P_2$ equals that of $X_{P_1} \times X_{P_2} \to \mathbb{P}^{s-1}$ above.

(iii) $X_{P_1 \times P_2} \cong X_{P_1} \times X_{P_2}$.

Proof. (i) is clear, and (iii) is an immediate consequence of (ii). For (ii), let T_{N_i} be the torus of X_{P_i}. The torus $T_{N_1} \times T_{N_2}$ is Zariski dense in $X_{P_1} \times X_{P_2}$, so by composing with the Segre embedding, we see $X_{P_1} \times X_{P_2}$ is the closure of the image of $X_{P_1} \times X_{P_2} \to \mathbb{P}^{s-1}$, which is given by the characters $\chi^m \chi^{m'}$, where $(m, m') \in (P_1 \times P_2) \cap (M_1 \times M_2)$. When we identify $T_{N_1} \times T_{N_2}$ with $T_{N_1 \times N_2}$, the characters $\chi^m \chi^{m'}$ becomes $\chi^{(m,m')}$, so the above map coincides with $T_{N_1 \times N_2} \to \mathbb{P}^{s-1}$ coming from the polytope $P_1 \times P_2$. This finishes the proof. □

In view of the theorem, we have

$$X_{P_1} \times X_{P_2} = \left(\bigcup_{v_1} U_{\sigma_{v_1}} \right) \times \left(\bigcup_{v_2} U_{\sigma_{v_2}} \right) = \bigcup_{v_1, v_2} U_{\sigma_{v_1}} \times U_{\sigma_{v_2}} = \bigcup_{(v_1, v_2)} U_{\sigma_{(v_1, v_2)}} = X_{P_1 \times P_2}.$$

In the third equality we use

$$\mathbb{C}[\sigma_1^\vee \cap M_1] \otimes_{\mathbb{C}} \mathbb{C}[\sigma_2^\vee \cap M_2] = \mathbb{C}[(\sigma_1 \times \sigma_2)^\vee \cap (M_1 \times M_2)].$$

and 10.35.(i).

10.3 Fans and toric varieties

10.37 Definition. Let N be a free $\mathbb{Z}$-module of rank n. A **fan** Σ in $N_{\mathbb{R}}$ is a finite collection of strongly convex N-rational polyhedral cones σ in $N_{\mathbb{R}}$ such that

1. for any $\sigma \in \Sigma$, all faces of σ lie in Σ;
2. for any $\sigma_1, \sigma_2 \in \Sigma$, the intersection $\sigma_1 \cap \sigma_2$ is a face of each.

The **support** of a fan Σ is defined as

$$|\Sigma| := \bigcup_{\sigma \in \Sigma} \sigma \subseteq N_{\mathbb{R}}.$$

For $r \in \mathbb{Z}_{\geqslant 0}$, let $\Sigma(r)$ be the set of all r-dimensional cones in Σ.

The normal fan defined in Theorem 9.24 is certainly a fan in the above sense.

10.38 Lemma. Let Σ be a fan and $\sigma_1, \sigma_2 \in \Sigma$. If $\tau = \sigma_1 \cap \sigma_2$, then

$$S_\tau = S_{\sigma_1} + S_{\sigma_2}.$$

Proof. First note that

$$\sigma_1^\vee + \sigma_2^\vee = (\sigma_1 \cap \sigma_2)^\vee = \tau^\vee.$$

This implies $S_{\sigma_1} + S_{\sigma_2} \subseteq S_\tau$. For the opposite inclusion, take $p \in S_\tau$ and pick $m \in \sigma_1^\vee \cap (-\sigma_2)^\vee \cap M$ such that $\sigma_1 \cap m^\perp = \tau = \sigma_2 \cap m^\perp$, which exists by Lemma 9.13. Then 10.23 implies that $S_\tau = S_{\sigma_1} + \mathbb{Z}_{\geqslant 0}(-m)$, so $p = q + \ell(-m)$ for some $q \in S_{\sigma_1}$ and $\ell \in \mathbb{Z}_{\geqslant 0}$. But $-m \in \sigma_2^\vee$ implies $-m \in S_{\sigma_2}$, so $p \in S_{\sigma_1} + S_{\sigma_2}$. $\square$

10.39 Toric variety from a fan. Let Σ be a fan in $N_{\mathbb{R}}$. By 10.12, each cone σ in Σ gives rise to an affine toric variety U_σ with coordinate ring $\mathbb{C}[\sigma^\vee \cap M]$. If $\tau = \sigma \cap m^\perp$ is a face of σ, then in 10.23 we see that $U_\tau = (U_\sigma)_{\chi^m}$ is an affine open subset of U_σ. If σ_1, $\sigma_2 \in \Sigma$ with $\tau = \sigma_1 \cap \sigma_2$, then Lemma 9.13 says that

$$\sigma_1 \cap m^\perp = \tau = \sigma_2 \cap m^\perp$$

for some $m \in \sigma_1^\vee \cap (-\sigma_2)^\vee \cap M$, which implies that

$$\mathbb{C}[\sigma_1^\vee \cap M]_{\chi^m} = \mathbb{C}[\tau^\vee \cap M] = \mathbb{C}[\sigma_2^\vee \cap M]_{\chi^{-m}}.$$

Let $g_{\sigma_1,\sigma_2} : (U_{\sigma_1})_{\chi^m} \to (U_{\sigma_2})_{\chi^{-m}}$ be the resulting isomorphism of affine varieties. We check $\{g_{\sigma_1,\sigma_2}\}_{\sigma_1,\,\sigma_2 \in \Sigma}$ satisfies the condition in (1.15). In fact, it suffices to check the conditions on topological spaces, for the morphisms on sheaves are completely determined by the maps of topological spaces. At this stage it is clear that both conditions in (1.15) hold. Hence $\{U_\sigma\}_{\sigma \in \Sigma}$ glues to a well-defined variety, denoted by X_Σ. We claim that X_Σ is a normal separated toric variety.

Proof. Since each cone σ in Σ is strongly convex, by (10.13) the torus in each affine toric variety U_σ is $T_N = U_{\{0\}}$ for any $\sigma \in \Sigma$. These T_N in U_σ are identified by the gluing, so that $T_N \subseteq X_\Sigma$ as a dense open subspace. In particular, X_Σ is irreducible. By (1.17), the action maps $T_N \times U_\sigma \to U_\sigma$ glue to an algebraic action $T_N \times X_\Sigma \to X_\Sigma$. Indeed, the action map corresponds to the homomorphism $\mathbb{C}[S_\sigma] \to \mathbb{C}[S_\sigma] \otimes \mathbb{C}[M]$, and on $U_{\sigma_1} \cap U_{\sigma_2} = U_\tau$, the two action maps agree (or are compatible with the g_{σ_1,σ_2}), for they are all $\mathbb{C}[S_\tau] \to \mathbb{C}[S_\tau] \otimes \mathbb{C}[M]$. Thus, this proves X_Σ is a toric variety.

Since each σ is strongly convex, each U_σ is normal by Theorem 10.19.(iii), and thus X_Σ is a normal variety. To see X_Σ is separated, we must show the diagonal morphism $X_\Sigma \to X_\Sigma \times X_\Sigma$ is a closed immersion, and in view of Proposition 1.36, it suffices to show that for any cones $\sigma_1,\ \sigma_2 \in \Sigma$, the diagonal morphism

$$\Delta : U_\tau \to U_{\sigma_1} \times U_{\sigma_2}$$

(where $\tau = \sigma_1 \cap \sigma_2$) is a closed immersion. On coordinate rings the diagonal morphism reads

$$\begin{aligned} \Delta^* : \mathbb{C}[S_{\sigma_1}] \times \mathbb{C}[S_{\sigma_2}] &\longrightarrow \mathbb{C}[S_\tau] \\ \chi^m \otimes \chi^{m'} &\longmapsto \chi^{m+m'}. \end{aligned}$$

By Lemma 10.38, Δ^* is surjective, showing that Δ is a closed immersion in turn. □

For a full dimensional lattice polytope $P \subseteq M_\mathbb{R}$, the description of the projective toric variety X_P in (10.26) shows that X_P is isomorphic to the toric variety X_{Σ_P} associated to the normal fan Σ_P.

10.40 Product of fans. Let N_i $(i = 1, 2)$ be a free $\mathbb{Z}$-module of rank n_i and let Σ_i be a fan in $(N_i)_\mathbb{R}$. Then the product

$$\Sigma_1 \times \Sigma_2 = \{\sigma_1 \times \sigma_2 \mid \sigma_i \in \Sigma_i,\ i = 1,\ 2\}$$

is a fan in $(N_1)_\mathbb{R} \times (N_2)_\mathbb{R} = (N_1 \times N_2)_\mathbb{R}$. Indeed, each $\sigma_1 \times \sigma_2$ is strongly convex $(N_1 \times N_2)$-rational polyhedral cone, and 10.35.(ii) implies $\Sigma_1 \times \Sigma_2$ is a fan. Therefore we may construct the toric variety $X_{\Sigma_1 \times \Sigma_2}$. A result is that

$$X_{\Sigma_1 \times \Sigma_2} \cong X_{\Sigma_1} \times X_{\Sigma_2}.$$

10.41 Example; the blowup of $\mathbb{C}^n$ at the origin. Let $N = \mathbb{Z}^n$ and $e_1, \dots, e_n$ be the standard basis. Set $e_0 = e_1 + \cdots + e_n$. Consider the fan Σ of $N_\mathbb{R}$ consisting of the cones generated by all subset of $\{e_0, \dots, e_n\}$ not containing $\{e_1, \dots, e_n\}$. Then the toric variety X_Σ is isomorphic to the blowup $\mathrm{Bl}_0(\mathbb{C}^n)$ of $\mathbb{C}^n$ at the origin.

To see this, for $1 \leqslant i \leqslant n$, let $\sigma_i = \mathrm{cone}\{e_0, e_1, \dots, \widehat{e_i}, \dots, e_n\} \subseteq N_\mathbb{R}$. Its dual in $M_\mathbb{R}$ is

$$\sigma_i^\vee = \mathrm{cone}\{e_i, e_1 - e_i, \dots, e_{i-1} - e_i, e_{i+1} - e_i, \dots, e_n - e_i\} \subseteq M_\mathbb{R}.$$

Note that the generating set of $\sigma_i^\vee$ entirely lies in M, so

$$S_{\sigma_i} = \sigma_i^\vee \cap M = \mathbb{Z}_{\geqslant 0}\{e_i, e_1 - e_i, \dots, e_{i-1} - e_i, e_{i+1} - e_i, \dots, e_n - e_i\}$$

and thus

$$U_{\sigma_i} = \mathrm{mSpec}\, \mathbb{C}\left[x_i, \frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right].$$

The separating hyperplane of σ_i and σ_j, where $1 \leqslant i < j \leqslant n$, is the hyperplane with normal vector $e_i - e_j$; note that $e_i - e_j \in \sigma_j^\vee$ and $e_j - e_i \in \sigma_i^\vee$. Thus the gluing data is the isomorphism

$$\mathbb{C}\left[x_i, \frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right]_{\frac{x_j}{x_i}} = \mathbb{C}\left[x_j, \frac{x_1}{x_j}, \dots, \frac{x_{j-1}}{x_j}, \frac{x_{j+1}}{x_j}, \dots, \frac{x_n}{x_j}\right]_{\frac{x_i}{x_j}}$$

By the change of coordinates $x_i \mapsto s_i$ and $\dfrac{x_j}{x_i} \mapsto \dfrac{t_j}{t_i}$, we see

$$U_{\sigma_i} = \operatorname{mSpec} \mathbb{C}\left[\frac{t_1}{t_i}, \dots, \frac{t_{i-1}}{t_i}, \frac{t_{i+1}}{t_i}, \dots, \frac{t_n}{t_i}, s_i\right]$$

and the gluing isomorphism becomes the map defined by $s_i \mapsto \dfrac{t_i}{t_j} s_j$ and $\dfrac{t_k}{t_i} \mapsto \dfrac{t_k}{t_i} = \dfrac{t_k}{t_j}\dfrac{t_j}{t_i}$. This is the same as the blowup

$$\mathrm{Bl}_0(\mathbb{C}^n) = V(t_i s_j - t_j s_i \mid 1 \leqslant i, j \leqslant n) \subseteq \mathbb{P}^{n-1} \times \mathbb{C}^n$$

at the origin; on the product we use the coordinates $([t_1 : \cdots : t_n], s_1, \dots, s_n)$.

10.42 Example; Hirzebruch surfaces. Let $r \in \mathbb{Z}_{\geqslant 0}$ and consider the fan consisting of the four cone

$$\sigma_1 = \mathrm{cone}\{(1,0),(0,1)\} \qquad \sigma_2 = \mathrm{cone}\{(1,0),(0,-1)\}$$
$$\sigma_3 = \mathrm{cone}\{(-1,r),(0,-1)\} \qquad \sigma_4 = \mathrm{cone}\{(-1,r),(0,1)\}$$

together with all of their faces. In 10.29 we consider the rational normal scroll $S_{a,b}$ constructed from the polytope $P_{a,b}$ $(1 \leqslant a \leqslant b)$. The normal fan $\Sigma_{P_{a,b}}$ is the fan Σ_{b-a} defines above, so $X_{\Sigma_{b-a}} \cong S_{a,b}$ as $\mathbb{C}$-varieties. Generally, the toric variety X_{Σ_r} is called the **Hirzebruch surface**, and by definition it is covered by the four affine varieties

$$U_{\sigma_1} = \operatorname{mSpec} \mathbb{C}[x,y] \qquad U_{\sigma_2} = \operatorname{mSpec} \mathbb{C}[x,y^{-1}]$$
$$U_{\sigma_3} = \operatorname{mSpec} \mathbb{C}[x^{-1}, x^{-r}y^{-1}] \qquad U_{\sigma_4} = \operatorname{mSpec} \mathbb{C}[x^{-1}, x^r y]$$

each isomorphic to the affine space $\mathbb{A}^2_{\mathbb{C}}$.

10.43 Definition. Let Σ be a fan in $N_{\mathbb{R}}$.

1. Σ is **smooth** if every cone in Σ is smooth.
2. Σ is **simplicial** if every cone in Σ is simplicial (10.20).
3. Σ is **complete** if its support $|\Sigma|$ is the whole $N_{\mathbb{R}}$.

10.44 Theorem. Let X_Σ be the toric variety defined by a fan $\Sigma \subseteq N_{\mathbb{R}}$.

(i) X_Σ is a smooth variety if and only if Σ is smooth.

(ii) X_Σ is a orbifold if and only if Σ is simplicial.

(iii) X_Σ is compact in the classical topology (3.26) if and only if Σ is complete.

Proof. Being smooth is a local property, so X_Σ is smooth if and only if each U_σ is smooth. At this stage (i) simply follows from (10.22). We will not prove (ii) here. The proof of (iii) will be given in the sequel. $\square$

10.3.1 The orbit-cone correspondence

10.45 Retain the notations in (10.12). Consider the map $\gamma_\sigma : S_\sigma \to \mathbb{C}$ defined by

$$m \mapsto \begin{cases} 1 & \text{, if } m \in S_\sigma \cap \sigma^\perp = \sigma^\perp \cap M \\ 0 & \text{, otherwise} \end{cases}$$

This is a monoid homomorphism, as $\sigma^\vee \cap \sigma^\perp = \sigma^*$ is a face of $\sigma^\vee$ (9.10) : if $m + m' \in S_\sigma$ and $m + m' \in S_\sigma \cap \sigma^\perp$, then $m, m' \in S_\sigma \cap \sigma^\perp$ by (9.3). The corresponding point (10.15) of γ_σ is again denoted by $\gamma_\sigma \in U_\sigma$. We call this the **distinguished point corresponding to** σ. By (10.18), γ_σ is fixed under the T_N-action if and only if $\dim \sigma = \dim_{\mathbb{R}} N_{\mathbb{R}}$.

If τ is a face of σ, then $\gamma_\tau \in U_\tau \subseteq U_\sigma$. More precisely, there is a commutative diagram

$$\begin{tikzcd} U_\tau \arrow[r, "\text{inclusion}"] \arrow[d, "\wr"] & U_\sigma \arrow[d, "\wr"] \\ \mathrm{Hom}_{\mathbf{Monoid}}(\tau^\vee \cap M, \mathbb{C}) \arrow[r, "\text{restriction}"] & \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{C}) \end{tikzcd}$$

and $\gamma_\tau|_{\sigma^\vee \cap M} \in \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{C})$.

10.46 Lemma. Let $\sigma \subseteq N_{\mathbb{R}}$ be a strongly convex rational polyhedral cone and $u \in N$. Then

$$u \in \sigma \quad \Leftrightarrow \quad \lim_{s \to 0} \lambda^u(s) \text{ exists in } U_\sigma.$$

Here the limit is taken in the classical topology of U_σ (3.26). Moreover, $u \in \mathrm{Relint}(\sigma) \Rightarrow \lim_{s \to 0} \lambda^u(s) = \gamma_\sigma$.

Proof. Say $U_\sigma = Y_A$ for some finite set $A = \{m_1, \ldots, m_s\} \subseteq S_\sigma \backslash \{0\}$. Since σ is strongly convex, the torus of Y_A is T_N (10.13). If we view T_N as a subgroup of $(\mathbb{C}^\times)^s$ by $t \mapsto (m_1(t), \ldots, m_s(t))$, the characters $m_i \in A$ are simply the projections to i-th component.

$$\begin{aligned} \lim_{s \to 0} \lambda^u(s) \text{ exists in } U_\sigma &\Leftrightarrow \lim_{s \to 0} \chi^m(\lambda^u(s)) \text{ exists in } \mathbb{C} \text{ for any } m \in S_\sigma \\ &\Leftrightarrow \lim_{s \to 0} s^{\langle m, u \rangle} \text{ exists in } \mathbb{C} \text{ for any } m \in S_\sigma \\ &\Leftrightarrow \langle m, u \rangle \geqslant 0 \text{ for any } m \in \sigma^\vee \cap M \\ &\Leftrightarrow u \in (\sigma^\vee)^\vee = \sigma. \end{aligned}$$

This proves the first part of the lemma. If $u \in \sigma \cap N$, since $m_i = \chi^{m_i}$ is simply the i-th projection as said above, from the explicit expression in (10.15) we see $\lim_{s\to 0} \lambda^u(t)$ is the point corresponding to the monoid homomorphism $S_\sigma \to \mathbb{C}$ defined by

$$\sigma^\vee \cap M \ni m \mapsto \lim_{s\to\infty} s^{\langle m,u\rangle}.$$

If $u \in \mathrm{Relint}(\sigma)$, then $\langle m, u\rangle > 0$ for all $m \in \sigma^\vee \backslash \sigma^\perp$, and $\langle m, u\rangle = 0$ if and only if $m \in \sigma^\vee \cap \sigma^\perp$. Thus this monoid homomorphism is the same as γ_σ defined in (10.45). □

10.47 Let σ be a strongly convex rational polyhedral cone in $N_\mathbb{R}$. Let N_σ be the sublattice of N generated by $\sigma \cap N$, and let $N(\sigma) = N/N_\sigma$. The perfect pairing $\langle\, ,\, \rangle : M \times N \to \mathbb{Z}$ induces a nondegenerate pairing

$$\langle\, ,\, \rangle : (\sigma^\perp \cap M) \times N(\sigma) \to \mathbb{Z}.$$

In fact, this is a perfect pairing. To see this, from the short exact sequence

$$0 \longrightarrow N_\sigma \longrightarrow N \longrightarrow N(\sigma) \longrightarrow 0$$

we obtain

$$\mathrm{Hom}_\mathbb{Z}(N(\sigma), \mathbb{Z}) = \{\varphi \in \mathrm{Hom}_\mathbb{Z}(N, \mathbb{Z}) \mid \varphi|_{N_\sigma} = 0\}$$

Using the identification $M = \mathrm{Hom}_\mathbb{Z}(N, \mathbb{Z})$, it becomes

$$\mathrm{Hom}_\mathbb{Z}(N(\sigma), \mathbb{Z}) = \{m \in M \mid \langle m, N_\sigma\rangle = 0\} = (N_\sigma)^\perp = \sigma^\perp \cap M$$

10.48 The torus orbits abstractly. Let Σ be a fan in $N_\mathbb{R} \cong \mathbb{R}^n$. Recall that $\tau \preceq \sigma$ means τ is a face of σ. For $\sigma \in \Sigma$, define the torus

$$O(\sigma) = T_{N(\sigma)} := N(\sigma) \otimes_\mathbb{Z} \mathbb{C}^\times.$$

Clearly, $\dim O(\sigma) = n - \dim \sigma$, and T_N acts on $T_{N(\sigma)}$ transitively via the projection $T_N \to T_{N(\sigma)}$.

Define the **star** of a cone $\tau \in \Sigma$ to be

$$\mathrm{Star}(\tau) = \{\overline{\sigma} = \sigma \bmod (N_\tau)_\mathbb{R} \subseteq N(\tau)_\mathbb{R} \mid \tau \preceq \sigma \in \Sigma\}.$$

This is the collection of cones $\sigma \in \Sigma$ that contain τ as a face.

Lemma. $\mathrm{Star}(\tau)$ is a fan in $N(\tau)_\mathbb{R}$

Proof. It is clear that each $\overline{\sigma}$ is a $N(\tau)$-rational convex polyhedral cone. The dual $\overline{\sigma}^\vee ee = \sigma^\vee \cap \tau^\perp \subseteq \tau^\perp$ has maximal dimension, as $\sigma^\vee \subseteq M_\mathbb{R}$ has, so $\overline{\sigma}$ is strongly convex.

If $\tau \preceq \sigma' \preceq \sigma$, then $\overline{\sigma'}$ is a face of $\overline{\sigma}$. Indeed, if $\sigma' = \sigma \cap u^\perp$ for some $u \in \sigma^\vee$, then $\tau \subseteq u^\perp$ so that $u \in \tau^\perp \cap \sigma^\vee = \overline{\sigma}^\vee$. We claim that

$$\overline{\sigma'} = \overline{\sigma} \cap u^\perp \subseteq N(\sigma)_\mathbb{R}$$

If $x \in \overline{\sigma'}$, then $x - x' \in (N_\sigma)_\mathbb{R}$ for some $x' \in \sigma'$, and thus

$$0 = \langle x - x', u\rangle = \langle x, u\rangle - \langle x', u\rangle = \langle x, u\rangle.$$

If $x \in \overline{\sigma} \cap u^\perp$, write $x - x' \in (N_\sigma)_\mathbb{R}$ for some $x' \in \sigma$, and $0 = \langle x' - x, u\rangle = \langle x', u\rangle$ so that $x' \in \sigma \cap u^\perp = \sigma'$.

Now let C be a face of $\overline{\sigma}$. Then $C^\perp \cap \overline{\sigma}^\vee$ is a face of $\overline{\sigma}^\vee = \sigma^\vee \cap \tau^\perp$, and thus by (9.10) it has the form $\sigma^\vee \cap (\sigma')^\perp$ for some face $\tau \preceq \sigma' \preceq \sigma$. In sum, $C^\perp \cap \overline{\sigma}^\vee = \overline{\sigma'}^\perp \cap \overline{\sigma}^\vee$, and since $\overline{\sigma'} \preceq \overline{\sigma}$, so by (9.10) again we have $C = \overline{\sigma}$. This shows every face of $\overline{\sigma}$ has the form $\overline{\sigma'}$ for some $\tau \preceq \sigma' \preceq \sigma$.

Let $\tau \preceq \sigma, \sigma'$. Then

$$\overline{\sigma \cap \sigma'}^\vee = (\sigma \cap \sigma')^\vee \cap \tau^\perp = \sigma^\vee \cap (\sigma')^\vee \cap \tau^\perp = \overline{\sigma}^\vee \cap \overline{\sigma'}^\vee \subseteq \tau^\perp$$

so by duality we have $\overline{\sigma \cap \sigma'} = \overline{\sigma} \cap \overline{\sigma'}$ in $N(\sigma)_\mathbb{R}$. □

Put $V(\tau) = X_{\mathrm{Star}(\tau)}$. Note that the torus of $V(\tau)$ is $T_{N(\tau)} = O(\tau)$. By (10.47), the toric variety $V(\tau)$ has an affine open cover consisting of

$$U_\tau(\sigma) := \mathrm{mSpec}\, \mathbb{C}[\sigma^\vee \cap \tau^\perp \cap M]$$

where σ runs over all faces in Σ with $\tau \preceq \sigma$. Note here that $\sigma^\vee \cap \tau^\perp$ is a face of $\sigma^\vee$ (9.10), and $U_\tau(\tau) = O(\tau)$ is dense in $V(\tau)$.

10.49 Lemma. There exists a closed immersion $V(\tau) \subseteq X_\Sigma$.

Proof. We prove this by constructing compatible closed immersion $U_\tau(\sigma) \subseteq U_\sigma$ for any $\tau \preceq \sigma \in \Sigma$. On coordinate rings there is an obvious projection

$$\mathbb{C}[\sigma^\vee \cap M] \longrightarrow \mathbb{C}[\sigma^\vee \cap \tau^\perp \cap M]$$

defined by $\chi^m \mapsto \begin{cases} \chi^m & \text{, if } m \in \sigma^\vee \cap \tau^\perp \cap M \\ 0 & \text{, otherwise} \end{cases}$. This gives a closed immersion $U_\tau(\sigma) \subseteq U(\sigma)$. There morphisms are compatible, for if $\tau \preceq \sigma \preceq \sigma'$, the diagram

$$\begin{tikzcd} U_\tau(\sigma) \arrow[r] \arrow[d] & U_\tau(\sigma') \arrow[d] \\ U_\sigma \arrow[r] & U_{\sigma'} \end{tikzcd}$$

corresponds to the diagram on coordinate rings

$$\begin{tikzcd} \mathbb{C}[(\sigma')^\vee \cap \tau^\perp \cap M]_{\chi^m} \arrow[r, equal] & \mathbb{C}[\sigma^\vee \cap \tau^\perp \cap M] & \mathbb{C}[(\sigma')^\vee \cap \tau^\perp \cap M] \arrow[l] \\ \mathbb{C}[(\sigma')^\vee \cap M]_{\chi^m} \arrow[r, equal] & \mathbb{C}[\sigma^\vee \cap M] \arrow[u] & \mathbb{C}[(\sigma')^\vee \cap M] \arrow[l] \arrow[u] \end{tikzcd}.$$

where $m \in (\sigma')^\vee$ is such that $\sigma = \sigma' \cap m^\perp$, and this is clearly commuting. Thus, these $U_\tau(\sigma) \to U(\sigma)$ glue to a closed immersion $V(\tau) \subseteq X_\Sigma$. $\square$

For the later use, we can also write $U_\tau(\sigma) \subseteq U_\sigma$ in terms of monoid homomorphisms. It corresponds to

$$\mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap \tau^\perp \cap M, \mathbb{C}) \longrightarrow \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{C})$$

defined by extending $\gamma \in \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap \tau^\perp \cap M, \mathbb{C})$ to $\sigma^\vee \cap M$ by zero, i.e, by setting $\gamma(m) = 0$ if $m \in (\sigma^\vee \cap M)\backslash\tau^\perp$. In particular, the inclusion $T_{N(\tau)} \subseteq U_\tau$ corresponds to

$$\mathrm{Hom}_{\mathbb{Z}}(\tau^\perp \cap M, \mathbb{C}^\times) \longrightarrow \mathrm{Hom}_{\mathbf{Monoid}}(\tau^\vee \cap M, \mathbb{C}).$$

10.50 If $\tau \preceq \tau' \in \Sigma$, we have a closed immersion $V(\tau') \subseteq V(\tau)$, which on each affine U_σ, $\sigma \in \mathrm{Star}(\tau')$ is given by the projection

$$\mathbb{C}[\sigma^\vee \cap (\tau')^\perp \cap M] \longleftarrow \mathbb{C}[\sigma^\vee \cap \tau^\perp \cap M]$$

Also, from the proof (10.49) we see the defining ideal of $U_\tau(\sigma) = V(\tau) \cap U_\sigma \subseteq U_\sigma$ is

$$\langle \chi^m \mid m \in S_\sigma \backslash \tau^\perp \rangle \subseteq \mathbb{C}[\sigma^\vee \cap M].$$

10.51 The torus orbits. The domain and codomain of the bijection

$$T_{N(\sigma)} \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\tau^\perp \cap M, \mathbb{C}^\times)$$

admit T_N-actions; the action on the left is from the projection $T_N \to T_{N(\sigma)}$, and the action on the right is from the inclusion

$$\mathrm{Hom}_{\mathbb{Z}}(\tau^\perp \cap M, \mathbb{C}^\times) \longrightarrow \mathrm{Hom}_{\mathbf{Monoid}}(\tau^\vee \cap M, \mathbb{C}).$$

and (10.16). In fact, this bijection intertwines the T_N-actions. To see this, let $N \otimes_{\mathbb{Z}} \mathbb{C}^\times \ni u \otimes s = t \in T_N$ and $u' \otimes s' \in N(\sigma) \otimes_{\mathbb{Z}} \mathbb{C}^\times$. The element $u' \otimes s'$ maps to $\gamma' \in \mathrm{Hom}_{\mathbb{Z}}(\sigma^\perp \cap M, \mathbb{C}^\times)$ defined by $\gamma'(m) = s'^{\langle m, u' \rangle}$. The element $t.\gamma' = (u \otimes s)(u' \otimes s')$ then corresponds to (c.f. (10.16) and (10.2))

$$m \mapsto s^{\langle m, u \rangle}\gamma'(m) = \chi^m(\lambda^u(s))\gamma'(m) = \chi^m(t)\gamma'(m) = t.\gamma'(m),$$

so the T_N-action on both sides are compatible.

In other word, the inclusion $O(\tau) \subseteq U_\tau$ is T_N-equivariant. Since T_N acts on $O(\tau)$ transitively, it follows that $O(\tau)$ is an T_N-orbit in U_τ. Moreover, the distinguished point $\gamma_\tau \in U_\tau$ lies in $O(\tau)$, so that

$$O(\tau) = T_N.\gamma_\tau \subseteq X_\Sigma.$$

It suffices to show that $\gamma_\tau \in \mathrm{Hom}_{\mathbf{Monoid}}(\tau^\vee \cap M, \mathbb{C})$ restricts to a nonvanishing map on $\tau^\perp \cap M$, and it is clear from its definition (10.45).

10.52 Lemma. Let σ be a rational convex polyhedral cone in $M_{\mathbb{R}}$, and $\gamma \in \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{C})$. Then there exists a face τ of σ such that

$$\gamma^{-1}(\mathbb{C}^\times) = \sigma^\vee \cap \tau^\perp \cap M.$$

Proof. Put $T = \gamma^{-1}(\mathbb{C}^\times)$. This is a finitely generated sub monoid of S_σ, so $\mathrm{cone}(T) \subseteq \sigma^\vee$ is a convex polyhedral cone. Note that we have

$$\gamma^{-1}(\mathbb{C}^\times) = \mathrm{cone}(T) \cap M$$

To see $\mathrm{cone}(T) \cap M \subseteq T$, we may replace $\mathrm{cone}(T)$ by $\mathrm{cone}_{\mathbb{Q}}(T) = \mathrm{cone}(T) \cap M_{\mathbb{Q}}$. If $v \in \mathrm{cone}_{\mathbb{Q}}(T) \cap M \subseteq S_\sigma$, we can find an integer $n \gg 0$ so that $nv \in T \cap M \subseteq S_\sigma$ and thus $0 \neq \gamma(nv) = \gamma(v)^n$.

For $x, y \in \sigma^\vee \cap M_{\mathbb{Q}}$ such that $x + y \in \mathrm{cone}_{\mathbb{Q}}(T)$ for some $\theta \in (0,1) \cap \mathbb{Q}$, take an integer $n \gg 0$ such that $nx, ny \in S_\sigma$ (possible as $\sigma^\vee$ is rational), and $nx + ny = n(x+y) \in T$. Then nx, $ny \in T$, and thus $x, y \in \mathrm{cone}_{\mathbb{Q}}(T)$. By continuity we see $\mathrm{cone}(T) \subseteq \sigma^\vee$ is a convex subset satisfying the condition in Lemma 9.4. Thus $\mathrm{cone}(T)$ is a face of $\sigma^\vee$, which must be of the form $\mathrm{cone}(T) = \sigma^\vee \cap \tau^\perp$ for some face τ of σ by Lemma 9.10.

□

10.53 Orbit-Cone correspondence. Let X_Σ be the toric variety of the fan Σ in $N_{\mathbb{R}}$.

(a) There is a bijection

$$\begin{aligned} \Sigma &\longrightarrow T_N \backslash X_\Sigma \\ \sigma &\longmapsto O(\sigma) = \mathrm{Hom}_{\mathbb{Z}}(\sigma^\perp \cap M, \mathbb{C}^\times). \end{aligned}$$

(b)

$$U_\sigma = \bigcup_{\tau \preceq \sigma} O(\tau),$$

(c) $\tau \preceq \sigma$ if and only if $O(\sigma) \subseteq V(\tau)$, and

$$V(\tau) = \bigcup_{\tau \preceq \sigma \in \Sigma} O(\sigma) = \overline{O(\tau)}^{C}.$$

Here $\overline{O(\tau)}^{C}$ denotes the closure in the classical topology.

Proof. Let O be a T_N-orbit in X_Σ. Since each U_σ is T_N-invariant, O is contained in U_σ for some $\sigma \in \Sigma$. Moreover, as $U_{\sigma_1} \cap U_{\sigma_2} = U_{\sigma_1 \cap \sigma_2}$ for any $\sigma_1, \sigma_2 \in \Sigma$, there exists a unique minimal cone $\sigma \in \Sigma$ with $O \subseteq U_\sigma$. We contend that $O = O(\sigma)$. Note that (a) follows from this contention.

To prove $O = O(\sigma)$, it suffices to show $O \cap O(\sigma) \neq \varnothing$. Let $\gamma \in O$. By Lemma 10.52, there exists a face τ of σ such that

$$\gamma^{-1}(\mathbb{C}^\times) = \sigma^\vee \cap \tau^\perp \cap M.$$

Write $\tau = \sigma \cap m^\perp$ for some $m \in \sigma^\vee \cap M$; then $U_\tau = (U_\sigma)_{\chi^m}$ (10.23). As in (10.15), we have $\chi^m(\gamma) = \gamma(m)$, and since $m \in \sigma^\vee \cap \tau^\perp \cap M$, we see $\chi^m(\gamma) \neq 0$, or $\gamma \in U_\tau$. By minimality of σ it forces that $\tau = \sigma$, and hence

$$\gamma^{-1}(\mathbb{C}^\times) = \sigma^\perp \cap M.$$

so that $\gamma|_{\sigma^\perp \cap M} \in \mathrm{Hom}_{\mathbb{Z}}(\sigma^\perp \cap M, \mathbb{C}^\times) = O(\sigma)$. Hence $(\gamma \in) O \cap O(\sigma)$ is nonempty.

For (b), since U_σ is T_N-invariant, it is a union of T_N-orbits. If O is an orbit, the contention above implies that $O = O(\tau)$, where τ is the minimal cone in Σ such that $O \subseteq U_\tau$. It follows that $\tau \preceq \sigma$, since $\sigma \cap \tau$ is a face of each with $U_{\sigma\cap\tau} = U_\tau \cap U_\sigma$, by minimality it must be the case $\sigma \cap \tau = \tau$, implying $\tau \preceq \sigma$. This proves (b).

It remains to prove (c). We prove the first statement with $V(\tau)$ replaced by $\overline{O(\tau)}^C$ first. Since $\overline{O(\tau)}^C$ is T_N-invariant, it is a union of T_N-orbits. If $O(\sigma) \in \overline{O(\tau)}^C$, then $O(\tau) \subseteq U_\sigma$, since for otherwise $O(\tau) \cap U_\sigma = \varnothing$, which implies $\overline{O(\tau)}^C \cap U_\sigma = \varnothing$ as U_σ is classically open. With $O(\tau) \subseteq U_\sigma$, it follows from (b) that $\tau \preceq \sigma$. Conversely, suppose $\tau \preceq \sigma$. Let $u \in \mathrm{Relint}(\sigma)$ and consider the curve $\gamma : \mathbb{C}^\times \to X_\Sigma$ defined by $\gamma(s) = \lambda^u(s).\gamma_\tau$; note that $\gamma(s) \in O(\tau) \subseteq U_\tau \subseteq U_\sigma$. As a monoid homomorphism, we have, for $m \in \tau^\vee \cap M$,

$$\gamma(s)(m) = \chi^m(\lambda^u(s))\gamma_\tau(m) = s^{\langle m,u\rangle}\gamma_\tau(m)$$

Since $u \in \mathrm{Relint}(\sigma)$, we have $\langle m, u\rangle > 0$ if $m \in \sigma^\vee \backslash \sigma^\perp$ and $\langle m, u\rangle = 0$ if $m \in \sigma^\perp$; it follows that

$$\lim_{s\to 0} \gamma(s)(m) = \begin{cases} 1 & \text{, if } m \in (\sigma^\vee \cap M)\backslash\sigma^\perp \\ 0 & \text{, if } m \in \sigma^\perp \cap M \end{cases} = \gamma_\sigma(m)$$

and thus $\lim_{s\to 0} \gamma(s)$ exists in U_σ and is equal to $\gamma_\sigma \in O(\sigma)$. In particular, this implies $\overline{O(\tau)}^C \cap O(\sigma) \neq \varnothing$, i.e., $O(\sigma) \subseteq \overline{O(\tau)}^C$. The equality

$$\overline{O(\tau)}^C = \bigcup_{\tau \preceq \sigma \in \Sigma} O(\sigma)$$

follows immediately from the first statement in (c), so it remains to show $V(\tau) = \overline{O(\tau)}^C$. From (b) and (c) we see

$$\overline{O(\tau)}^C \cap U_\sigma = \bigcup_{\tau \preceq \sigma'} O(\sigma') \cap \bigcup_{\sigma' \preceq \sigma} O(\sigma') = \bigcup_{\tau \preceq \sigma' \preceq \sigma} O(\sigma') \subseteq U_\sigma.$$

On the other hand we have the closed subvariety $U_\tau(\sigma)$, which, as monoid homomorphisms, is given by (10.49)

$$\mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap \tau^\perp \cap M, \mathbb{C}) \longrightarrow \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{C})$$

Since $U_\tau(\sigma)$ is the Zariski closure of $O(\tau)$ in U_σ, we see $\overline{O(\tau)}^C \cap U_\sigma \subseteq U_\tau(\sigma)$. Conversely, if $\gamma \in \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap \tau^\perp \cap M, \mathbb{C})$, by Lemma 10.52 there exists $\sigma' \preceq \sigma$ with, viewed γ as a map $\gamma : \sigma^\vee \cap M \to \mathbb{C}$,

$$\gamma^{-1}(\mathbb{C}^\times) = \sigma^\vee \cap (\sigma')^\perp \cap M$$

which implies $\sigma^\vee \cap \tau^\perp \supseteq \sigma^\vee \cap (\sigma')^\vee$, so that by Lemma 9.10 we obtain $\tau \preceq \sigma'$. Thus $\gamma \in \overline{O(\tau)}^C \cap U_\sigma$, proving the opposite inclusion. $\square$

10.3.2 Toric morphisms

10.54 Definition. Let N_1, N_2 be two free $\mathbb{Z}$-modules of finite rank and let $\Sigma_i \subseteq (N_i)_{\mathbb{R}}$ $(i = 1, 2)$ be a fan.

1. A $\mathbb{Z}$-linear morphism $\overline{\phi} : N_1 \to N_2$ is **compatible** with the fans Σ_1 and Σ_2 if for every cone $\sigma_1 \in \Sigma_1$, there exits a cone $\sigma \in \Sigma_2$ such that $\overline{\phi}_{\mathbb{R}}(\sigma_1) \subseteq \sigma_2$.
2. A morphism $\phi : X_{\Sigma_1} \to X_{\Sigma_2}$ is **toric** if $\phi(T_{N_1}) \subseteq T_{N_2}$ and $\phi|_{T_{N_1}} : T_{N_1} \to T_{N_2}$ is a group homomorphism.

10.55 Theorem. Let N_1, N_2 be two free $\mathbb{Z}$-modules of finite rank and let $\Sigma_i \subseteq (N_i)_{\mathbb{R}}$ $(i = 1, 2)$ be a fan.

(i) If $\overline{\phi} : N_1 \to N_2$ is a $\mathbb{Z}$-linear map compatible with Σ_1 and Σ_2, then there is a toric morphism $\phi : X_{\Sigma_1} \to X_{\Sigma_2}$ such that $\phi|_{T_{N_1}}$ is the map

$$\overline{\phi} \otimes \mathrm{id}_{\mathbb{C}^\times} : N_1 \otimes_{\mathbb{Z}} \mathbb{C}^\times \longrightarrow N_2 \otimes_{\mathbb{Z}} \mathbb{C}^\times.$$

(ii) If $\phi : X_{\Sigma_1} \to X_{\Sigma_2}$ is a toric morphism, then ϕ induces a $\mathbb{Z}$-linear map $\overline{\phi} : N_1 \to N_2$ compatible with the fans Σ_1 and Σ_2.

10.56 Proposition. Let $N' \leqslant N$ be a sublattice with $\dim_{\mathbb{R}} N_{\mathbb{R}} = n$ and $\dim_{\mathbb{R}} N'_{\mathbb{R}} = k$. Let Σ be a fan in $N'_{\mathbb{R}}$, which can be viewed as a fan in $N_{\mathbb{R}}$. Then

(i) If N' is spanned by a subset of basis of N, then there is an isomorphism

$$X_{\Sigma,N} \cong X_{\Sigma,N'} \times T_{N/N'}.$$

(ii) In general, a basis for N' can be extended to a sublattice $N'' \leqslant N$ of finite index. Then $X_{\Sigma,N}$ is isomorphic to the quotient of

$$X_{\Sigma,N''} \cong X_{\Sigma'} \times T_{N''/N'}$$

by the finite abelian group N/N''.

10.57 Fibre bundle. Let N, N' be two free $\mathbb{Z}$-modules of finite rank and let $\overline{\phi} : N \to N'$ be a surjective $\mathbb{Z}$-linear map. Let Σ, Σ' be fans in $N_{\mathbb{R}}$ and $N'_{\mathbb{R}}$, respectively, that are compatible with $\overline{\phi}$. Then there is a corresponding toric morphism

$$\phi : X_{\Sigma} \longrightarrow X_{\Sigma'}.$$

Let $N_0 = \ker \overline{\phi}$; note that N_0 is spanned by a subset of basis of N. The collection

$$\Sigma_0 = \{\sigma \in \Sigma \mid \sigma \subseteq (N_0)_{\mathbb{R}}\}$$

is a fan in $(N_0)_{\mathbb{R}}$. By (10.56).(i) we have an isomorphism

$$X_{\Sigma_0,N} \cong X_{\Sigma_0,N_0} \times T_{N'}$$

as $N' \cong N/N_0$. Furthermore, $\overline{\phi}$ is compatible with Σ_0 in $N_{\mathbb{R}}$ and the trivial fan $\{0\}$ in $N'_{\mathbb{R}}$, so it gives the toric morphism

$$\phi|_{X_{\Sigma_0,N}} : X_{\Sigma_0,N} \longrightarrow T_{N'}.$$

Moreover,

$$\phi^{-1}(T_{N'}) = X_{\Sigma_0,N} \cong X_{\Sigma_0,N_0} \times T_{N'}$$

so X_Σ is a fibre bundle over $T_{N'}$ with fibre X_{Σ_0,N_0}.

10.58 Lemma. Let V_1, V_2 be two finite dimensional real vector spaces and let $\sigma_i \subseteq V_i$ be a convex polyhedral cone. Let $\phi : V_1 \to V_2$ be an $\mathbb{R}$-linear map such that $\phi(\sigma_1) \subseteq \sigma_2$. If

$$\phi(\mathrm{Relint}(\sigma_1)) \cap (\sigma_2 \backslash \mathrm{Relint}(\sigma_2)) \neq \varnothing,$$

then $\phi(\sigma_1)$ is contained in a proper face of σ_2.

Proof. Let $v_0 \in \mathrm{Relint}(\sigma_1)$ such that $\phi(v_0) \in \sigma_2 \backslash \mathrm{Relint}(\sigma_2)$. By Lemma 9.6, we see $\phi(v_0) \in \tau'$ for some proper face τ' of σ_2. Take any $v \in \sigma_1$; as $v_0 \in \mathrm{Relint}(\sigma_1)$, we have $v_0 - \varepsilon v \in \sigma_1$ if $\varepsilon > 0$ is small enough. Then

$$\phi(v_0) = \phi(v_0 - \varepsilon v) + \phi(\varepsilon v)$$

with the two summands in σ_2. Since τ' is a face, Lemma 9.3 implies $\phi(\varepsilon v) \in \tau'$, and thus $\phi(v) \in \tau'$. □

10.59 Images of Distinguished points. Let $\phi : X_\Sigma \to X_{\Sigma'}$ be a toric morphism coming from a map $\overline{\phi} : N \to N'$ compatible with the fans Σ, Σ'. Let σ be a given cone in Σ, and let σ' be the minimal cone in Σ' containing $\overline{\phi}_{\mathbb{R}}(\sigma)$. Then

(i) $\phi(\gamma_\sigma) = \gamma_{\sigma'}$.

(ii) $\phi(O(\sigma)) \subseteq O(\sigma')$, and $\sigma(V(\sigma)) \subseteq V(\sigma')$.

(iii) The induced map $\phi|_{V(\sigma)} : V(\sigma) \to V(\sigma')$ is toric.

Proof.

(i) By Lemma 10.58 and the minimality of σ', we have $\phi(\mathrm{Relint}(\sigma)) \subseteq \mathrm{Relint}(\sigma')$.

□

10.3.3 Complete and proper

10.60 Theorem. Let X_Σ be the toric variety of the fan Σ in $N_{\mathbb{R}}$. TFAE :

(i) X_Σ is a complete variety.

(ii) X_Σ is compact in the classical topology.

(iii) The limit $\lim_{s \to 0} \lambda^u(s)$ exists in X_Σ for all $u \in N$.

(iv) Σ is a **complete fan**, i.e., $|\Sigma| = N_{\mathbb{R}}$.

Proof. (i)$\Leftrightarrow$(ii) is a general result (3.32) on varieties over $\mathbb{C}$. For the others, note that by (10.39) and (3.31), X_Σ is Hausdorff in the classical topology. Since the classical topology on each U_σ is metrizable, X_Σ is compact if and only if it is sequentially compact (both classically).

We first show (ii)$\Rightarrow$(iii). Assume X_Σ is compact and fix a $u \in N$. Let $(t_k)_k$ be a sequence in $\mathbb{C}^\times$ converging to 0. Applying λ^u, we obtain a sequence $(\lambda^u(t_k))_k$ in X_Σ. Since X_Σ is compact, by passing to a convergent subsequence we may assume $\lambda^u(t_k) \to \gamma \in X_\Sigma$. Since X_Σ is covered by the U_σ, and these are classically open, we may assume $\lambda^u(t_k) \in U_\sigma \ni \gamma$ for any k. Take $m \in \sigma^\vee \cap M$. Since χ^m is a morphism, it is a classical continuous function on U_σ, so that

$$\chi^m(\gamma) = \lim_{k\to\infty} \chi^m(\lambda^u(t_k)) = \lim_{k\to\infty} t_k^{\langle m,u\rangle}.$$

Since $t_k \to 0$, it must be $\langle m, u\rangle \geqslant 0$ for any $m \in \sigma^\vee \cap M$. This implies $\langle m, u\rangle \geqslant 0$ for any $m \in \sigma^\vee$, so that $u \in (\sigma^\vee)^\vee = \sigma$. Now Lemma 10.46 implies that $\lim_{s\to 0} \lambda^u(s)$ exists in U_σ, and hence in X_Σ.

Now consider (iii)$\Rightarrow$(iv). Take $u \in N$ and consider the limit $\lim_{s\to 0} \lambda^u(s)$. This lies in some U_σ, so Lemma 10.46 implies $u \in \sigma \cap N$. Thus $N \subseteq |\Sigma|$, and hence $N_\mathbb{R} = |\Sigma|$.

Finally we show (iv)$\Rightarrow$(ii). We prove this by induction on $n = \dim_\mathbb{R} N_\mathbb{R}$. For $n = 1$, assume $N = \mathbb{Z}$. The only complete fan in $\mathbb{R}$ is $\Sigma = \{\{0\}, \mathbb{R}_{\geqslant 0}, \mathbb{R}_{\leqslant 0}\}$. The corresponding toric variety is $\mathbb{P}^1_\mathbb{C}$, which is classically homeomorphic to S^2, the unit sphere in $\mathbb{R}^3$, and hence is compact.

Assume the statement is true for all complete fans of dimension $< n$, and let Σ be a complete fan in $N_\mathbb{R} \cong \mathbb{R}^n$. Let $(\gamma_k)_k$ be a sequence in X_Σ. We are going to show that $(\gamma_k)_k$ has a convergent subsequence.

Since X_Σ is a finite union of the $O(\tau)$ $(\tau \in \Sigma)$ (10.53), by passing to a subsequence we may assume $(\gamma_k)_k \subseteq O(\tau)$ for some $\tau \in \Sigma$. If $\tau \neq \{0\}$, then the orbit closure $V(\tau) = X_{\mathrm{Star}(\tau)}$ is a toric variety of dimension $\leqslant n-1$.

Lemma. If Σ is a complete fan, then for any $\tau \in \Sigma$, then $\mathrm{Star}(\tau)$ is a complete fan in $N(\tau)_\mathbb{R}$.

Proof. This amounts to show that for $x \in N_\mathbb{R} \backslash (N_\tau)_\mathbb{R}$, there exists a cone $\tau \preceq \sigma \in \Sigma$ such that $x \in \sigma + (N_\tau)_\mathbb{R}$. □

Since Σ is complete, the fan $\mathrm{Star}(\tau)$ is complete in $N(\tau)_\mathbb{R}$. By induction hypothesis, $(\gamma_k)_k$ has a convergent subsequence in $V(\tau)$, and we are done. It remains to deal with the case $\tau = \{0\}$; in other words, we may assume $(\gamma_k)_k$ lies in $O(\{0\}) \subseteq T_N \subseteq X_\Sigma$. To proceed, we introduce a key tool in the next paragraph. □

10.61 In this paragraph we introduce the **logarithm map** $L : T_N \to N_\mathbb{R}$ and discuss some of its important properties. Let $p \in T_N$ and let $\gamma : M \to \mathbb{C}^\times$ be the corresponding (semi)group homomorphism. Define $L(\gamma) : M \to \mathbb{R}$ by the formula

$$L(\gamma)(m) = \log|\gamma(m)|$$

Then $L(\gamma) \in \mathrm{Hom}_\mathbb{Z}(M, \mathbb{R}) \cong N_\mathbb{R}$, and we define $L(p)$ to be the corresponding vector in $N_\mathbb{R}$.

Lemma. If $L(p) \in -\sigma$ for some $\sigma \in \Sigma$, then $|\gamma(m)| \leqslant 1$ for all $m \in \sigma^\vee \cap M$.

Proof. Let $p \in T_N$ be such that $L(p) \in -\sigma$ for some $\sigma \in \Sigma$. If $m \in \sigma^\vee \cap M$, from the definition we have

$$\log|\gamma(m)| = L(\gamma)(m) = \langle m, L(p)\rangle.$$

Since $L(p) \in -\sigma$, it follows that $\log|\gamma(m)| \leqslant 0$, i.e. $|\gamma(m)| \leqslant 1$. □

10.62 Finish of the proof. We apply the logarithm map $L : T_N \to N_{\mathbb{R}}$ to our sequence $(\gamma_k)_k \subseteq T_N$, so we obtain a sequence $(L(\gamma_k))_k$ in $N_{\mathbb{R}}$. Since Σ is complete, we have $N_{\mathbb{R}} = \bigcup_{\sigma\in\Sigma}(-\sigma)$, so by passing to subsequence we may assume there is a $\sigma \in \Sigma$ with $L(\gamma_k) \in -\sigma$ for any k. By (10.61) we conclude $|\gamma_k(m)| \leqslant 1$ for all $m \in \sigma^\vee \cap M$ and k. Let A be a finite generating set of the monoid $\sigma^\vee \cap M$. Denote by $\mathbb{D}$ the closed unit ball in $\mathbb{C}$. Then there is an injection

$$\begin{aligned} B := \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{D}) &\longrightarrow \mathbb{D}^A \\ \gamma &\longmapsto (\gamma(m))_{m\in A}. \end{aligned}$$

We topologize the left hand side by the subspace topology inherited from $\mathbb{D}^A$. Then B is a closed subspace of $\mathbb{D}^A$. Since $\mathbb{D}^A$ is compact, so is B. Thus by passing to a convergent subsequence, we may assume that $\gamma_k \to \gamma \in B = \mathrm{Hom}_{\mathbf{Monoid}}(\sigma^\vee \cap M, \mathbb{D})$. This corresponds to a point γ in U_σ, and it remains to show $\gamma_k \to \gamma$ in the classical topology of U_σ. For this it suffices to note that $\chi^m(\gamma_k) \to \chi^m(\gamma)$ in $\mathbb{C}$ for all $m \in \sigma^\vee \cap M$.

10.63 Next we prove the relative version of Theorem 10.60.

Theorem. Let $\phi : X_\Sigma \to X_{\Sigma'}$ be the toric morphism corresponding to a homomorphism $\overline{\phi} : N \to N'$ that is compatible with fans Σ in $N_{\mathbb{R}}$ and Σ' in $N'_{\mathbb{R}}$. TFAE :

(i) ϕ is classically proper.

(ii) ϕ is a proper morphism.

(iii) If $u \in N$ and $\lim_{s\to 0} \lambda^{\overline{\phi}(u)}(s)$ exists in $X_{\Sigma'}$, then $\lim_{s\to 0} \lambda^u(s)$ exists in X_Σ.

(iv) $(\overline{\phi}_{\mathbb{R}})^{-1}(|\Sigma'|) = |\Sigma|$.

Proof. (i)⇔(ii) is a general result (3.33) for varieties over $\mathbb{C}$. To prove (ii)⇒(iii), let $u \in N$ and suppose $\lim_{s\to 0} \lambda^{\overline{\phi}(u)}(s) = \gamma' \in X_{\Sigma'}$. First consider the case that $\overline{\phi}(u) \neq 0$. By Theorem 3.23, the image $\lambda^u(\mathbb{C}^\times)$ is construtible in X_Σ. By (3.29), the Zariski closure $\overline{\lambda^u(\mathbb{C}^\times)}$ in X_Σ is the same as the classical closure. Since ϕ is Zariski proper, $\phi(\overline{\lambda^u(\mathbb{C}^\times)}) \subseteq X_{\Sigma'}$ is Zariski closed, hence classically, closed. It follows that

$$\overline{\lambda^{\overline{\phi}(u)}(\mathbb{C}^\times)} \subseteq \phi(\overline{\lambda^u(\mathbb{C}^\times)})$$

and thus we can find $\gamma \in \overline{\lambda^u(\mathbb{C}^\times)}$ such that $\phi(\gamma) = \gamma'$. Take $(t_k)_k \subseteq \mathbb{C}^\times$ with $\lambda^u(t_k) \to \gamma$. Then

$$\lim_{s\to 0} \lambda^{\overline{\phi}(u)}(s) = \gamma' = \phi(\gamma) = \lim_{k\to\infty} \phi(\lambda^u(t_k)) = \lim_{k\to\infty} \lambda^{\overline{\phi}(u)}(t_k)$$

Assume $\gamma' \in U_{\sigma'}$ for some $\sigma' \in \Sigma'$. Since $U_{\sigma'}$ is classically open, we can further assume that $\lambda^{\overline{\phi}(u)}(t_k) \in U_{\sigma'}$ for all k and $\lambda^{\overline{\phi}(u)}(s) \in U_{\sigma'}$ for s sufficiently small. As $\overline{\phi}(u) \neq 0$, the cocharacter $\lambda^{\overline{\phi}(u)} : \mathbb{C}^\times \to T_{N'} \subseteq U_{\sigma'}$ is nontrivial, so there must be $m \in (\sigma')^\vee \cap M'$ with $\langle m, \overline{\phi}(u)\rangle \neq 0$. But

$$\lim_{s\to 0} s^{\langle m,\overline{\phi}(u)\rangle} = \chi^m(\gamma') = \lim_{k\to\infty} t_k^{\langle m,\overline{\phi}(u)\rangle}$$

exists, we must have $\langle m, \overline{\phi}(u)\rangle > 0$ and thus $t_k \to 0$. Now the argument in (10.60).(ii)⇒(iii) shows that $\lim_{s\to 0} \lambda^u(s)$ exists in X_Σ.

Now drop the restriction $\overline{\phi}(u) \neq 0$. Consider the map $(\phi, \mathrm{id}_{\mathbb{C}}) : X_\Sigma \times \mathbb{A}^1_{\mathbb{C}} \to X_{\Sigma'} \times \mathbb{A}^1_{\mathbb{C}}$, which is proper as ϕ is. This is the toric morphism corresponding to $(\phi, \mathrm{id}_{\mathbb{Z}}) : N \times \mathbb{Z} \to N' \times \mathbb{Z}$. Now applying the argument above to $(u, 1) \in N \times \mathbb{Z}$ shows that $\lim_{s\to 0} \lambda^u(s)$ exists in X_Σ.

For (iii)⇒(iv), first note that since $\overline{\phi}$ is compatible with Σ and Σ', the inclusion

$$|\Sigma| \subseteq \overline{\phi}_{\mathbb{R}}^{-1}(|\Sigma'|)$$

is automatic. For the opposite inclusion, let $u \in \overline{\phi}_{\mathbb{R}}^{-1}(|\Sigma'|)$. Then $\overline{\phi}(u) \in |\Sigma'|$, so by Lemma 10.46 the limit $\lim_{s\to 0} \lambda^{\overline{\phi}(u)}(s)$ exists in $X_{\Sigma'}$. By assumption, $\lim_{s\to 0} \lambda^u(s)$ exists in X_Σ, and Lemma 10.46 again implies $u \in \sigma \cap N$ for some $\sigma \in \Sigma$. Since all the cones are rational, this implies $\overline{\phi}_{\mathbb{R}}^{-1}(|\Sigma'|) \subseteq |\Sigma|$.

Finally we prove (iv)⇒(i). We begin with two special cases. Suppose that a toric morphism $\phi : X_\Sigma \to T_{N'}$ satisfies (iv) and has the additional property that $\overline{\phi} : N \to N'$ is surjective. The fan of $T_{N'}$ consists of the trivial cone $\{0\}$, so that (iv) implies

$$|\Sigma| = (\overline{\phi}_{\mathbb{R}})^{-1}(0) = \ker(\overline{\phi}_{\mathbb{R}}) = (\ker \overline{\phi})_{\mathbb{R}}.$$

We can think of Σ as a fan Σ'' in $\ker(\overline{\phi}_{\mathbb{R}})$. As in (10.57), there is an isomorphism

$$X_\Sigma \cong X_{\Sigma''} \times T_{N'},$$

and ϕ corresponds to the projection $X_{\Sigma''} \times T_{N'} \to T_{N'}$. The fan Σ'' is complete in $\ker(\overline{\phi}_{\mathbb{R}})$, so by (10.60) $X_{\Sigma''}$ is compact, implying $X_{\Sigma''} \times T_{N'} \to T_{N'}$ is proper. The second case is discussed in the next paragraph. □

10.64 Suppose that a homomorphism of tori $\phi : T_N \to T_{N'}$ has the additional property that $\overline{\phi} : N \to N'$ is injective. The condition (iv) is then clear for ϕ. We prove directly that ϕ is proper.

Let $\overline{\phi}^* : M' \to M$ be the adjoint of ϕ; note that $(\overline{\phi}^*)_{\mathbb{R}} = (\overline{\phi}_{\mathbb{R}})^*$. Since $\overline{\phi}$ is injective, $(\overline{\phi}^*)_{\mathbb{R}}$ is surjective, and thus $\overline{\phi}^*(M') \leqslant M$ has finite index. Pick any integer $d > 0$ such that $dM \subseteq \overline{\phi}^*(M')$.

Let $(\gamma_k)_k \subseteq T_N$ be a sequence such that $(\phi(\gamma_k))_k$ converges in $T_{N'}$. We are going to show $(\gamma_k)_k$ admits a convergent subsequence in T_N. Since $\chi^{\overline{\phi}*(m')} = \chi^{m'} \circ \phi$ for any $m' \in M'$, we see $\chi^m(\gamma_k)$ converges in T_N for all $m \in \overline{\phi}^*(M')$, and hence $\chi^m(\gamma_k^d)$ converges in T_N for all $m \in M$. Choose a basis of $M \cong \mathbb{Z}^n$ and identify T_N with $(\mathbb{C}^\times)^n$. If we write $\gamma_k = (\gamma_{1k}, \ldots, \gamma_{nk})$, then each γ_{1k}^d converges to some point $\widetilde{\gamma_1} \in \mathbb{C}^\times$, and hence

$$\gamma_k^d = (\gamma_{1k}^d, \ldots, \gamma_{nk}^d) \to (\widetilde{\gamma_1}, \ldots, \widetilde{\gamma_n}) \in (\mathbb{C}^\times)^n.$$

As each $\tilde{\gamma}_i \in \mathbb{C}^\times$, it follows that we can choose the d-th roots $\tilde{\gamma}_i^{1/d}$ in a way that, by passing to subsequences, $\gamma_{ik} \to \tilde{\gamma}_i^{1/d}$. This proves γ_k converges in T_N.

10.65 Proof of (iv)⇒(i). Consider a general toric morphism $\phi : X_\Sigma \to X_{\Sigma'}$ satisfying (iv). Assume $(\gamma_k)_k$ is a sequence in X_Σ such that $(\phi(\gamma_k))_k$ converges in $X_{\Sigma'}$. Our goal is to show that $(\gamma_k)_k$ admits a convergent subsequence in X_Σ.

Since X_Σ has only finitely many T_N-orbits, we may assume $(\gamma_k)_k \subseteq O(\sigma)$ for some $\sigma \in \Sigma$. Let σ' be the minimal cone of Σ' containing $\overline{\phi}_\mathbb{R}(\sigma)$. By (10.59).(iii), the restriction $\phi|_{V(\sigma)} : V(\sigma) \to V(\sigma')$ is toric, and it corresponds to the map $N(\sigma) \to N(\sigma')$ induced by $\overline{\phi}$. This map satisfies the condition (iv) with the fans $\mathrm{Star}(\sigma)$, $\mathrm{Star}(\sigma')$. By (10.59).(ii), we thus can assume $\gamma_k \in T_N$ and $\phi(\gamma_k) \in T_{N'}$ for all k.

The limit $\gamma' = \lim_{k\to\infty} \phi(\gamma_k)$ lies in some $O(\tau')$ $(\tau' \in \Sigma)$; we then can assume the $\phi(\gamma_k)$ and γ' lie in $U_{\tau'}$. Note that (iv) implies

$$(\overline{\phi}_\mathbb{R})^{-1}(\tau') = \bigcup_{\overline{\phi}_\mathbb{R}(\sigma)\subseteq\tau'} \sigma$$

(If $u \in (\overline{\phi}_\mathbb{R})^{-1}(\tau')$, let σ be the minimal cone in Σ containing u, and σ' the minimal cone in Σ' containing $\overline{\phi}_\mathbb{R}(\sigma)$. Since σ is minimal, we have $u \in \mathrm{Relint}(\sigma)$. Since σ' is minimal, by Lemma 10.58 we have $\overline{\phi}_\mathbb{R}(u) \in \mathrm{Relint}(\sigma') \cap \tau'$. Since Σ' is a fan, $\sigma' \cap \tau'$ is a face of σ', so $\mathrm{Relint}(\sigma')$ does not meet $\sigma' \cap \tau$ unless $\sigma' \cap \tau' = \sigma'$, i.e., $\sigma' \preceq \tau'$. Thus $\overline{\phi}_\mathbb{R}(\sigma) \subseteq \sigma' \subseteq \tau'$). Also, the construction of ϕ from $\overline{\phi}_\mathbb{R}$ (10.55) implies that $\phi^{-1}(U_{\tau'})$ is the toric variety given by the fan $\{\sigma \in \Sigma \mid \overline{\phi}_\mathbb{R}(\sigma) \subseteq \tau'\}$. Therefore, we may assume $X_\Sigma = U_{\tau'}$ and $(\overline{\phi}_\mathbb{R})^{-1}(\tau') = |\Sigma|$.

If $\tau' = \{0\}$, then $O(\tau') = U_{\tau'} = T_{N'}$. By writing $\overline{\phi}$ as the composition $N \to \overline{\phi}(N) \to N'$, we see $\phi : X_\Sigma \to T_{N'}$ factors as

$$X_\Sigma \longrightarrow T_{\overline{\phi}(N)} \longrightarrow T_{N'}.$$

Since the composition of proper maps remains proper, we conclude ϕ is proper from the above two special cases.

It remains to deal with the case $\tau' \neq \{0\}$. Recall

$$\gamma' \in O(\tau') = \mathrm{Hom}_\mathbb{Z}((\tau')^\perp \cap M', \mathbb{C}^\times) \subseteq \mathrm{Hom}_{\mathbf{Monoid}}((\tau')^\vee \cap M', \mathbb{C}).$$

As $\phi(\gamma_k) \to \gamma'$ and $(\tau')^\vee \cap M'$ is a finitely generated monoid, by passing to a subsequence we may assume

$$|\phi(\gamma_k)(m')| \leqslant 1 \text{ for all } k \text{ and } m' \in (\tau')^\vee \cap M \backslash (\tau')^\perp$$

The logarithm map defined in (10.61) gives maps $L_N : T_N \to N_\mathbb{R}$ and $L_{N'} : T_{N'} \to N'_\mathbb{R}$, and they fit into a commutative square

$$\begin{tikzcd} T_N \arrow[r, "L_N"] \arrow[d, "\phi|_{T_N}"'] & N_\mathbb{R} \arrow[d, "\overline{\phi}_\mathbb{R}"] \\ T_{N'} \arrow[r, "L_{N'}"'] & N'_\mathbb{R}. \end{tikzcd}$$

This is commutative by (10.55).(i). Let $\overline{\phi}^* : M' \to M$ be the adjoint of $\overline{\phi} : N \to N'$. Then for $m' \in (\tau')^\vee \cap M \backslash (\tau')^\perp$ and all k, we have

$$\begin{aligned}\langle \overline{\phi}^*(m'), L_N(\gamma_k)\rangle &= \langle m', \overline{\phi}_{\mathbb{R}}(L_N(\gamma_k))\rangle \\ &= \langle m', L_{N'}(\phi(\gamma_k))\rangle = \log|\phi(\gamma_k)(m')| \leqslant 0. \end{aligned} \tag{$\clubsuit$}$$

On the other hand, we have the equivalences :

$$\begin{aligned} u \in (\overline{\phi}_{\mathbb{R}})^{-1}(\tau') &\Leftrightarrow \overline{\phi}_{\mathbb{R}}(u) \in \tau' \\ &\Leftrightarrow \langle m', \overline{\phi}_{\mathbb{R}}(u)\rangle \geqslant 0 \text{ for all } m' \in (\tau')^\vee \cap M' \\ &\Leftrightarrow \langle \overline{\phi}^*(m'), u\rangle \geqslant 0 \text{ for all } m' \in (\tau')^\vee \cap M' \end{aligned}$$

To proceed further, we need a lemma.

10.66 Lemma. Let $\sigma \neq \{0\}$ be a strongly convex rational polyhedral cone in $N_{\mathbb{R}}$. Then

$$u \in \sigma \text{ if and only if } \langle m, u\rangle \geqslant 0 \text{ for all } m \in \sigma^\vee \cap M \backslash \sigma^\perp$$

10.67 Finish of the proof. The preceding lemma establishes the equivalence

$$u \in (\overline{\phi}_{\mathbb{R}})^{-1}(\tau') \Leftrightarrow \langle \overline{\phi}^*(m'), u\rangle \geqslant 0 \text{ for all } m' \in (\tau')^\vee \cap M' \backslash (\tau')^\perp.$$

From ($\clubsuit$) we conclude that $-L_N(\gamma_k) \in (\overline{\phi}_{\mathbb{R}})^{-1}(\tau')$ for all k. But we are assuming $(\overline{\phi}_{\mathbb{R}})^{-1}(\tau') = |\Sigma|$, so

$$-L_N(\gamma_k) \in |\Sigma|$$

for all k. By passing to a subsequence we may assume $L_N(\gamma_k) \in -\sigma$ for some $\sigma \in \Sigma$. The argument in (10.62) then implies γ_k admits a convergent subsequence in $U_\sigma \subseteq X_\Sigma$. This finishes the proof.

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